DCI Hollow Metal

Technical Data

DCI HOLLOW METAL
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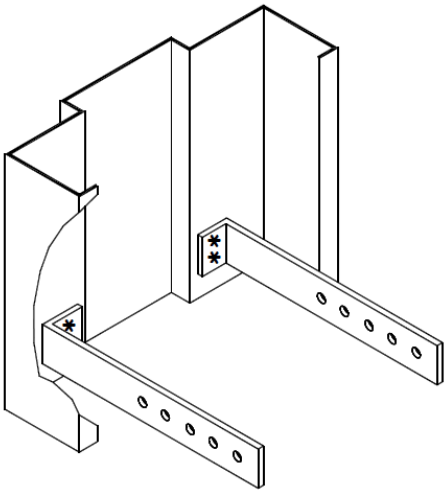
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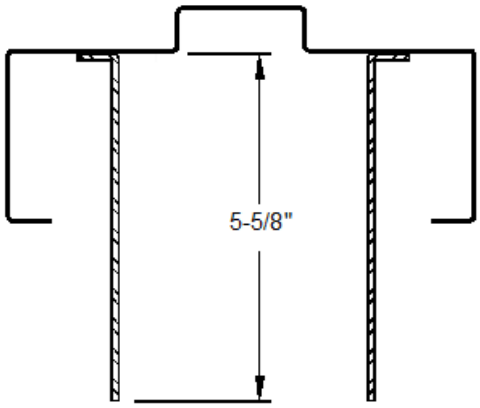
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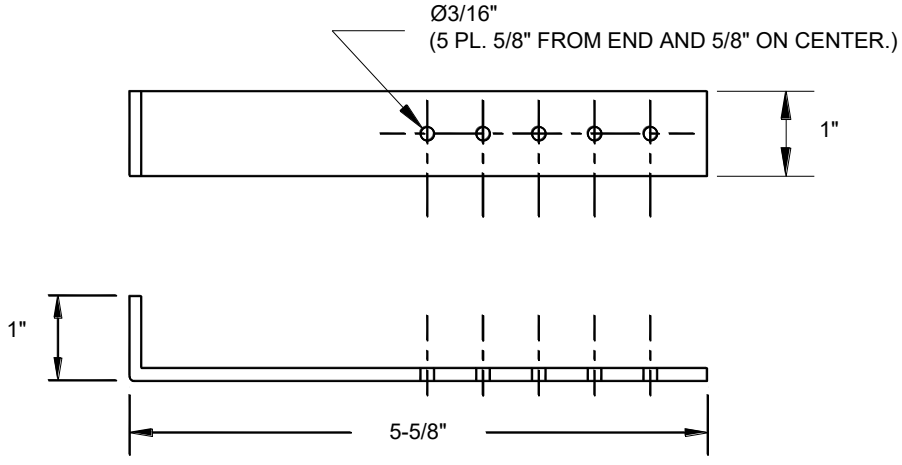
A1 STANDARD STRAP ANCHOR

MATERIAL: GALVANNEAL
GAUGE: 18
TWO SPOTWELDS PER STRAP





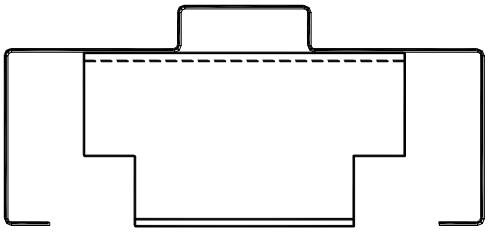
NAILS OR SCREWS FOR WALL ATTACHMENT
NOT INCLUDED



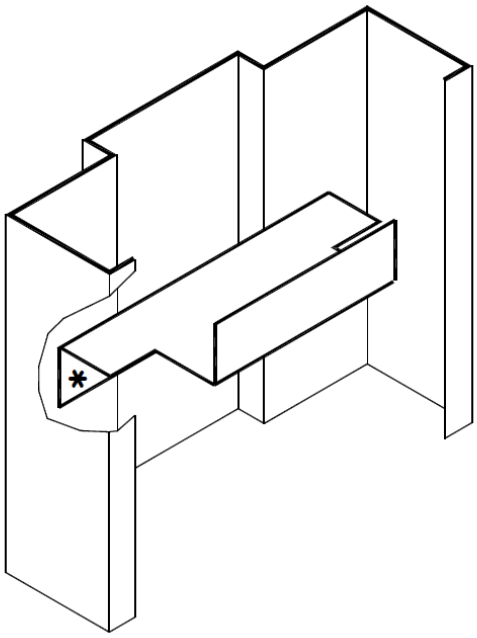
1.) WHEN REQUESTED TO BE USED AS A BASE ANCHOR, THE A1 BASE ANCHOR WILL BE LOCATED WITHIN THE BOTTOM 6” OF THE OPENING

A2 FLUSH METAL STUD ANCHOR

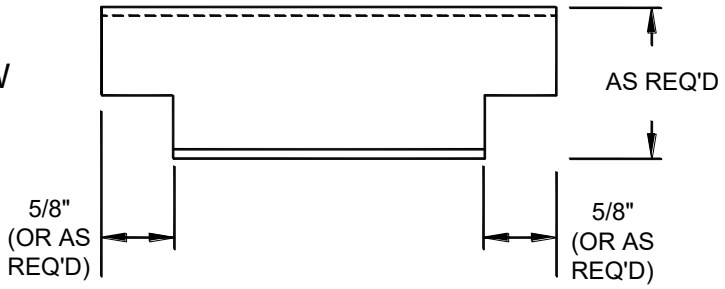
MATERIAL: GALVANNEAL
GAUGE: 16
TWO SPOTWELDS PER END



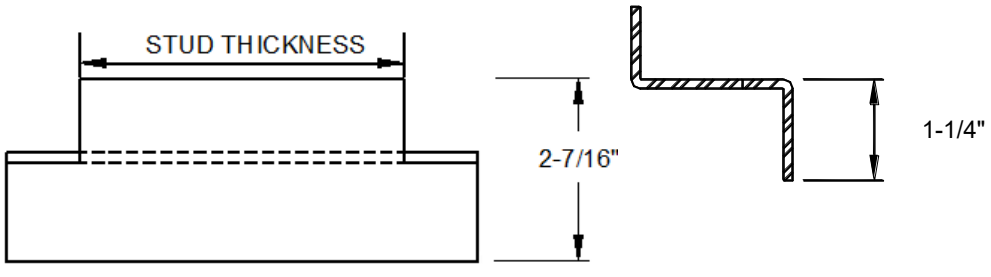
SCREWS USED TO FASTEN TO STUD
NOT INCLUDED



TOP VIEW



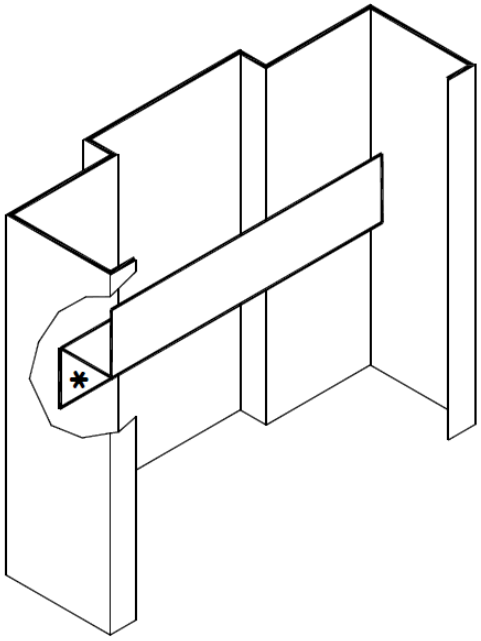
FRONT VIEW

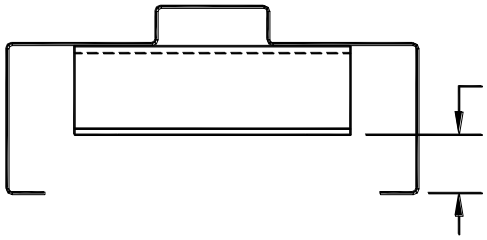


1.) WHEN REQUESTED TO BE USED AS A BASE ANCHOR, THE A2 BASE ANCHOR WILL BE LOCATED WITHIN THE BOTTOM 6” OF THE OPENING
2.) SPACER USED FOR SINGLE RABBET AND DOUBLE EGRESS PROFILES

A3 RECESSED METAL STUD ANCHOR

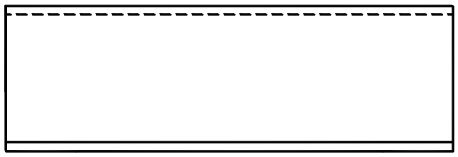
MATERIAL: GALVANNEAL
GAUGE: 16
TWO SPOTWELDS PER END



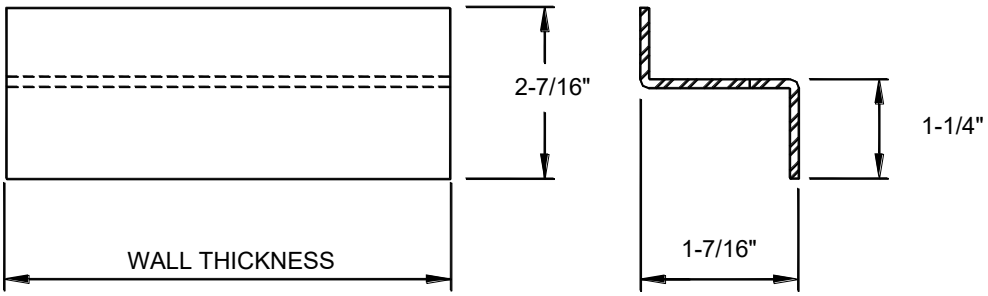


SCREWS USED TO FASTEN TO STUD
ARE NOT INCLUDED

TOP VIEW



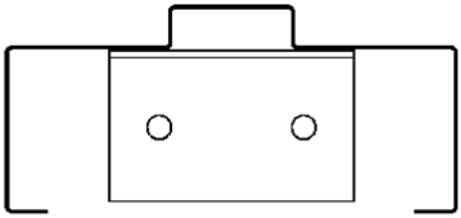
FRONT VIEW



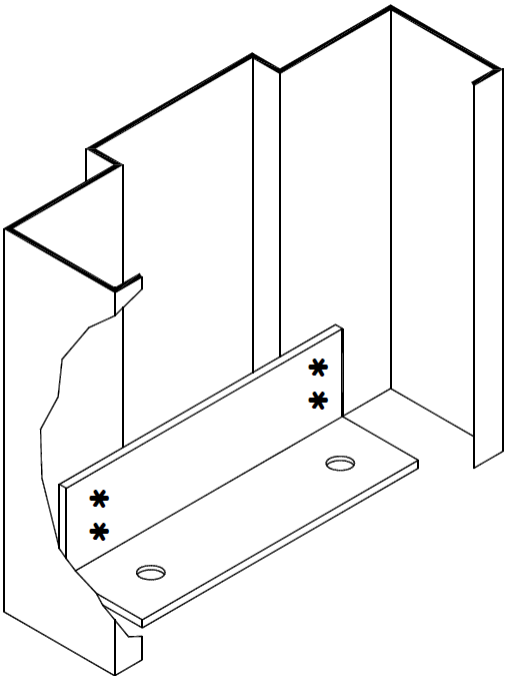
1.) WHEN REQUESTED TO BE USED AS A BASE ANCHOR, THE A3 BASE ANCHOR WILL BE LOCATED WITHIN THE BOTTOM 6” OF THE OPENING
2.) SPACER USED FOR SINGLE RABBET AND DOUBLE EGRESS PROFILES

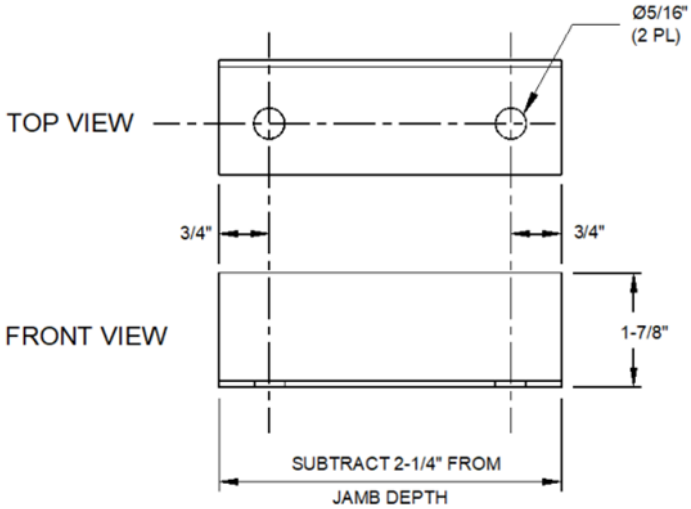
A4 STANDARD BASE ANCHOR

MATERIAL: GALVANNEAL
GAUGE: 16
TWO SPOTWELDS PER END

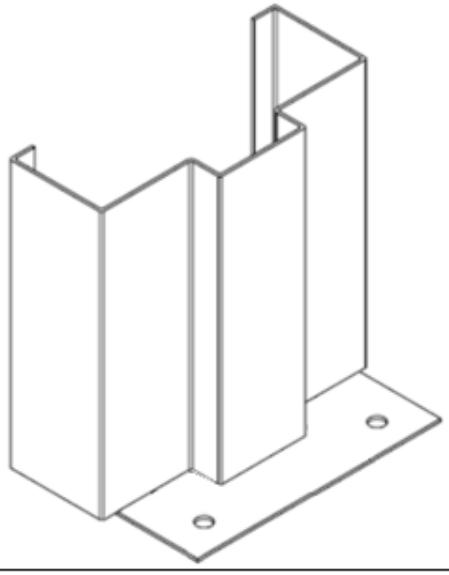


ANCHOR BOLTS FOR FLOOR
ATTACHMENT NOT INCLUDED



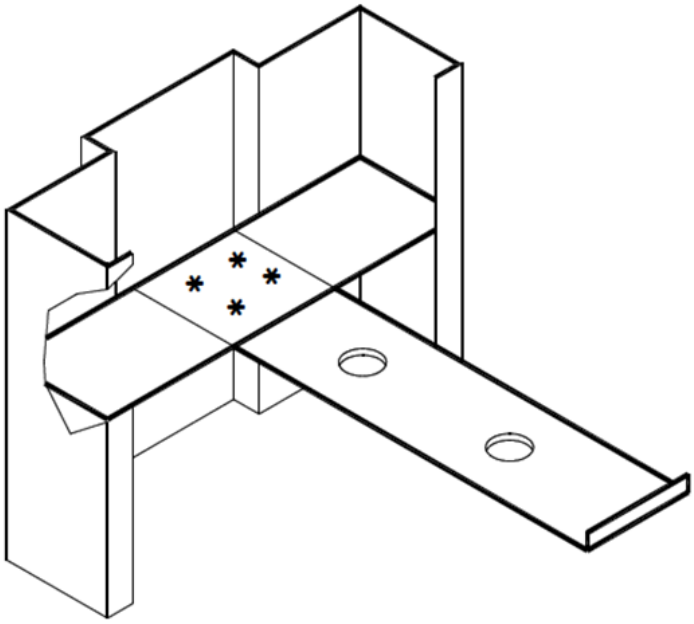


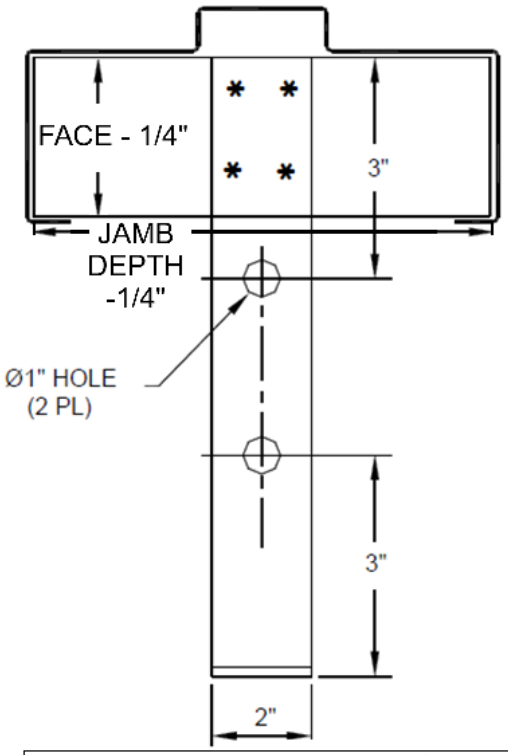
ALTERNATE
A4R REVERSE BASE ANCHOR

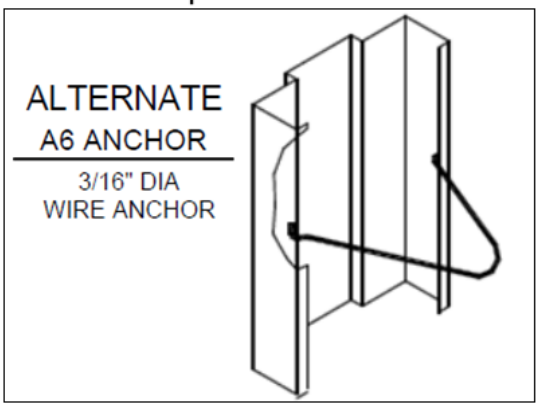


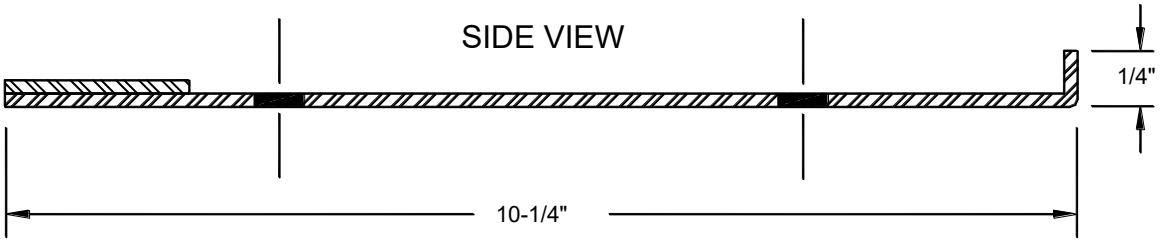
A5 MASONRY “TEE” ANCHOR
A6 MASONRY WIRE ANCHOR

MATERIAL: GALVANNEAL
GAUGE: 16
WELDED: (4) SPOTWELDS
THIS ANCHOR SHIPS LOOSE







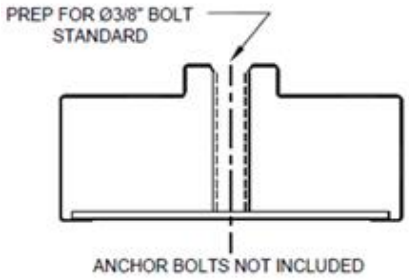


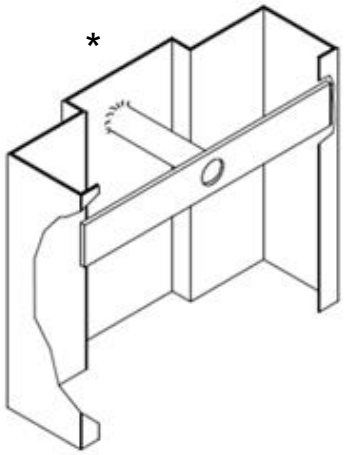
A7 PUNCH & DIMPLE MASONRY ANCHORS

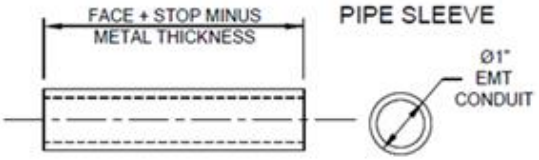
A7P ANCHOR
"PIPE AND STRAP"

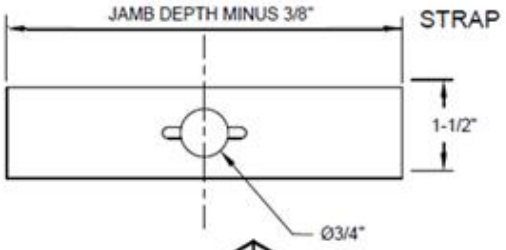
16 GAUGE – 2 TACK WELDS PER END

STANDARD ANCHOR WHEN P&D ANCHORS REQUESTED





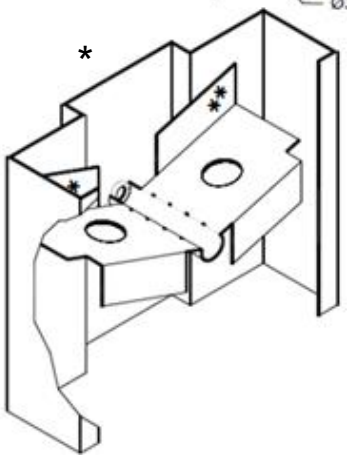




A7B ALTERNATE
"BUTTERFLY ANCHOR"

18 GAUGE – 2 SPOT WELDS PER SIDE

AVAILABLE FOR JAMBS WITH 2" FACES AND UNDER 9" JAMB DEPTH



If expansion bolt anchors are to be used, the soffit shall be provided with 7/16" dimpled holes located not more than **6" from the top and bottom** with intermediate holes spaced no more than **26" on center**. The installer is to provide each hole with a minimum 3/8" roundhead bolt.

If 1/4" sheet metal screws are to be used, then the dimpled holes are to be located not more than **6" from the top and bottom** with intermediate holes spaced no more than **24" on center**. The installer is to provide each hole with a minimum 1/4" sheet metal screw.

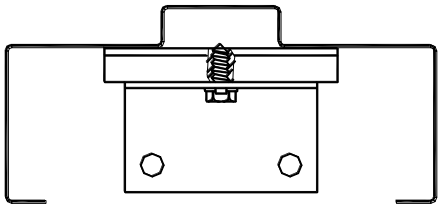
Wood lag screws are not approved for use with these anchors on fire rated frames.

DCI's standard is for the bottom bolt to act as the base anchor. However, adjustable type base anchors may also be used upon request.

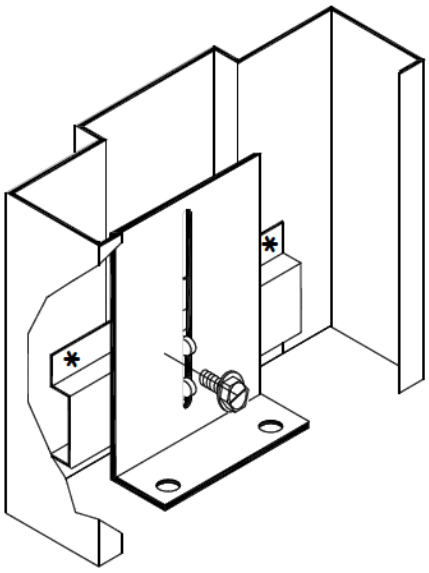
*Please consult factory if soffit width is less than 1-1/4"

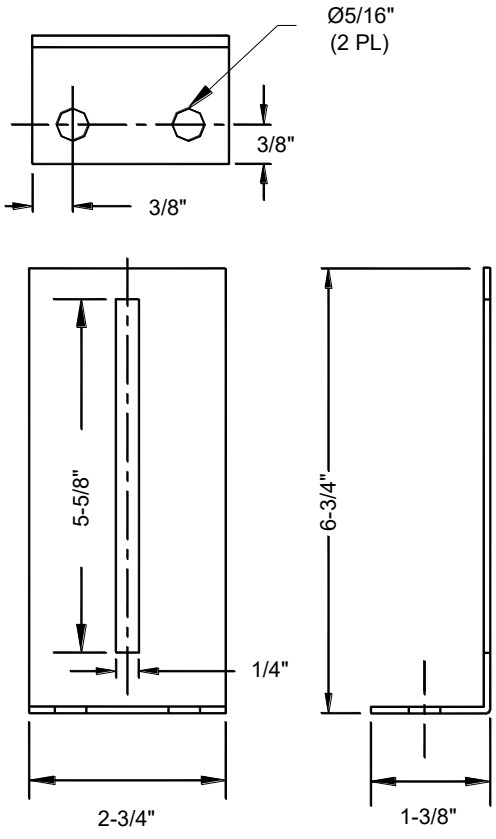
A8 ADJUSTABLE BASE ANCHOR

MATERIAL: GALVANNEAL
GAUGE: 16
TWO SPOT WELDS PER END

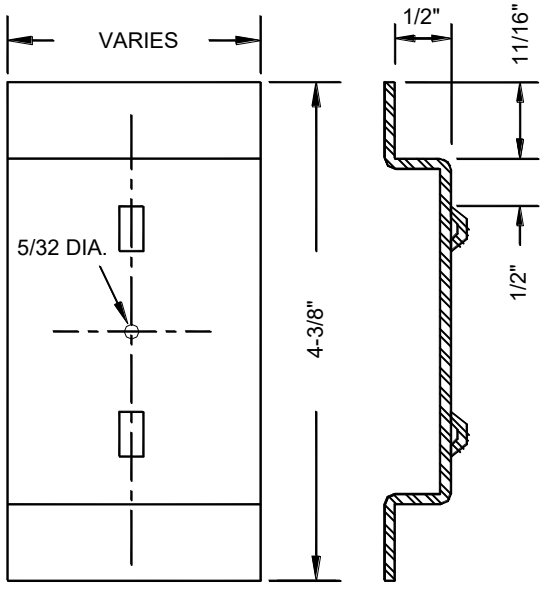


ANCHOR BOLTS USED FOR FLOOR CONNECTION ARE NOT INCLUDED





ADJUSTABLE ANGLE

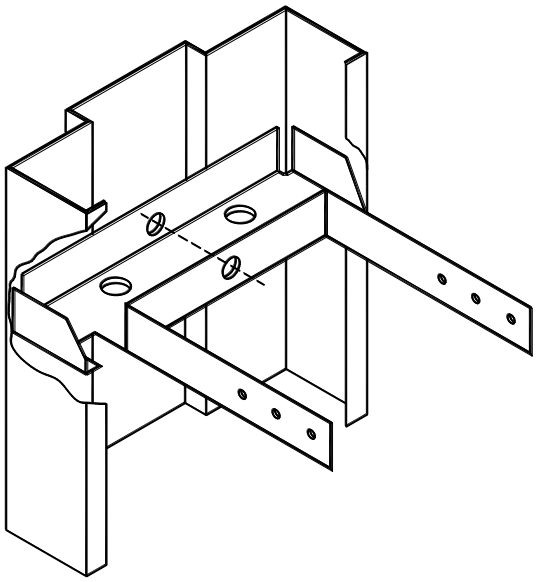


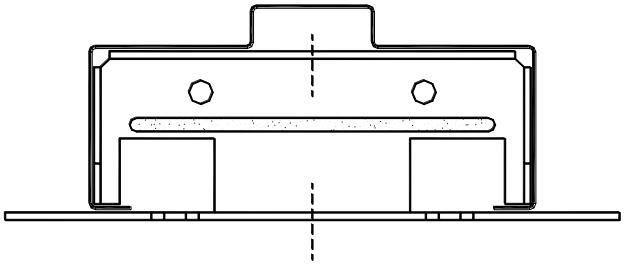
HAT STRAP

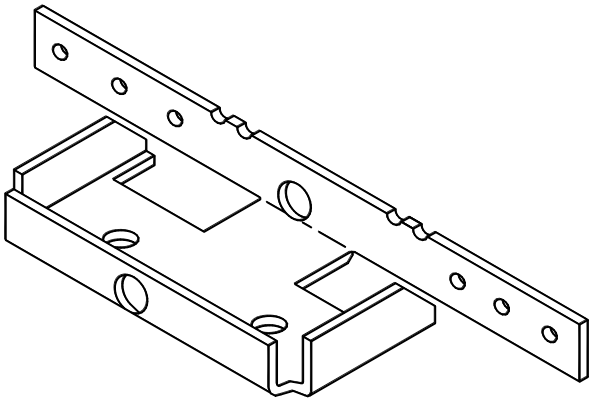
1.) RECOMMENDED MINIMUM 1-1/2" FACE FOR ANCHOR TO NOT EXTEND PAST RETURNS
2.) CUSTOM DESIGN WITH LIMITED ADJUSTABLE RANGE AVAILABLE WHEN USED WITH A7 ANCHORS
3.) FASTENERS BY OTHERS. REFER TO ANSI/SDI A250.11 OR HMMA 840
4.) SPACER USED FOR SINGLE RABBET AND DOUBLE EGRESS PROFILES

A9 UNIVERSAL STUD ANCHORS

A9T "TWIST-IN" ANCHOR
MATERIAL: GALVANNEAL
GAUGE: 16







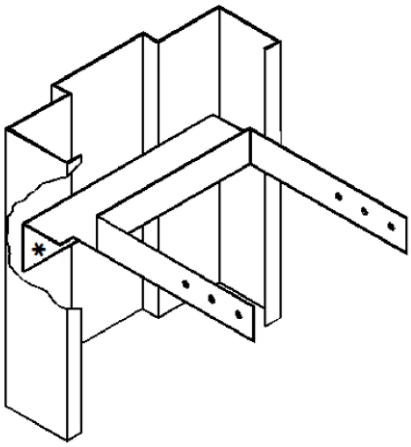
MAY BE MANUFACTURED FOR CUSTOM JAMB DEPTHS, FACES AND PROFILES.

STOCKED FOR THE FOLLOWING JAMB DEPTHS - SHIPS LOOSE.

JAMB DEPTH	FACE
4-3/4"	2"
5-3/4"	2"
6-1/4"	2"
6-3/4"	2"
7-1/4"	2"
7-3/4"	2"
8-1/4"	2"
8-3/4"	2"

ALTERNATE
A9W ANCHOR

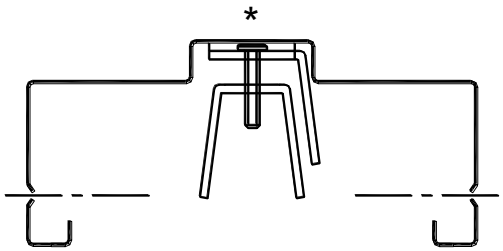
WELDED IN -
FOR CUSTOM PROFILES

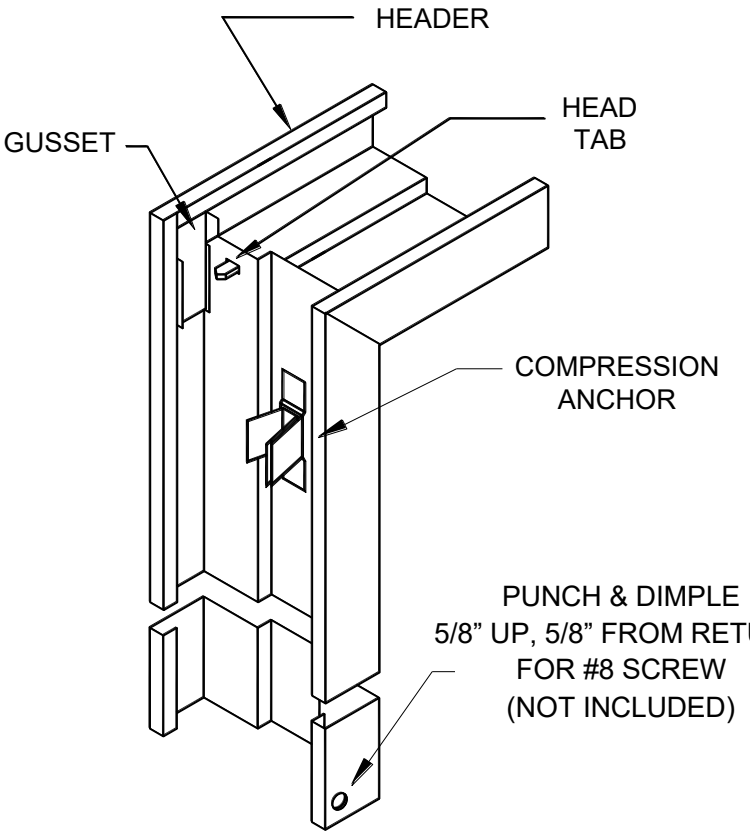


1.) FOR A9T ANCHORS, 5/8" DRYWALL SPACING ONLY

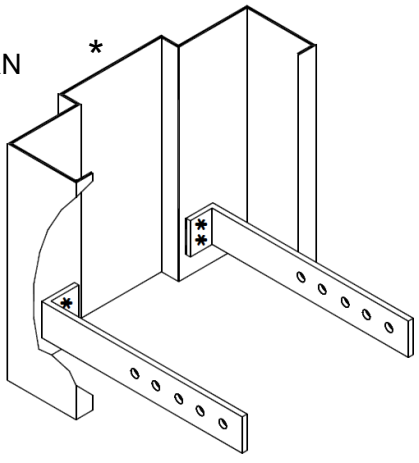
A10 KDDW COMPRESSION ANCHOR

MATERIAL: GALVANNEAL
GAUGE: 16
TWO SPOTWELDS PER END





STANDARD BASE ANCHOR



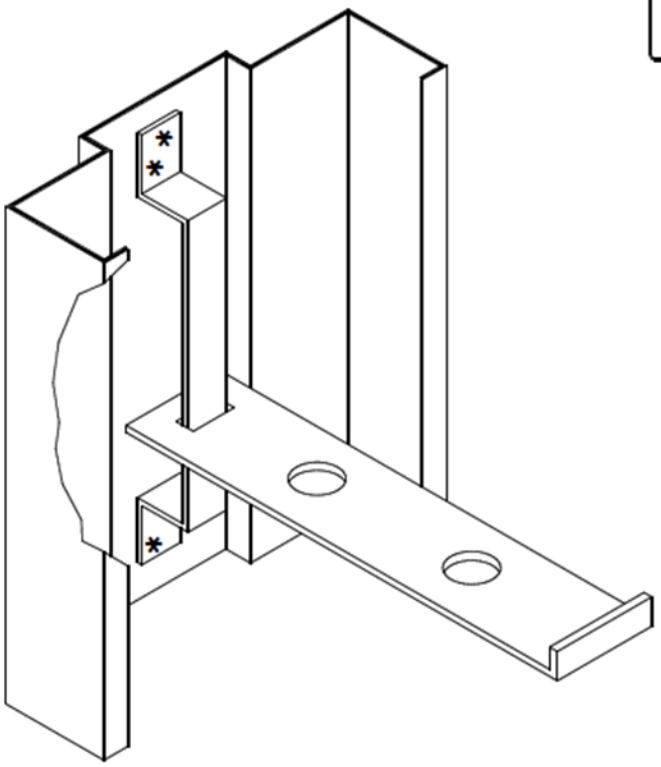
ALTERNATE BASE ANCHOR
INSTALLED TIGHT AGAINST RETURNS
(SEE PAGE 8 FOR DETAILS)

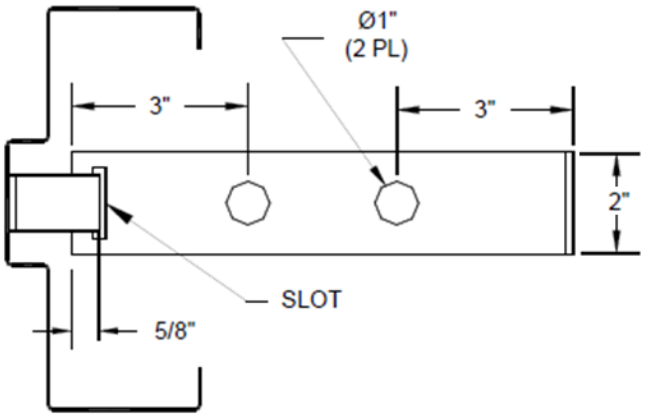
* Please consult factory if soffit width is less than 1-1/4".

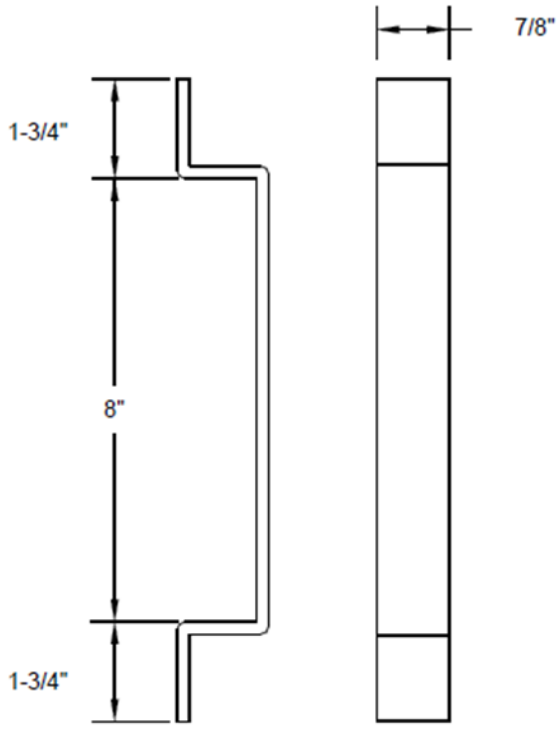
1) When soffit is at least 4", additional A10 anchor included at same height. When frame is taller than 96", additional A10 anchor included 48" up from bottom. (when both conditions are true, there will be 4 A10s per jamb)

A11 ADJUSTABLE MASONRY ANCHOR

MATERIAL: GALVANNEAL
GAUGE: 16
TWO SPOT WELDS PER END

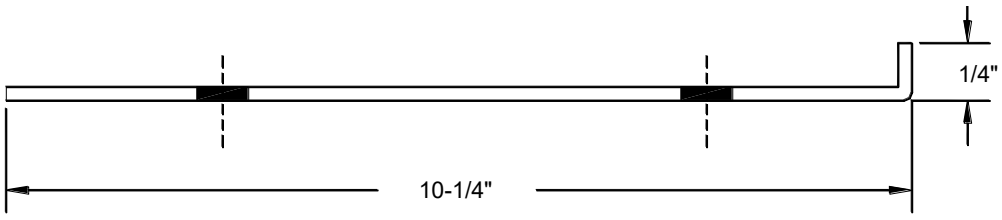






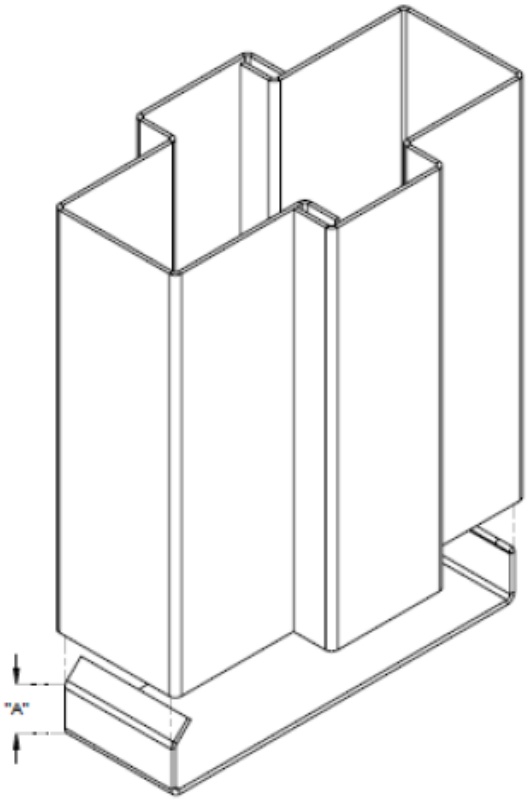
ADJUSTABLE HAT SECTION

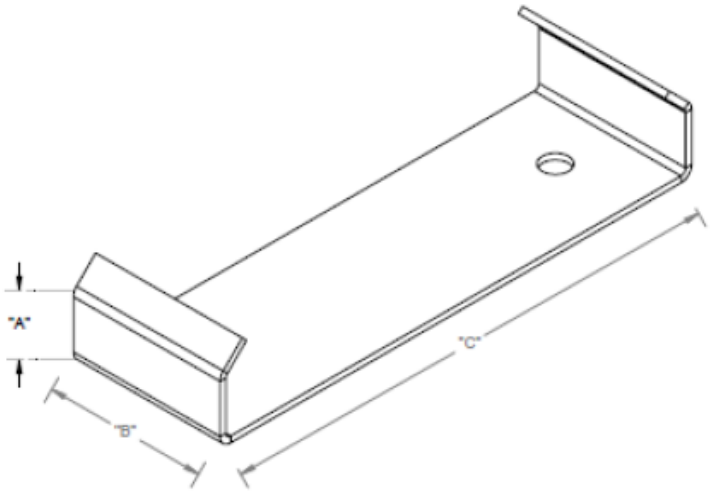
ANCHOR STRAP



A12 MULLION BASE ANCHOR

MATERIAL: GALVANNEAL
GAUGE: 14
THIS ANCHOR SHIPS LOOSE

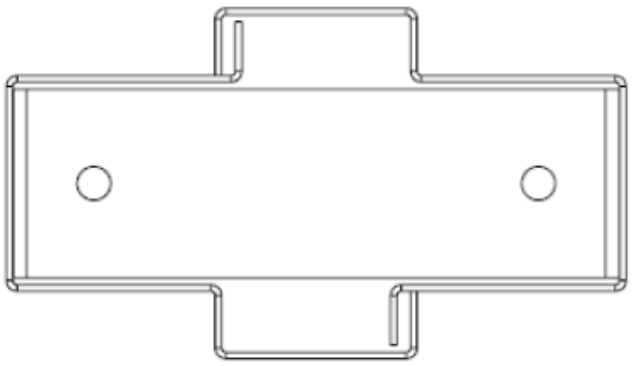




A = 3/4" STANDARD

B = TO FIT INSIDE MULLION FACE

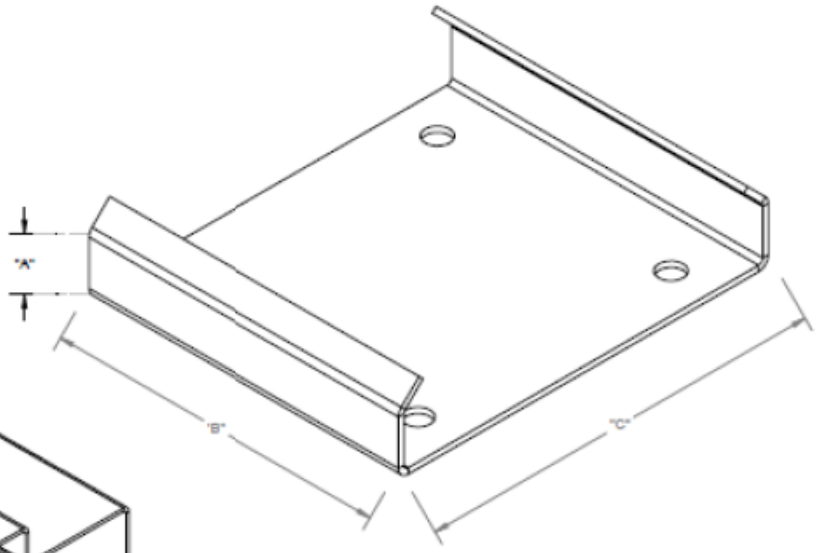
C = TO FIT INSIDE MULLION JAMB DEPTH

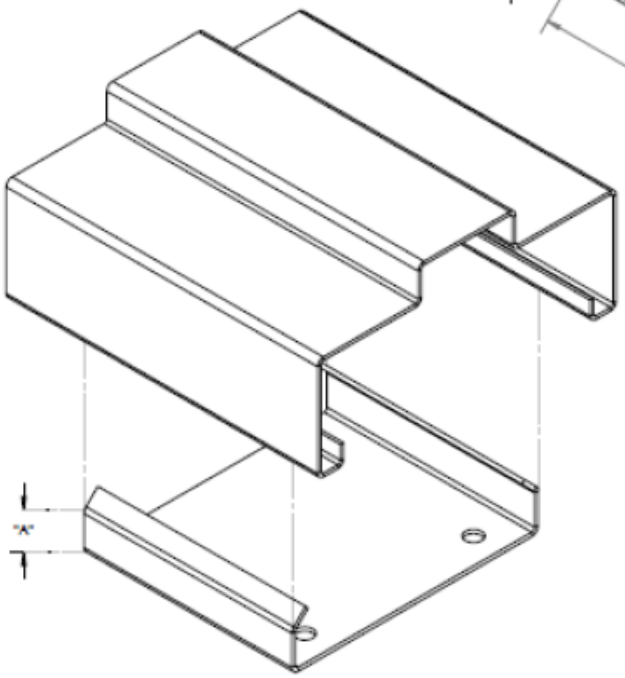


ANCHOR BOLTS FOR FLOOR
ATTACHMENT NOT INCLUDED

A13 SILL BASE ANCHOR

MATERIAL: GALVANNEAL
GAUGE: 14
THIS ANCHOR SHIPS LOOSE

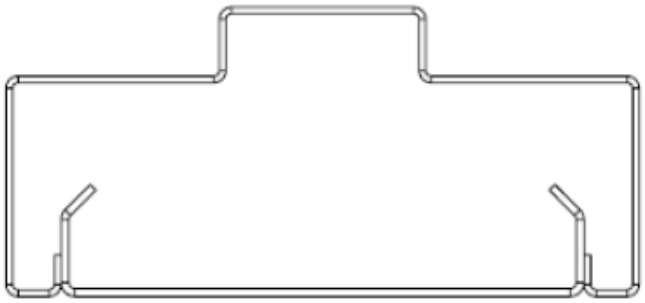




A = 3/4" STANDARD

B = 2" SHORTER THAN THE SILL *

C = TO FIT INSIDE SILL THROAT



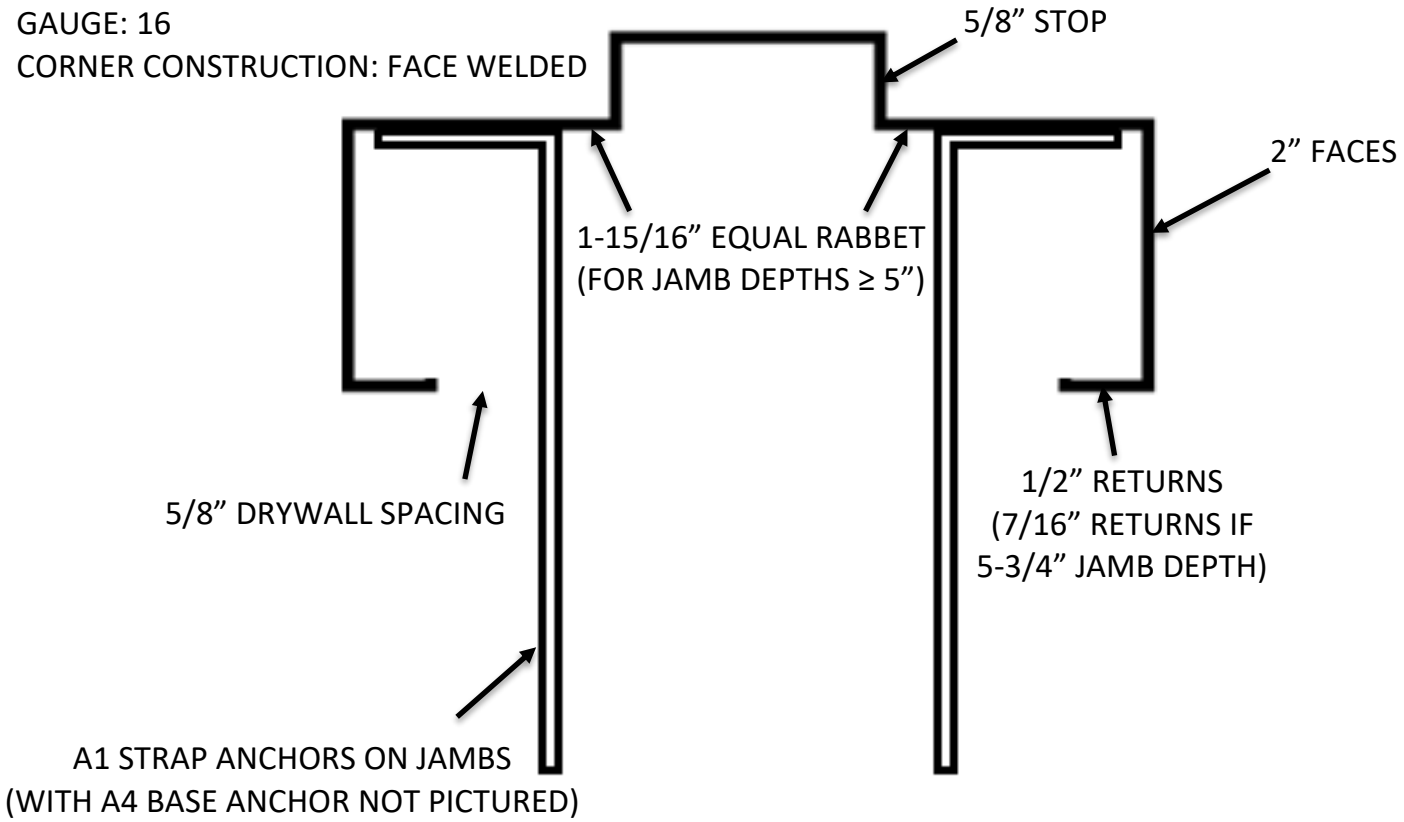
ANCHOR BOLTS FOR FLOOR
ATTACHMENT NOT INCLUDED

*Standard sill base anchor length is 2” shorter than the sill. Stop height on adjacent vertical sections might impact the length of the sill base anchor.

STANDARD FRAME AND DOOR DETAILS UNLESS OTHERWISE NOTED

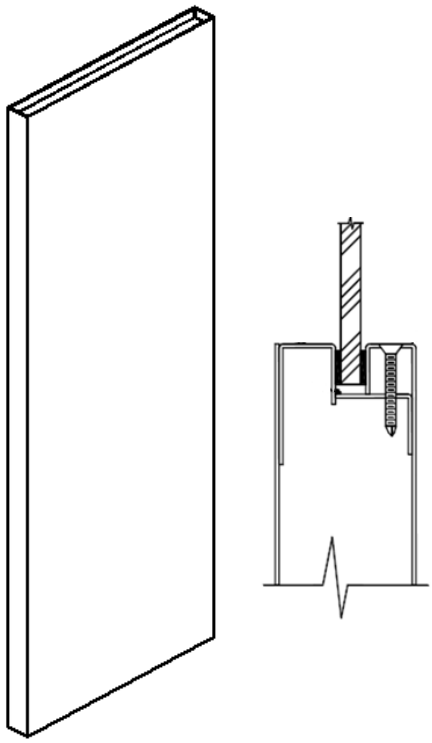
FRAMES:

MATERIAL: A40 GALVANNEAL WITH PRIME FINISH
GAUGE: 16
CORNER CONSTRUCTION: FACE WELDED



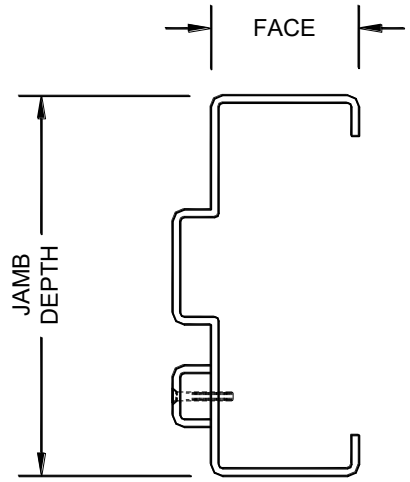
DOORS:

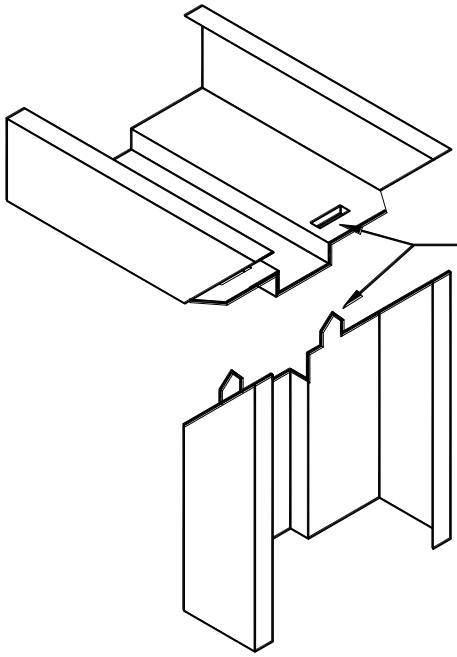
MATERIAL: A40 GALVANNEAL WITH PRIME FINISH
GAUGE: 18
EDGE CONSTRUCTION: SEAMLESS WELD; BEVELED EDGES
THICKNESS: 1-3/4”
TOP AND BOTTOM CHANNELS: RECESSED
UNDERSIZING: 1/8” AT TOP AND SIDES (BUTT HINGES ONLY);
3/4” UNDERCUT AT BOTTOM
WHEN INTEGRAL LITES REQUESTED: SIZED FOR 1/4” THICK GLASS



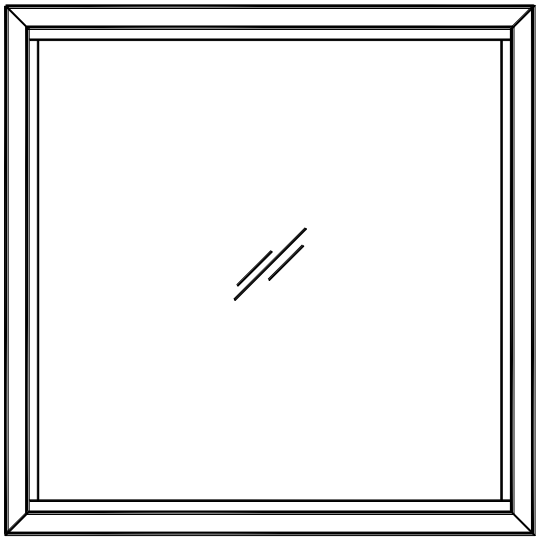
BORROWED LITE - FACE WELDED

CONSTRUCTION: FULLY WELDED
GLASS STOP: 5/8 x 5/8 x 18 GA.
ANCHORS: AS REQUIRED
FINISH: RUST INHIBITIVE BAKED-ON GREY PRIMER





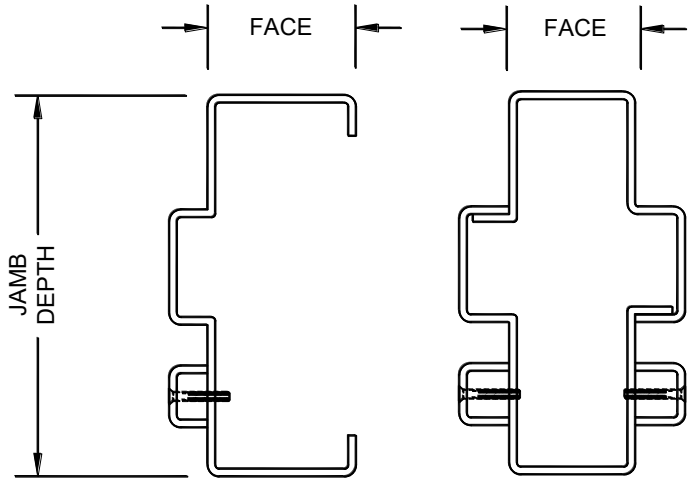
JAMB TABS INTERLOCK WITH SLOTS AT HEAD AND SILL.

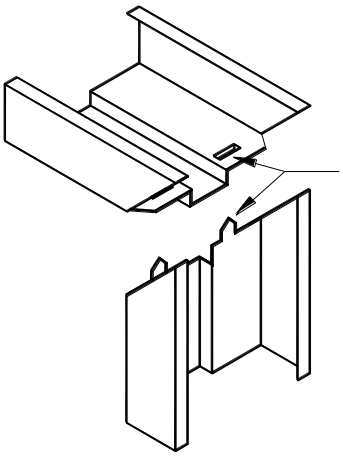


MACHINE-MITERED AND FULLY WELDED. (CONTINUOUSLY WELDED AT FACE AND RETURN. FACE GROUND SMOOTH AND FINISHED.)

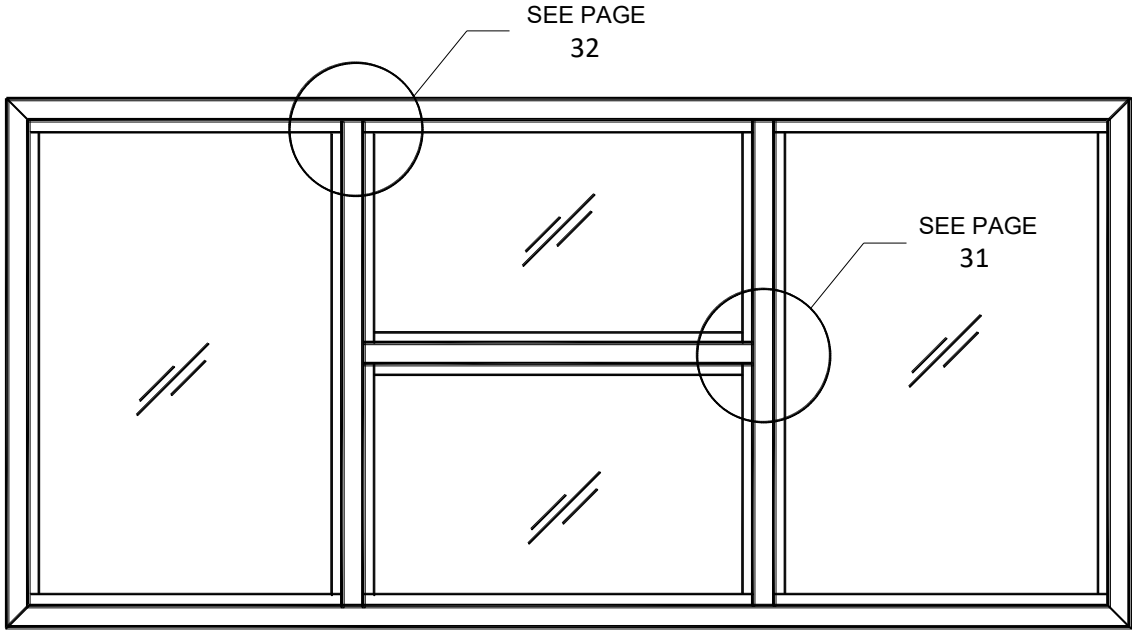
MULTI-LITE BORROWED LITE

CONSTRUCTION: FULLY WELDED
GLASS STOP: 5/8 x 5/8 x 18 GA.
ANCHORS: AS REQUIRED
FINISH: RUST INHIBITIVE BAKED-ON GREY PRIMER



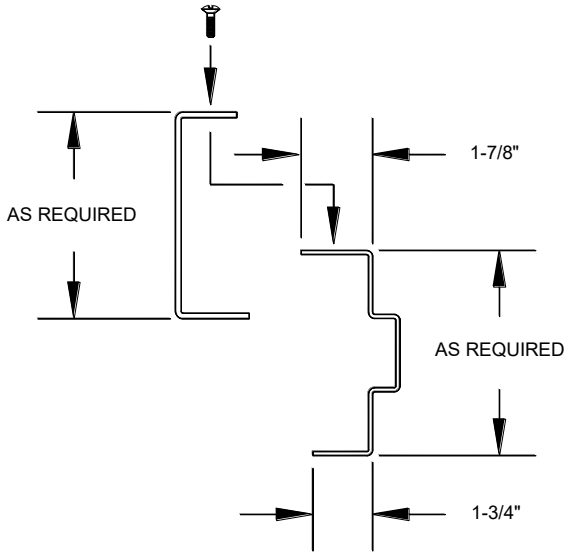


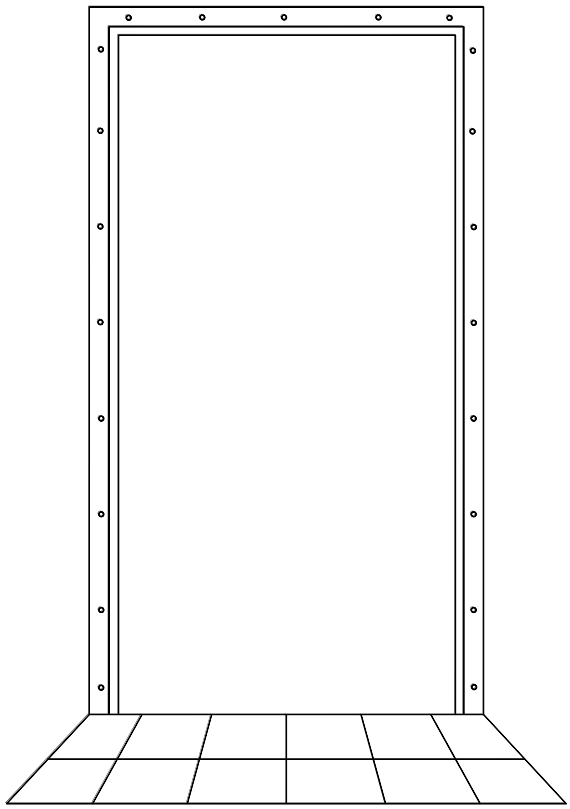
MACHINE-MITERED AND FULLY WELDED.
STANDARD: CONTINUOUSLY WELDED FACES
GRIND AND FINISH SMOOTH
OPTIONAL: FILLET WELD (TIG) HALF OR ENTIRE DEPTH

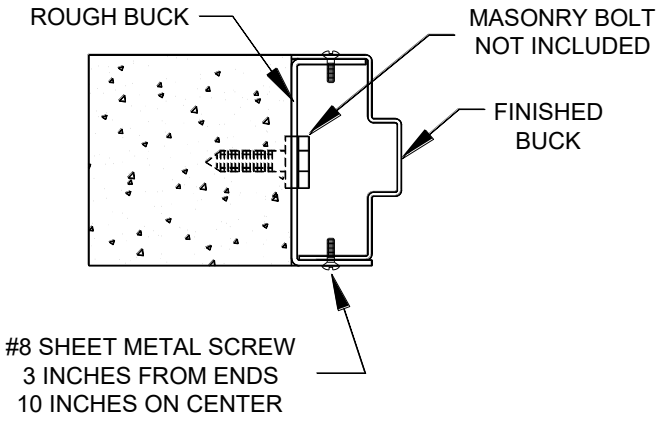


ROUGH BUCK

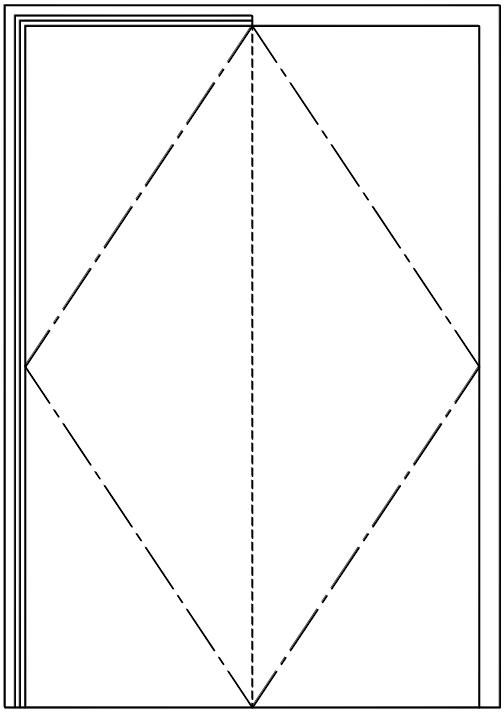
MATERIAL: GALVANNEAL
GAUGE: 16
CONSTRUCTION:
COMPOSED OF 3 LOOSE CHANNELS
WITH WELDED FINISHED BUCK

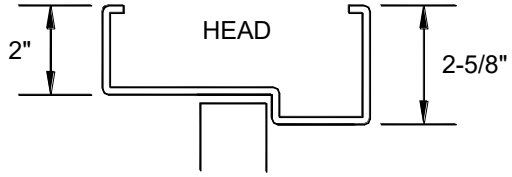


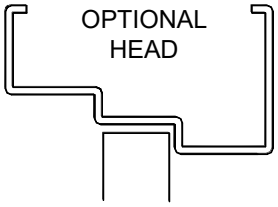


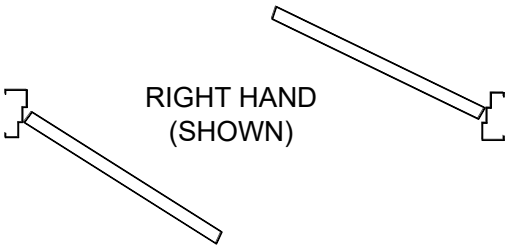


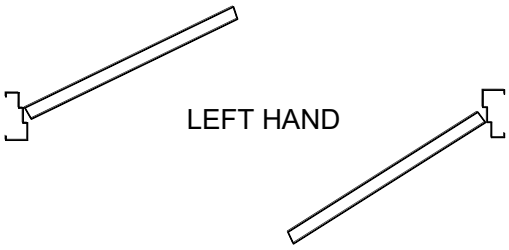
STANDARD DOUBLE EGRESS FRAME

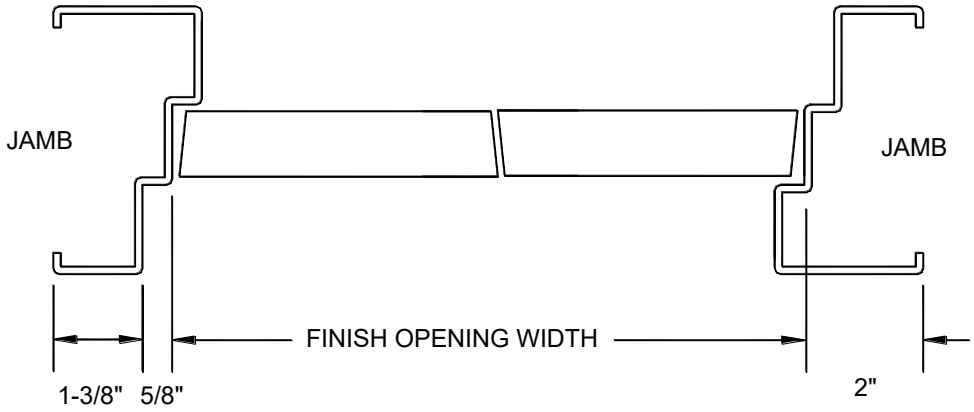








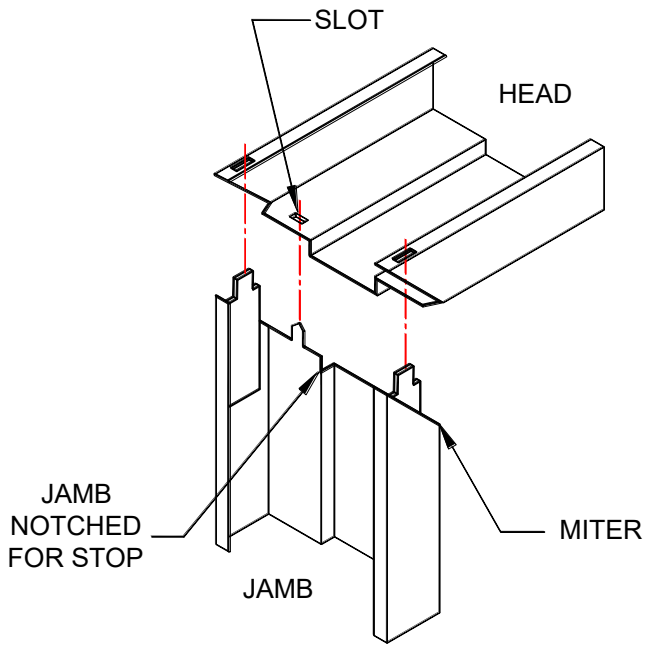


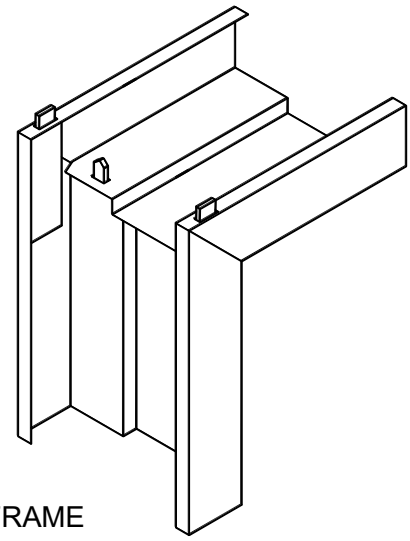


KD-COMPLETE FRAME CORNER

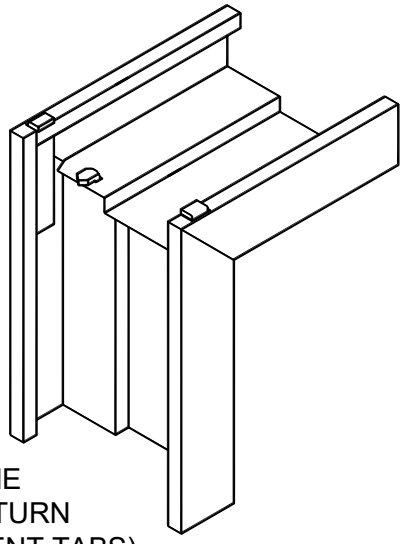
KD COMPLETE 3-SIDED FRAME
FIELD ASSEMBLED - JAMB AND GUSSET TABS TO BE BENT IN FIELD
AFTER ASSEMBLY, INSTALLATION SIMILAR TO WELDED FRAMES
INSTALLED WITH THE SAME HARDWARE AND ANCHORS AS WELDED FRAMES

WHI FIRE RATINGS AVAILABLE - SEE FIRE RATED FRAMES SECTION



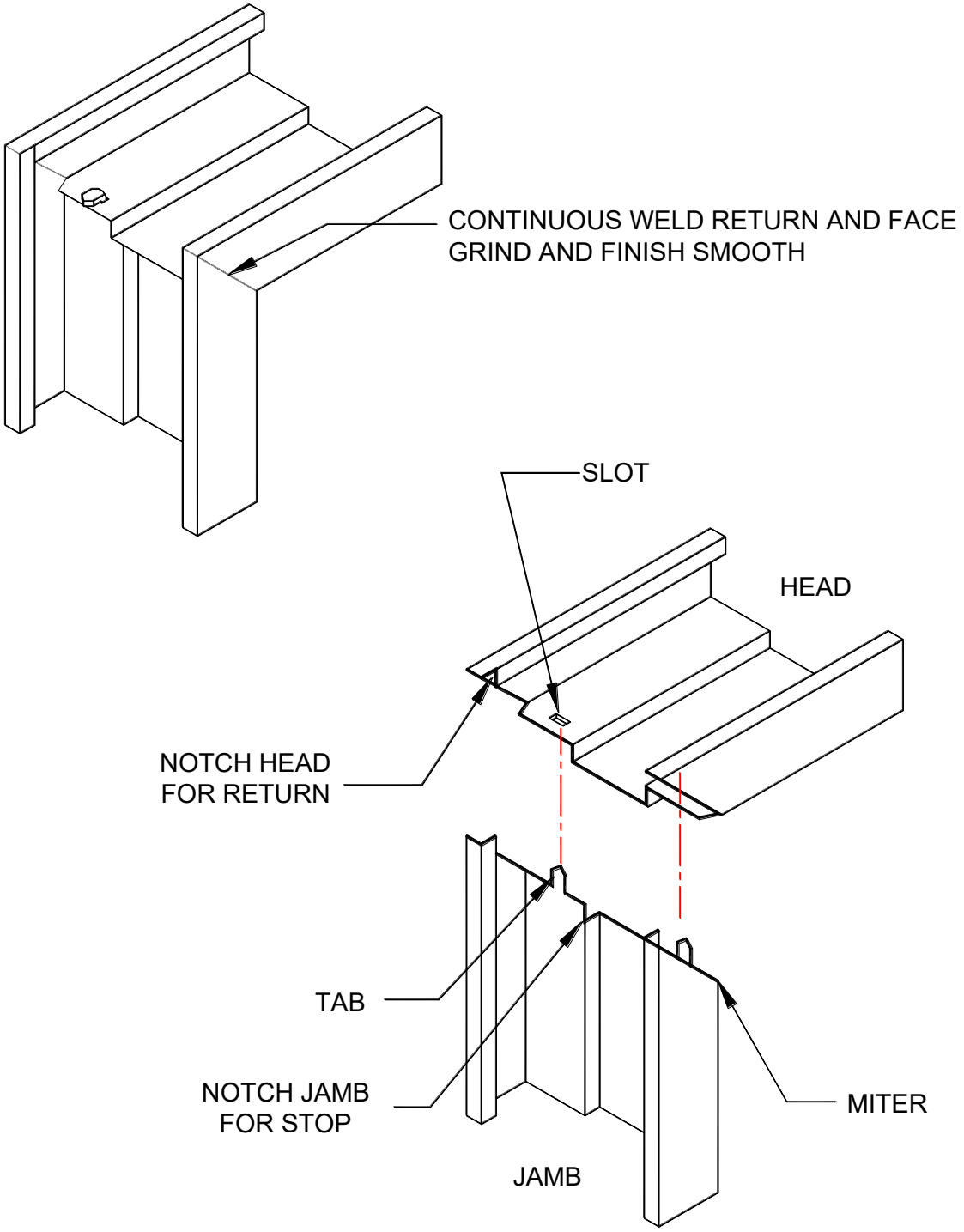


KD FRAME
SINGLE RETURN

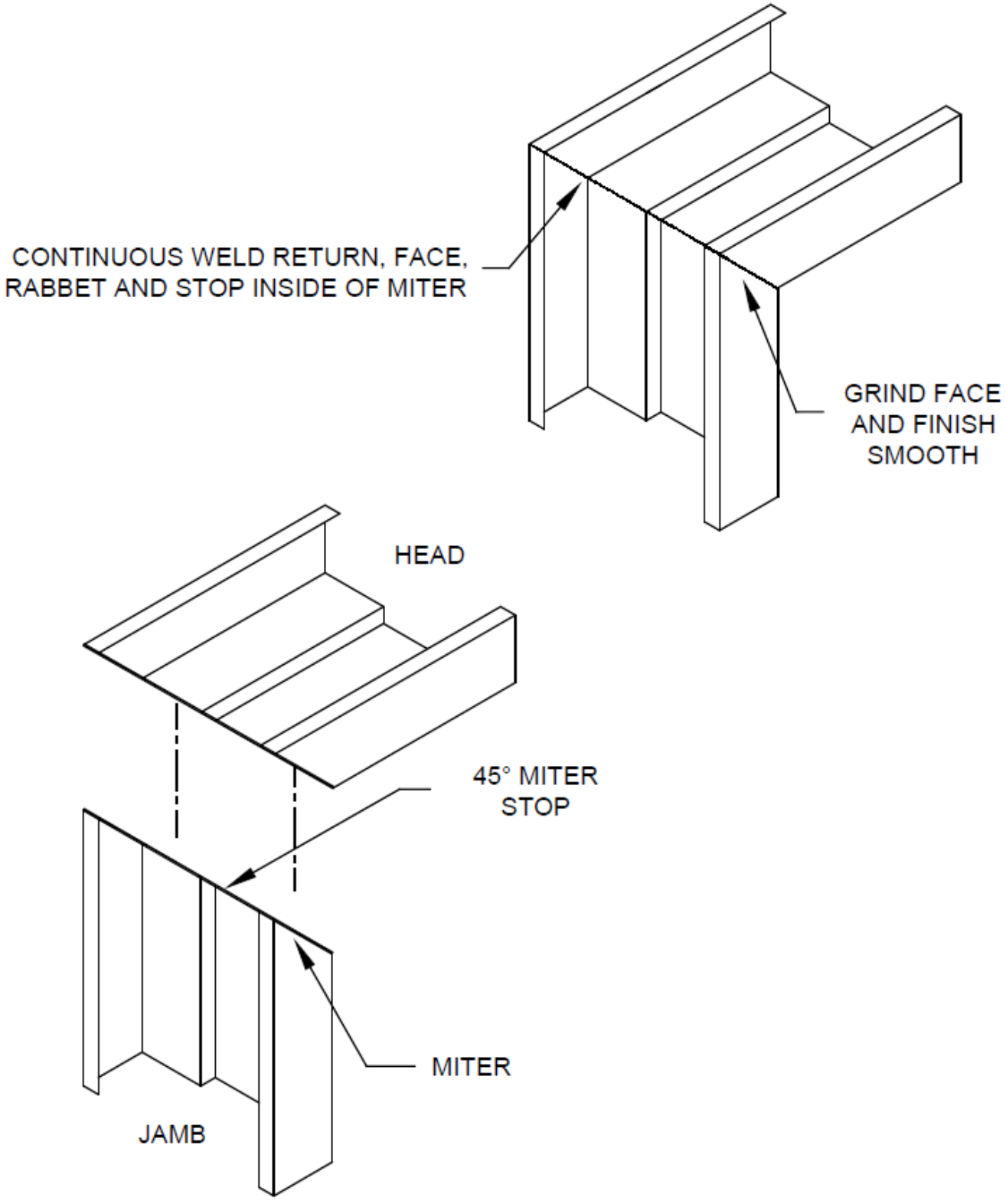


KD FRAME
DOUBLE RETURN
(SHOWN WITH BENT TABS)

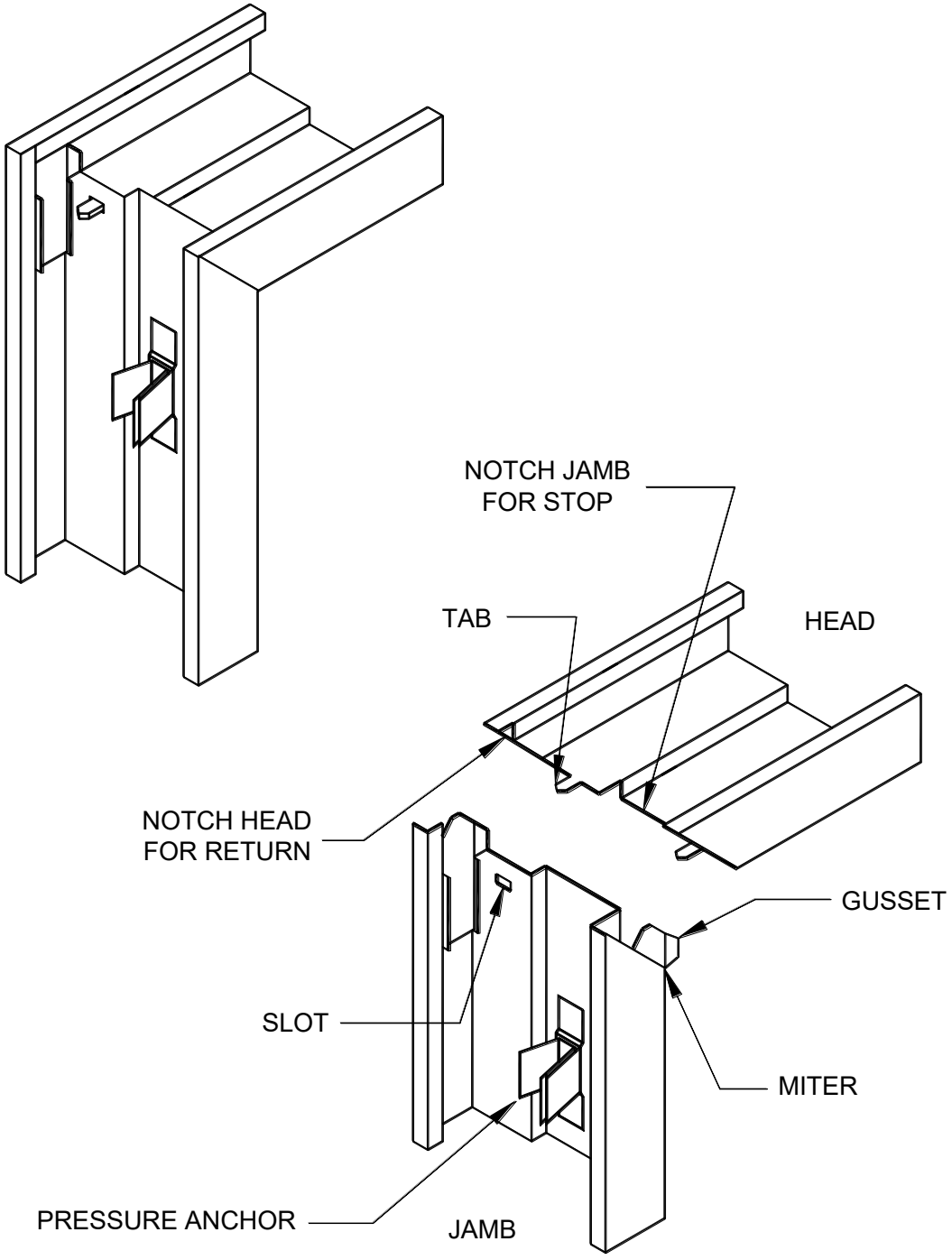
FACE WELDED CORNER



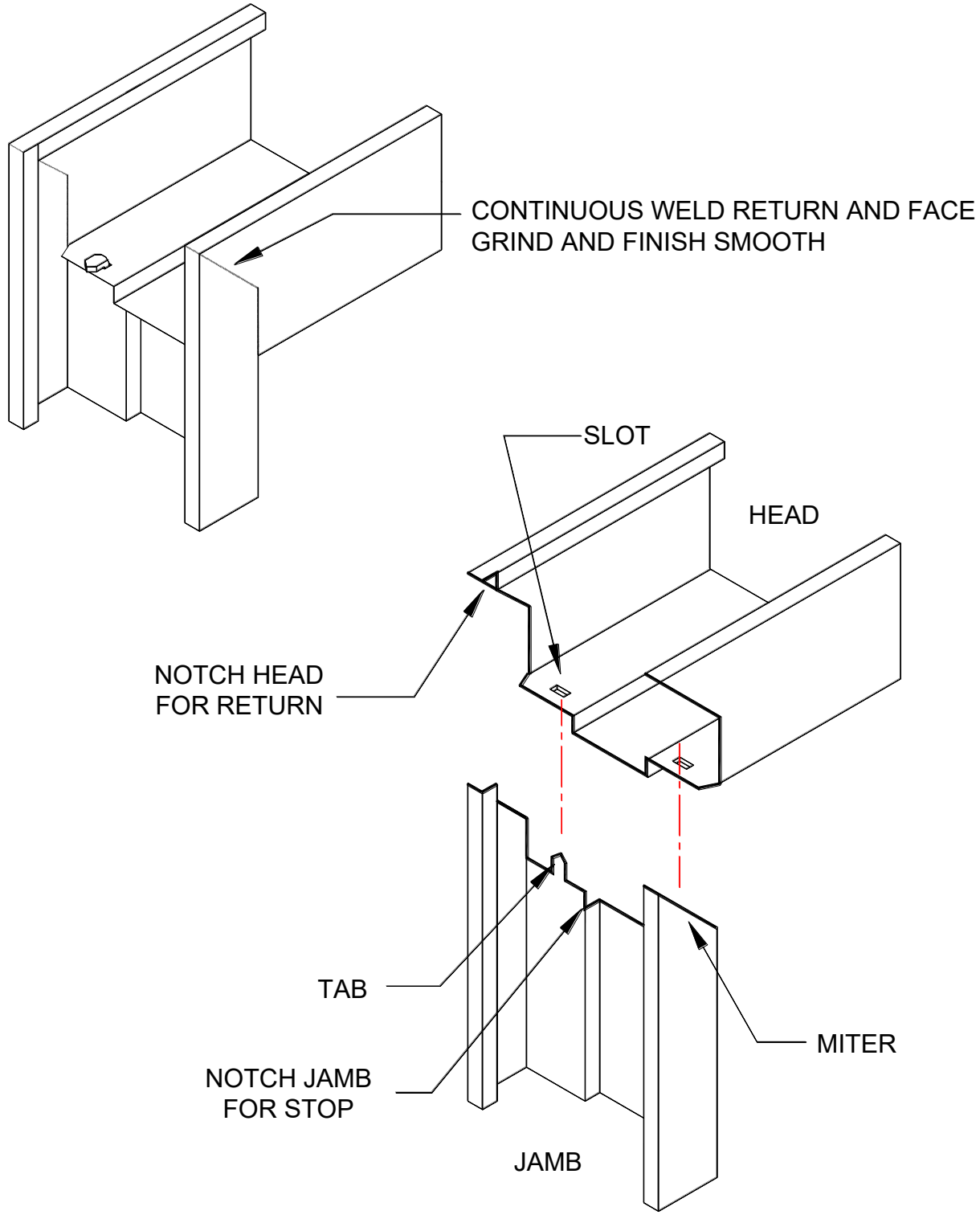
CONTINUOUSLY WELDED CORNER



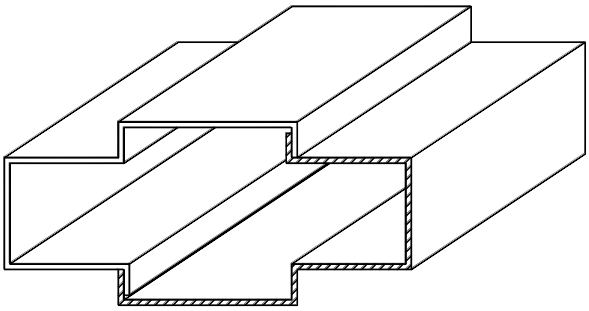
KDDW CORNER

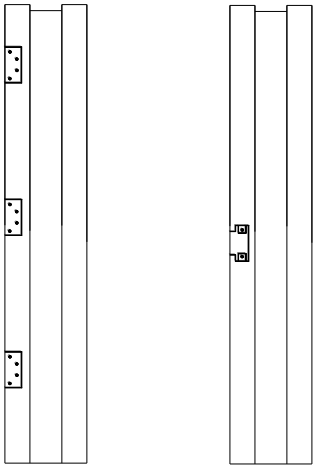

NOTCH JAMB
FOR STOP
TAB
HEAD
NOTCH HEAD
FOR RETURN
GUSSET
SLOT
MITER
PRESSURE ANCHOR
JAMB

LARGE-FACE-HEAD WELDED CORNER

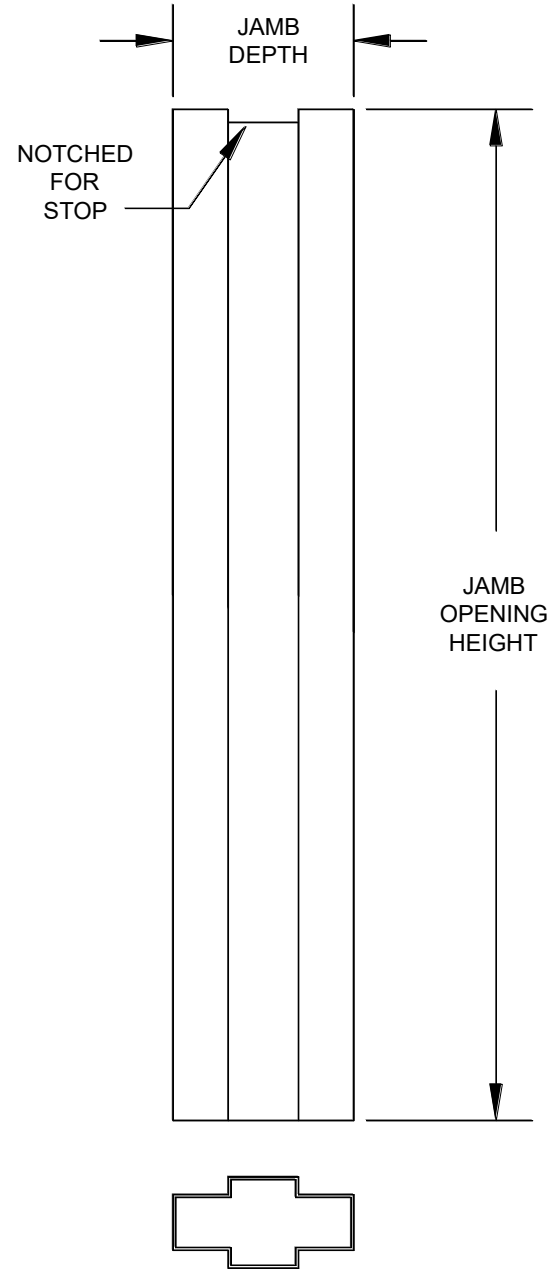


STANDARD MULLION

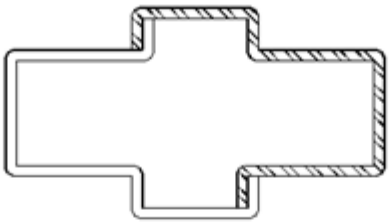


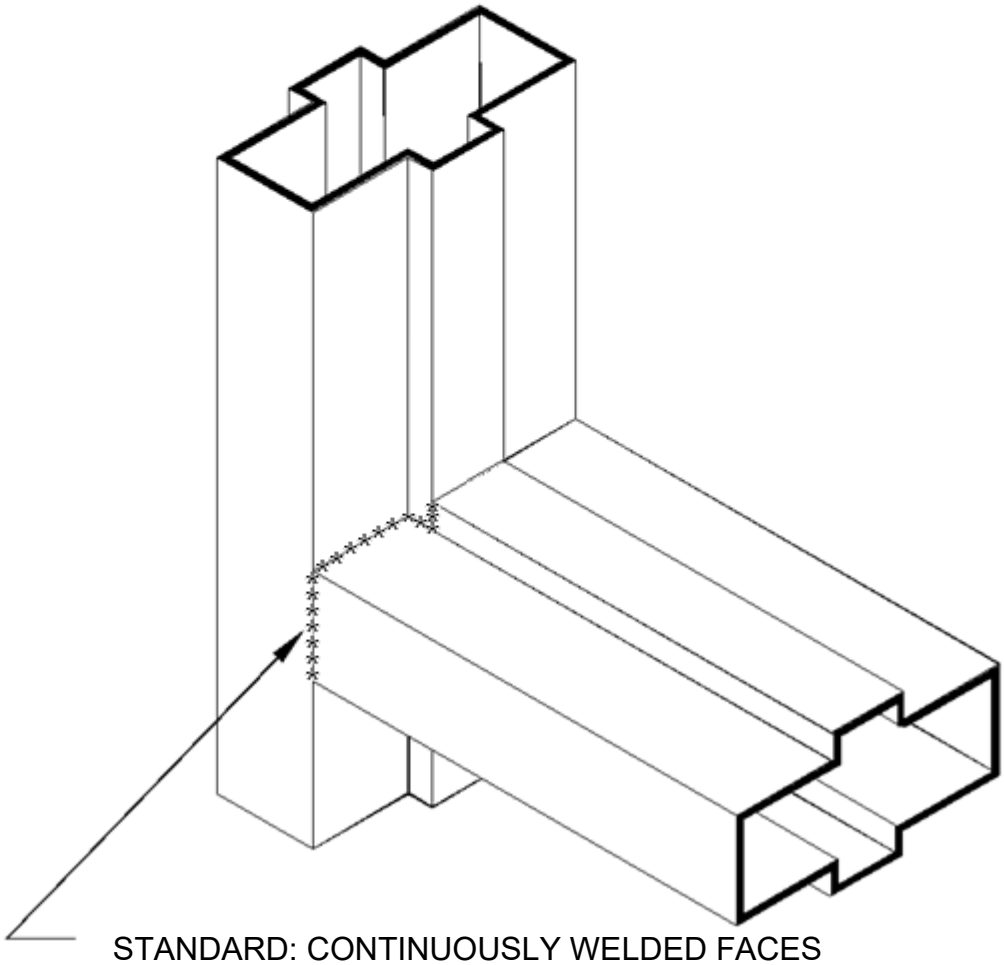


HARDWARE PREPS
AS REQUIRED



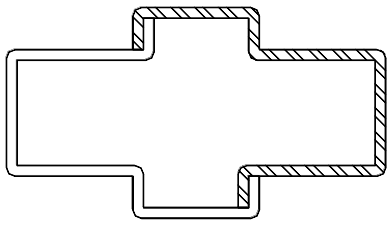
MULLION T-JOINT

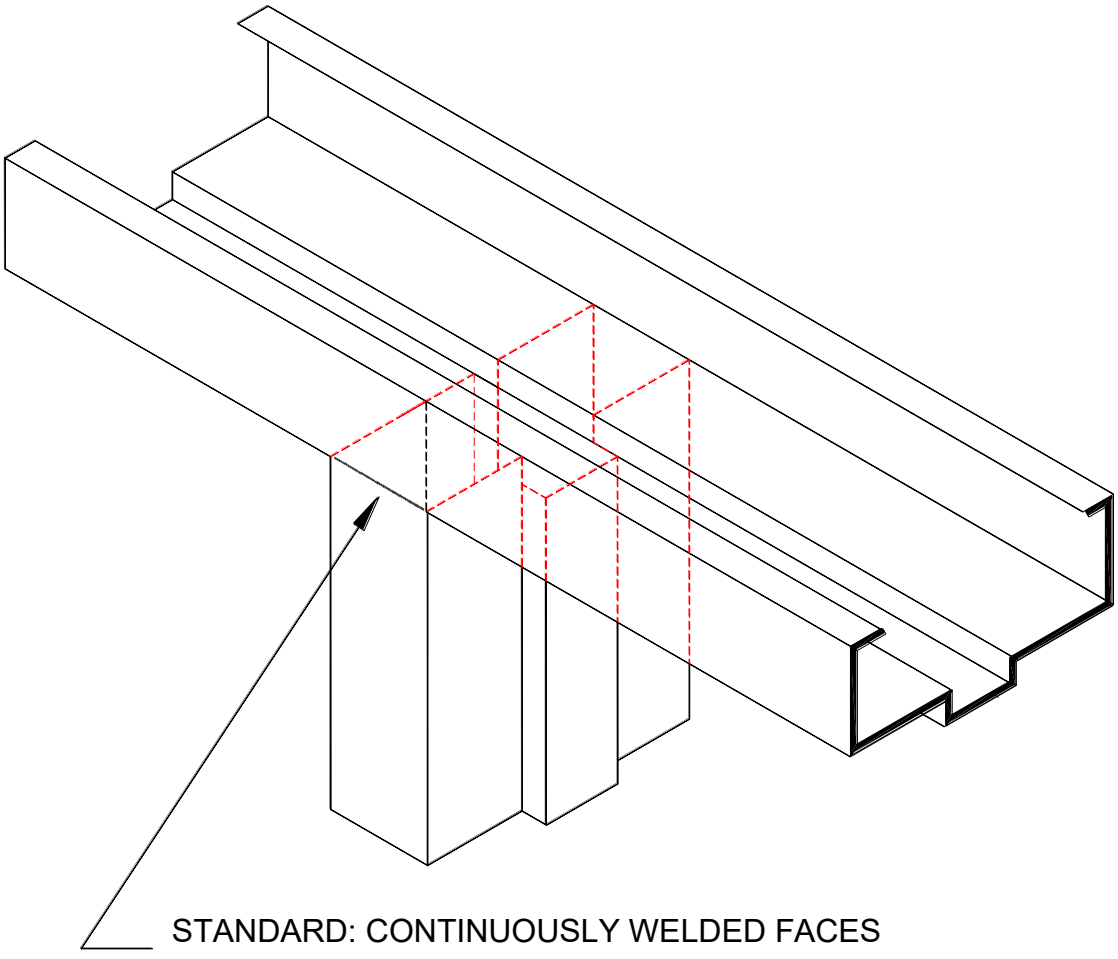




STANDARD: CONTINUOUSLY WELDED FACES
GRIND AND FINISH SMOOTH
OPTIONAL: FILLET WELD (TIG)

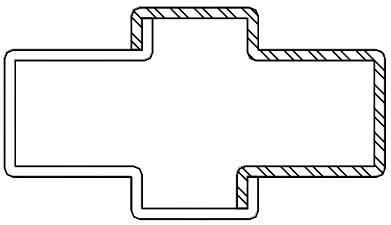
MULLION T-JOINT AT HEAD

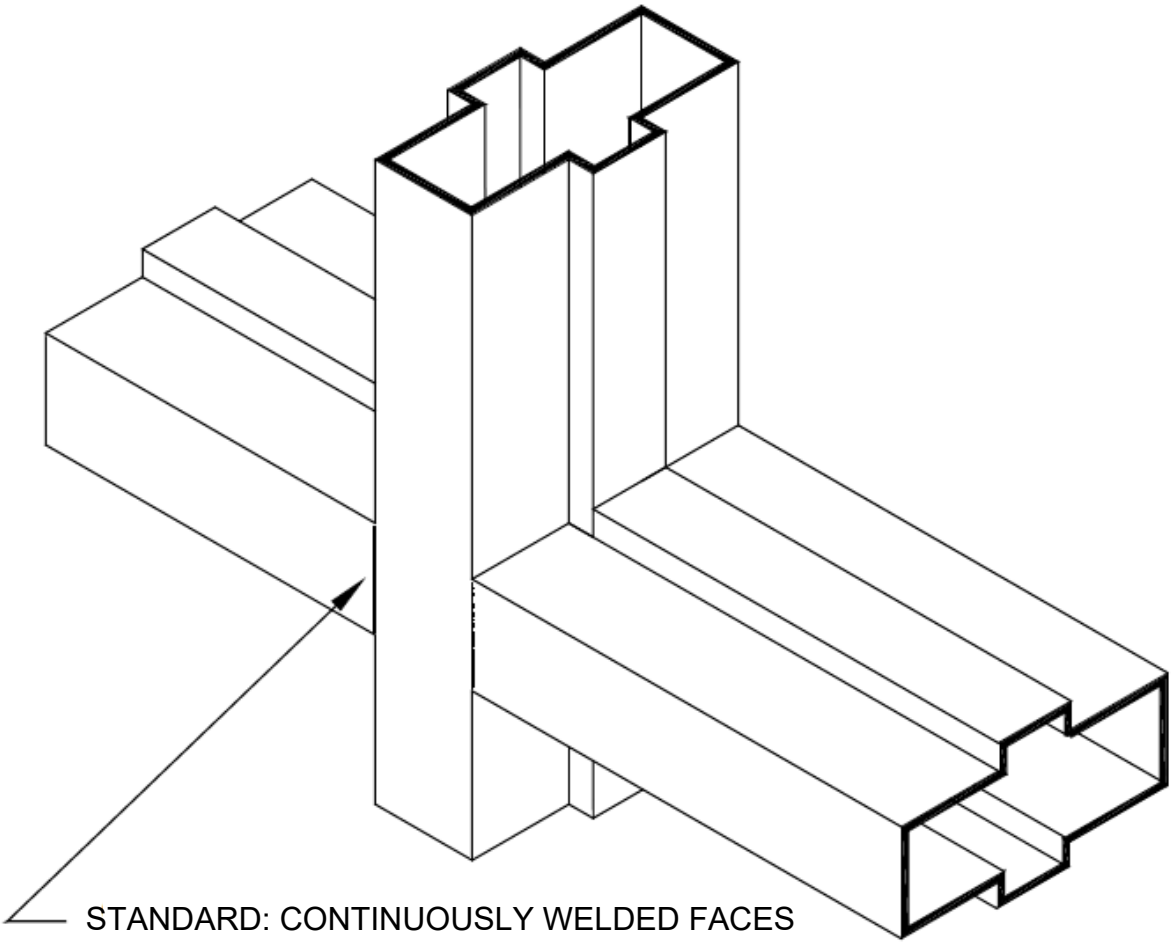




STANDARD: CONTINUOUSLY WELDED FACES
GRIND AND FINISH SMOOTH
OPTIONAL: FILLET WELD (TIG)

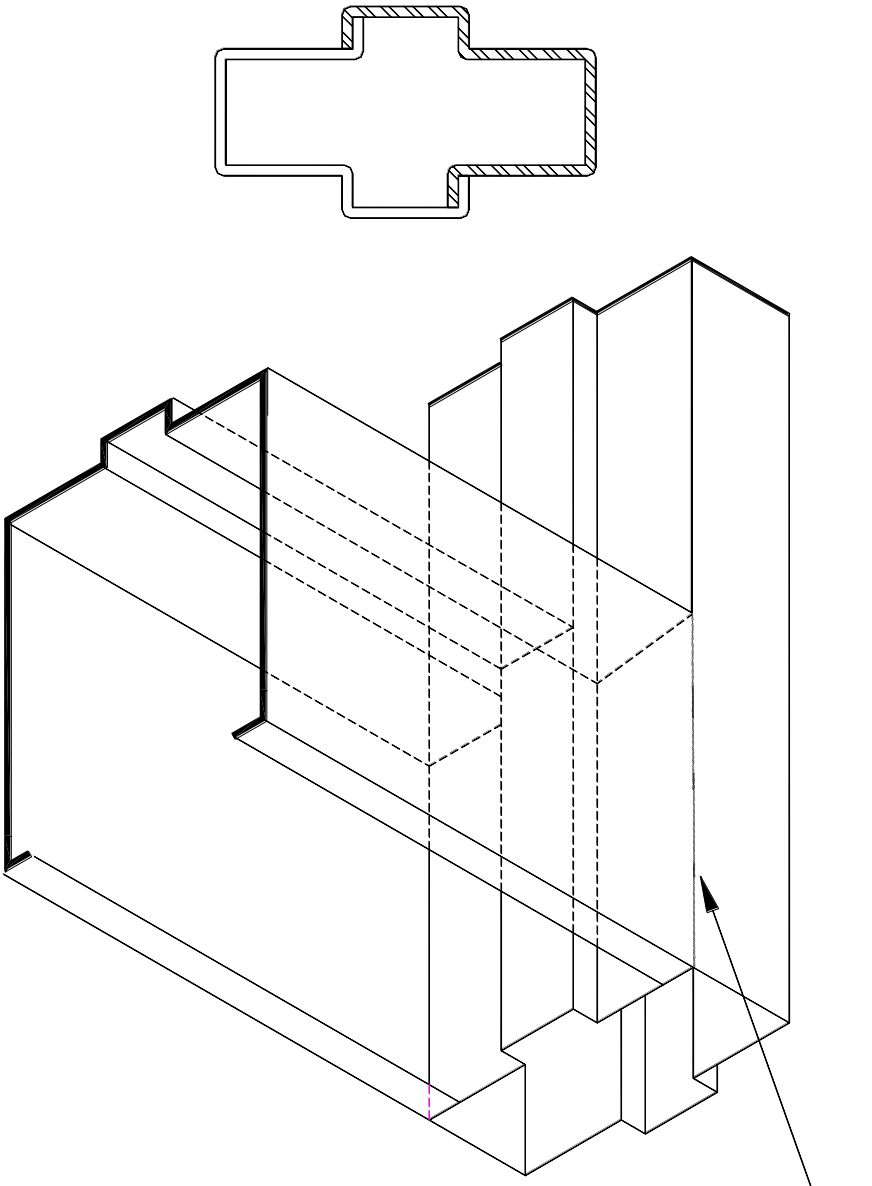
MULLION CROSS JOINT





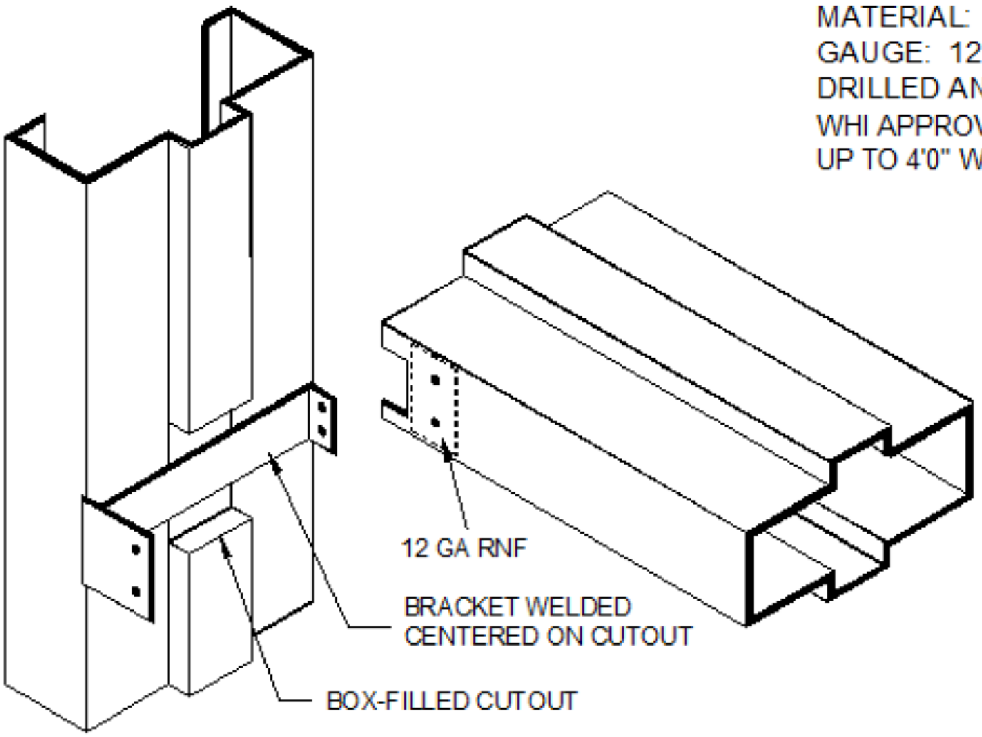
STANDARD: CONTINUOUSLY WELDED FACES
GRIND AND FINISH SMOOTH
OPTIONAL: FILLET WELD (TIG)

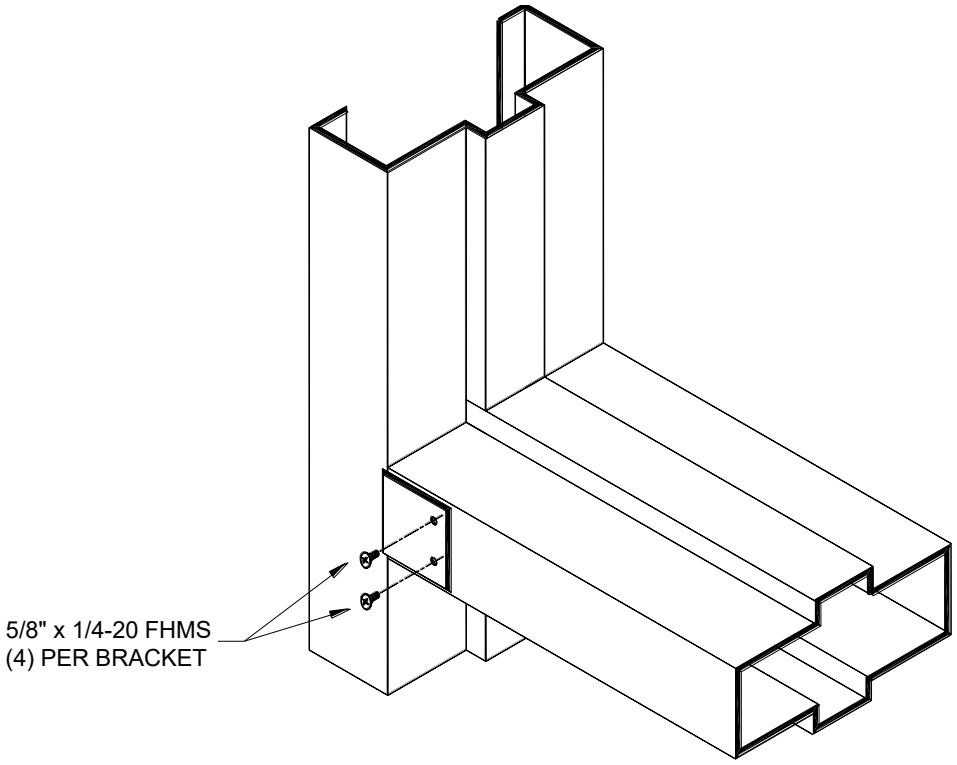
BUTTED SILL CORNER SECTION



REMOVABLE MULLION

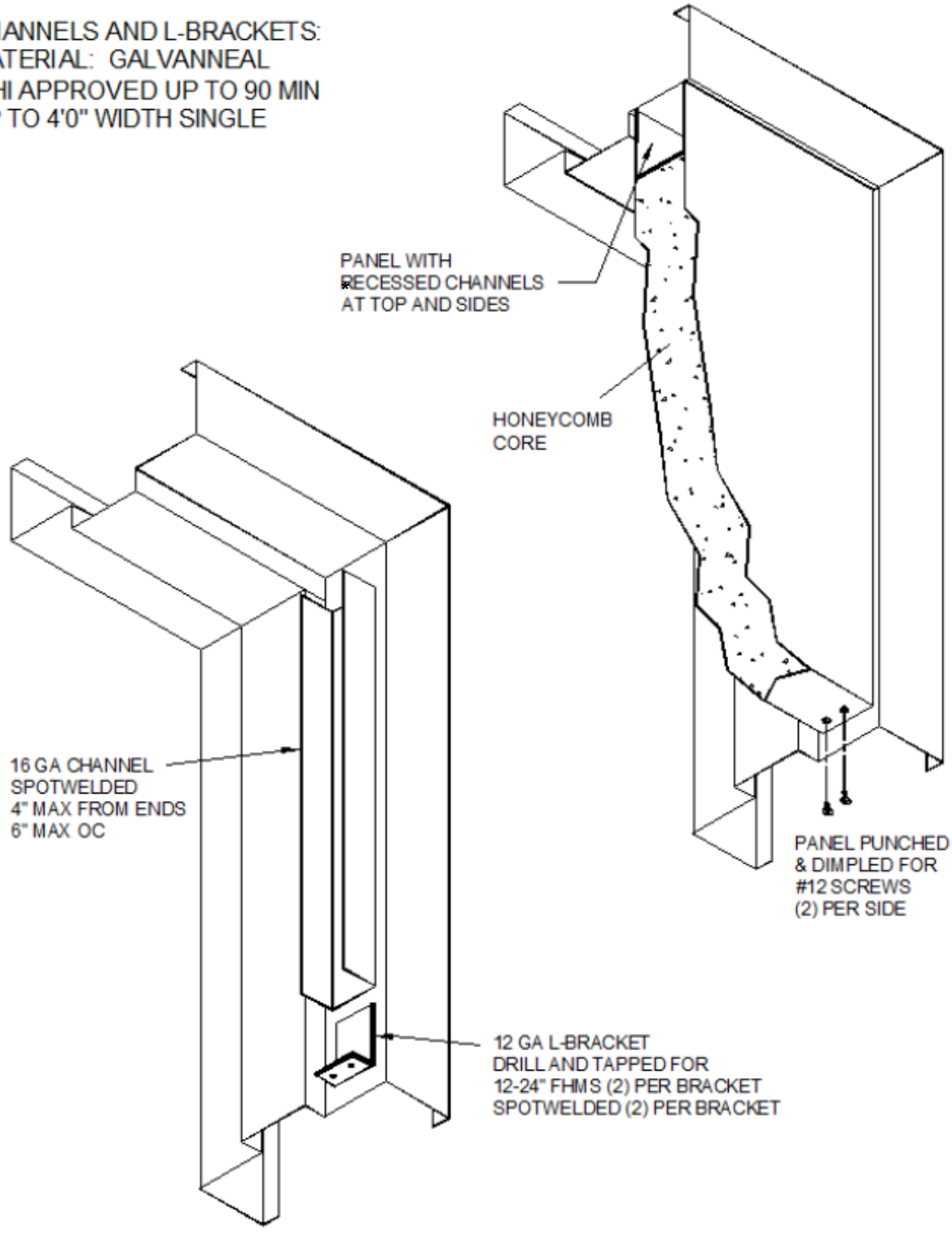
BRACKET & REINF PLATES:
MATERIAL: GALVANNEAL
GAUGE: 12
DRILLED AND TAPPED
WHI APPROVED UP TO 90 MIN
UP TO 4'0" WIDTH SINGLE





REMOVABLE TRANSOM PANEL

CHANNELS AND L-BRACKETS:
MATERIAL: GALVANNEAL
WHI APPROVED UP TO 90 MIN
UP TO 4'0" WIDTH SINGLE



* FLUSH TOP AND BOTTOM CAP RECOMMENDED IF DEBRIS IS CONCERN

POCKET SLIDER DOOR FRAME

NOTES:

1.) TRACK POCKET DESIGNED TO ACCEPT GRANT 1230 OR HAGER 9878HD ASSEMBLIES (BY OTHERS); DCI MUST BE INFORMED OF HARDWARE UP FRONT TO DESIGN THE FRAME TO ACCOMMODATE THE CORRECT HARDWARE

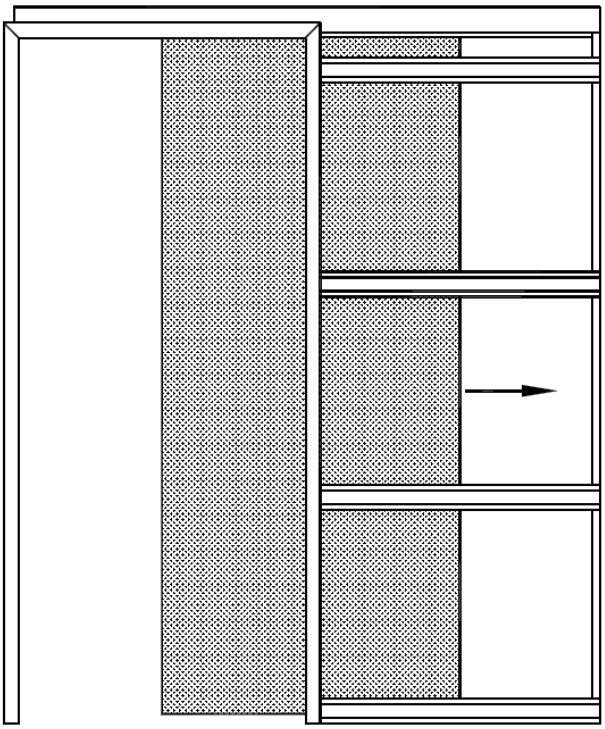
2.) POCKET SLIDER FOR 1-3/4"

3.) SINGLE ELEVATION SHOWN (PAIR SIMILAR)

4.) MATERIAL: 16GA A60 GALVANNEAL

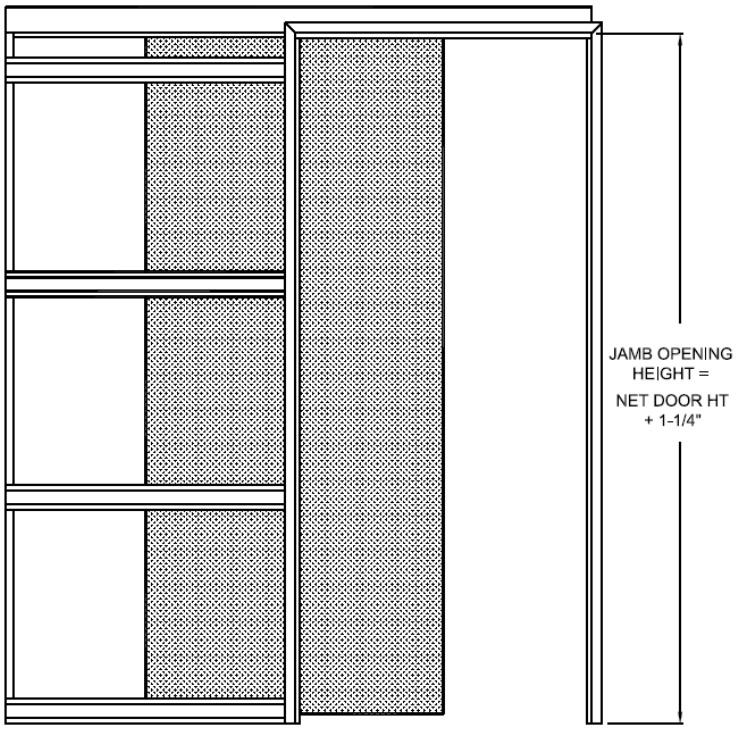
5.) SPECIFY STEEL STUD OR WOOD STUD ANCHORS

VIEW FROM EXTERIOR SIDE

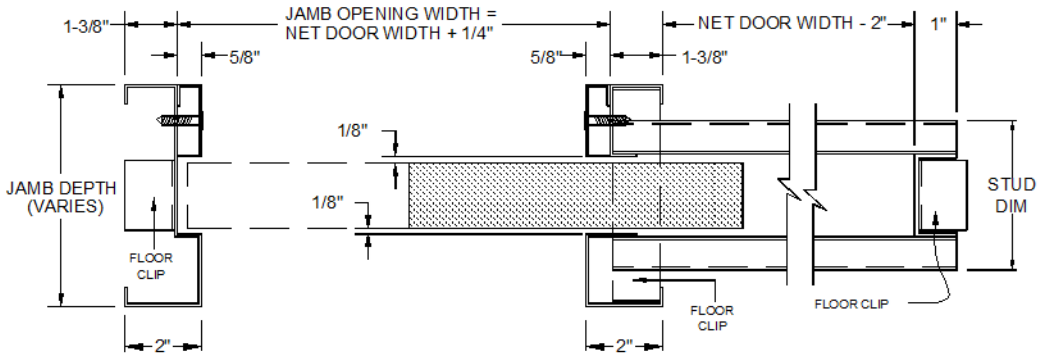


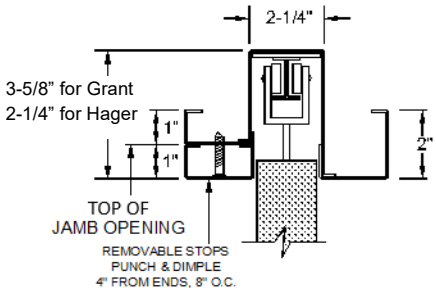
HANDING:
RIGHT HAND SHOWN
(OPENS TO THE **RIGHT** WHEN VIEWED FROM **EXTERIOR**)

VIEW FROM INTERIOR SIDE



HANDING:
RIGHT HAND SHOWN
(OPENS TO THE **LEFT** WHEN VIEWED FROM **INTERIOR**)

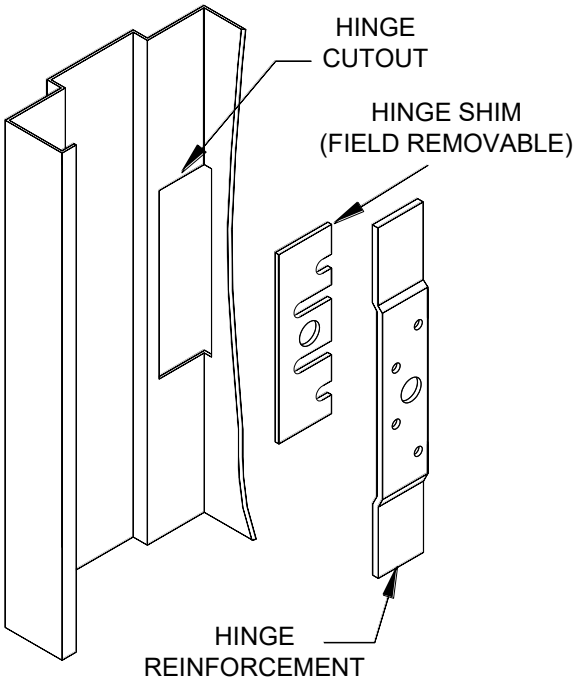


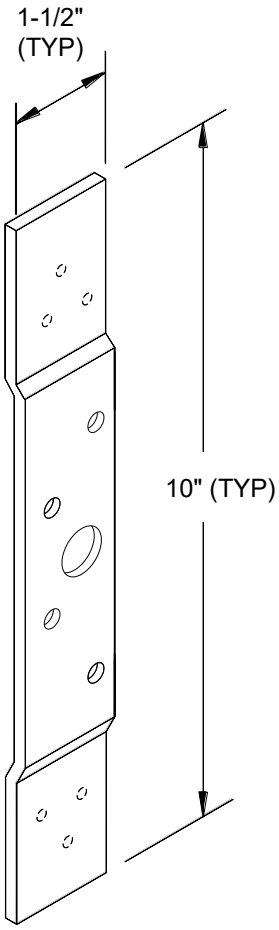


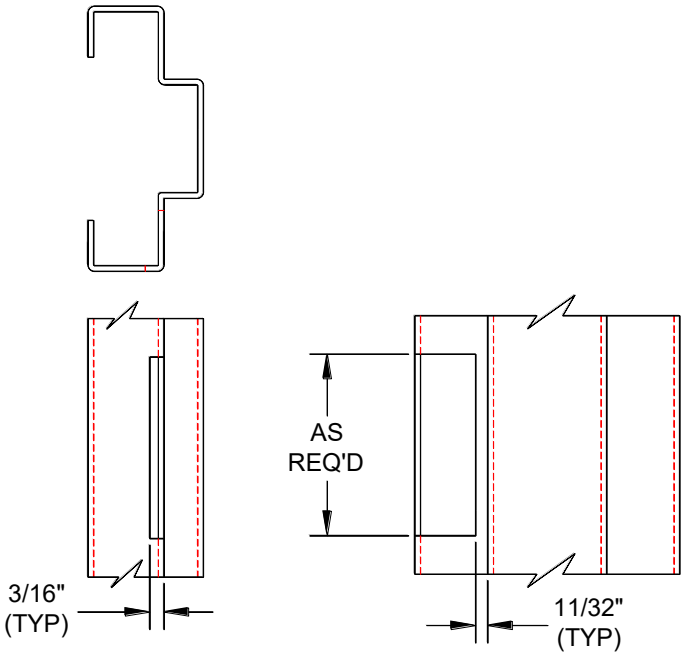
HEADER AND TRACK HANGING SYSTEM

STANDARD FRAME HINGE REINF

MATERIAL: HOT ROLLED STEEL
GAUGE: 7
THREE SPOTWELDS PER END
WHI APPROVED



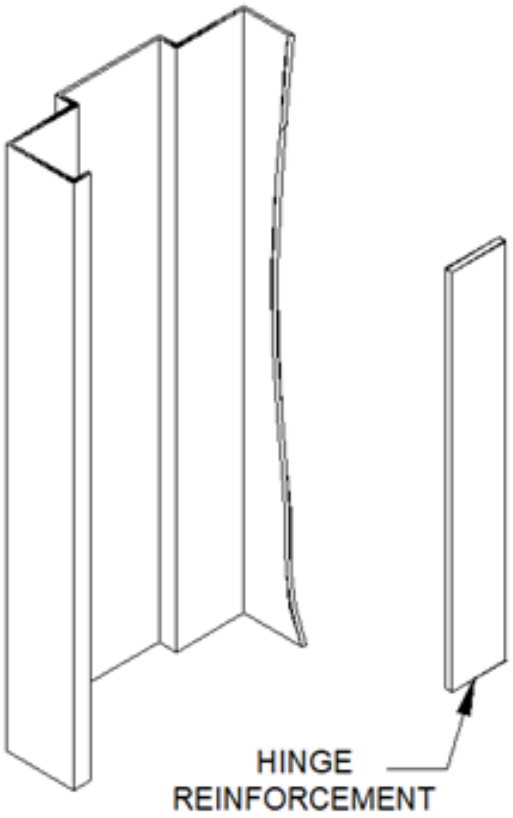


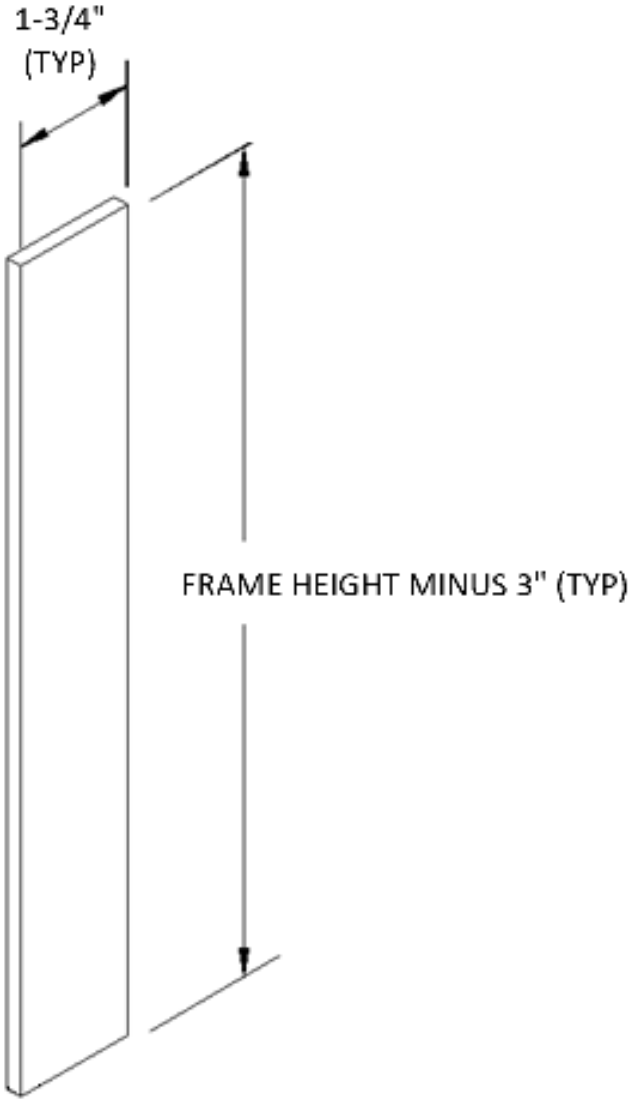


ALL DIMENSIONS
TYPICAL FOR 4",
4-1/2" AND 5" HINGES

FRAME CONTINUOUS HINGE REINF

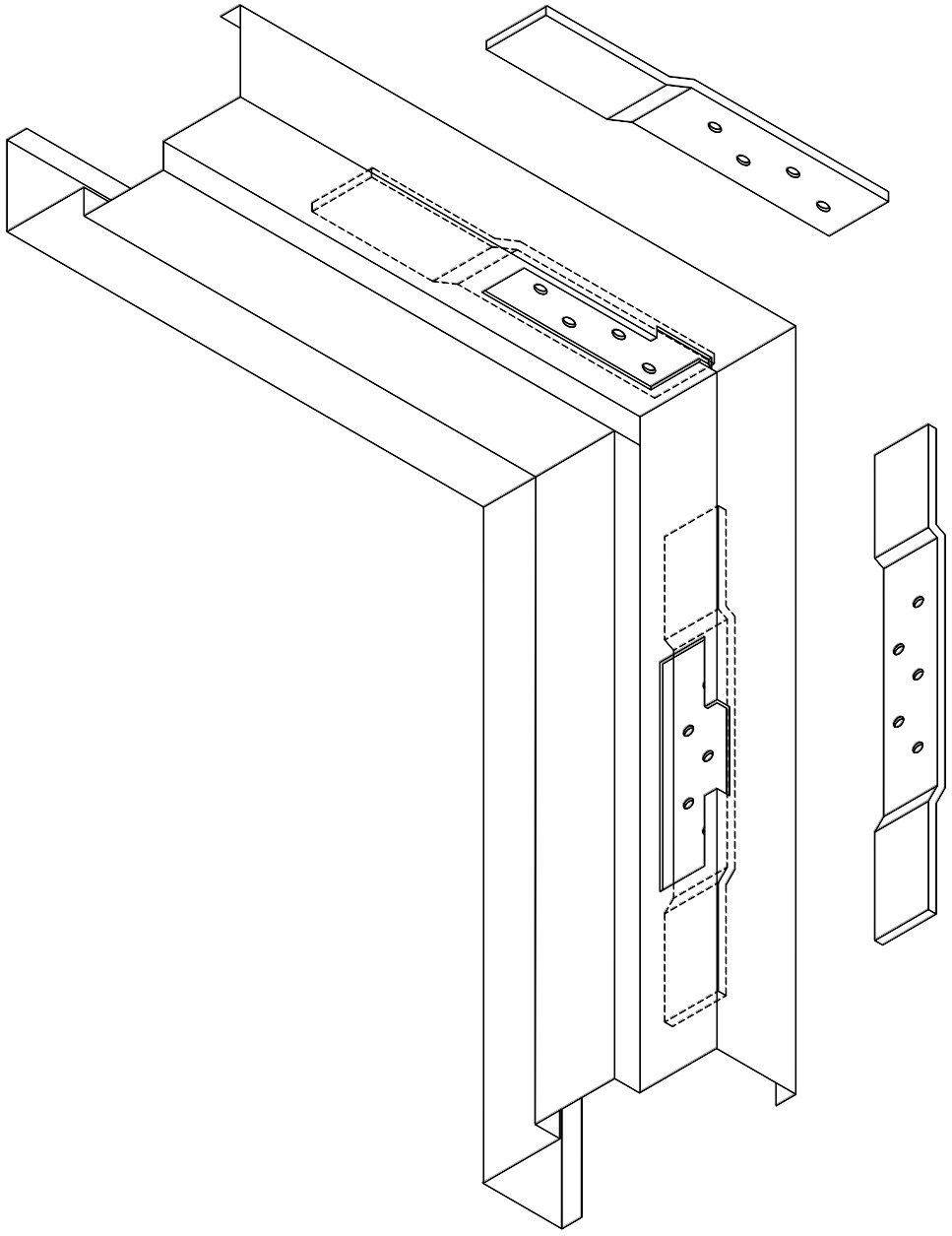
CONTINUOUS HINGE REINFORCEMENTS
ONLY ADDED UPON REQUEST
MATERIAL: GALVANNEAL
GAUGE: 16 (14, 12, 10 AVAILABLE)
WELDED TO FRAME
APPLIED TO RABBET OR FACE DEPENDING
ON CONTINUOUS HINGE TYPE
WHI APPROVED





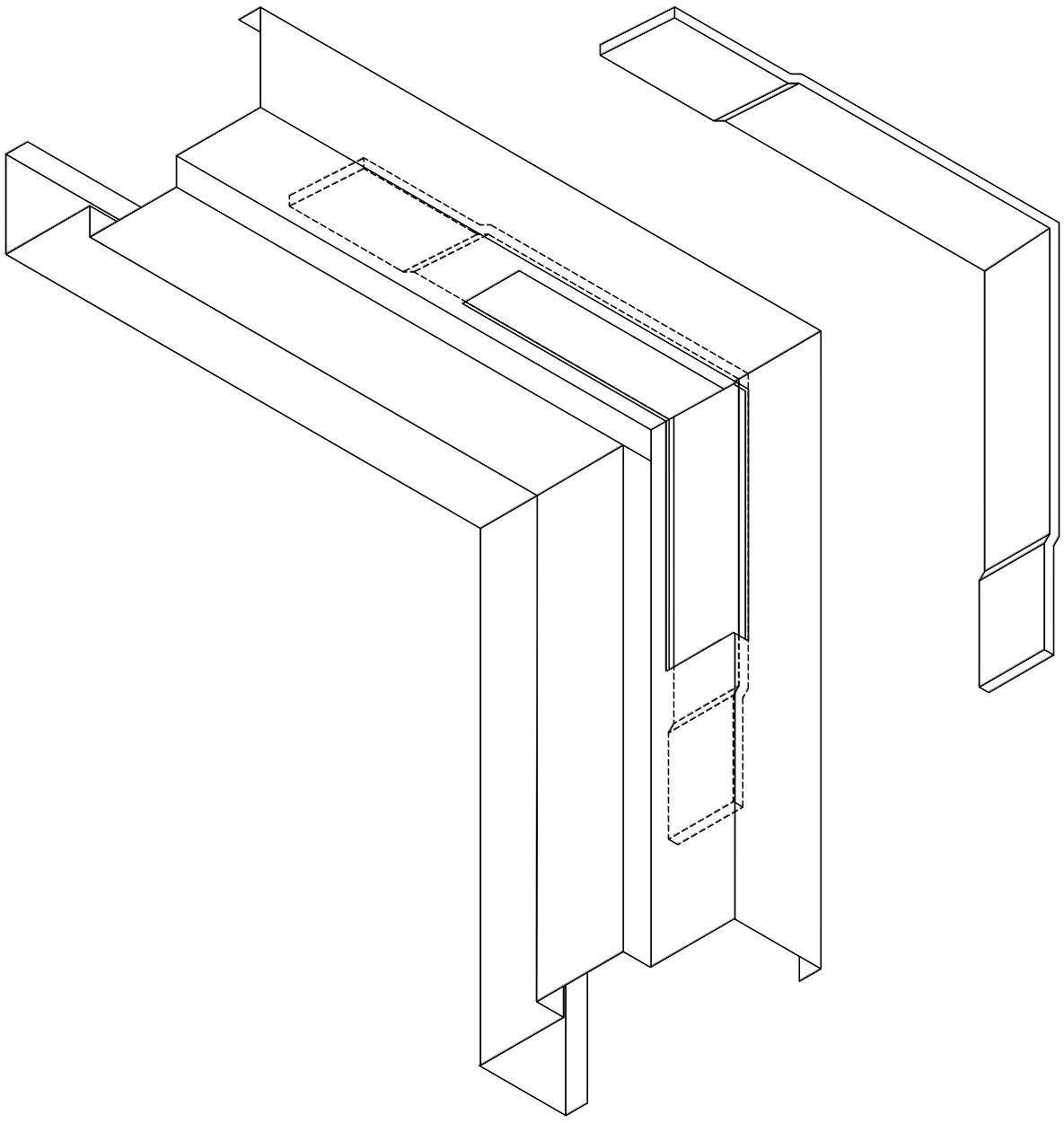
TOP/INTERMEDIATE PIVOTS

MATERIAL: HOT ROLLED STEEL
GAUGE: 7
THREE SPOTWELDS PER END
DRILLED AND TAPPED PER TEMPLATE
DUST BOXES OPTIONAL

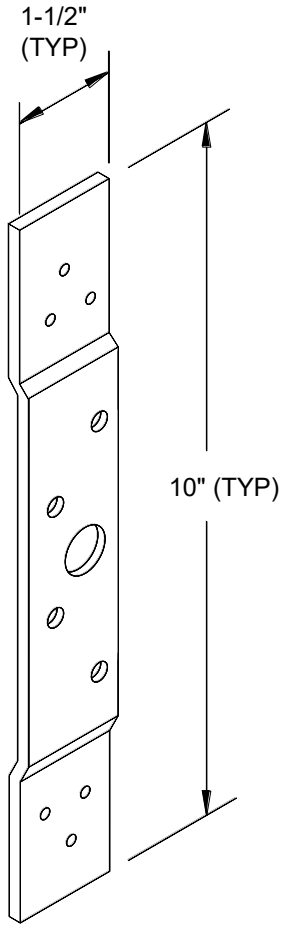


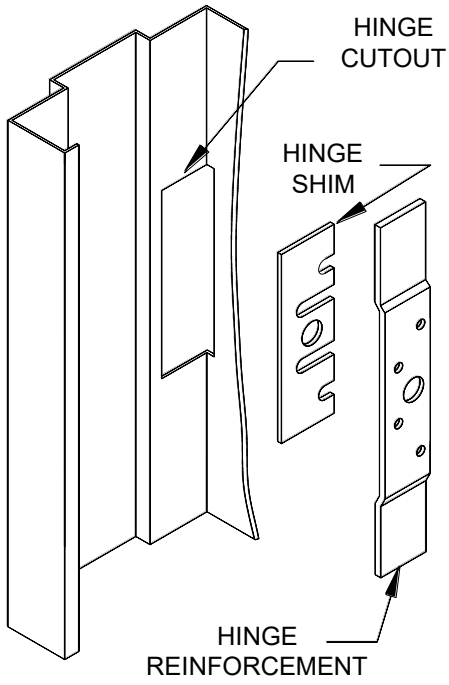
ANCHOR HINGE REINFORCEMENT

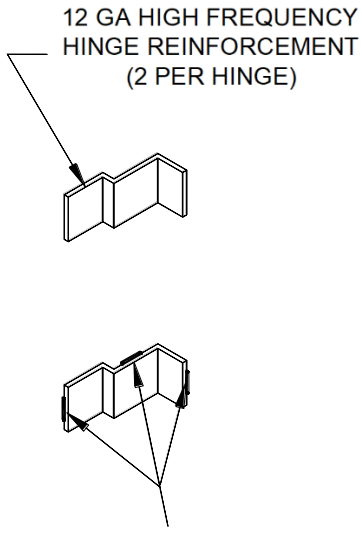
MATERIAL: HOT ROLLED STEEL
GAUGE: 7
THREE SPOTWELDS PER END
DRILLED AND TAPPED BY OTHERS

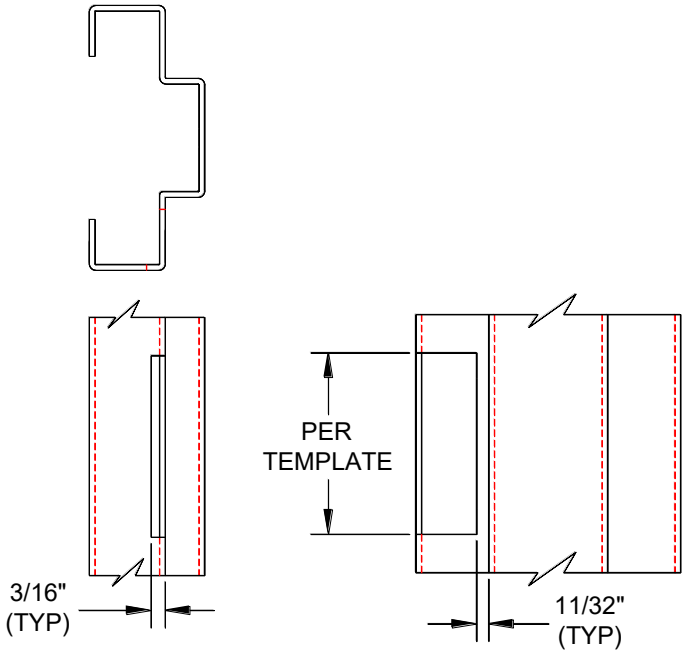


HIGH FREQUENCY HINGE REINF





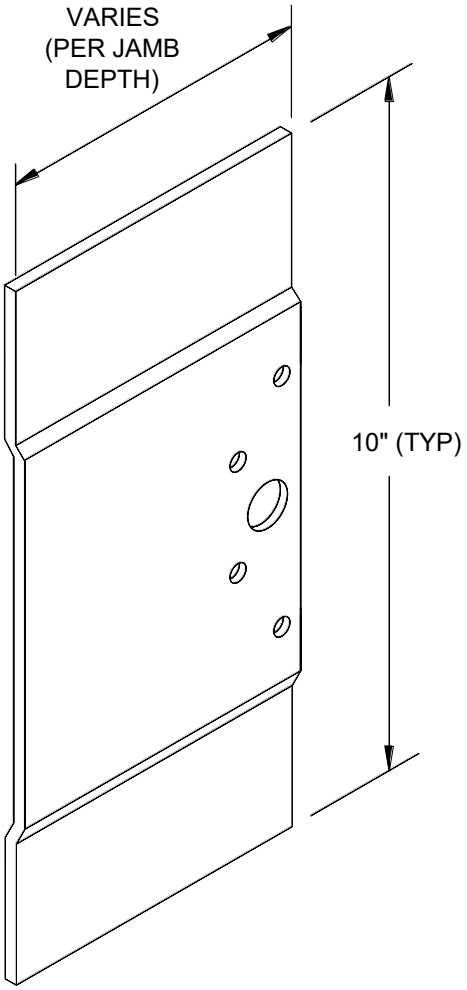


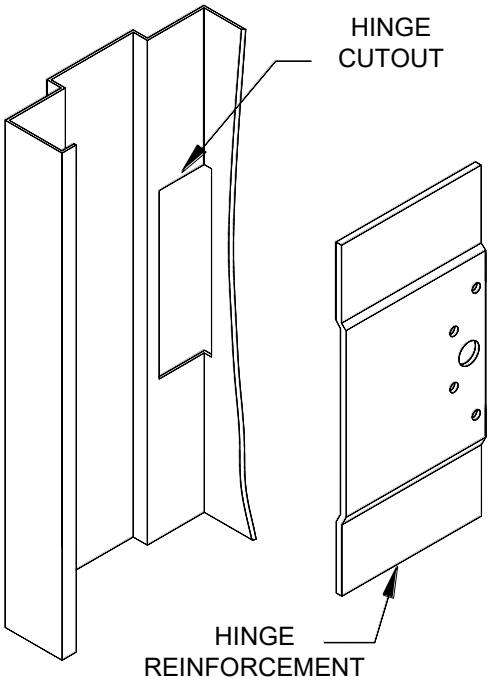


7 GA HINGE REINF
ALL DIMENSIONS
TYPICAL FOR 4-1/2"
AND 5" HINGES

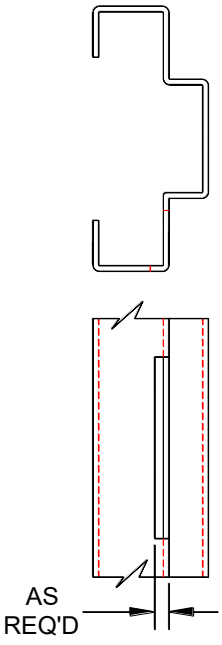
FULL WIDTH HINGE REINFORCEMENT

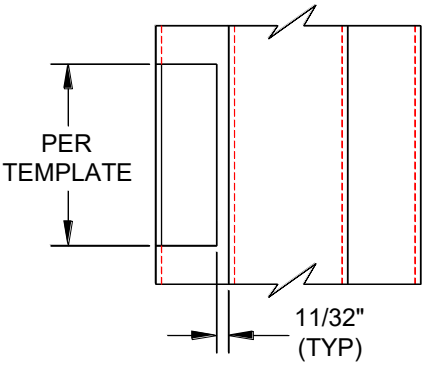
MATERIAL: HOT ROLLED STEEL
GAUGE: 7
WELDED TO BOTH RABBETS





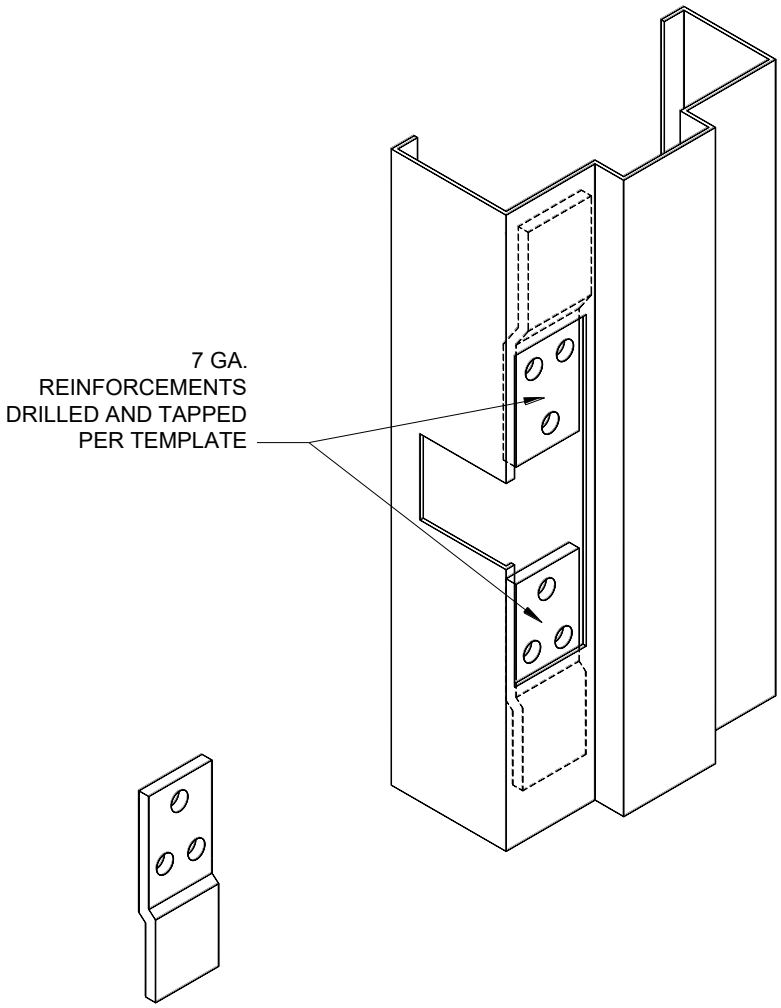
DIMENSIONS
TYPICAL FOR 4-1/2"
AND 5" HINGES





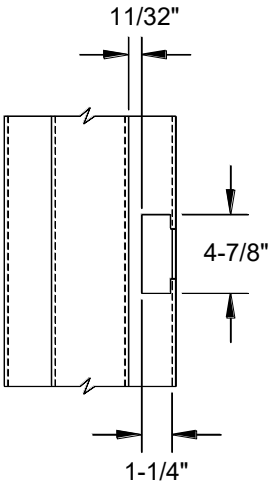
FRAME POCKET PIVOT PREP

MATERIAL: HOT ROLLED STEEL
GAUGE: 7
THREE SPOTWELDS PER END
2-1/2" MIN FACE DIM FOR FIRE RATED OPENINGS
DUST BOXES OPTIONAL



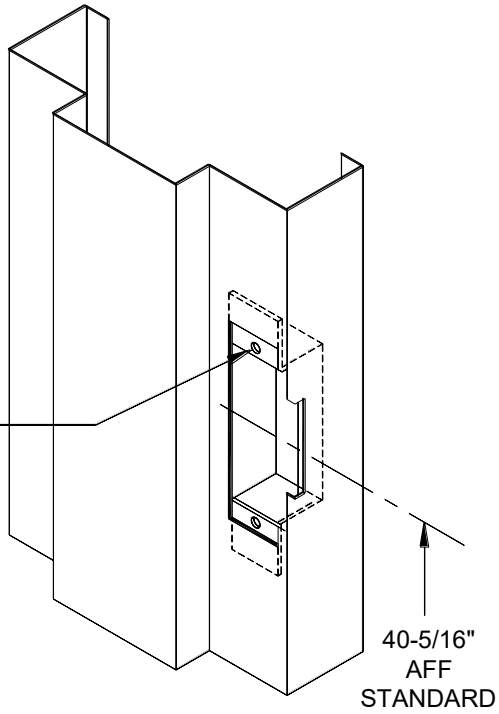
STANDARD STRIKE REINFORCEMENTS

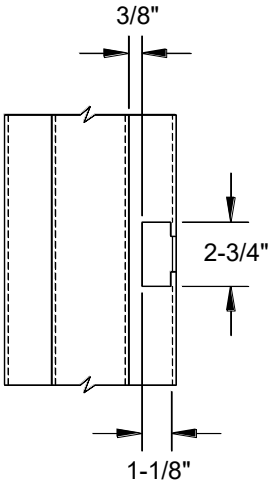
MATERIAL: GALVANNEAL
GAUGE: 16
TWO SPOTWELDS PER END



4-7/8" ASA STRIKE REINF
(ANSI A115.1 AND A115.2)

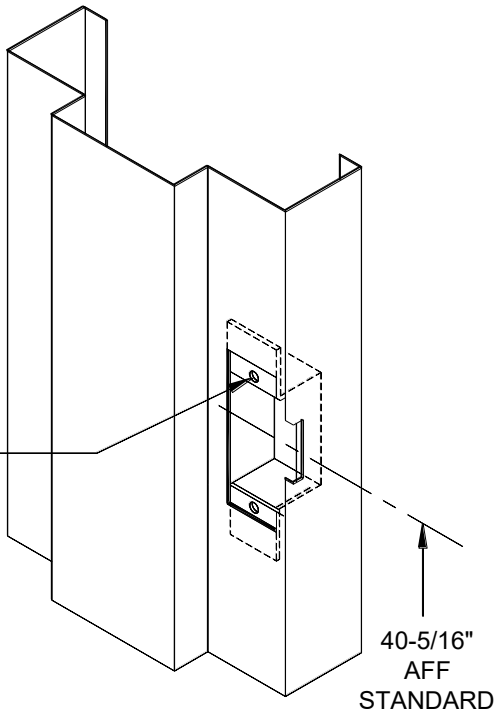
SCREW HOLES EXTRUDED TO ACCOMODATE 12-24 FHMS EQUAL TO 12 GA





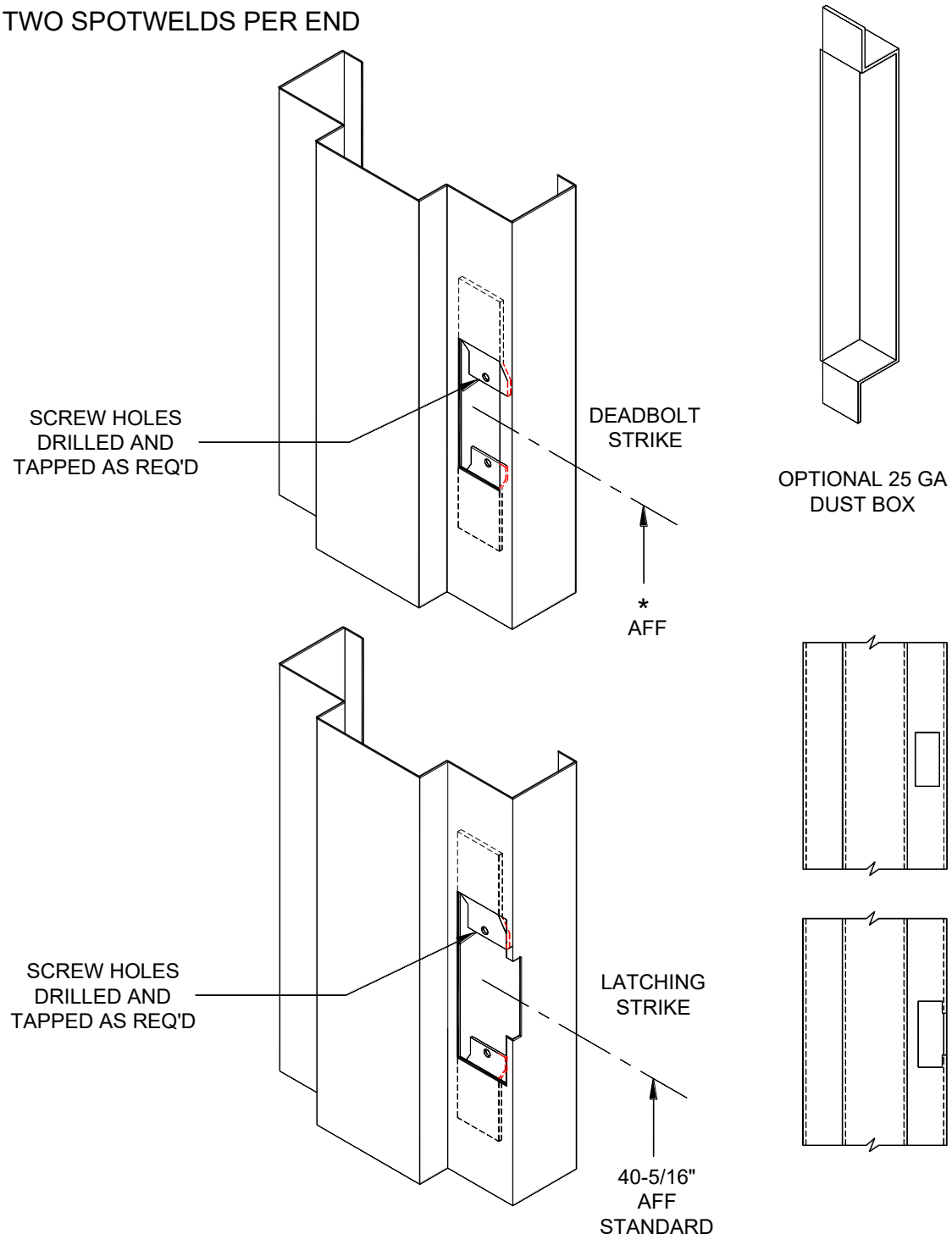
2-3/4" T-STRIKE REINF
(ANSI A115.3)

SCREW HOLES EXTRUDED TO ACCOMODATE 8-32 FHMS EQUAL TO 12 GA



OPTIONAL STRIKE REINFORCING TABS

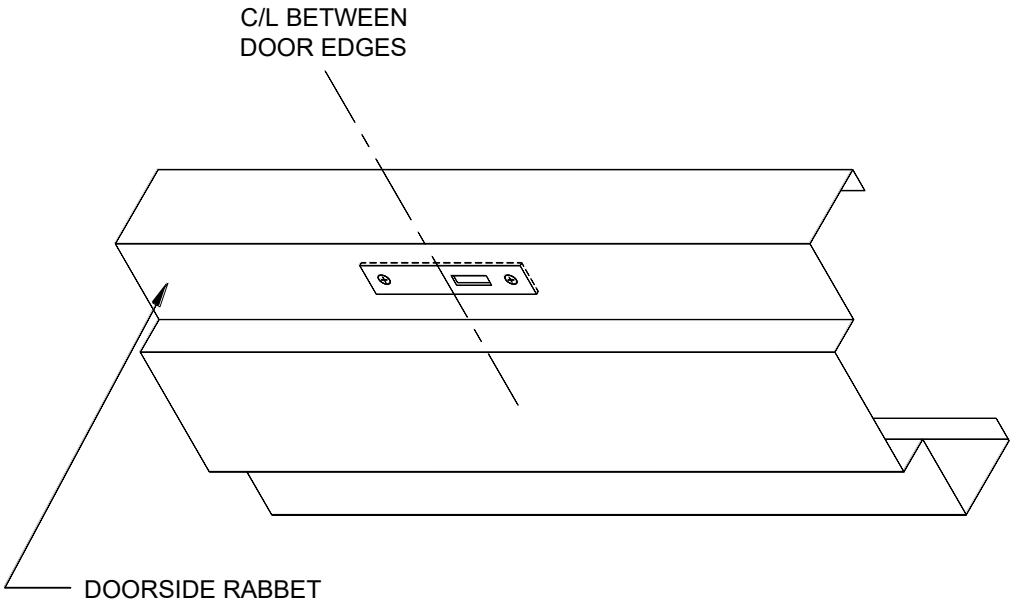
MATERIAL: HOT ROLLED BAR STOCK
GAUGE: 11
TWO SPOTWELDS PER END

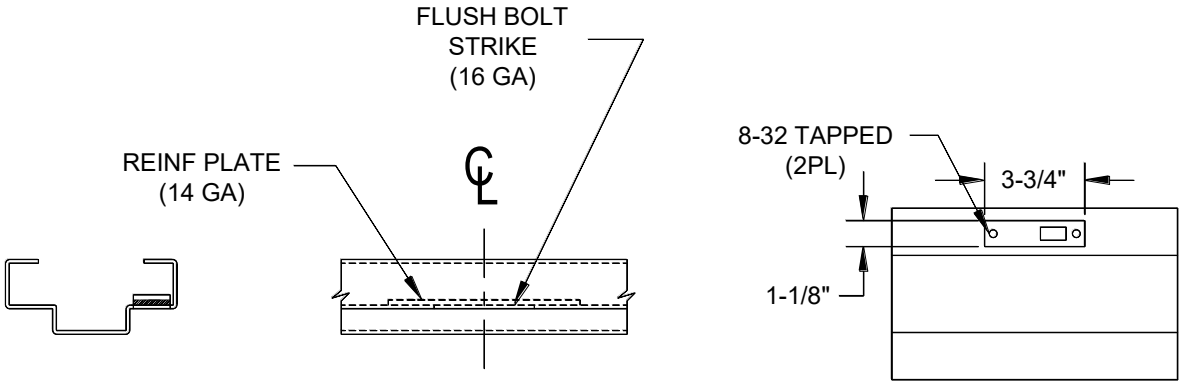


* DUE TO VARIATIONS IN STATE AND LOCAL CODES, DEADLOCK AND DEADLOCK STRIKE LOCATIONS MUST BE PROVIDED

REVERSIBLE FLUSH BOLT STRIKE

MATERIAL: GALVANNEAL
TWO SPOTWELDS PER END

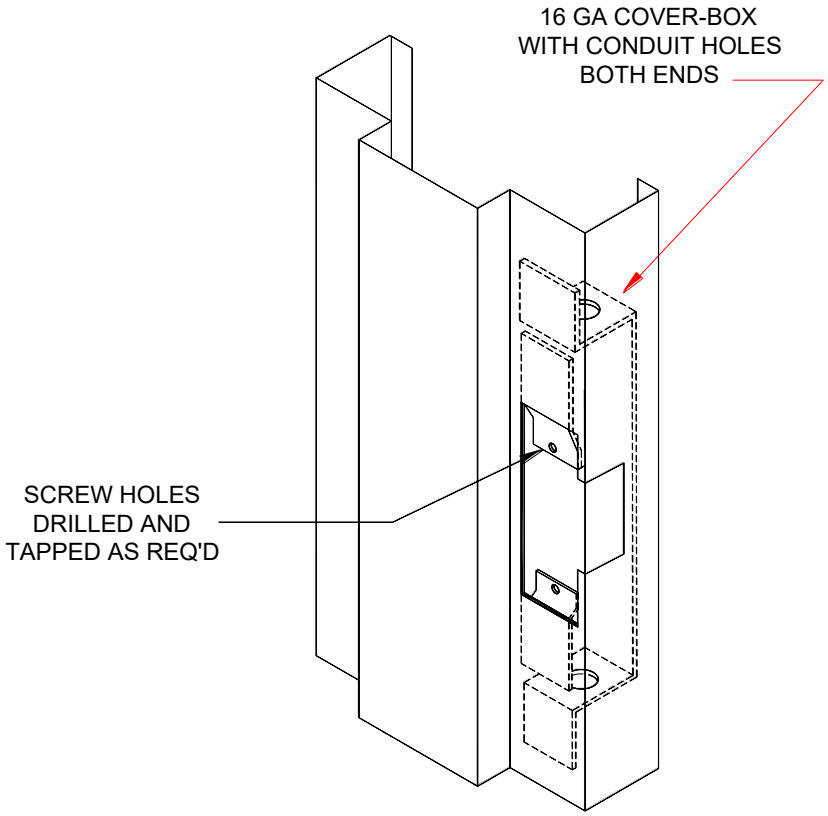


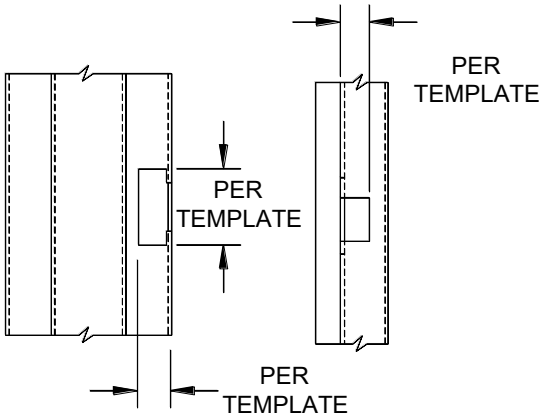


REVERSIBLE 505 FLUSH BOLT STRIKE SHOWN IS USED FOR FULLY AUTO FLUSH BOLTS AND MANUAL FLUSH BOLTS (CONSTANT/SELF LATCHING FLUSH BOLTS USE T-STRIKES)

ELECTRIC STRIKE

MATERIAL: HOT ROLLED STEEL
GAUGE: 11
TWO SPOTWELDS PER END

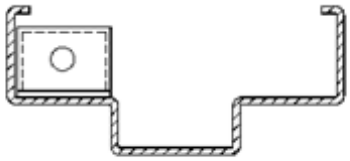




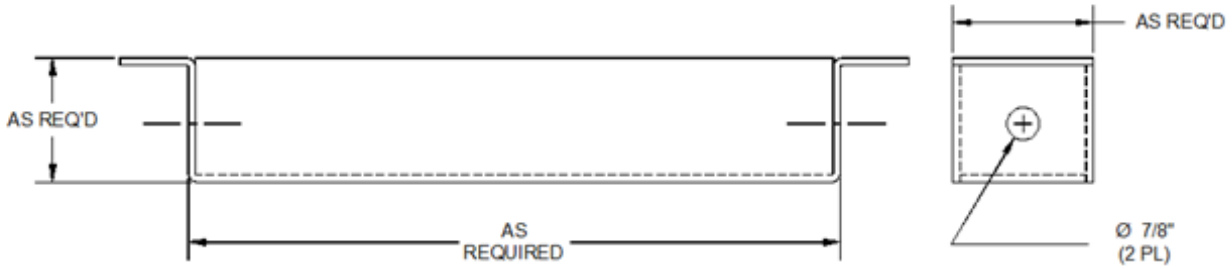
REVERSIBLE 505 FLUSH BOLT STRIKE SHOWN IS USED FOR FULLY AUTO FLUSH BOLTS AND MANUAL FLUSH BOLTS
(CONSTANT/SELF LATCHING FLUSH BOLTS USE T-STRIKES)

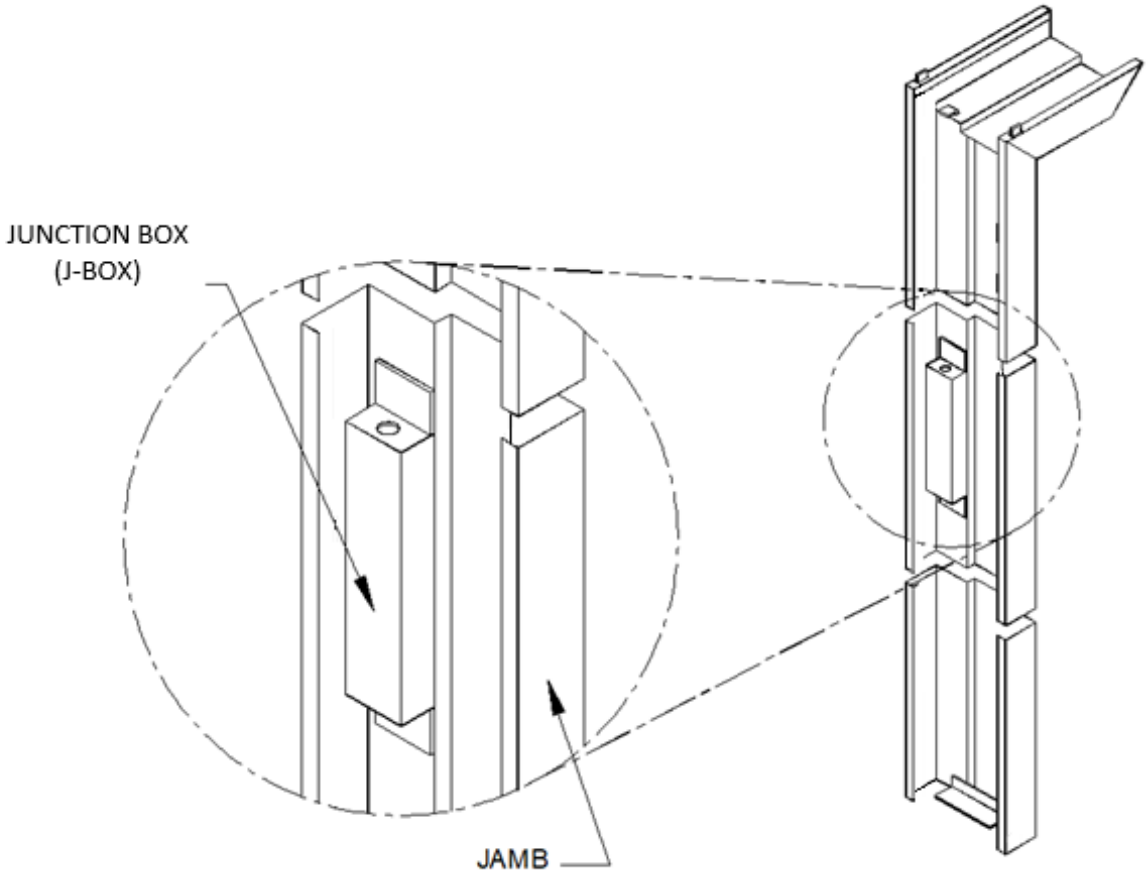
JUNCTION BOX (J-BOX)

MATERIAL: GALVANNEAL
GAUGE: 16
TWO SPOTWELDS PER END
TYPICAL AT ELECTRIC HINGE PREPS



TOP SECTION VIEW

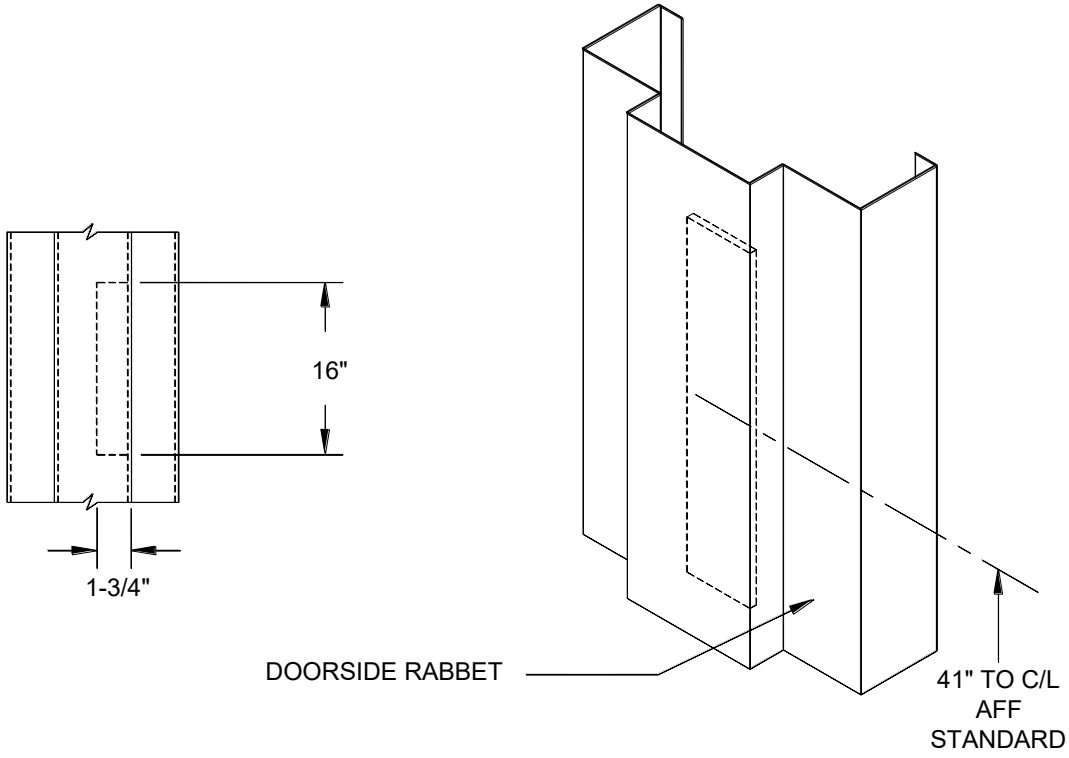




1) INCLUDED AUTOMATICALLY FOR FRAMES WITH MASONRY ANCHORS (A5, A6, A7P, A7B, A11); ONLY UPON REQUEST FOR OTHER FRAMES
2) NOT TYPICAL FOR KDDW APPLICATIONS DUE TO DRYWALL RETURNS AND RECOMMENDED WALL PENETRATION

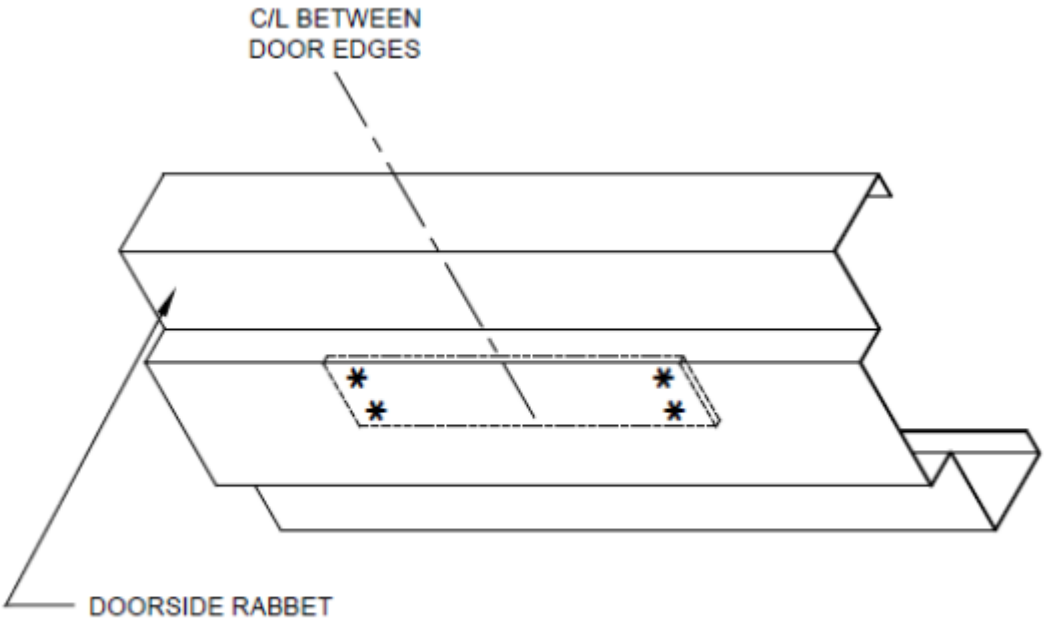
RIM PANIC REINFORCEMENT

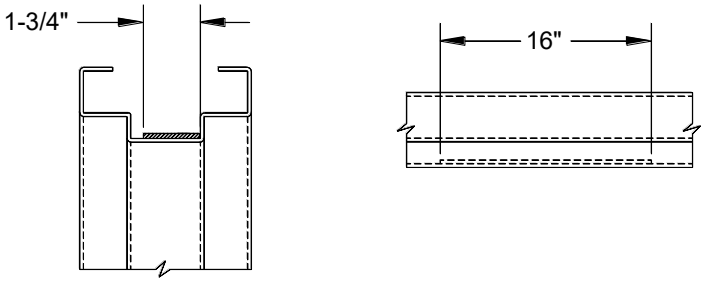
MATERIAL: GALVANNEAL
GAUGE: 12
TWO SPOTWELDS PER END



SURFACE VERTICAL ROD REINF

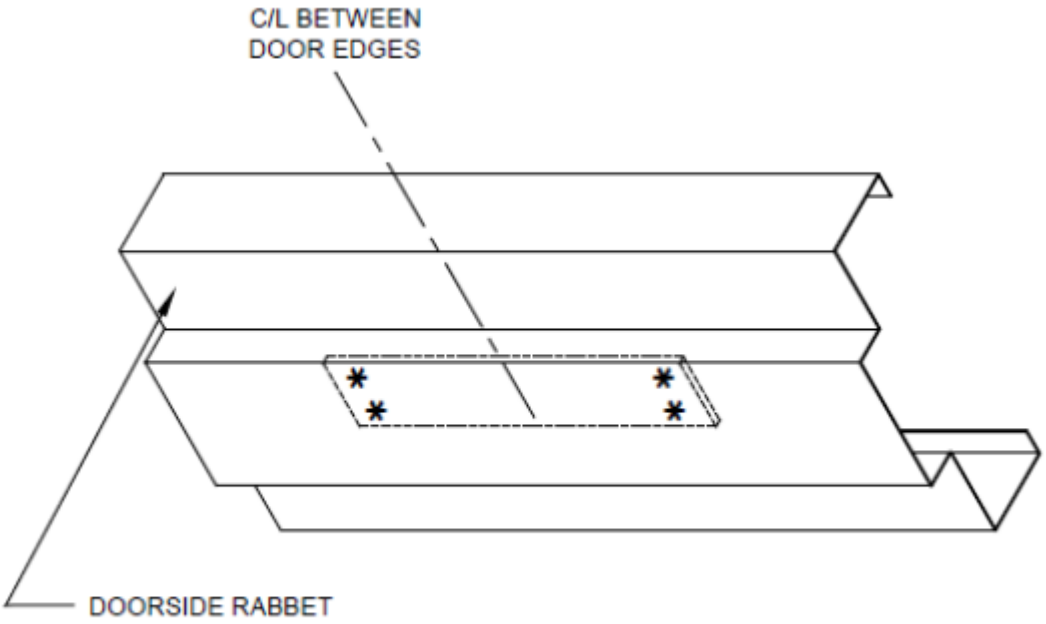
MATERIAL: GALVANNEAL
GAUGE: 12
TWO SPOTWELDS PER END

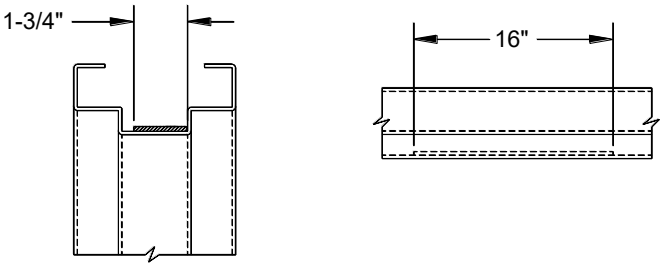




REMOVABLE HARDWARE MULLION REINF

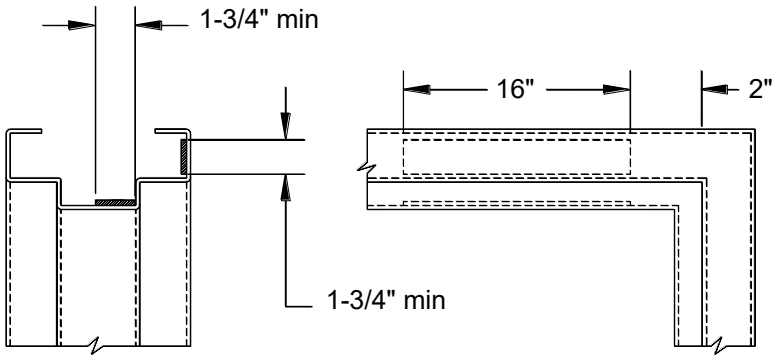
MATERIAL: GALVANNEAL
GAUGE: 12
TWO SPOTWELDS PER END



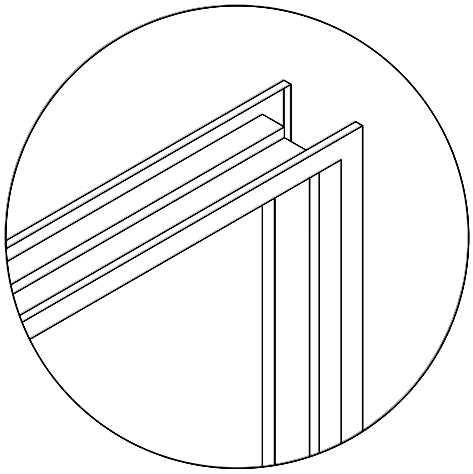


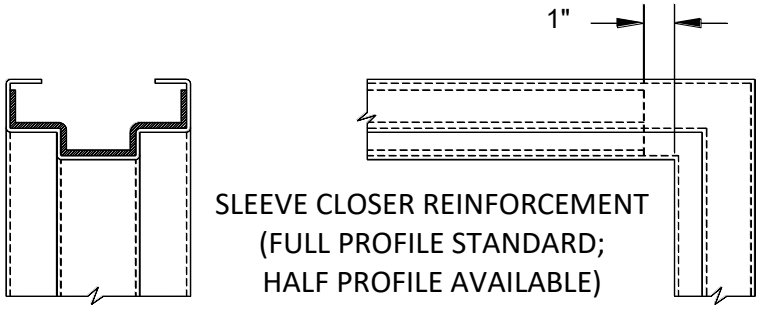
STANDARD CLOSER REINFORCEMENTS

MATERIAL: GALVANNEAL
GAUGE: 12
TWO SPOTWELDS PER END



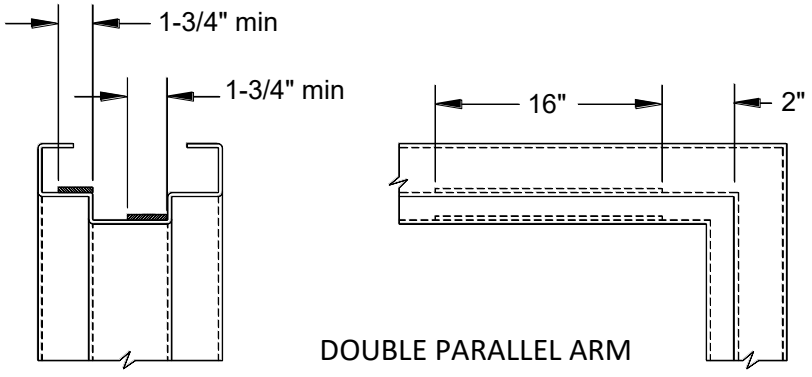
REGULAR ARM AND PARALLEL
ARM REINFORCEMENT
BOTH APPLIED STANDARD
(FULL WIDTH AVAILABLE)





SLEEVE CLOSER REINFORCEMENT
(FULL PROFILE STANDARD;
HALF PROFILE AVAILABLE)

FULL WIDTH STANDARD
(CUSTOM WIDTH AVAILABLE)

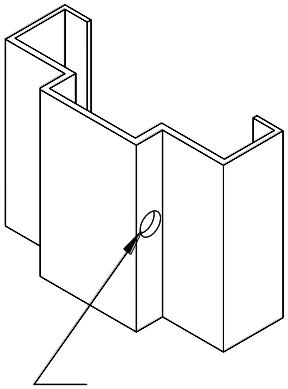


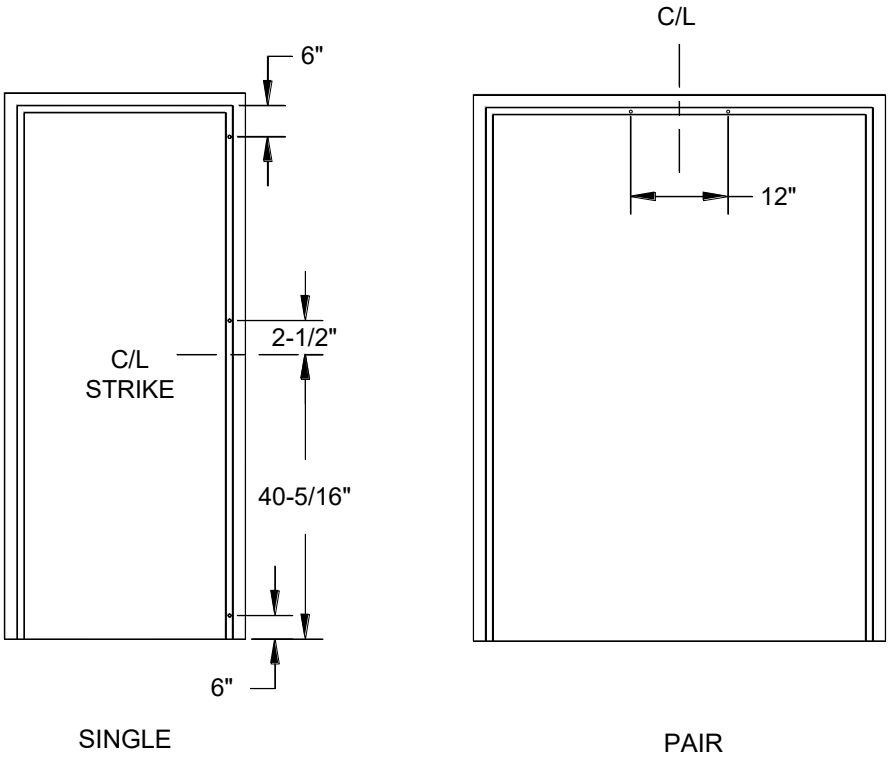
DOUBLE PARALLEL ARM
REINFORCEMENT

DOOR SILENCER PREPARATION

SILENCERS ARE PUNCHED
ONLY UPON REQUEST

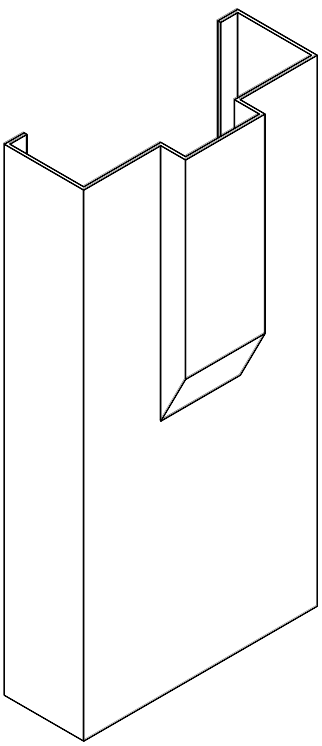
3 EA AT SINGLE SWING
2 EA AT PAIR SWING

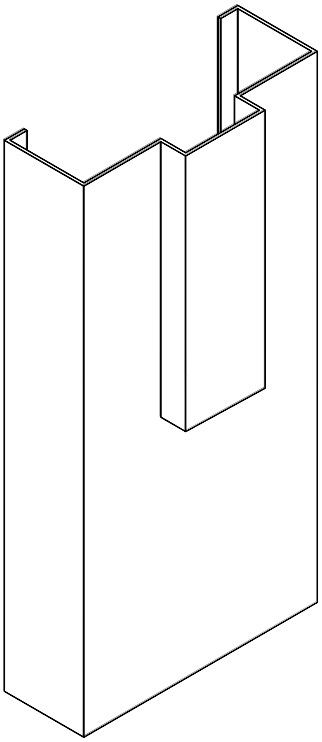


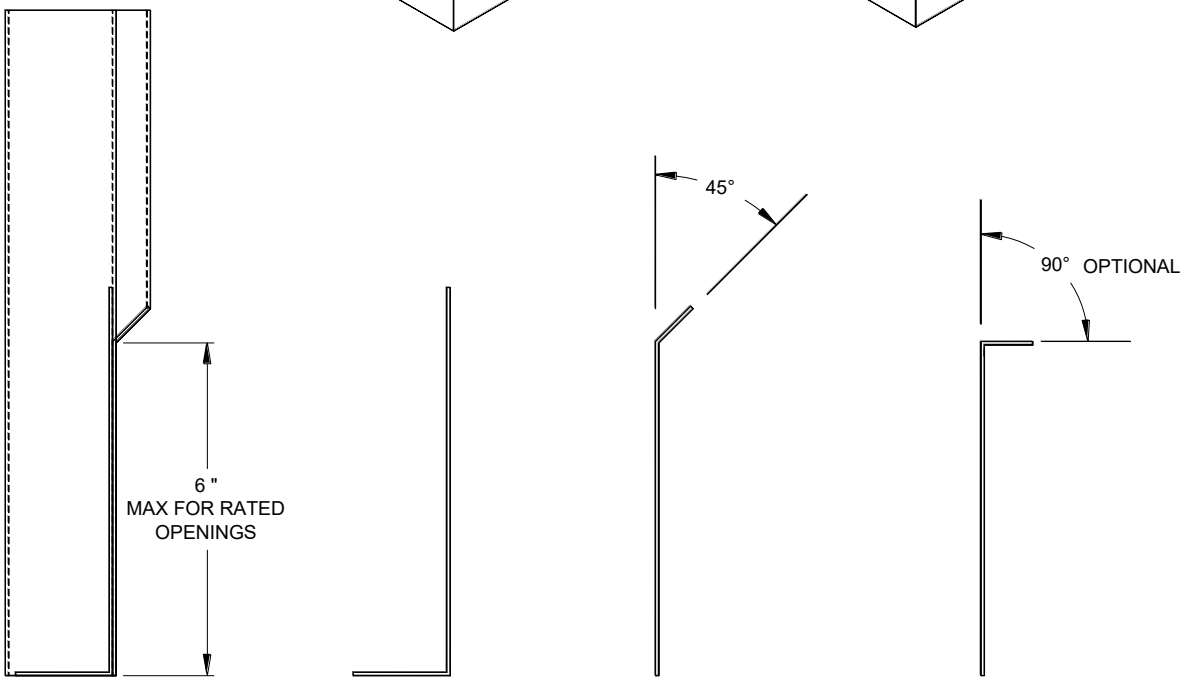


HOSPITAL STOP

GAUGE: 16
AVAILABLE FOR 45° OR 90°

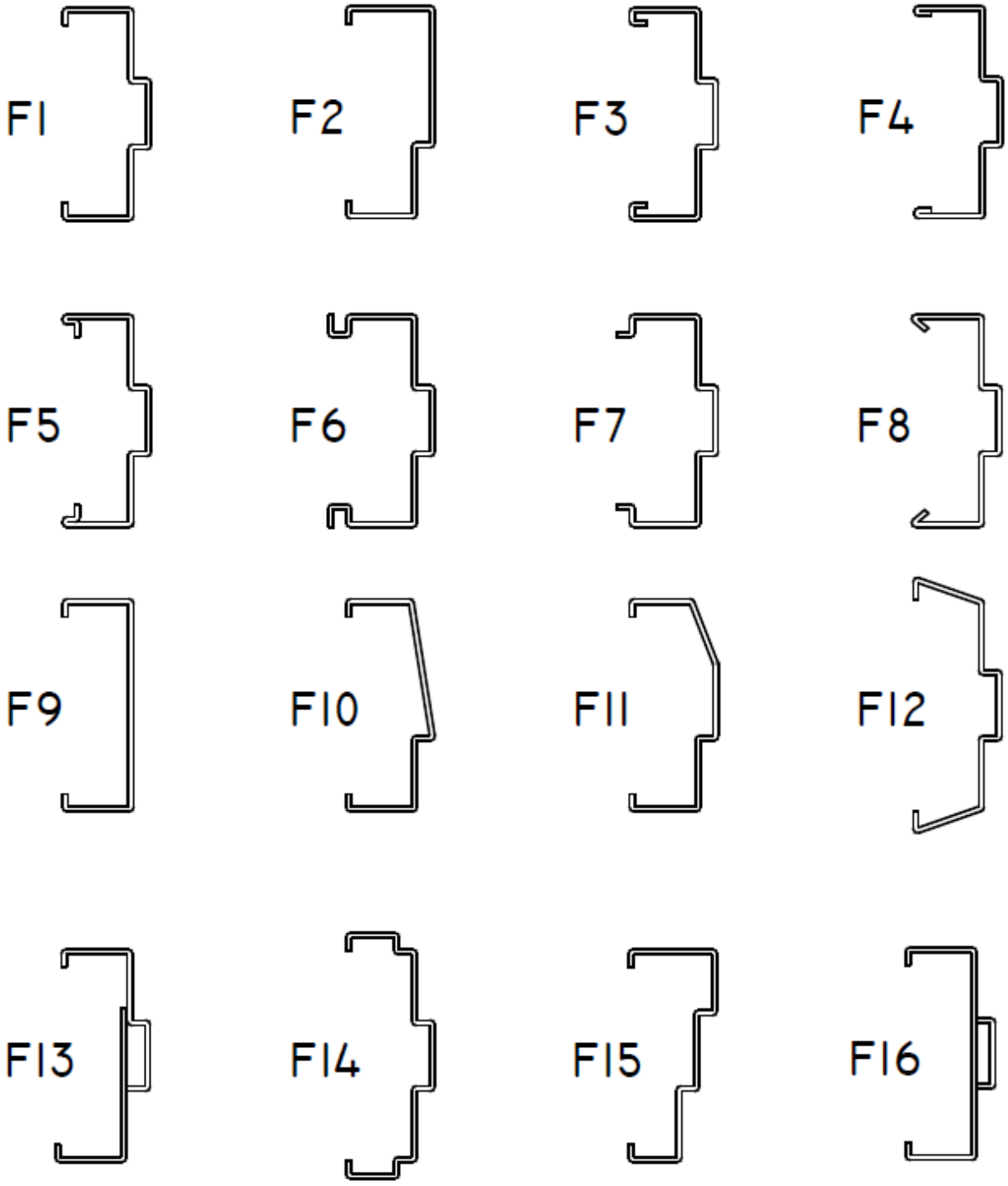






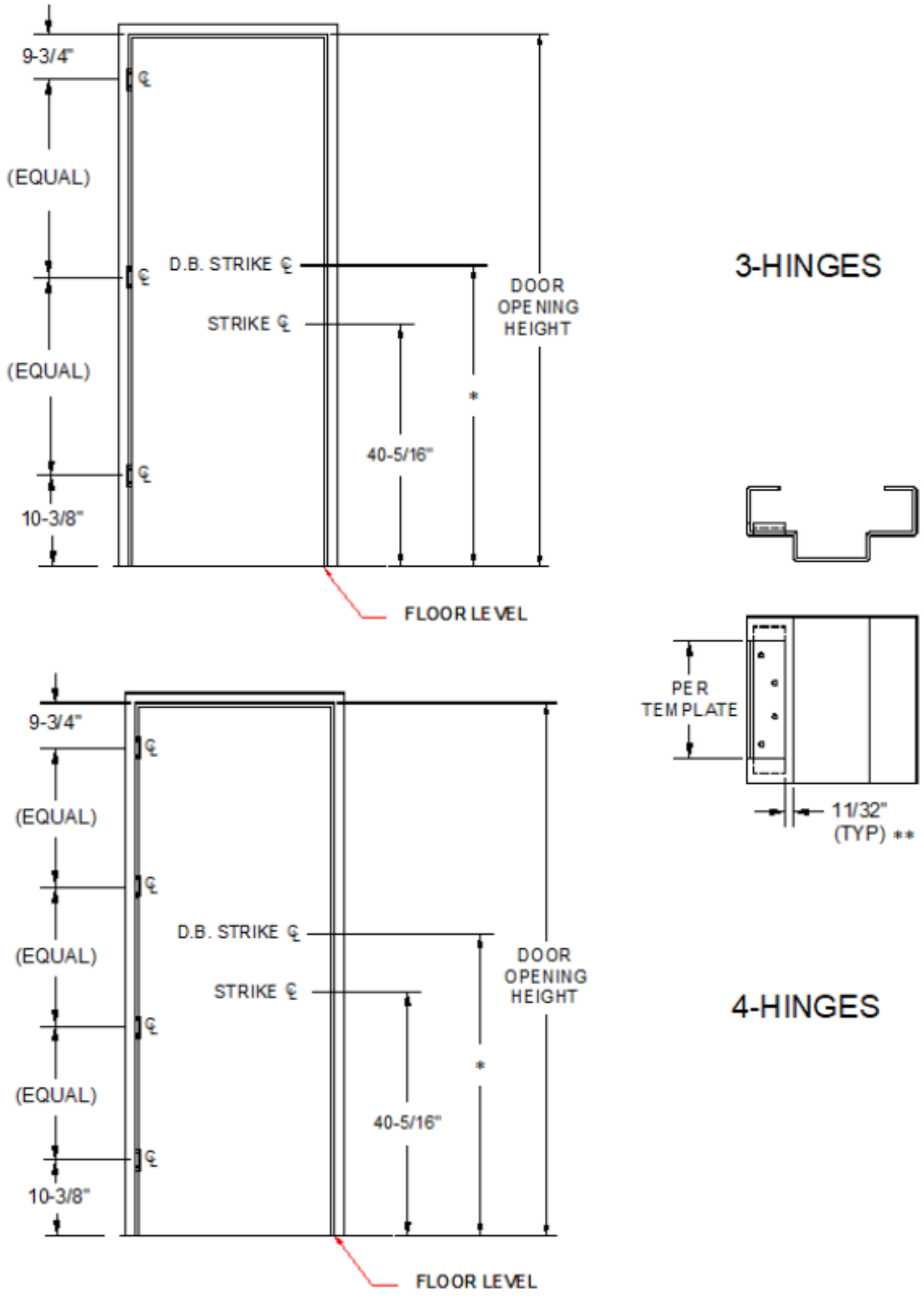
NOT PERMITTED FOR LEAD LINED FRAMES

CUSTOM FRAME PROFILES



FRAME PROFILES SHOWN MAY HAVE ANY BACKBEND STYLE OR COMBINATION OF BACKBEND STYLES

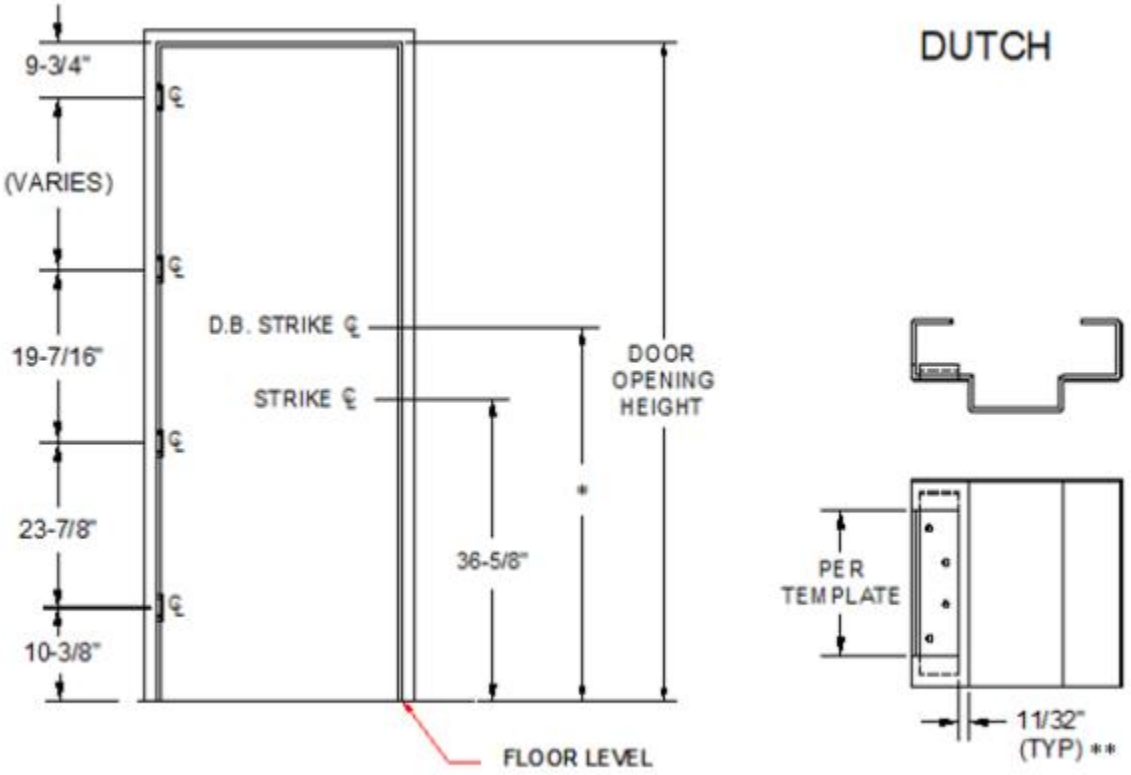
STANDARD FRAME HARDWARE LOCATIONS



* Due to variations in state and local codes, deadlock and deadlock strike locations must be provided to DCI.

** For frames receiving doors greater than 1-3/4" thickness, the hinge backset must be provided to DCI.

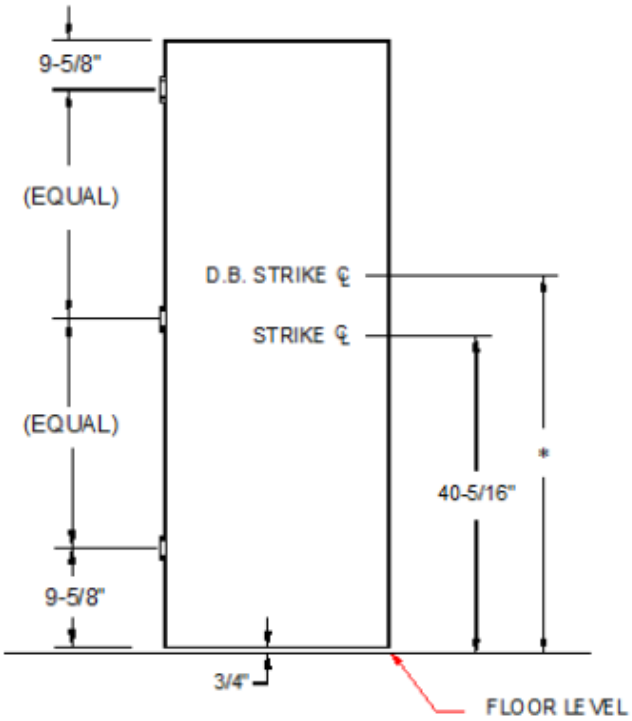
DUTCH FRAME HARDWARE LOCATIONS



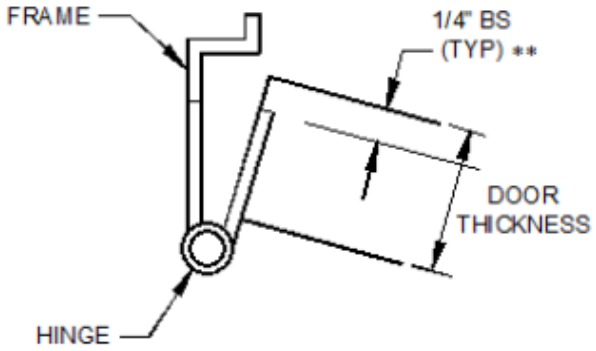
* Due to variations in state and local codes, deadlock and deadlock strike locations must be provided to DCI.

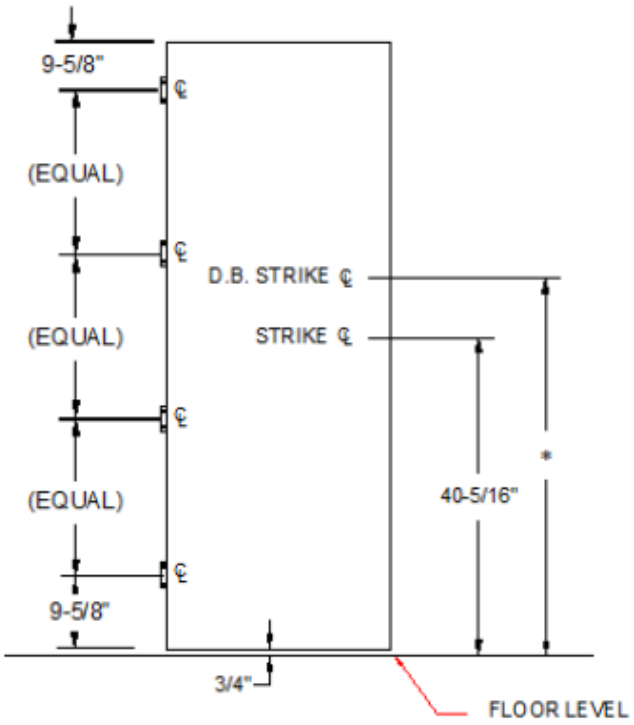
** For frames receiving doors greater than 1-3/4" thickness, the hinge backset must be provided to DCI.

STANDARD DOOR HARDWARE LOCATIONS



3-HINGES



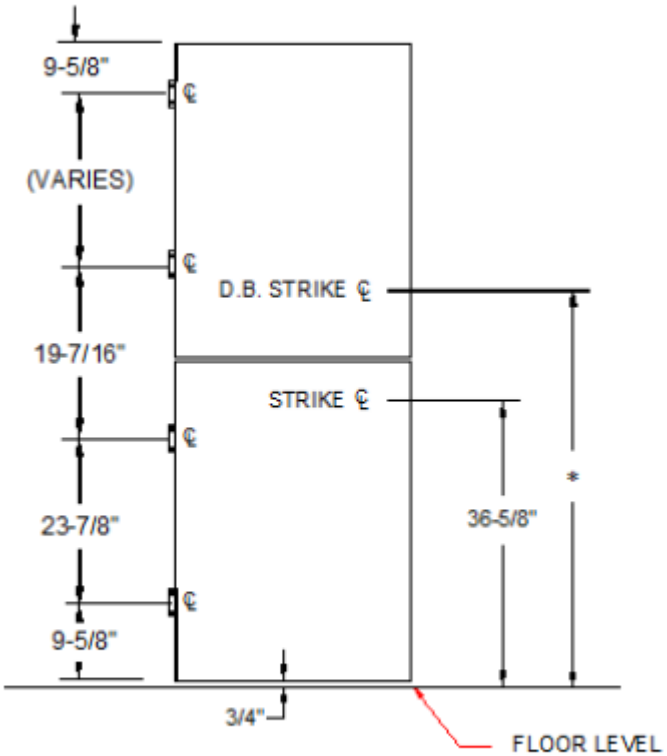


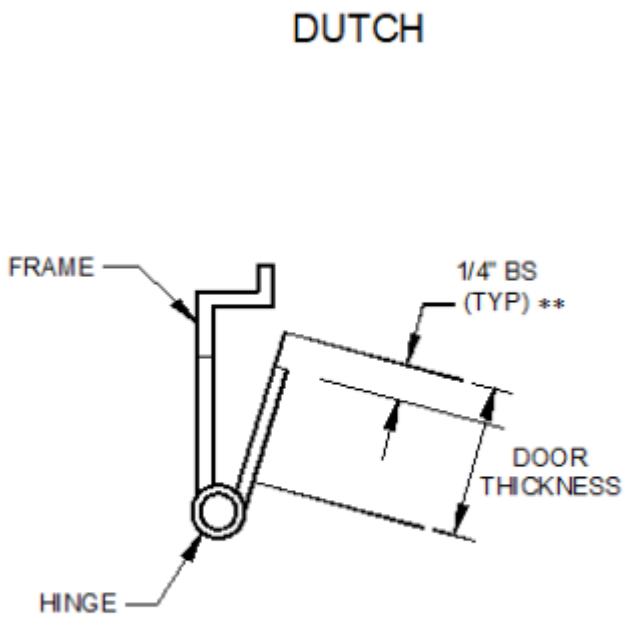
4-HINGES

* Due to variations in state and local codes, deadlock and deadlock strike locations must be provided to DCI.

** For frames receiving doors greater than 1-3/4" thickness, the hinge backset must be provided to DCI.

DUTCH DOOR HARDWARE LOCATIONS



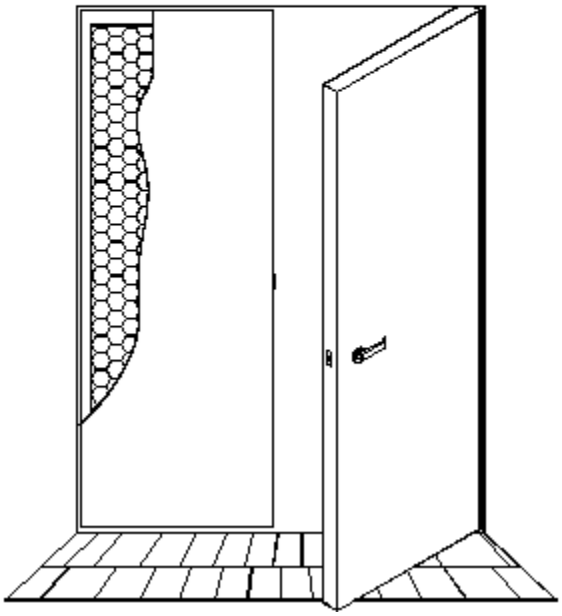


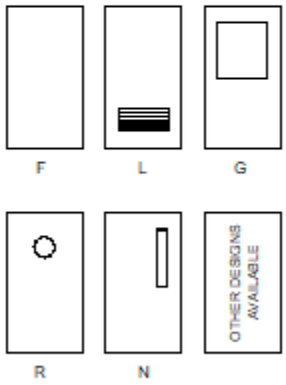
* Due to variations in state and local codes, deadlock and deadlock strike locations must be provided to DCI.

** For frames receiving doors greater than 1-3/4" thickness, the hinge backset must be provided to DCI.

HONEYCOMB CORE DOOR

STANDARD SPECIFICATION
MATERIAL: A40 GALVANNEAL
GAUGE: 18

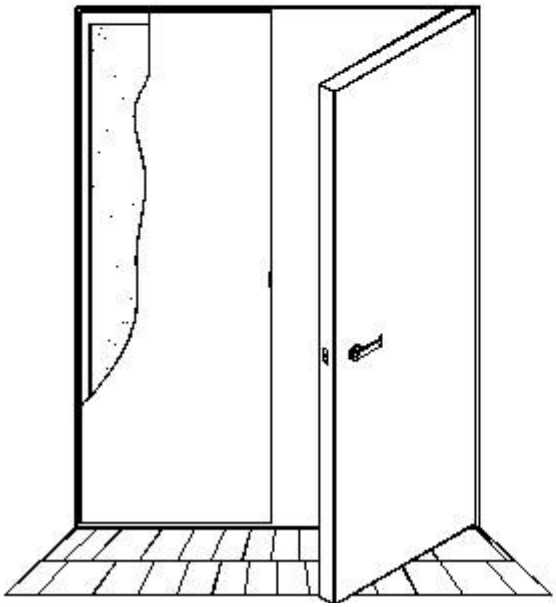


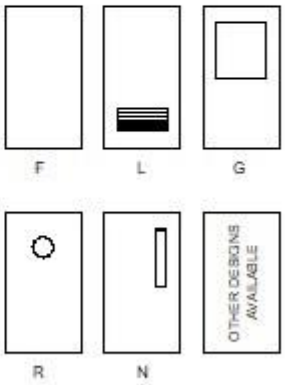


1.) 1-3/4" STANDARD DOOR THICKNESS. 1-3/8" THICKNESS AVAILABLE.
2.) A60 GALVANNEAL AND G90 GALVANIZED AVAILABLE.
3.) 16 AND 14 GAUGE AVAILABLE.
4.) STANDARD CORE: 1" HONEYCOMB HEX CELLS PERMANENTLY BONDED TO BOTH FACE SHEETS. IMPACT RESISTANT (HONEYCOMB IS CRUSH RESISTANT TO 45 PSI).
5.) RESIN IMPREGNATED HONEYCOMB OPTIONAL (CONSULT FACTORY FOR LEAD TIME)
6.) THERMAL PERFORMANCE FOR DOORS WITH STANDARD CORE TESTED PER ASTM C1363:
THERMAL TRANSMITTANCE U-FACTOR = 0.48
THERMAL RESISTANCE R-VALUE = 2.08
7.) 18 GA TOP AND BOTTOM CHANNELS SPOTWELDED TO FACE SHEETS. OTHER GAUGES AVAILABLE.
8.) CONTINUOUS WELD SEAMLESS EDGE STANDARD – VISIBLE EDGE SEAM AVAILABLE.
9.) HARDWARE REINFORCEMENTS:
3/16" FOR HINGES
12 GA FOR LOCKS
14 GA FOR CLOSER
10.) WHI LABELS AVAILABLE
11.) STC 33 PERFORMANCE TESTED PER ASTM E90. CONTACT FACTORY FOR DETAILS.

POLYSTYRENE CORE DOOR

STANDARD SPECIFICATION
MATERIAL: A40 GALVANNEAL
GAUGE: 18





1.) 1-3/4" STANDARD DOOR THICKNESS. 1-3/8" THICKNESS AVAILABLE (CONSULT FACTORY FOR LEAD TIME).
2.) A60 GALVANNEAL AND G90 GALVANIZED AVAILABLE.
3.) 16 AND 14 GAUGE AVAILABLE.
4.) STANDARD CORE: 0.7 LB DENSITY POLYSTYRENE CORE PERMANENTLY BONDED TO BOTH FACE SHEETS.
5.) 1.0 LB DENSITY CORE AVAILABLE (CONSULT FACTORY FOR LEAD TIME). OPTIONAL CORE IS NOT ELIGIBLE FOR FIRE LABELS.
6.) THERMAL PERFORMANCE FOR DOORS WITH STANDARD 0.7 LB DENSITY POLYSTYRENE CORE TESTED PER ASTM C1363:

THERMAL TRANSMITTANCE	U-FACTOR =	0.42
THERMAL RESISTANCE	R-VALUE =	2.39

7.) THERMAL PERFORMANCE FOR DOORS WITH OPTIONAL 1.0 LB DENSITY POLYSTYRENE CORE TESTED PER ASTM C1363:

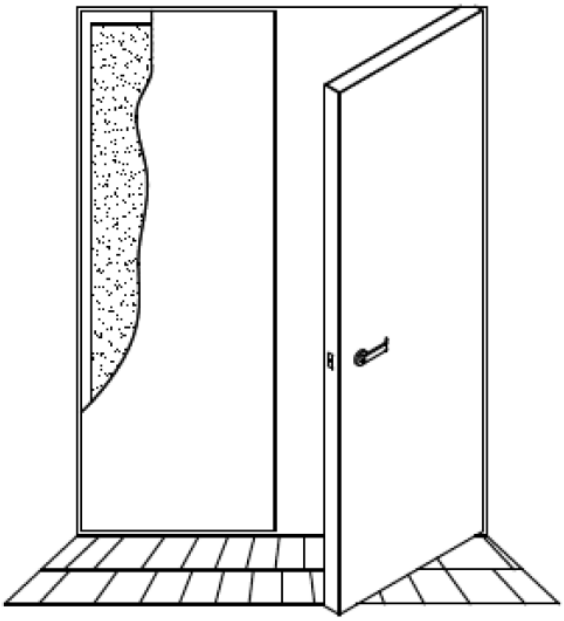
THERMAL TRANSMITTANCE	U-FACTOR =	0.39
THERMAL RESISTANCE	R-VALUE =	2.57

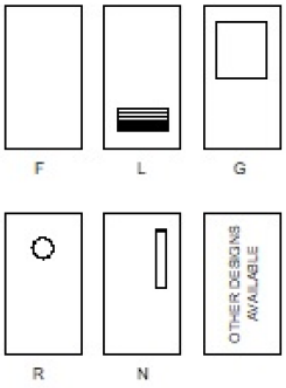
8.) 18 GA TOP AND BOTTOM CHANNELS SPOTWELDED TO FACE SHEETS. OTHER GAUGES AVAILABLE.
9.) CONTINUOUS WELD SEAMLESS EDGE STANDARD – VISIBLE EDGE SEAM AVAILABLE.
10.) HARDWARE REINFORCEMENTS:
- 3/16" FOR HINGES
- 12 GA FOR LOCKS
- 14 GA FOR CLOSER

11.) WHI LABELS AVAILABLE
12.) STC 28 PERFORMANCE TESTED PER ASTM E90. CONTACT FACTORY FOR DETAILS.

POLYURETHANE CORE DOOR

STANDARD SPECIFICATION
MATERIAL: A40 GALVANNEAL
GAUGE: 18

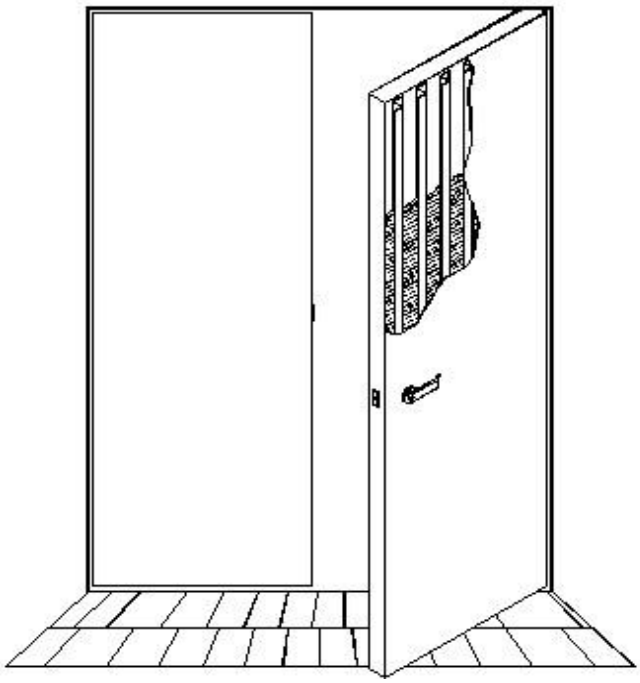


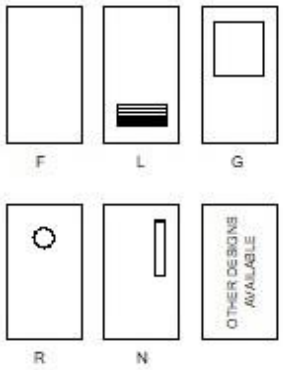


1.) 1-3/4" STANDARD DOOR THICKNESS.
2.) A60 GALVANNEAL AND G90 GALVANIZED AVAILABLE.
3.) 16 AND 14 GAUGE AVAILABLE.
4.) 2 LB DENSITY POLYURETHANE CORE PERMANENTLY BONDED TO BOTH FACE SHEETS.
5.) THERMAL PERFORMANCE FOR DOORS WITH STANDARD CORE TESTED PER
ASTM C1363: THERMAL TRANSMITTANCE U-FACTOR = 0.36
THERMAL RESISTANCE R-VALUE = 2.77
6.) 18 GA TOP AND BOTTOM CHANNELS SPOTWELDED TO FACE SHEETS. OTHER GAUGES AVAILABLE.
7.) CONTINUOUS WELD SEAMLESS EDGE STANDARD – VISIBLE EDGE SEAM AVAILABLE.
8.) HARDWARE REINFORCEMENTS:
3/16" FOR HINGES
12 GA FOR LOCKS
14 GA FOR CLOSER
9.) NOT AVAILABLE AS A FIRE RATED PRODUCT

STANDARD STEEL-STIFFENED CORE DOOR

STANDARD SPECIFICATION
MATERIAL: A40 GALVANNEAL
GAUGE: 18

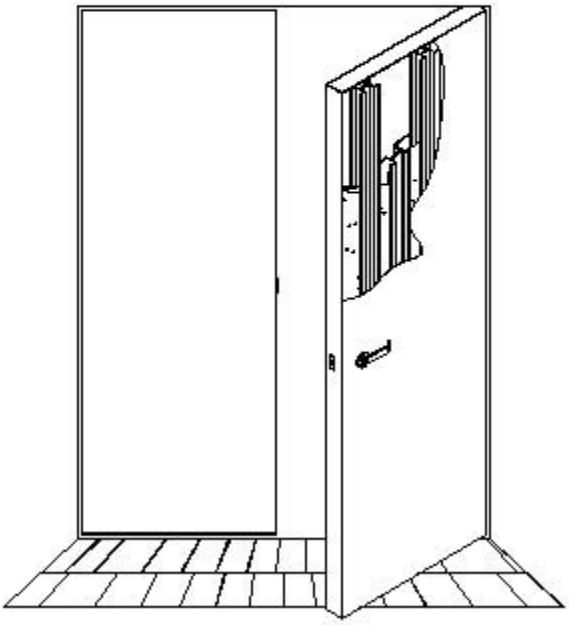


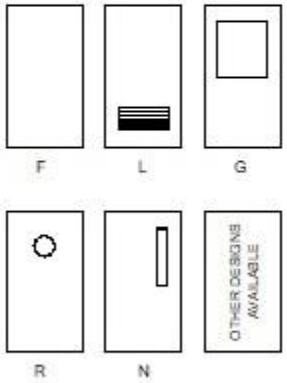


1.) 1-3/4" STANDARD DOOR THICKNESS. OTHER THICKNESSES AVAILABLE.
2.) A60 GALVANNEAL AND G90 GALVANIZED AVAILABLE.
3.) 16, 14, 12, AND 10 GAUGE AVAILABLE.
4.) 20 GA Z-CHANNEL STIFFENERS 6" ON CENTER WITH FIBERGLASS INSULATION BETWEEN (OTHER GAUGES AVAILABLE).
5.) THERMAL PERFORMANCE FOR DOORS WITH STANDARD CORE TESTED PER ASTM C1363:
THERMAL TRANSMITTANCE U-FACTOR = 0.53
THERMAL RESISTANCE R-VALUE = 1.90
6.) 18 GA TOP AND BOTTOM CHANNELS SPOTWELDED TO FACE SHEETS. OTHER GAUGES AVAILABLE.
7.) CONTINUOUS WELD SEAMLESS EDGE STANDARD.
8.) HARDWARE REINFORCEMENTS:
3/16" FOR HINGES
12 GA FOR LOCKS
14 GA FOR CLOSER
9.) WHI LABELS AVAILABLE
10.) STC 35 PERFORMANCE TESTED PER ASTM E90. CONTACT FACTORY FOR DETAILS.

HAT-STYLE STEEL-STIFFENED CORE DOOR

STANDARD SPECIFICATION
MATERIAL: A40 GALVANNEAL
GAUGE: 18

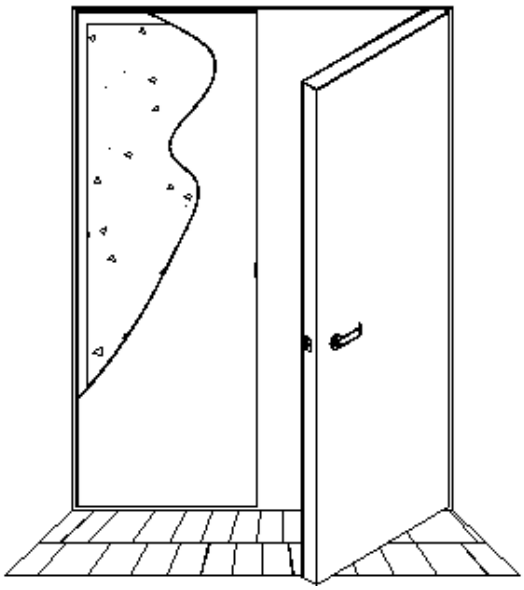


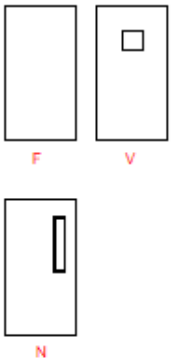


1.) 1-3/4" STANDARD DOOR THICKNESS. OTHER THICKNESSES AVAILABLE.
2.) A60 GALVANNEAL AND G90 GALVANIZED AVAILABLE.
3.) 16, 14, 12, AND 10 GAUGE AVAILABLE.
4.) 20 GA HAT-STYLE STIFFENERS 6" ON CENTER WITH FIBERGLASS INSULATION BETWEEN (OTHER GAUGES AVAILABLE).
5.) 18 GA TOP AND BOTTOM CHANNELS SPOTWELDED TO FACE SHEETS. OTHER GAUGES AVAILABLE.
6.) CONTINUOUS WELD SEAMLESS EDGE STANDARD.
7.) HARDWARE REINFORCEMENTS:
    3/16" FOR HINGES
    12 GA FOR LOCKS
    14 GA FOR CLOSER
8.) STANDARD CORE USED FOR LEAD-LINED DOORS (NO INTEGRAL LITES, LOUVERS, CVR, MORTISED DOOR BOTTOMS, CONCEALED CLOSERS, RECESSED PANIC DEVICES)
9.) WHI LABELS AVAILABLE

TEMPERATURE RISE CORE DOOR

STANDARD SPECIFICATION
MATERIAL: A40 GALVANNEAL
GAUGE: 18



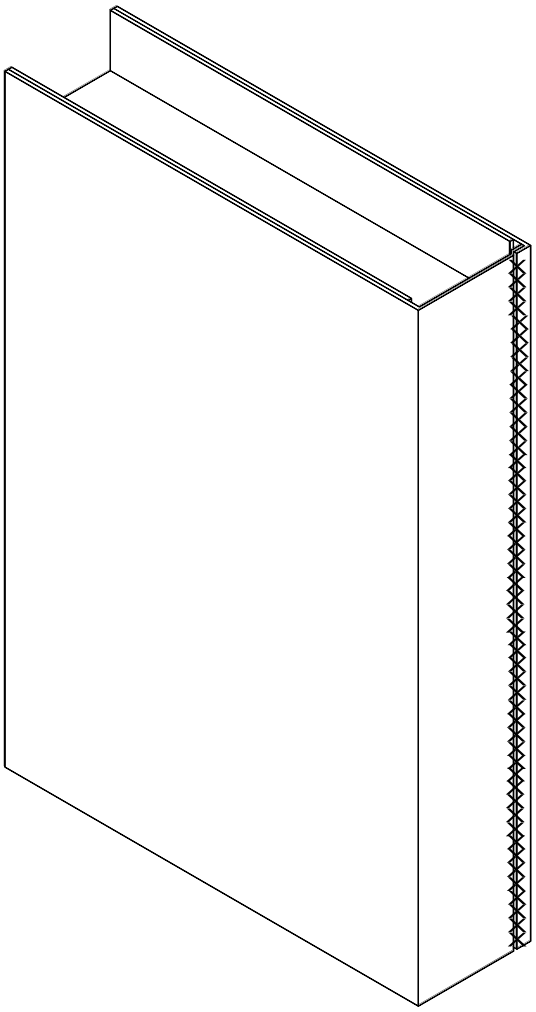


1.) 1-3/4" STANDARD DOOR THICKNESS.
2.) A60 GALVANNEAL AND G90 GALVANIZED AVAILABLE.
3.) 16 AND 14 GAUGE AVAILABLE.
4.) PROPRIETARY TEMP RISE CORE PERMANENTLY BONDED TO BOTH FACE SHEETS (APPLIES TO BOTH THE STANDARD TRG AND OPTIONAL TRP TEMP RISE CORES).
5.) OPTIONAL TRS TEMP RISE CORE INCLUDES 20 GA STIFFENERS 6" ON CENTER WITH INSULATION BETWEEN (OTHER GAUGES AVAILABLE).
6.) THERMAL PERFORMANCE FOR DOORS WITH STANDARD TRG CORE TESTED PER ASTM C1363:
THERMAL TRANSMITTANCE U-FACTOR = 0.53 THERMAL
RESISTANCE R-VALUE = 1.88
7.) THERMAL PERFORMANCE FOR DOORS WITH OPTIONAL TRP CORE TESTED PER ASTM C1363:
THERMAL TRANSMITTANCE U-FACTOR = 0.49 THERMAL
RESISTANCE R-VALUE = 2.03
8.) THERMAL PERFORMANCE FOR DOORS WITH OPTIONAL TRS CORE TESTED PER ASTM C1363:
THERMAL TRANSMITTANCE U-FACTOR = 0.49 THERMAL
RESISTANCE R-VALUE = 2.04
9.) 18 GA TOP AND BOTTOM CHANNELS SPOTWELDED TO FACE SHEETS. OTHER GAUGES AVAILABLE.
10.) CONTINUOUS WELD SEAMLESS EDGE STANDARD.
11.) HARDWARE REINFORCEMENTS:
3/16" FOR HINGES
12 GA FOR LOCKS
14 GA FOR CLOSER
12.) 60-180 MIN WHI LABELS AVAILABLE FOR 450° AND 250° TEMP RISE AT 30 MIN.
LITE KITS (UP TO 100 SQ INCHES OF VISIBLE GLASS) AVAILABLE FOR UP TO 90 MIN RATED DOORS

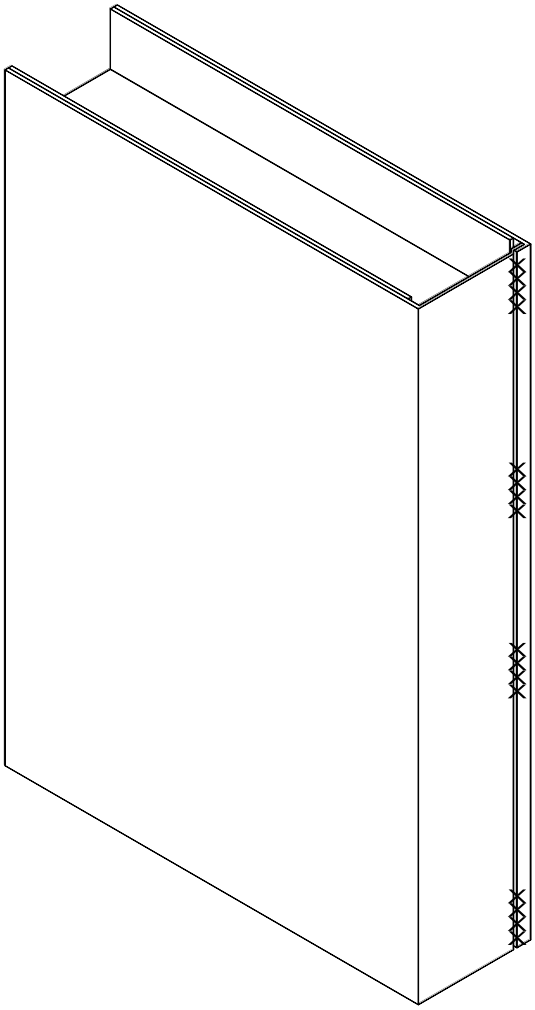
STANDARD DOOR EDGE SEAM



PROVIDED WITH BEVELED EDGES AT BOTH SIDES (STANDARD)
BEVELED AND SQUARE EDGES ARE AVAILABLE IN ANY COMBINATION



CONTINUOUSLY WELDED EDGE SEAM
DRESSED SMOOTH
(STANDARD)

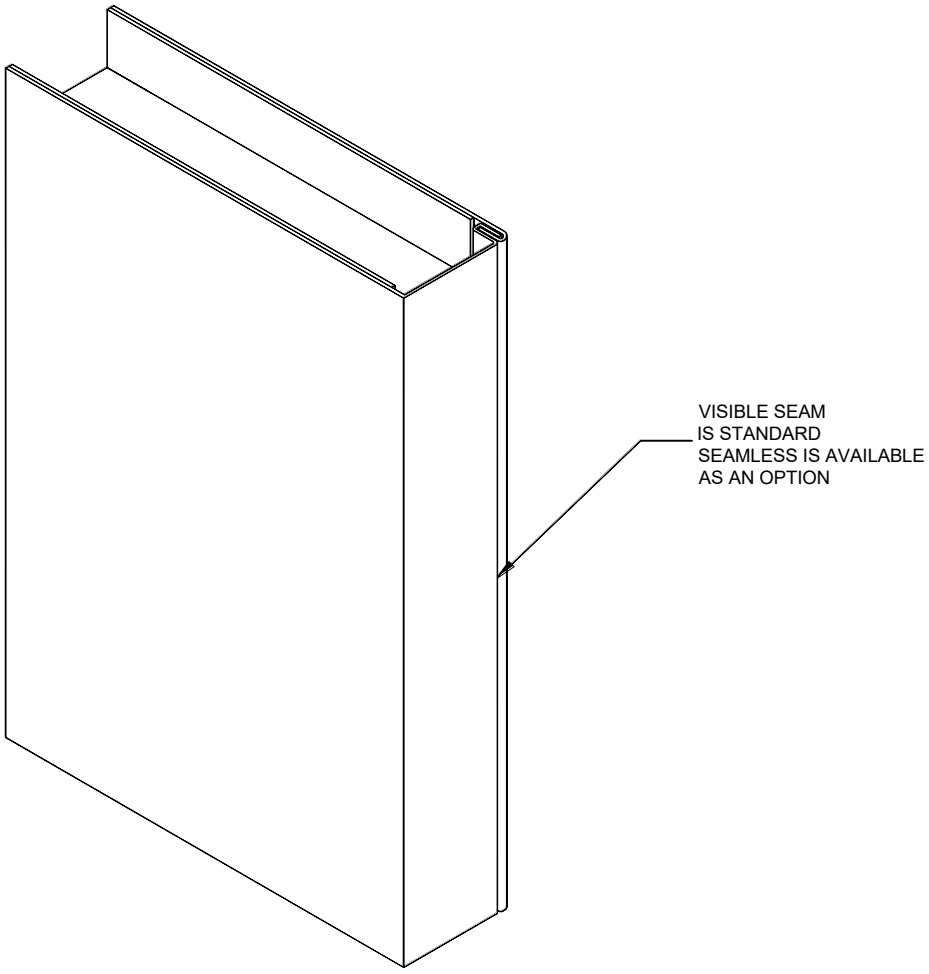


STITCH WELDED AT STRESS POINTS
DRESSED AND GROUND SMOOTH
(OPTIONAL)

VES DOOR EDGE SEAM

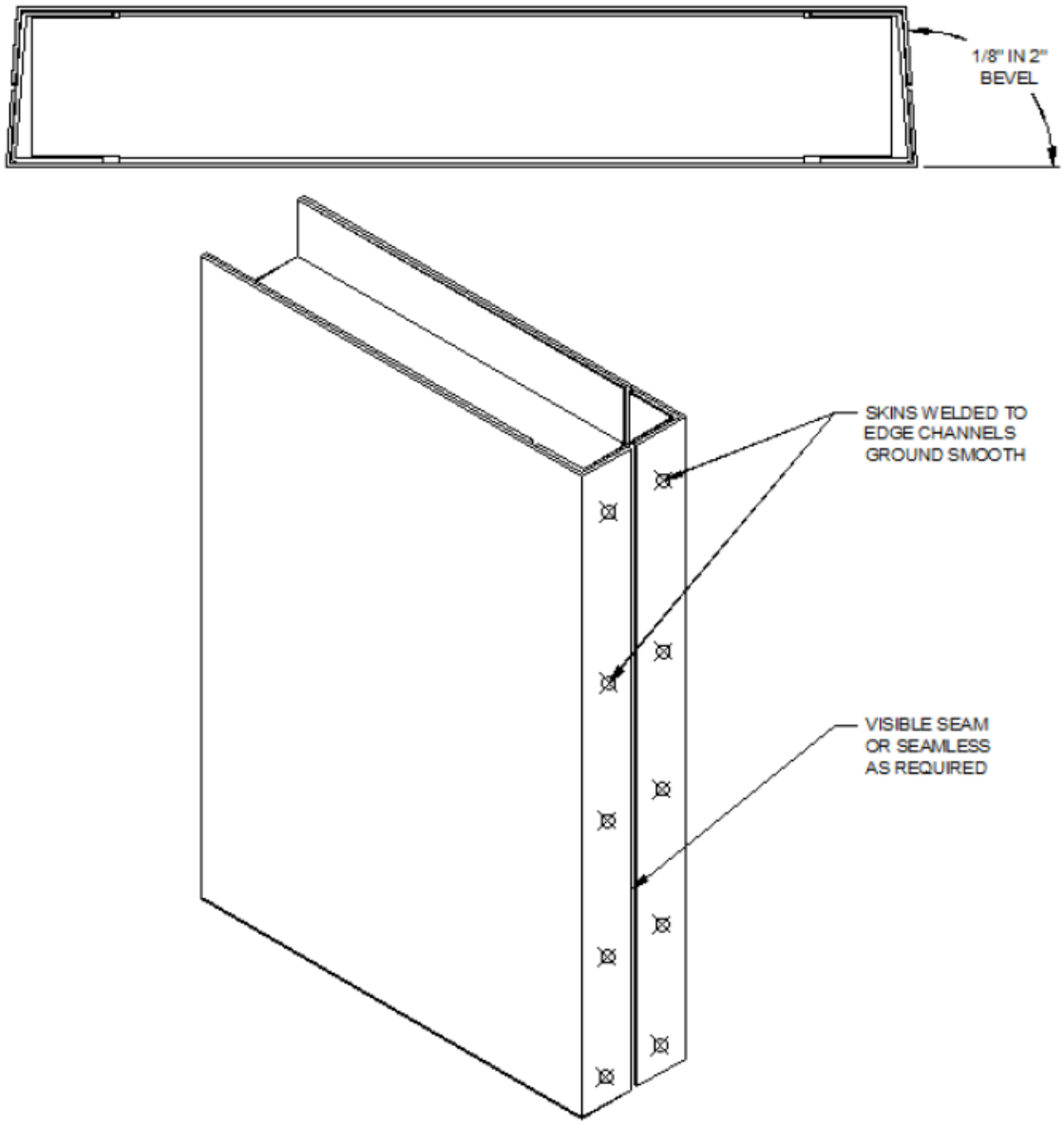


PROVIDED WITH BEVELED EDGES AT BOTH SIDES (STANDARD)
SQUARE EDGES AT BOTH SIDES IS OPTIONAL



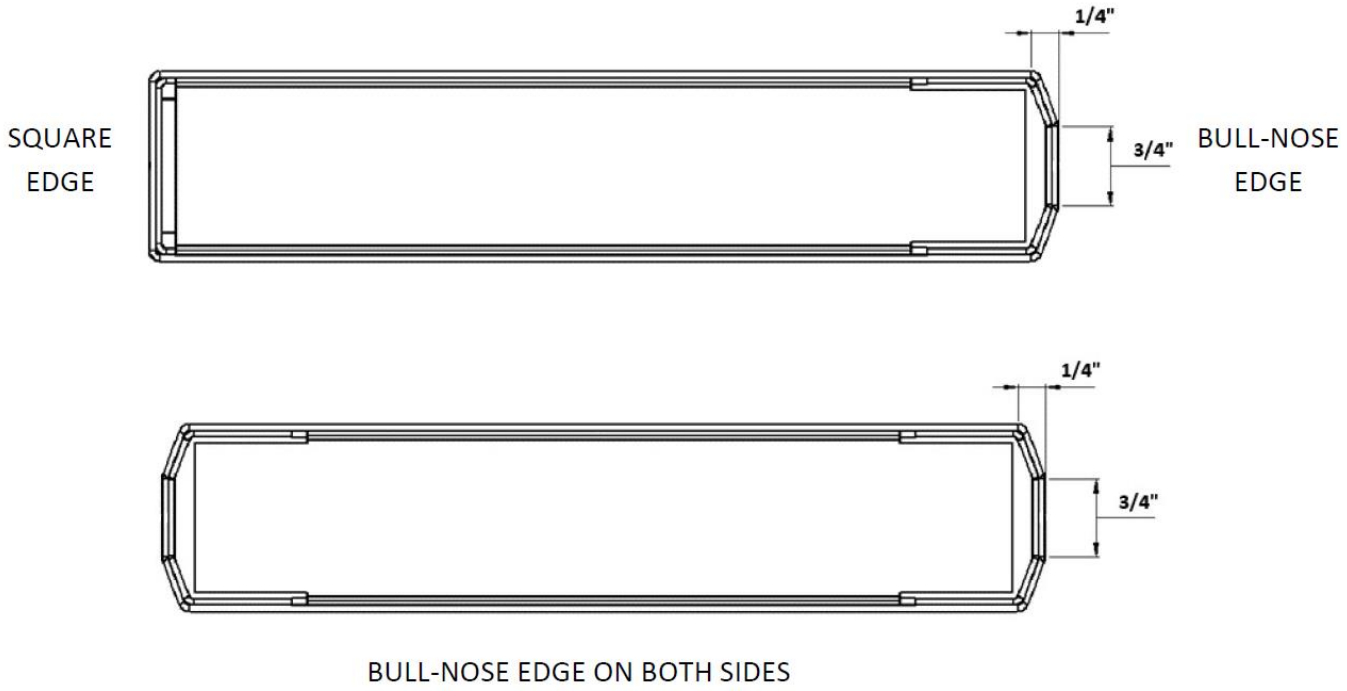
VES DOORS ARE ONLY AVAILABLE WITH 18 GAUGE FACE SHEETS

R-SERIES DOOR EDGE SEAM



1.) CONTINUOUSLY WELDED EDGE SEAM DRESSED SMOOTH STANDARD
2.) BEVELED, BULL-NOSED, AND SQUARE EDGE ARE AVAILABLE IN ANY COMBINATION
3.) INTERNAL EDGE CHANNEL 18 GA STANDARD (UP TO 10 GA AVAILABLE)

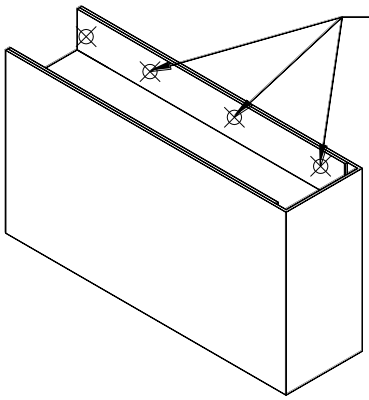
BULL-NOSE DOOR EDGE



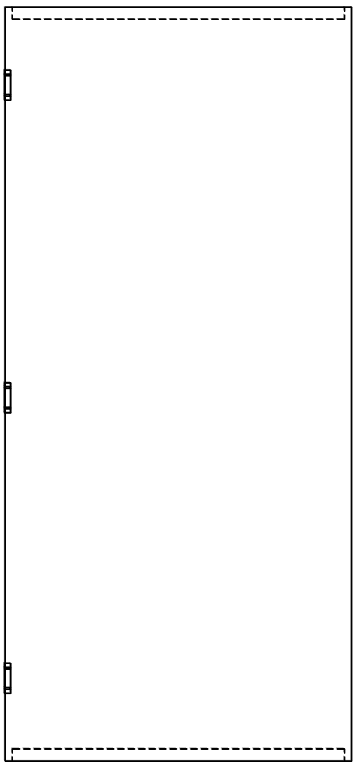
BULL-NOSE EDGE ON BOTH SIDES

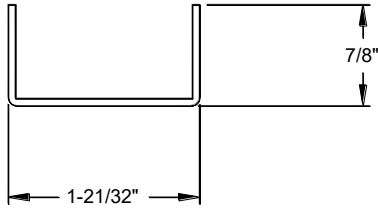
1.) BEVELED, BULL-NOSED, AND SQUARE EDGE ARE AVAILABLE IN ANY COMBINATION
2.) AVAILABLE ON LOCK AND/OR HINGE EDGE
3.) NOT AVAILABLE ON FIRE RATED DOORS

STANDARD TOP AND BOTTOM CHANNEL



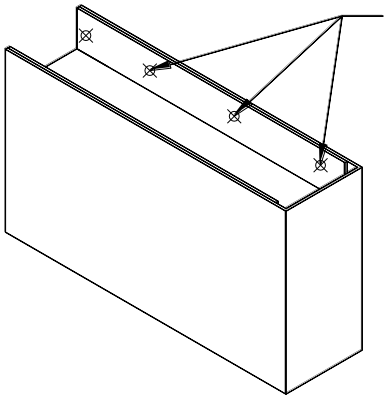
TOP AND BOTTOM CHANNELS SPOT WELDED
2" FROM ENDS 4" ON CENTER TO BOTH SKINS



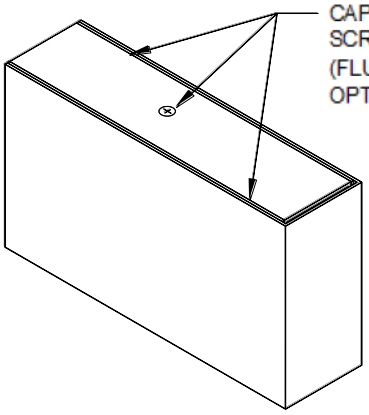


TOP AND BOTTOM CHANNELS
ARE 18 GA STANDARD
(OTHER GAUGES AVAILABLE
AS AN OPTION)

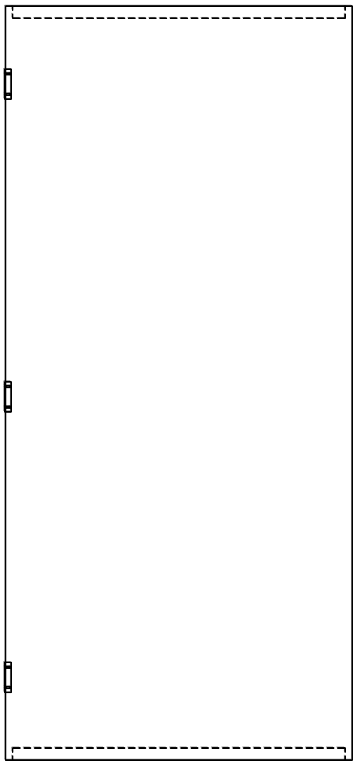
CAP TOP FLUSH

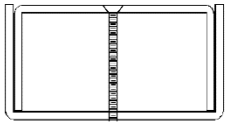


TOP AND BOTTOM CHANNELS SPOT WELDED 2" FROM ENDS 4" ON CENTER TO BOTH SKINS



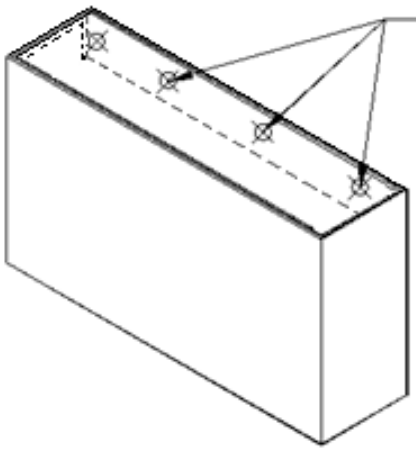
CAPPING CHANNEL SCREWED-IN (FLUSH AND SEALED OPTIONAL)





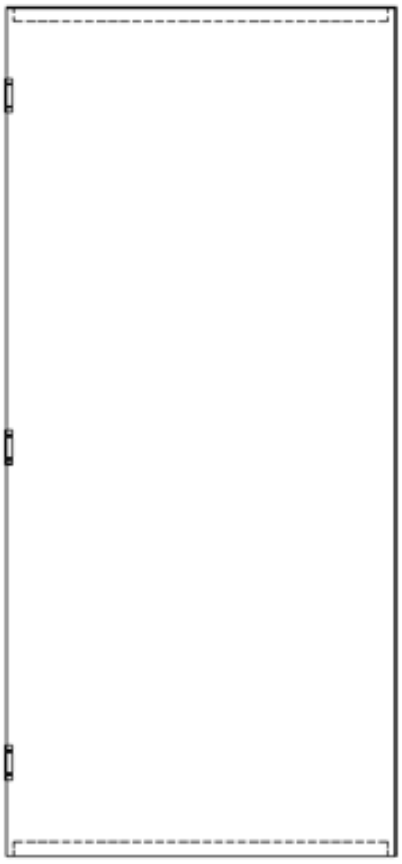
RECESSED AND CAPPING CHANNELS ARE 18 GA STANDARD (OTHER GAUGES AVAILABLE AS AN OPTION)

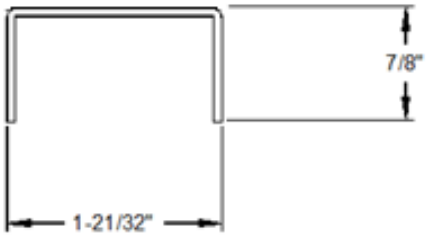
REVERSED TOP AND BOTTOM CHANNEL



TOP AND BOTTOM CHANNELS SPOT WELDED
2" FROM ENDS 4" ON CENTER TO BOTH SKINS

SAME AS STANDARD CHANNELS BUT FLIPPED
UPSIDE DOWN FOR FULLER CORE COVERAGE

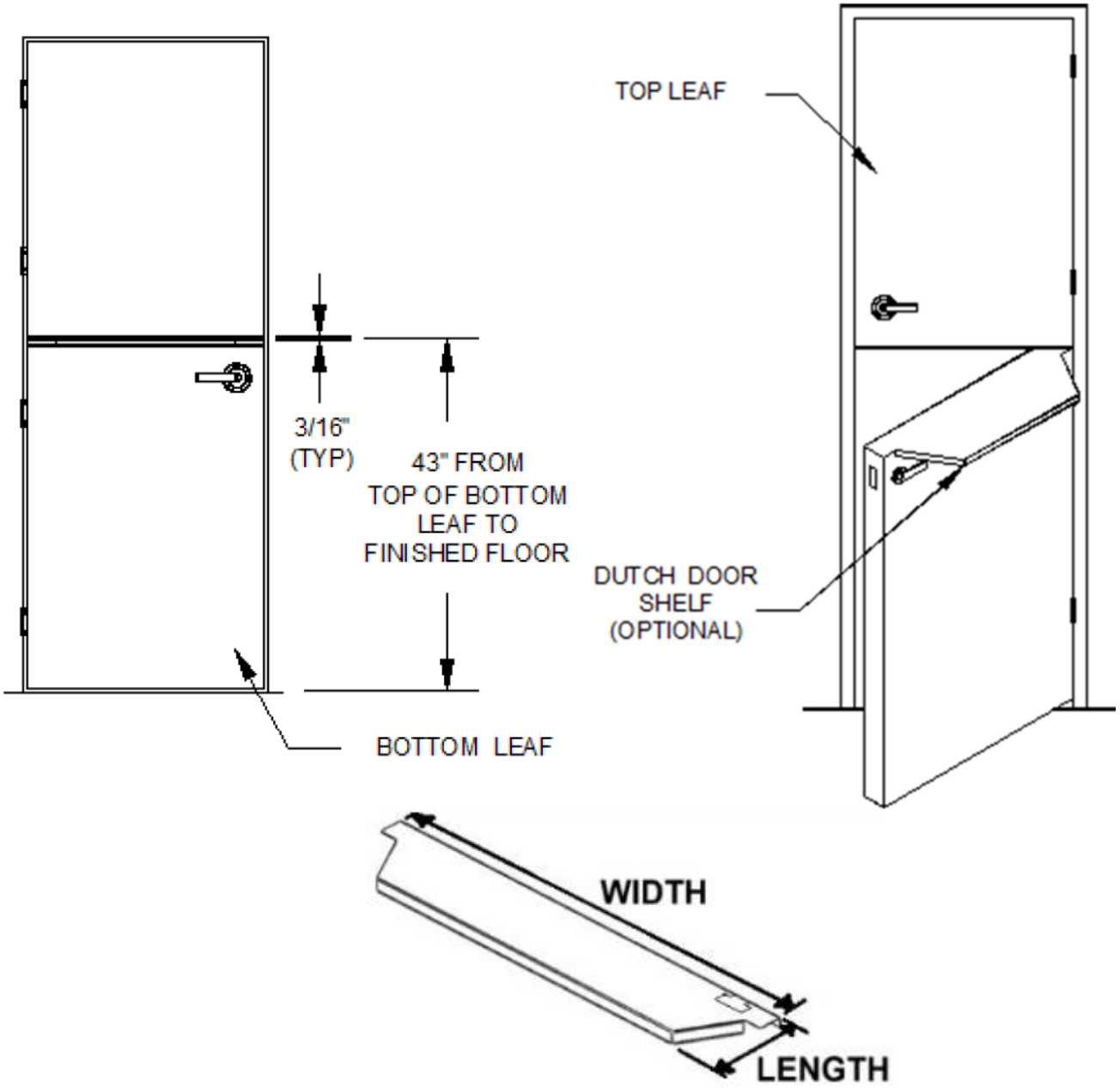




TOP AND BOTTOM CHANNELS
ARE 18 GA STANDARD
(OTHER GAUGES AVAILABLE
AS AN OPTION)

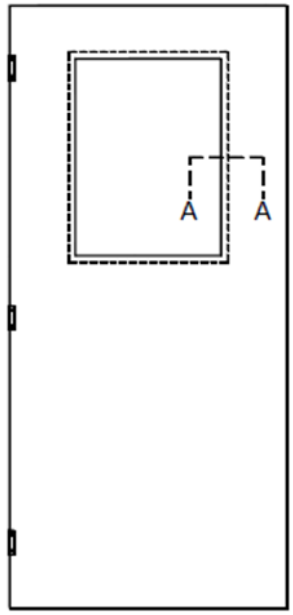
4.) REVERSED CHANNELS USED FOR LEAD-LINED, BULLET RESISTANT, AND STC DOORS TO ALLOW FOR FULLER CORE COVERAGE

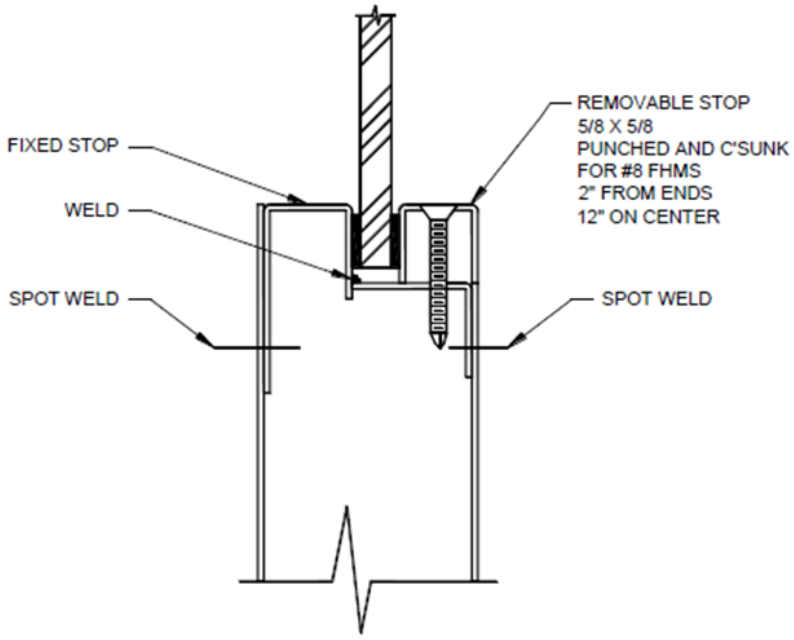
DUTCH DOOR



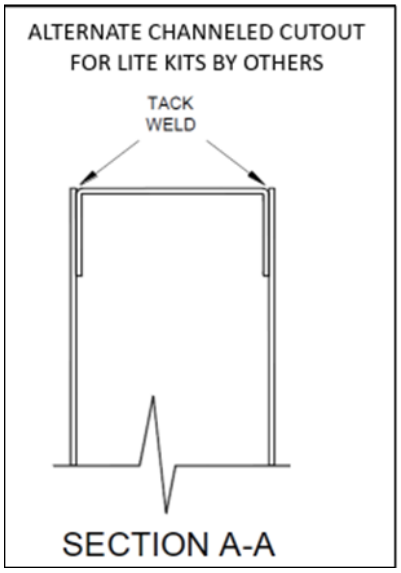
1.) 1-3/4" DOOR STANDARD. OTHER THICKNESSES AVAILABLE
2.) CAP TOP CHANNEL OF BOTTOM LEAF.
3.) ASTRAGAL OPTIONAL (REQUIRED WHEN FIRE RATED)
4.) STRIKE PREP IN TOP OF BOTTOM LEAF AND/OR SHELF
5.) STANDARD DUTCH DOOR SHELF: LENGTH: 9.75" (DOOR THICKNESS PLUS 8"), WIDTH: TO MATCH DOOR WIDTH. BRACKETS INCLUDED. ON ONE SIDE OF DOOR.
6.) CUSTOM DUTCH DOOR SHELF WIDTH/LENGTH AVAILABLE; SHELF CAN BE ON ONE SIDE OR BOTH SIDES (CAN ONLY BE ON NARROW/PUSH SIDE WHEN FIRE RATED)

INTEGRAL LITE

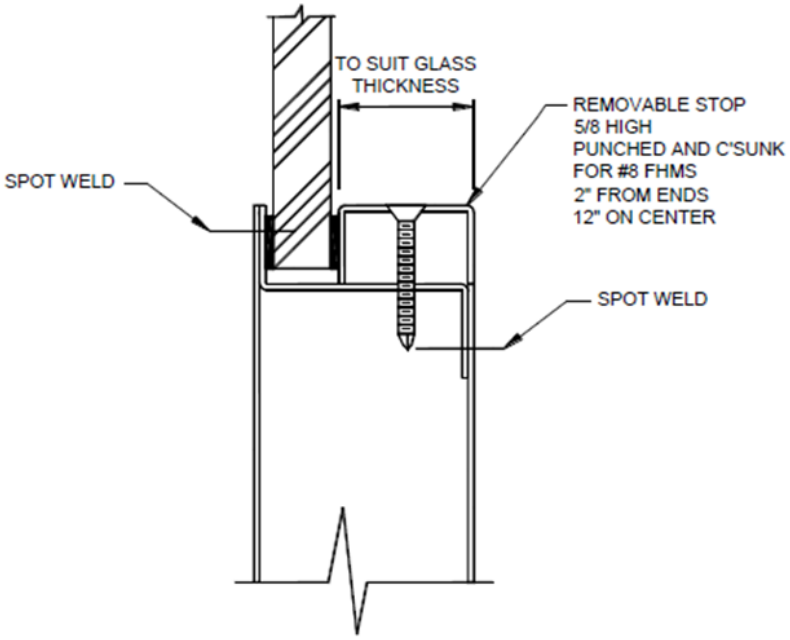




SECTION A-A
(STANDARD)



SECTION A-A

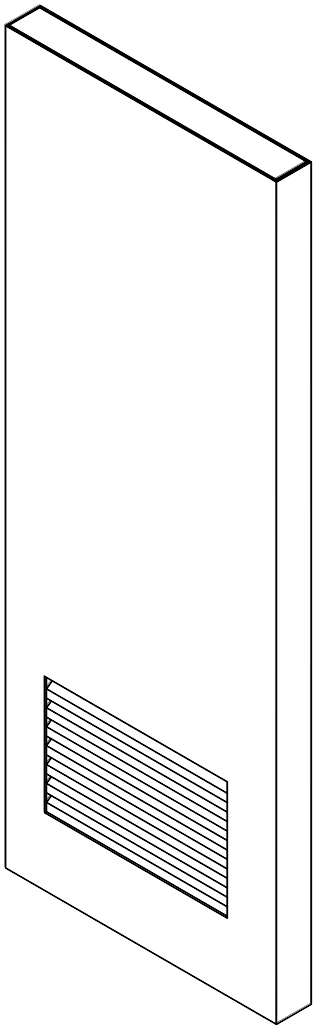


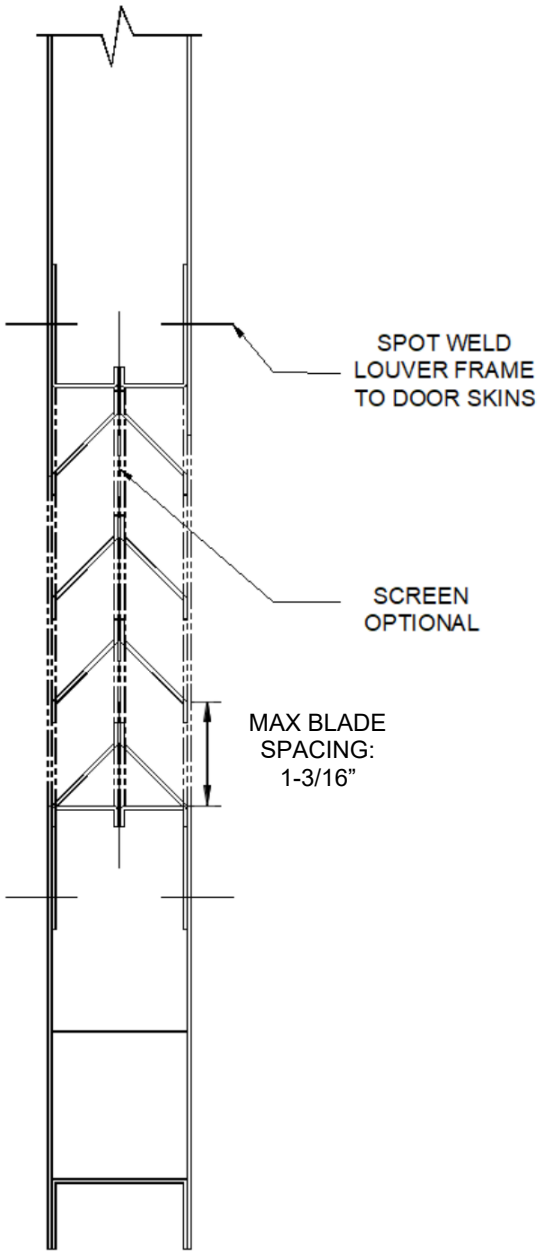
SECTION A-A
(OPTIONAL)

1.) 18 GA GLASS STOP (OTHER GAUGES AVAILABLE REMOVABLE STOP IS PUNCHED AND COUNTERSUNK 2" FROM ENDS AND 12" ON CENTER (MIN 2 PER SIDE)
2.) GLASS GAP 1/8" LARGER THAN GLASS TO ACCOMMODATE GLAZING TAPE

INTEGRAL DOOR LOUVER

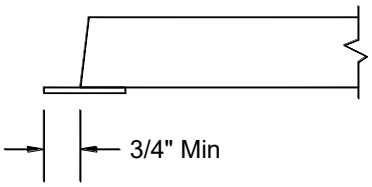
REINFORCEMENTS: GALVANNEAL
GAUGE: 18
SPOTWELDED TO DOOR SKIN
FREE AIR FLOW: 50% FREE AREA

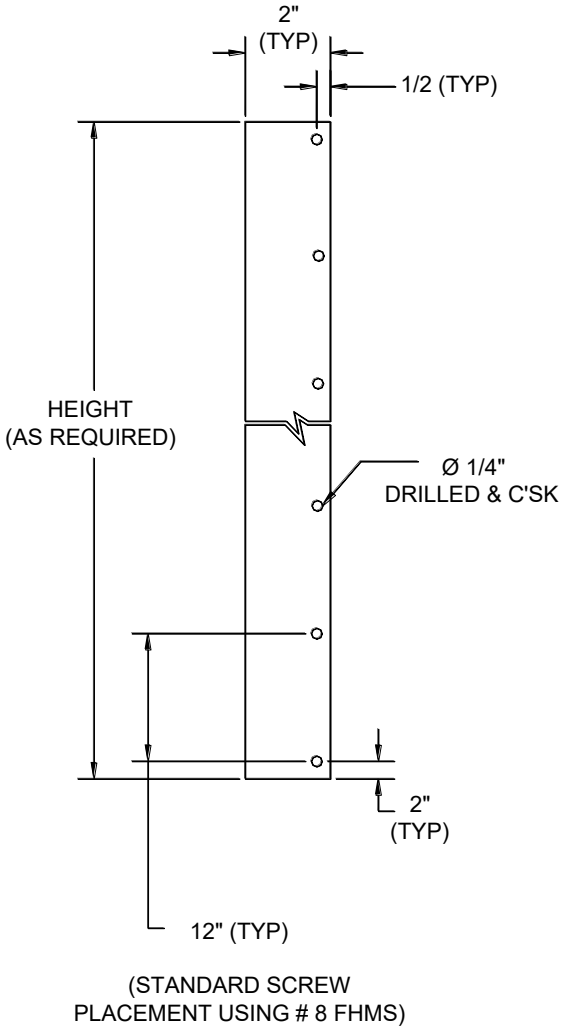




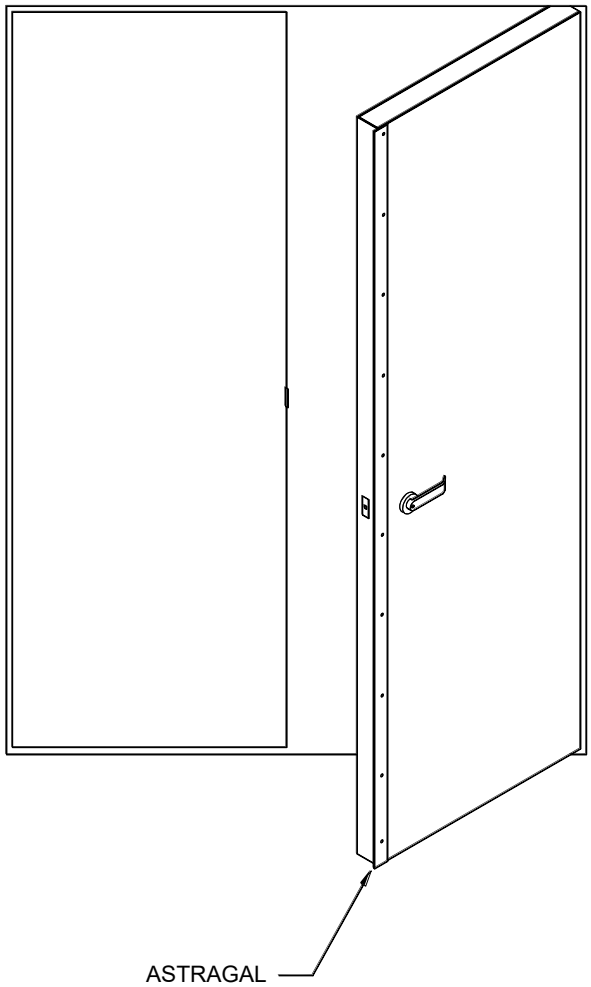
FLAT BAR ASTRAGAL

MATERIAL: GALVANNEAL
GAUGE: 12



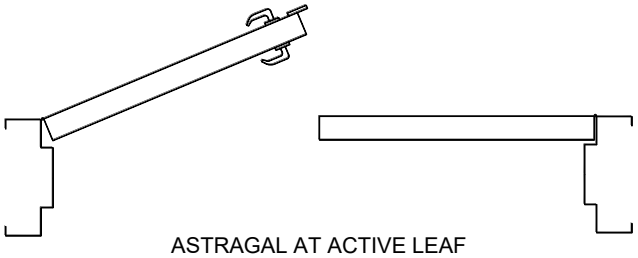


(STANDARD SCREW
PLACEMENT USING # 8 FHMS)

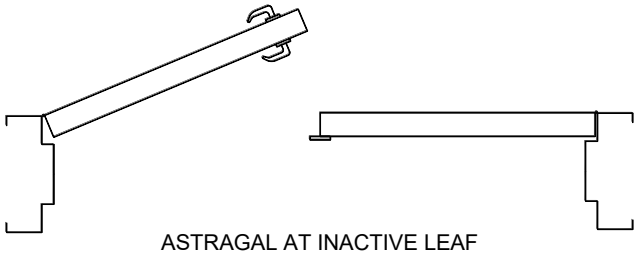


FLAT BAR ASTRAGAL APPLICATIONS

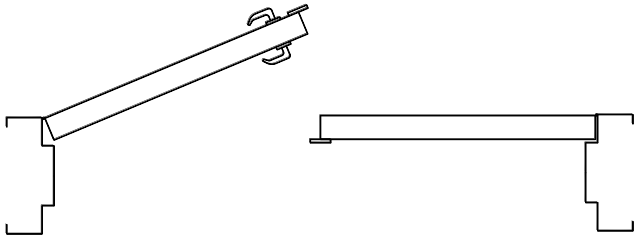
MATERIAL: GALVANNEAL
GAUGE: 12



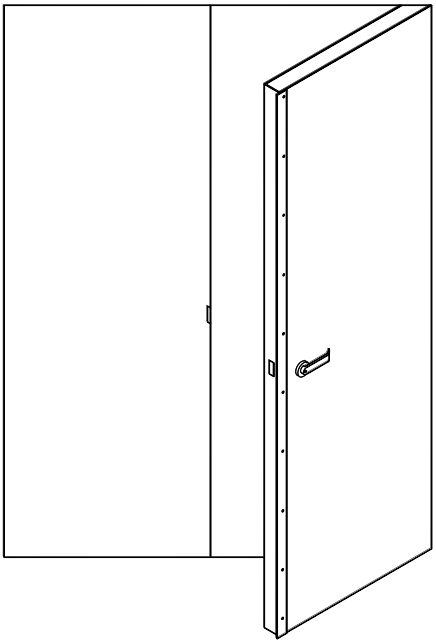
ASTRAGAL AT ACTIVE LEAF



ASTRAGAL AT INACTIVE LEAF

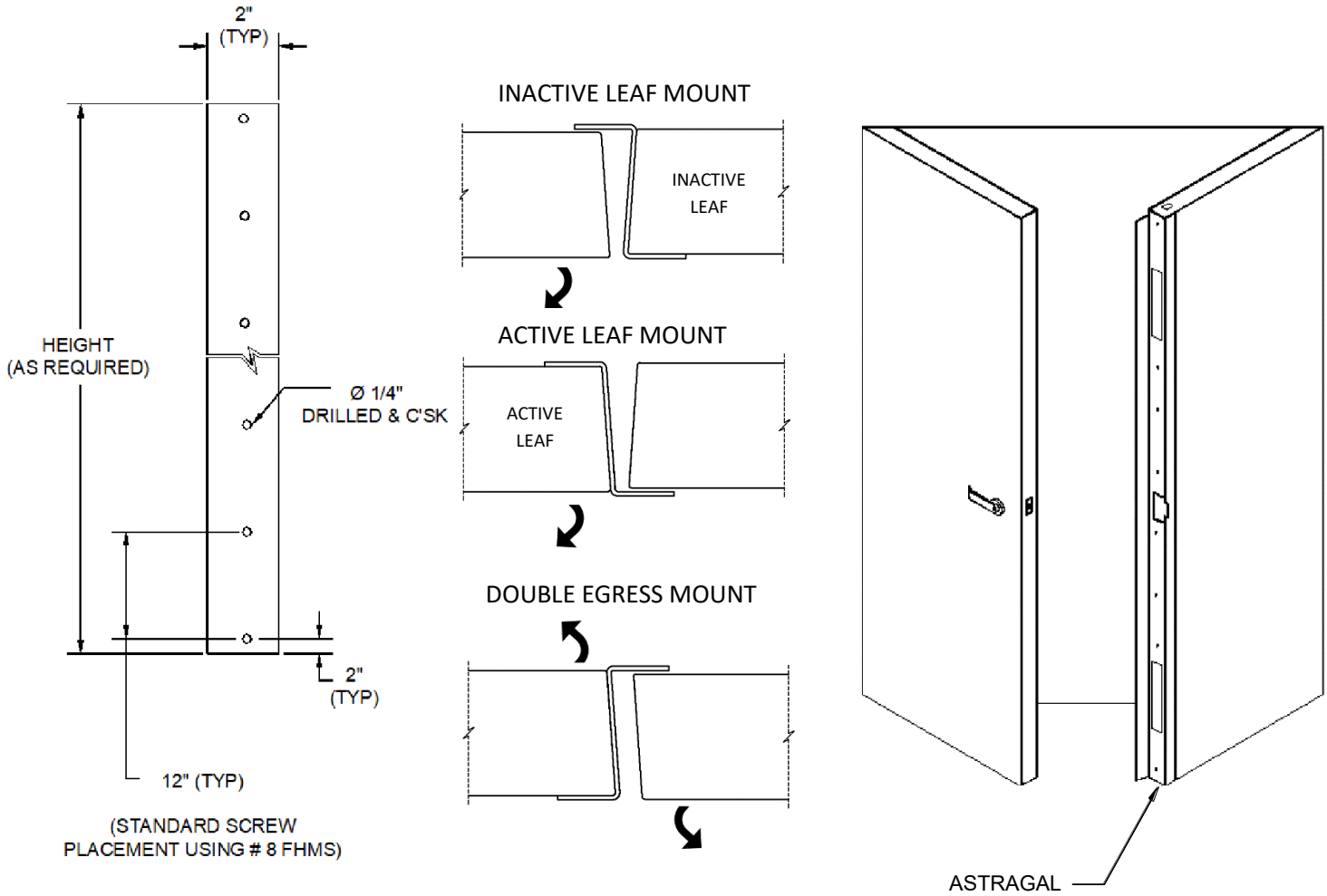


(OPTIONAL)
ASTRAGAL AT BOTH LEAVES



Z ASTRAGAL

MATERIAL: GALVANNEAL
GAUGE: 14

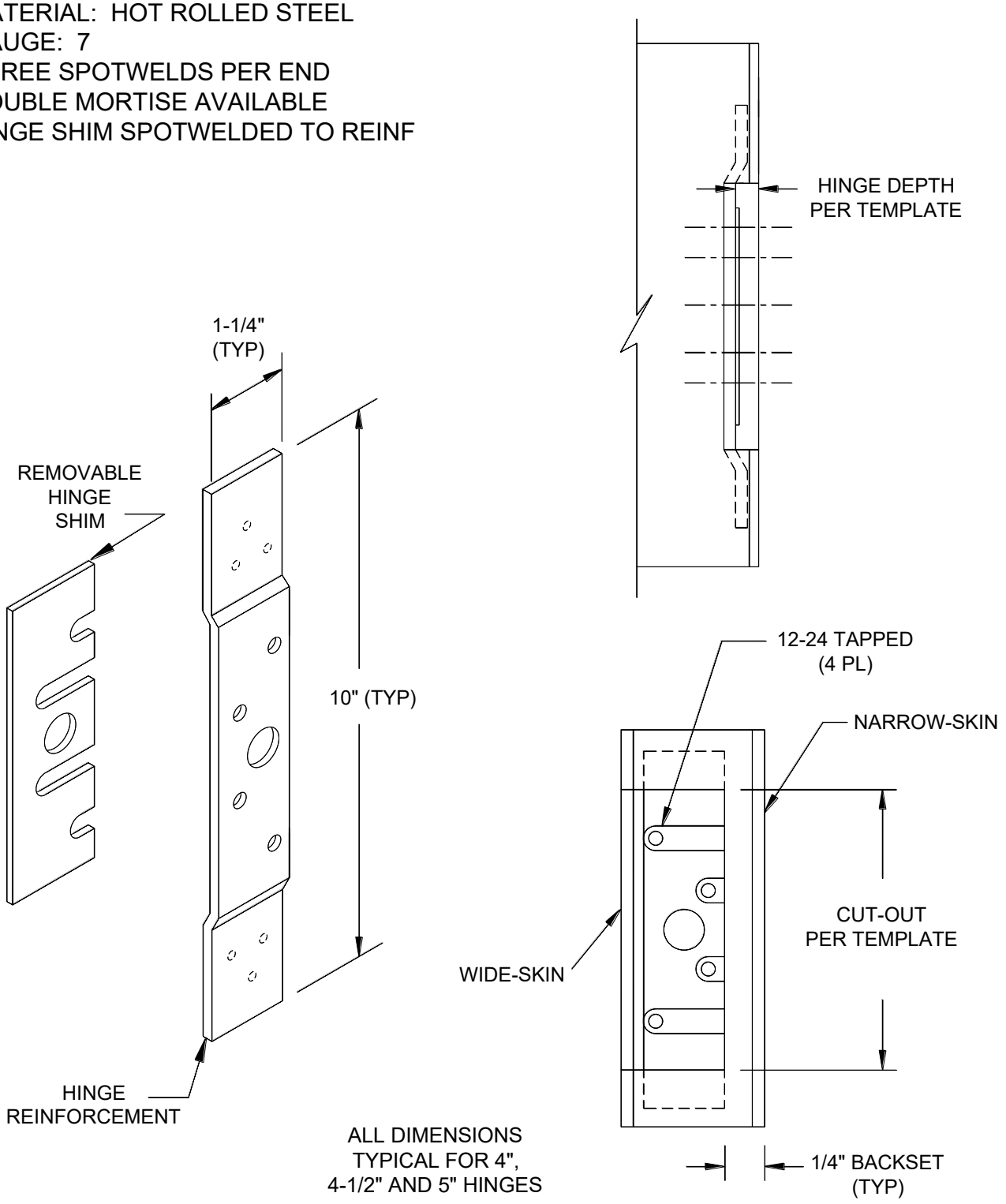


NOTE:

1. HARDWARE REINFORCEMENTS AND MOUNTING TABS ARE PREPPED IN DOOR EDGE, ASTRAGAL HAS HARDWARE CUTOUTS ONLY
2. TOP OF ASTRAGAL IS NOTCHED TO CLEAR FRAME STOP AT HEAD WHEN MOUNTED ON INACTIVE LEAF OR ON DOUBLE EGRESS LEAF
3. NET WIDTH OF ONE OR BOTH DOOR LEAVES MAY NEED TO BE UNDERSIZED TO PERMIT PROPER CLEARANCE BETWEEN DOORS

STANDARD DOOR HINGE REINF

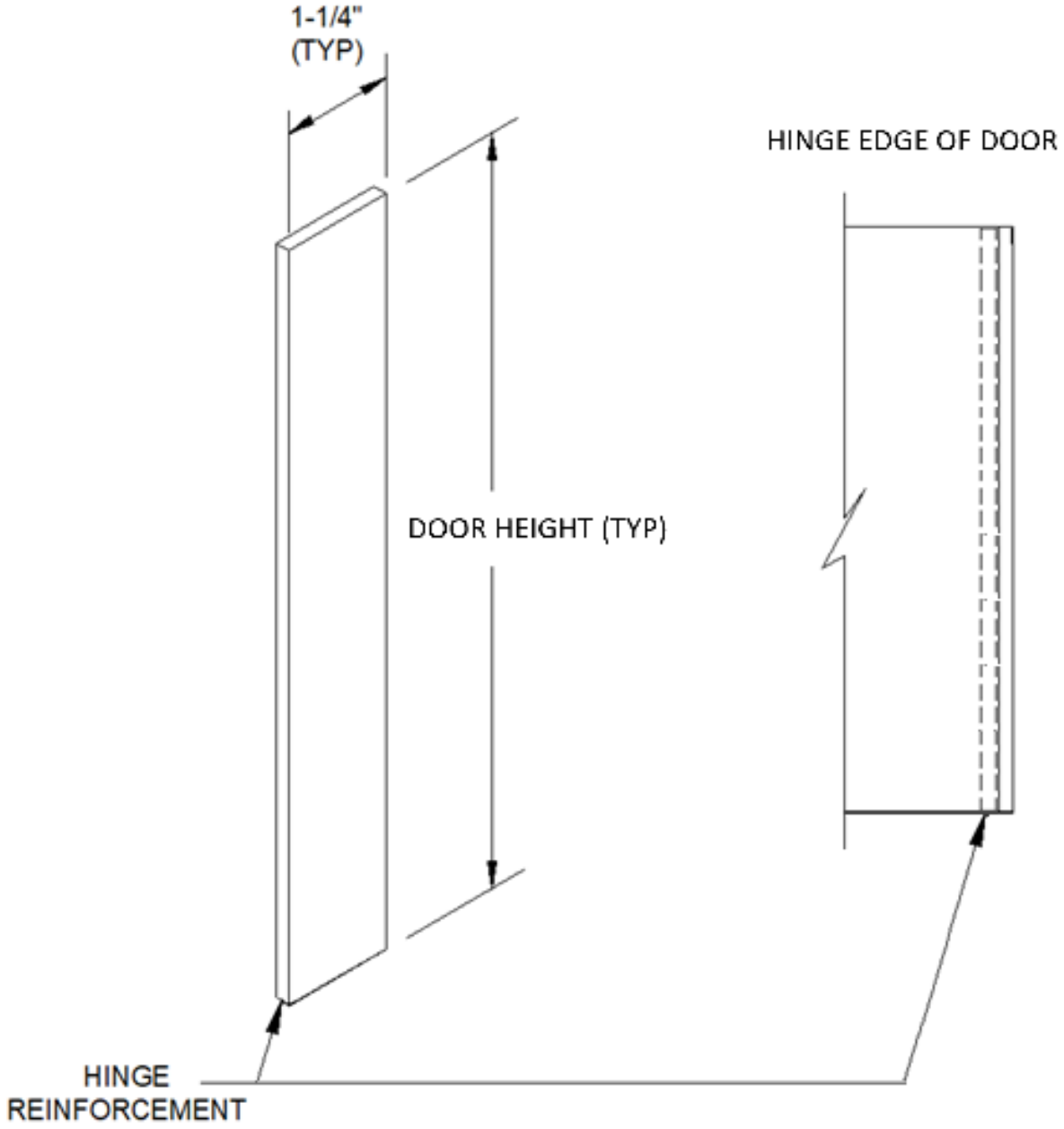
MATERIAL: HOT ROLLED STEEL
GAUGE: 7
THREE SPOTWELDS PER END
DOUBLE MORTISE AVAILABLE
HINGE SHIM SPOTWELDED TO REINF



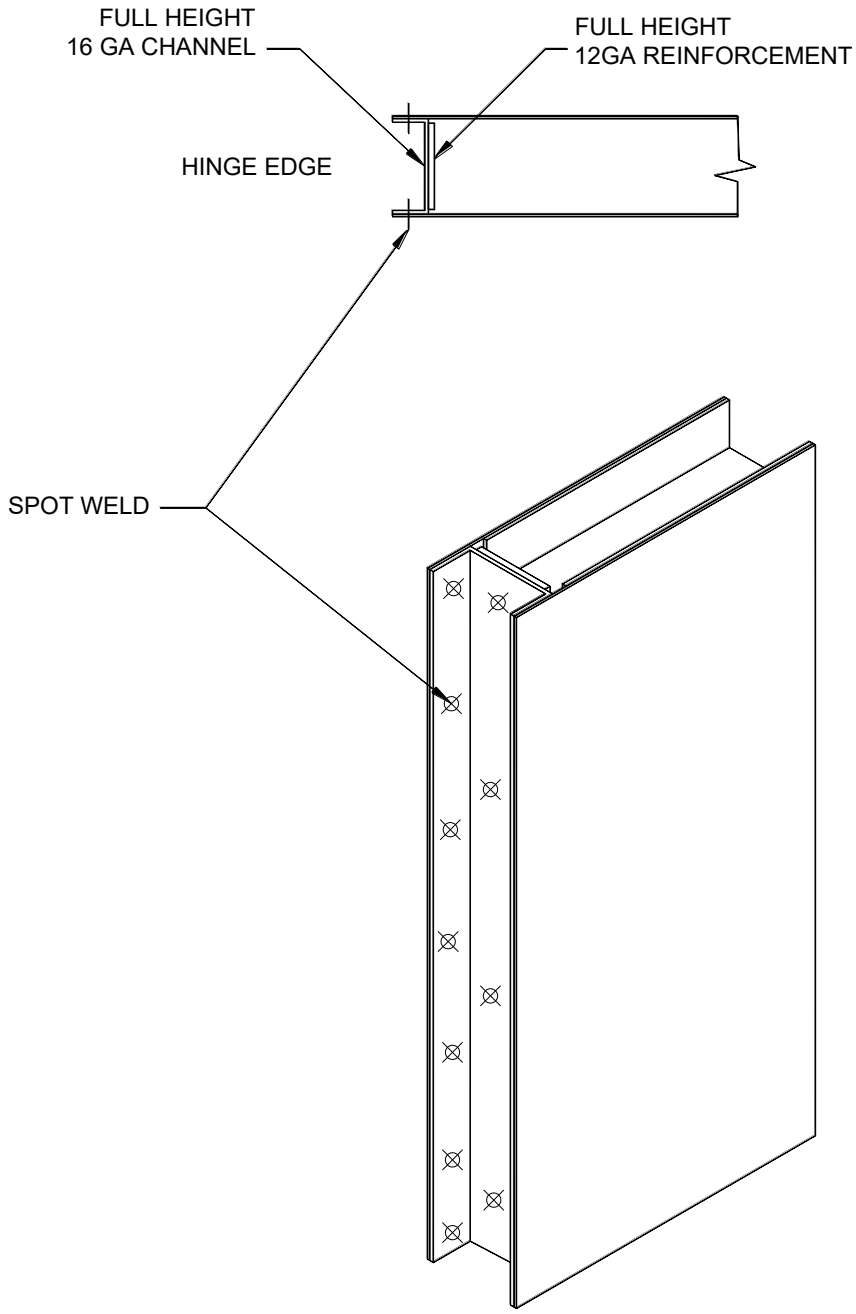
ALL DIMENSIONS
TYPICAL FOR 4",
4-1/2" AND 5" HINGES

DOOR CONTINUOUS HINGE REINF

CONTINUOUS HINGE REINFORCEMENTS
ONLY ADDED UPON REQUEST
MATERIAL: GALVANNEAL
GAUGE: 16 (14, 12, 10 AVAILABLE)
WELDED TO DOOR
APPLIED TO HINGE EDGE OR FACE DEPENDING
ON CONTINUOUS HINGE TYPE
WHI APPROVED

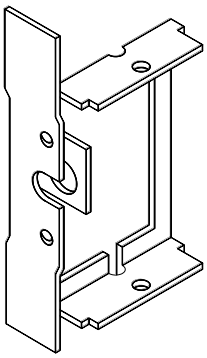


POCKET PIVOT DOOR REINFORCEMENT

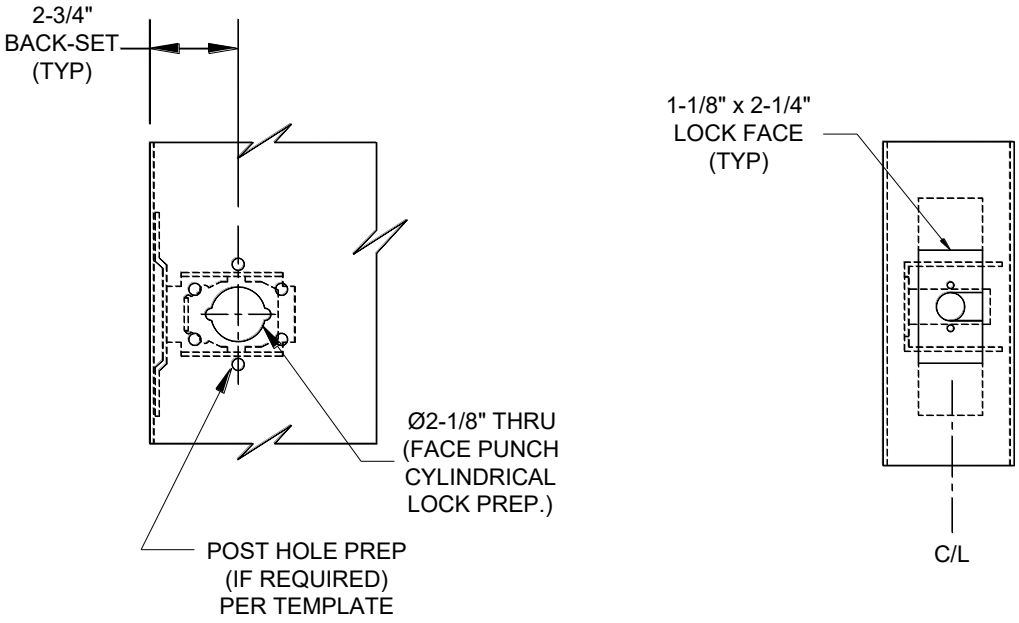


CYLINDRICAL LOCK REINFORCEMENT (161 LOCKS)

REINFORCEMENT: COLD ROLLED STEEL
GAUGE: 16 (EXTRUDED TO 12 GA AT SCREW HOLES)
TWO SPOTWELDS PER END

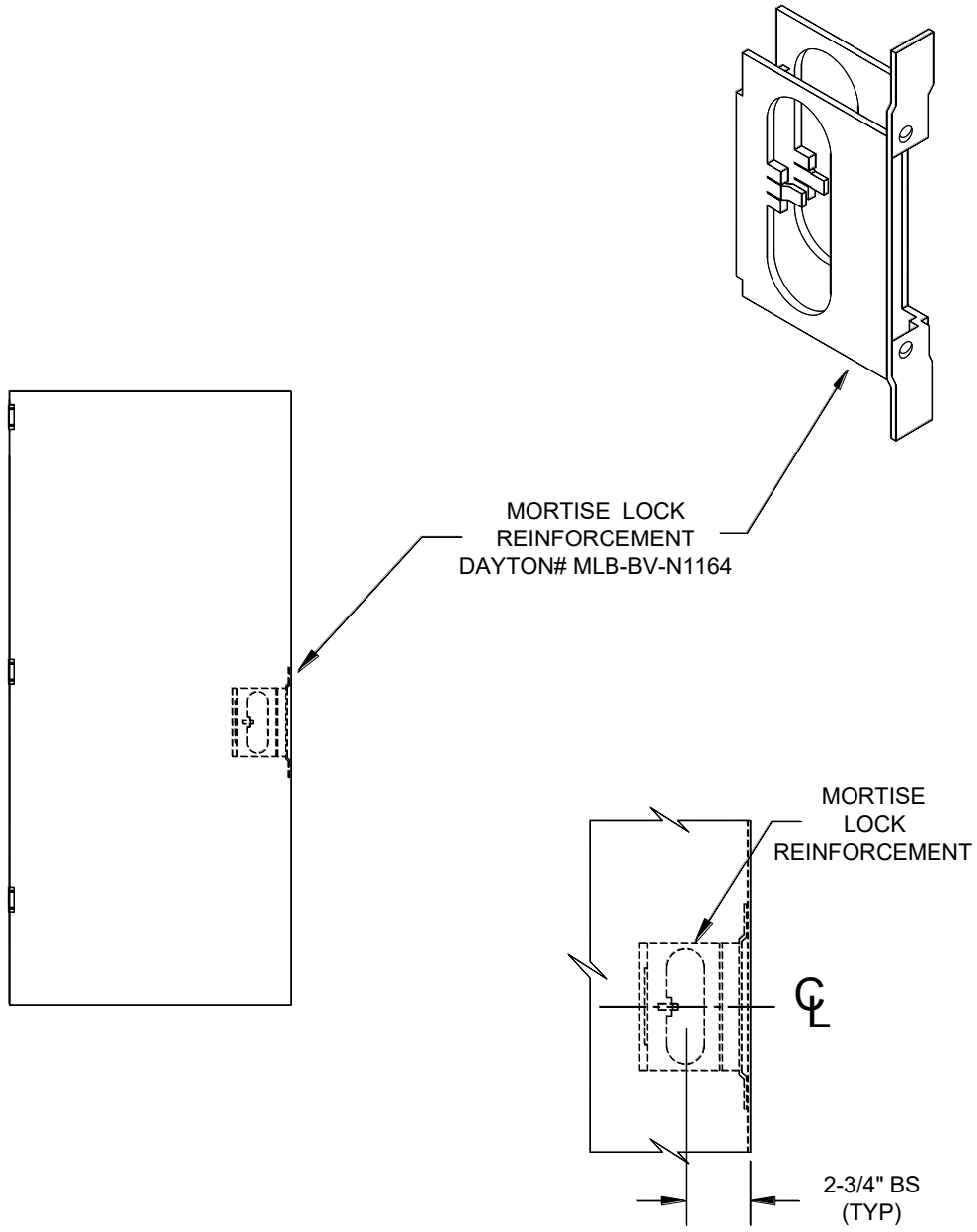


CYLINDRICAL LOCK
REINFORCEMENT
DAYTON# CLB-BV-N158



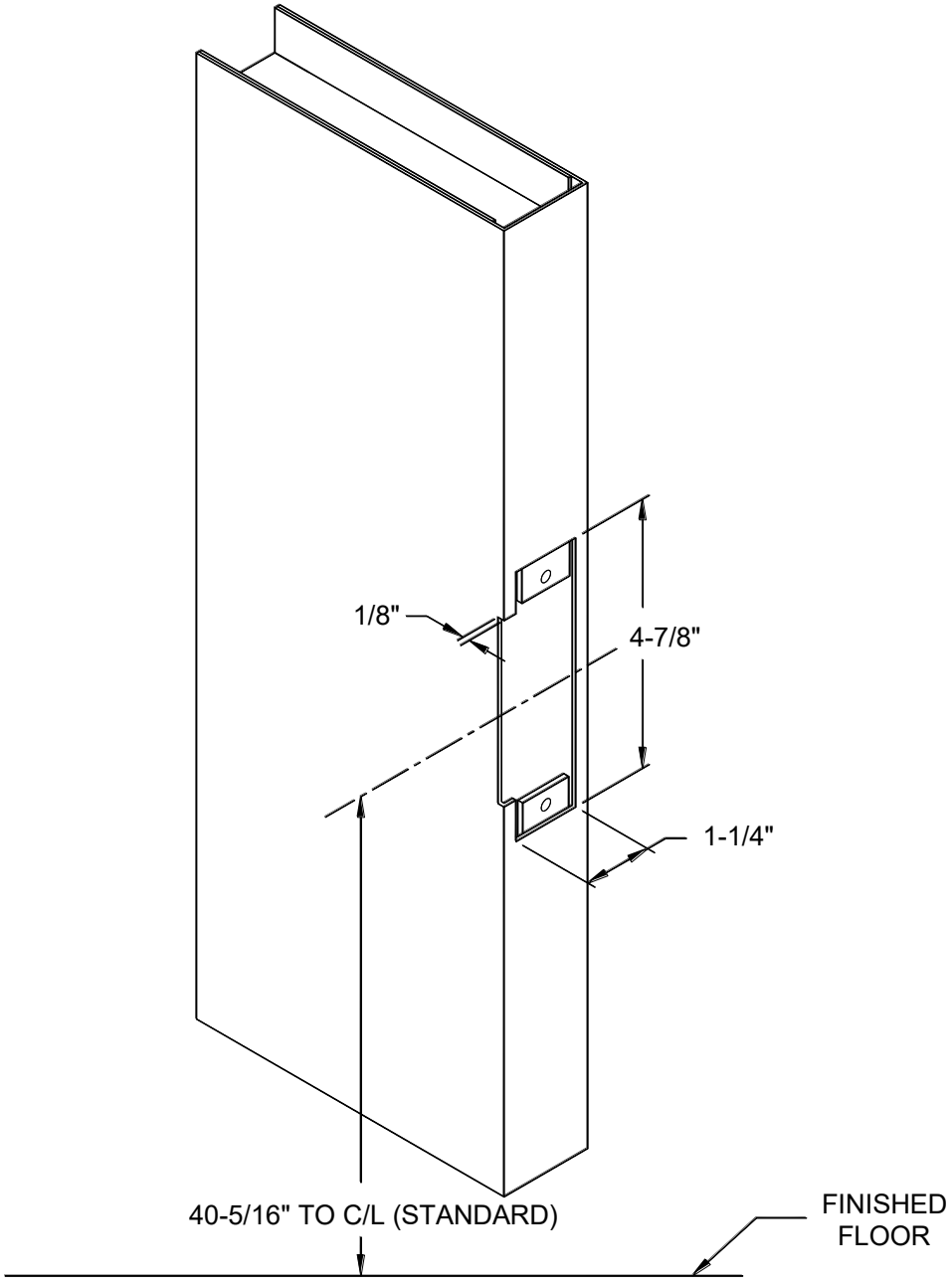
MORTISE LOCK REINFORCEMENT (86 LOCK)

REINFORCEMENTS: GALVANNEAL
GAUGE: 16 (EXTRUDED TO 12 GA AT SCREW HOLES)
TWO SPOTWELDS PER END
FACE PREP PER HARDWARE MANUFACTURER'S TEMPLATE



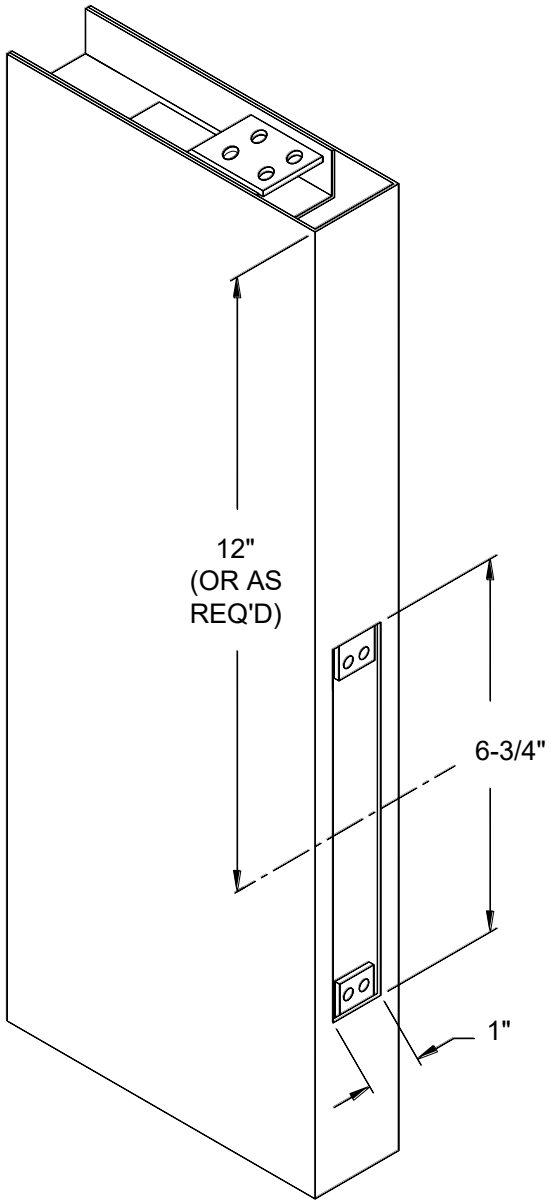
DOOR STRIKE PREPARATION

2-1/2" x 1" x 11 GAUGE
DRILLED AND TAPPED FOR 12-24 FHMS
TWO SPOTWELDS PER END



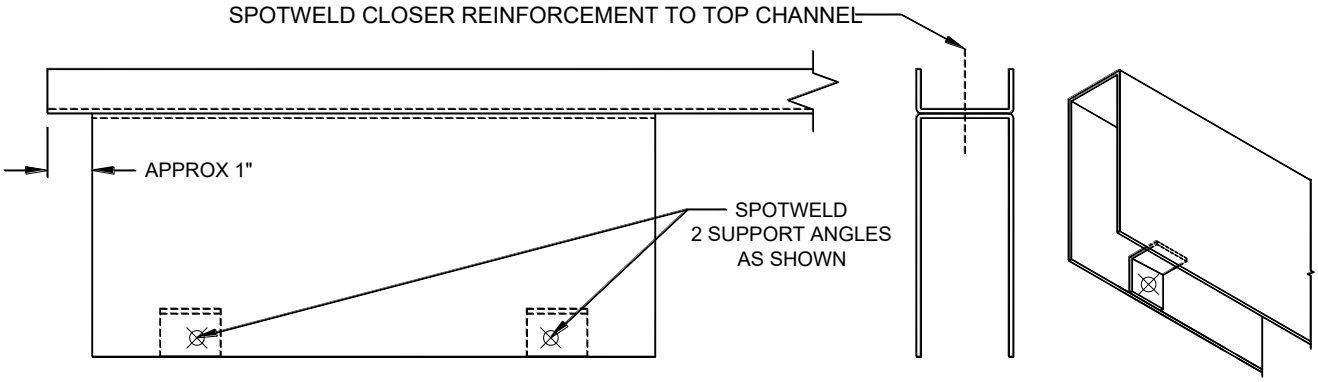
DOOR FLUSH BOLT PREPARATION

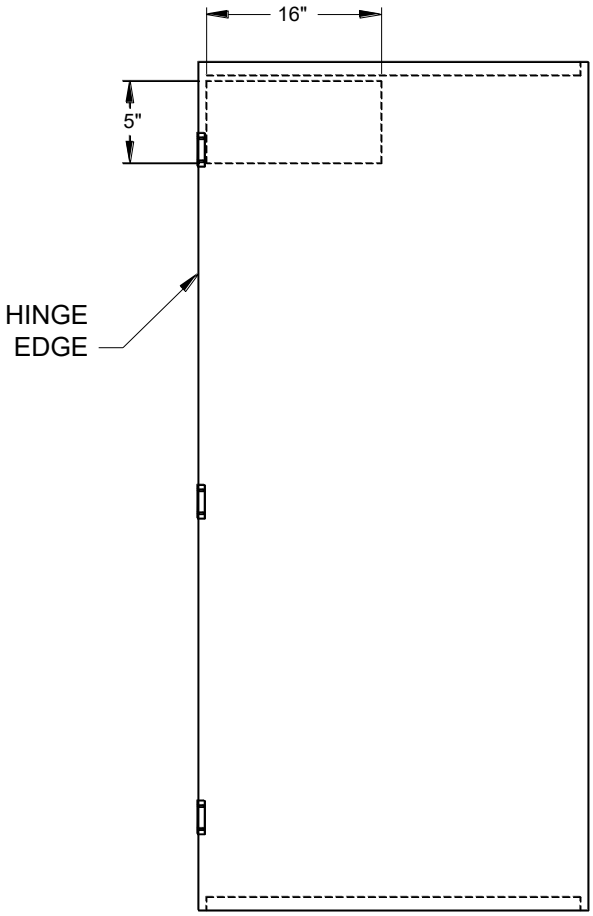
2-1/2" x 1" x 11 GAUGE
DRILLED AND TAPPED AS REQ'D
TWO SPOTWELDS PER END



DOOR CLOSER REINFORCEMENT

MATERIAL: GALVANNEAL
GAUGE: 14
WHI APPROVED

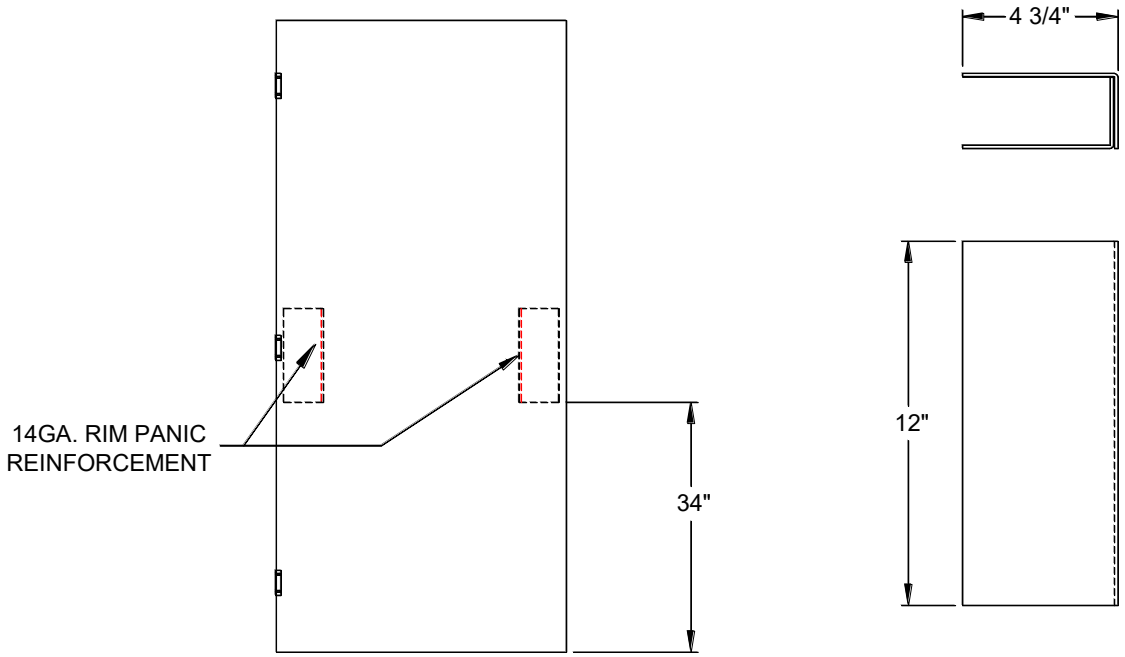




STANDARD CLOSER REINFORCEMENT AREA SHOWN; FULL WIDTH CLOSER AVAILABLE

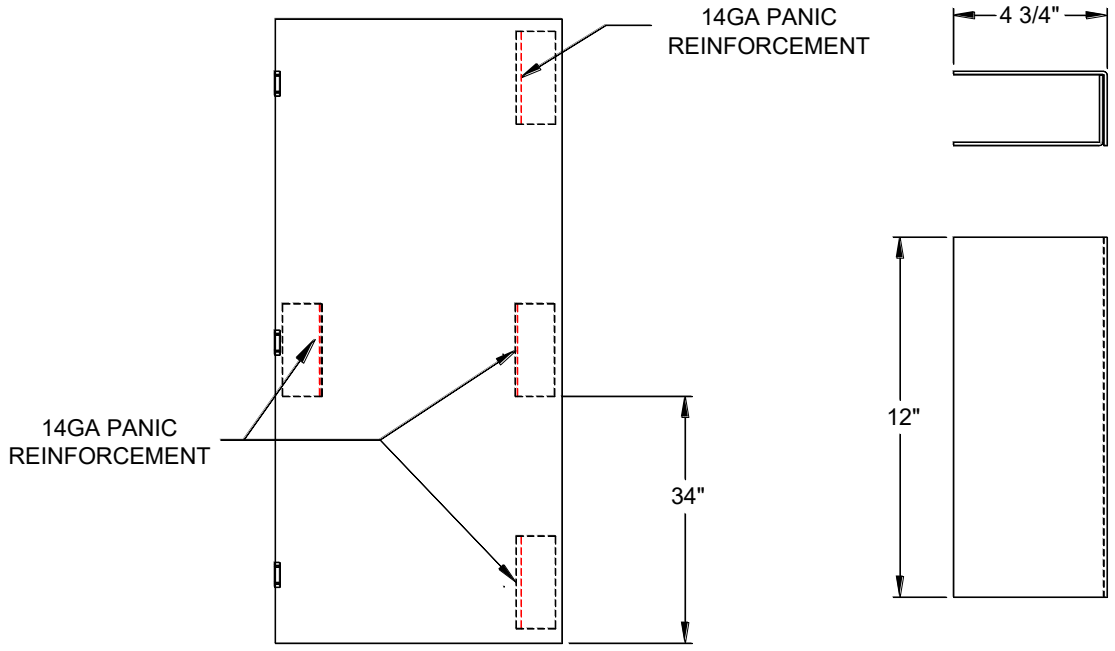
RIM PANIC REINFORCEMENT

REINFORCEMENTS: GALVANNEAL
GAUGE: 14
SPOTWELDED TO DOOR SKIN



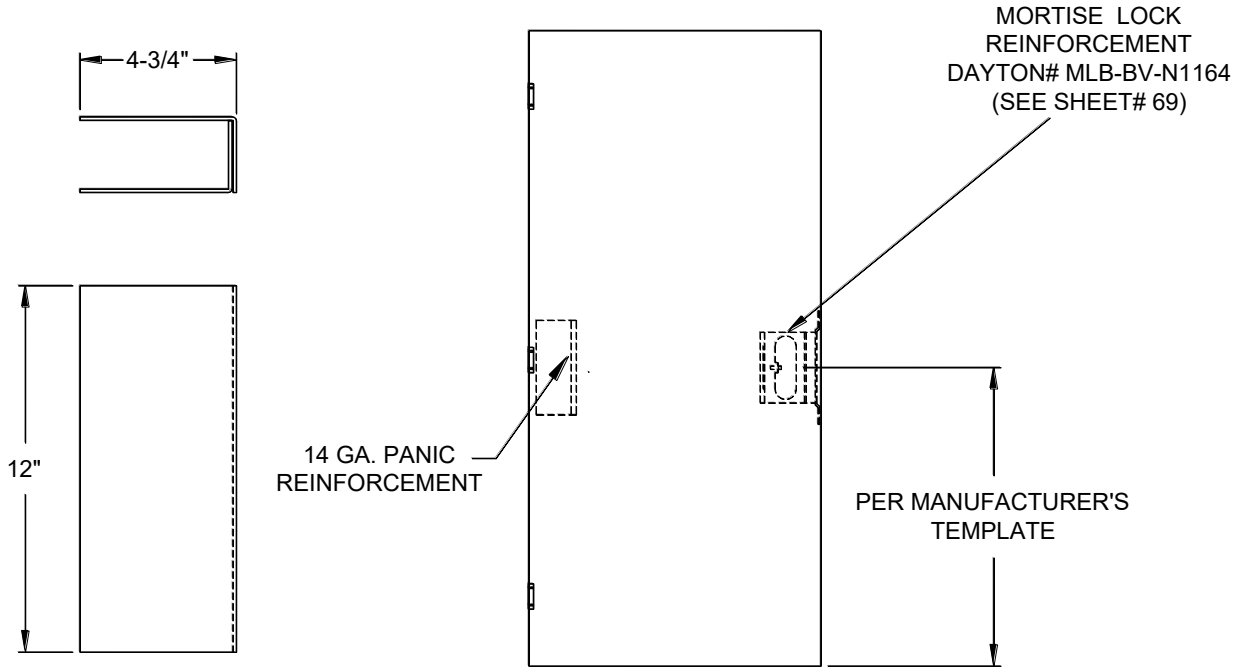
SURFACE VERTICAL ROD REINF

REINFORCEMENTS: GALVANNEAL
GAUGE: 14
SPOTWELDED TO DOOR SKIN



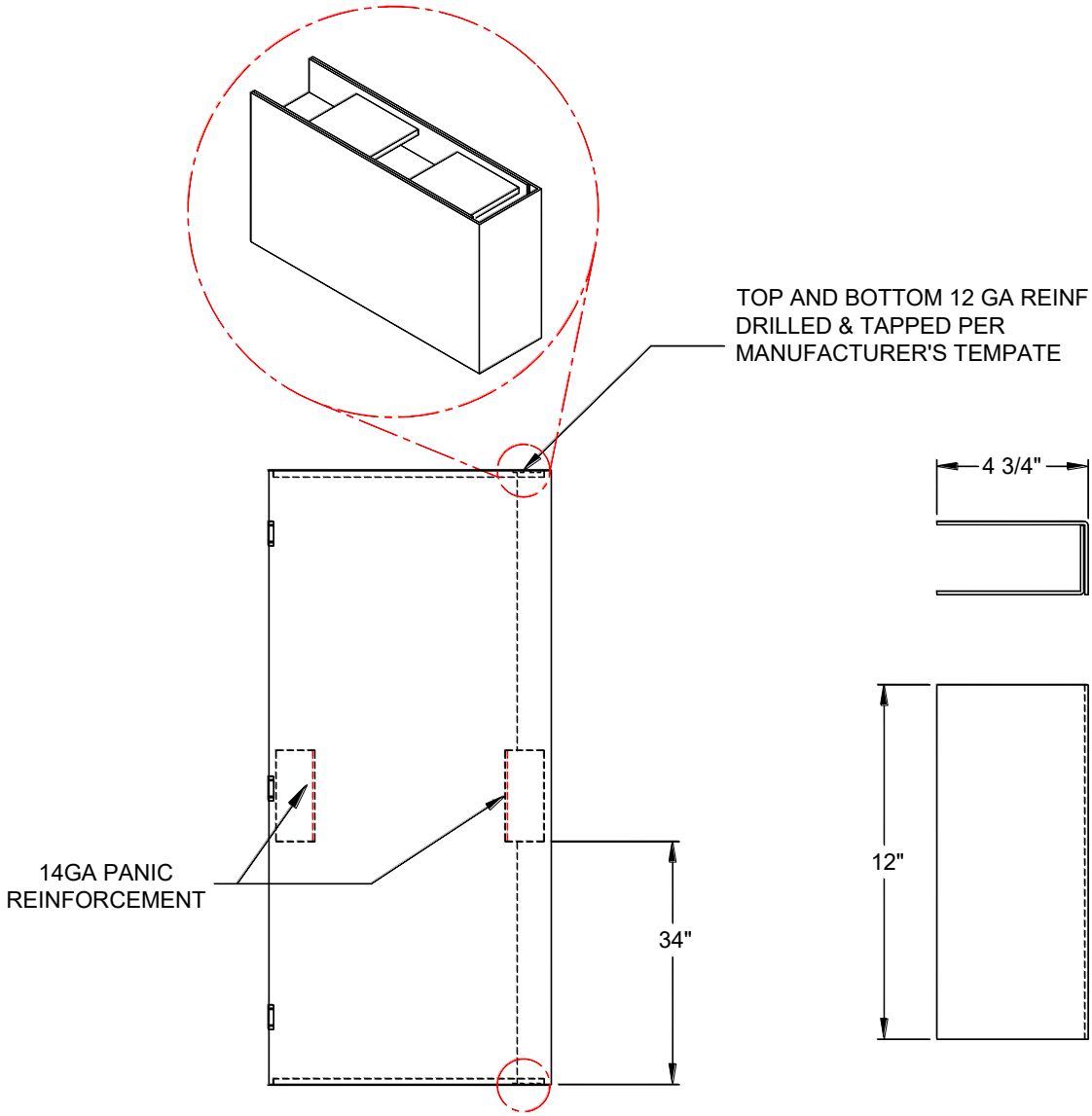
MORTISE PANIC REINFORCEMENT

PANIC REINFORCEMENT

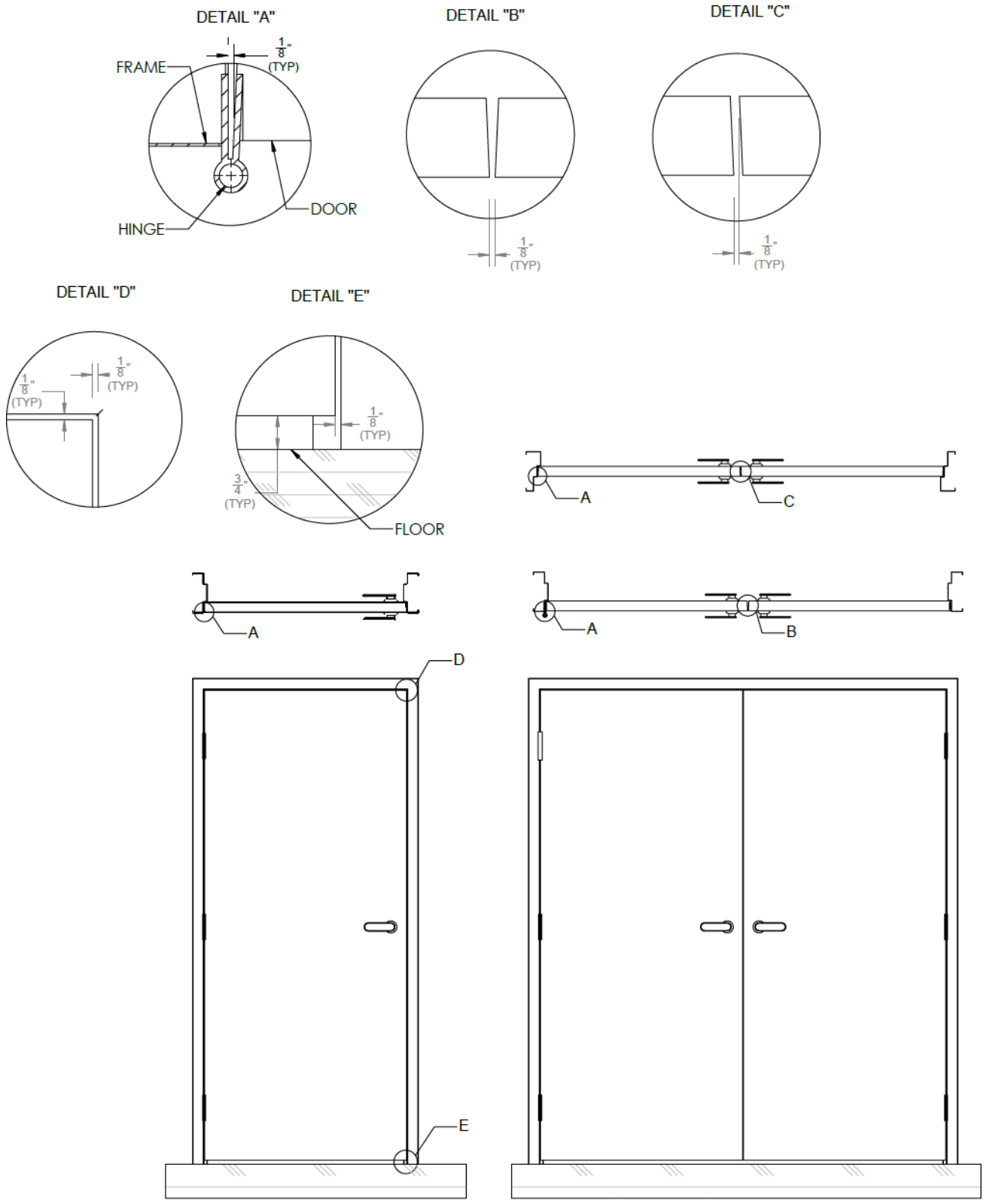


CONCEALED VERTICAL ROD REINF

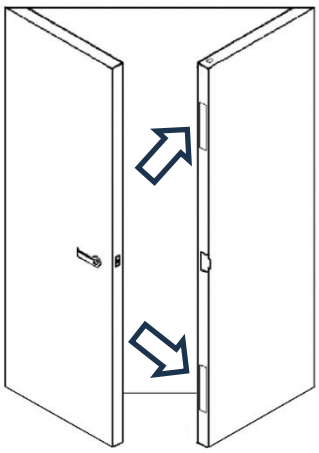
REINFORCEMENTS: GALVANNEAL
GAUGE: AS SHOWN BELOW
SPOTWELDED TO DOOR SKIN/
TACKWELDED INTO TOP CHANNEL

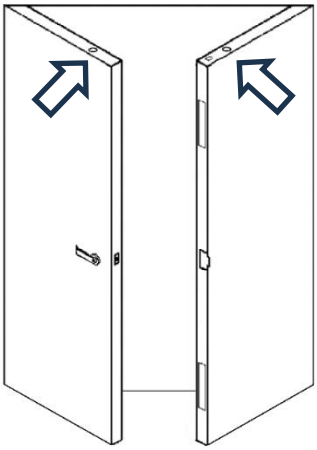


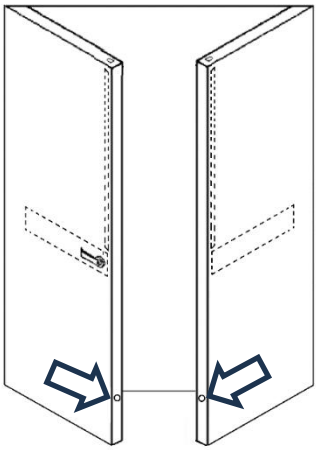
STANDARD DOOR CLEARANCES

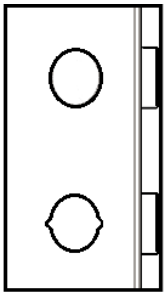


SUGGESTED HARDWARE AND LITE LOCATIONS*









Flush Bolts:

Top Flush bolt: 12” TTCL for doors up to 84”, for taller doors with non-auto flush bolts consider a larger value (like 24” or 36” TTCL) for accessibility
Bottom Flush bolt: 12” BTCL

DPS:

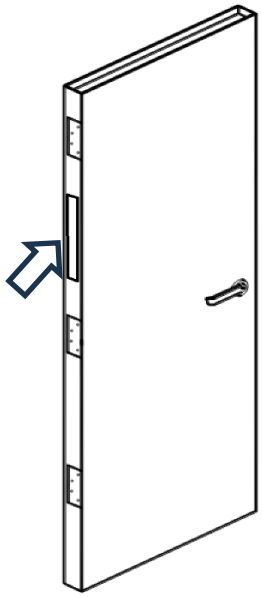
Singles: On top edge
5-7/8” from latch edge
(6” in on frame)
Pairs: On top edge
5-15/16” from latch edge

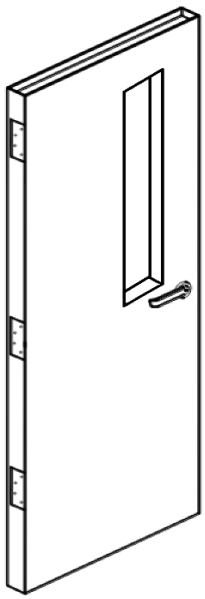
Fire Pin:

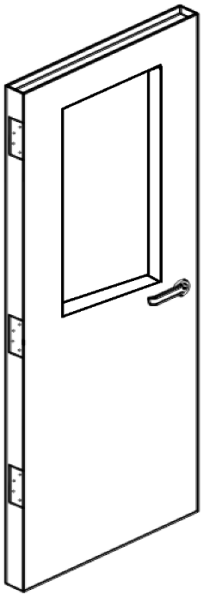
On rated doors with LBR/LBL (Less Bottom Rod/Latch) or without bottom flush bolt,
6” up from bottom

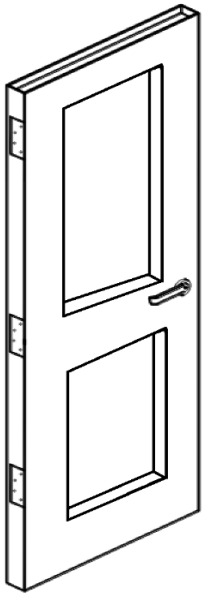
Double Bore:

For cylindrical lock with deadbolt, 5-1/2” center-to-center;
Use T-strike with Deadbolt Strike (ASA too big)









EPT:

3 hinges: Centered between 1st/2nd hinge
4 hinges: Centered between 2nd/3rd hinge
(Get EPT close to lock)

Narrow Lite:

6” in from latch edge
and minimum 6” top rail;
If ADA: Bottom of visible glass must be 43” max from floor
(Propose 40” up)

Half Lite:

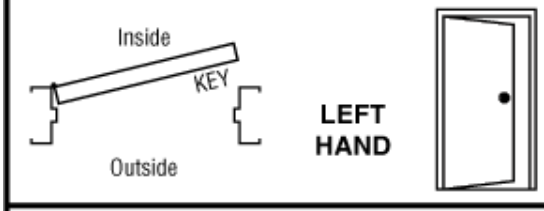
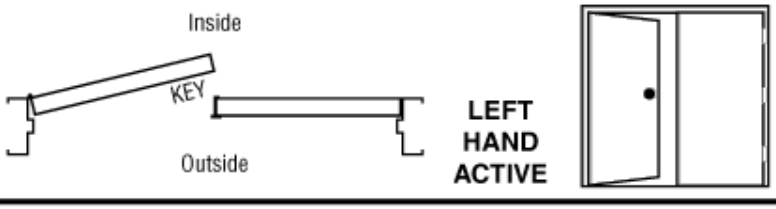
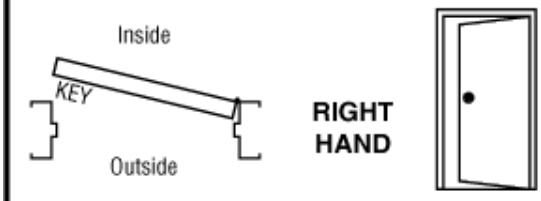
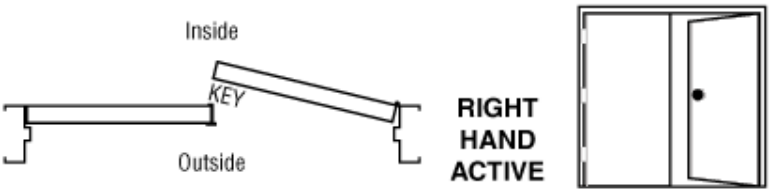
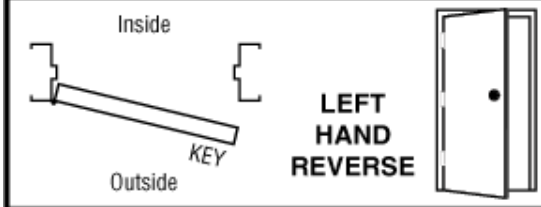
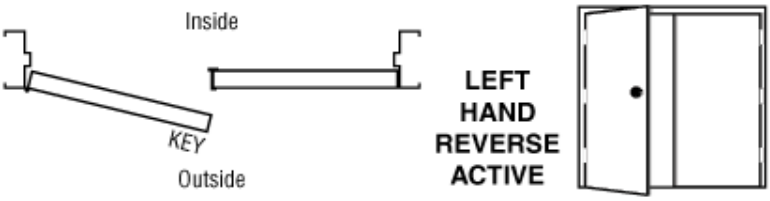
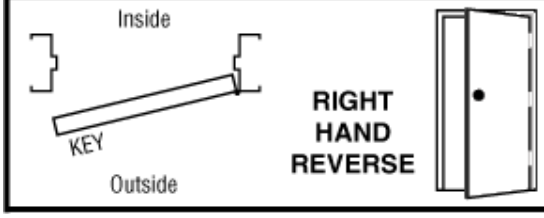
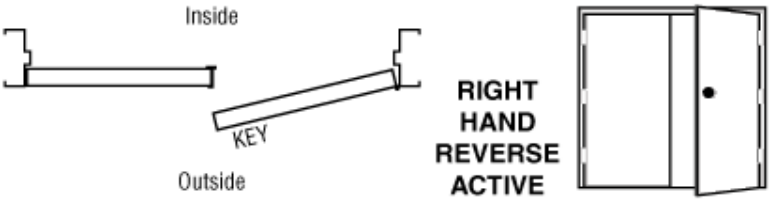
Centered with 6” stiles
and minimum 6” top rail;
If ADA: Bottom of visible glass must be 43” max from floor
(Propose 40” up)

Two Half Lites or Full Lite:

Centered with 6” stiles
and minimum 6” top/middle rails
(No middle rail for Full Lite)
If ADA: Bottom 10” of door must be flush

***Note that these are typical locations and do not cover all fire rating rules, manufacturing limitations, code compliance, or certain hardware prep limitations/combinations**

DOOR HANDING CHART

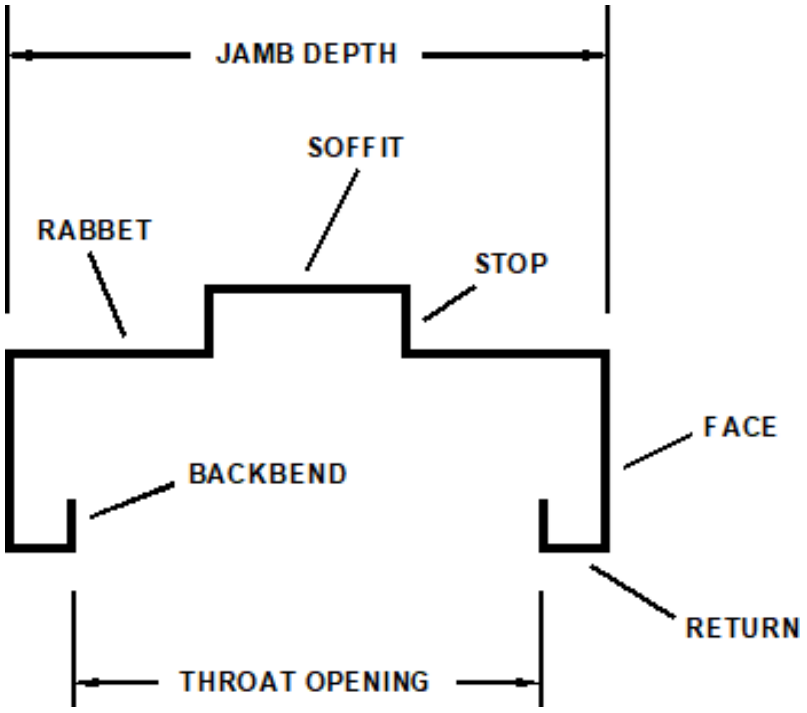
SINGLE DOOR	DOUBLE DOOR
 **LEFT HAND**	 **LEFT HAND ACTIVE**
 **RIGHT HAND**	 **RIGHT HAND ACTIVE**
 **LEFT HAND REVERSE**	 **LEFT HAND REVERSE ACTIVE**
 **RIGHT HAND REVERSE**	 **RIGHT HAND REVERSE ACTIVE**

FIRE RATED DOOR AND FRAME GENERAL REQUIREMENTS

These requirements apply to all fire rated doors, frames, and windows.

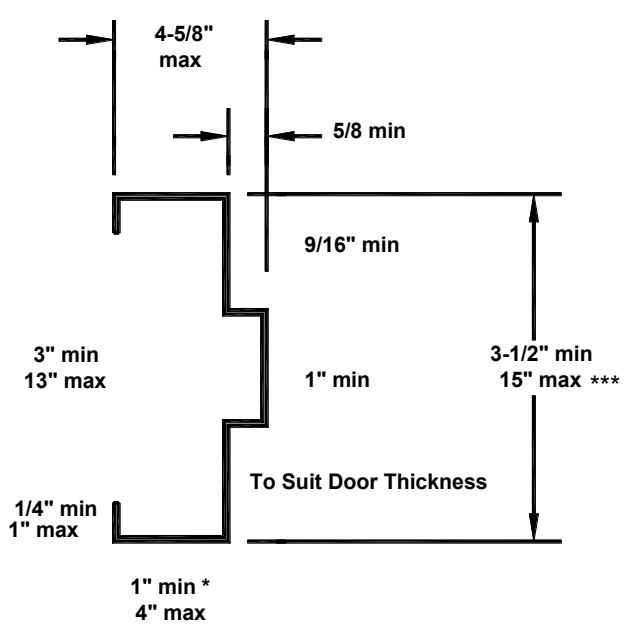
Requirements for the installation of these assemblies are included in the Standard of the National Fire Protection Association for Fire Doors and Fire Windows, NFPA-80 and DCI Hollow Metal's installation instructions.

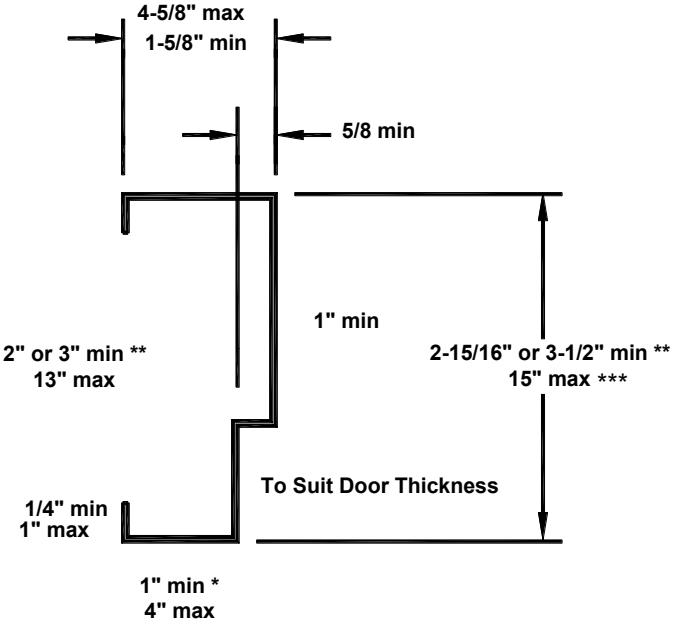
Terminology used in describing these frames follows as closely as possible the Glossary of Terms for Hollow Metal Doors and Frames (SDI-134).

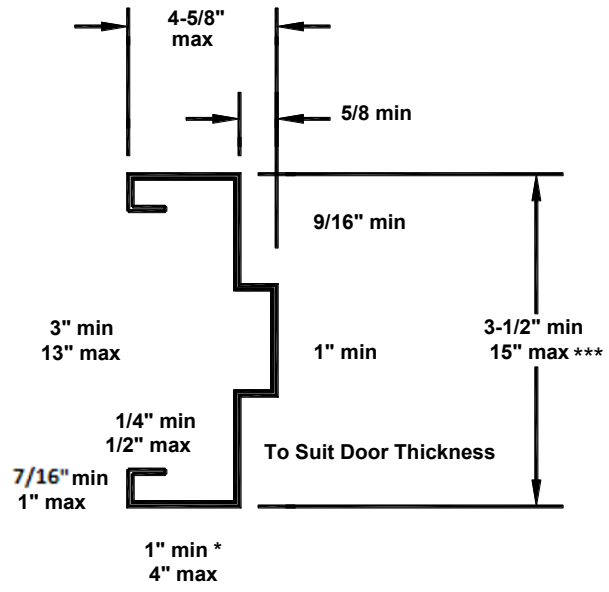


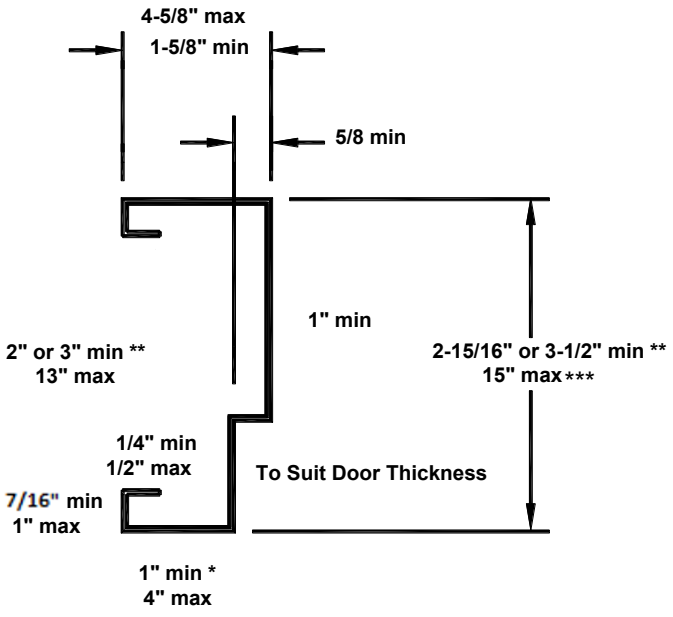
STANDARD FRAME PROFILES

The minimum and maximum dimensions shown below may vary based upon frame elevation, rating, wall type, forming limitations, and for frames other than 16 gauge.
Standard profile frames may have combinations of the return types as shown on pages 96 & 97.









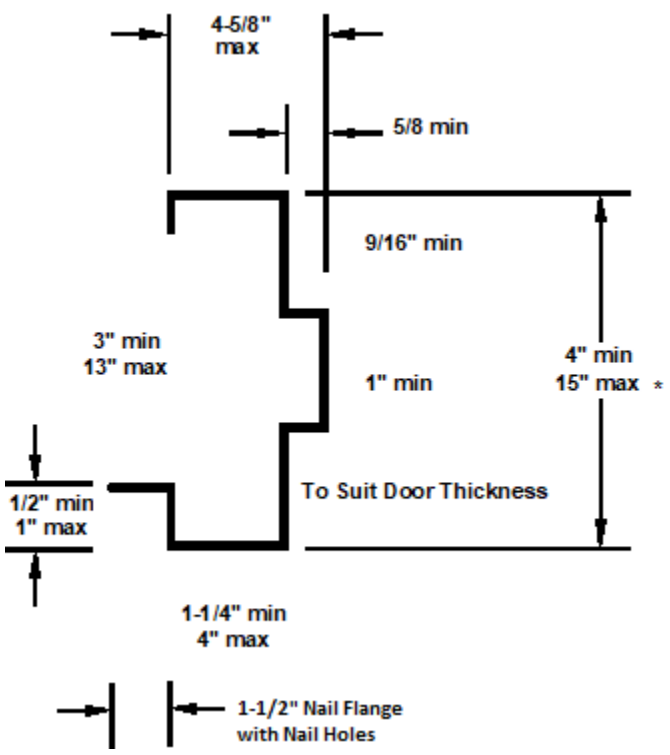
* 1" min face permitted at masonry, 1-1/4" min at drywall
** 2" throat permitted for a single rabbet profile when door side face is min 2" and opposite face is min 2-5/8". Minimum jamb depth is also dependent on variations used for the minimum throat and upon the minimum dimensions for the backbend types used on each side of the profile.
*** Jamb depths greater than 14" cannot achieve the 13" maximum throat without the backbends being larger than ½". Frames with mullions have a maximum jamb depth of 10-1/2" and a maximum throat of 9".

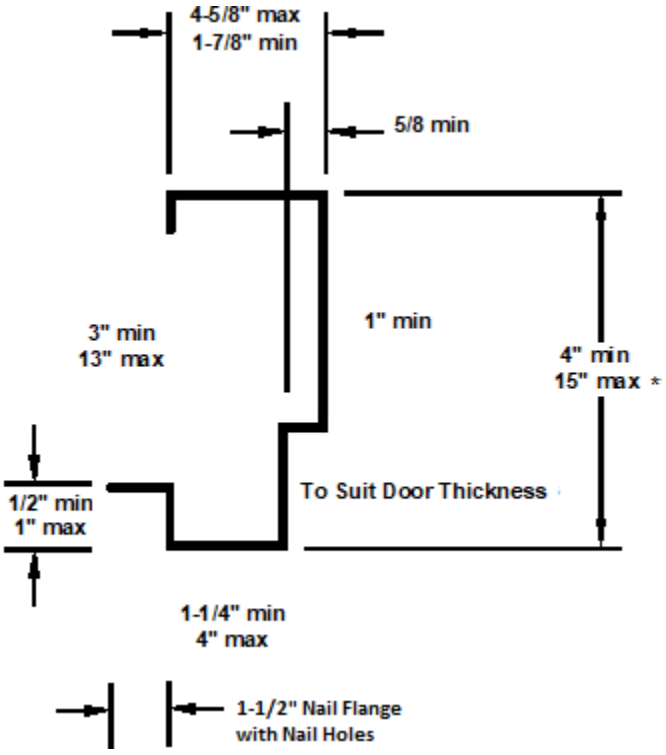
STANDARD FRAME PROFILES (NAIL FLANGE)

The minimum and maximum dimensions shown below may vary based upon frame elevation, rating, wall type, forming limitations, and for frames other than 16 gauge.

Standard profile frames may have combinations of the return types as shown on pages 96 & 97.

When nail flanges are present, the overall frame width & height dimensional limitations as listed for frame elevations (starting on page 105) will include the flange dimensions.





Nail flange as shown on the above profiles is to serve as the anchor on that side of the profile only.
Additional anchors required if only one side of the profile includes a nail flange.
Nail holes at flange are provided by DCI.

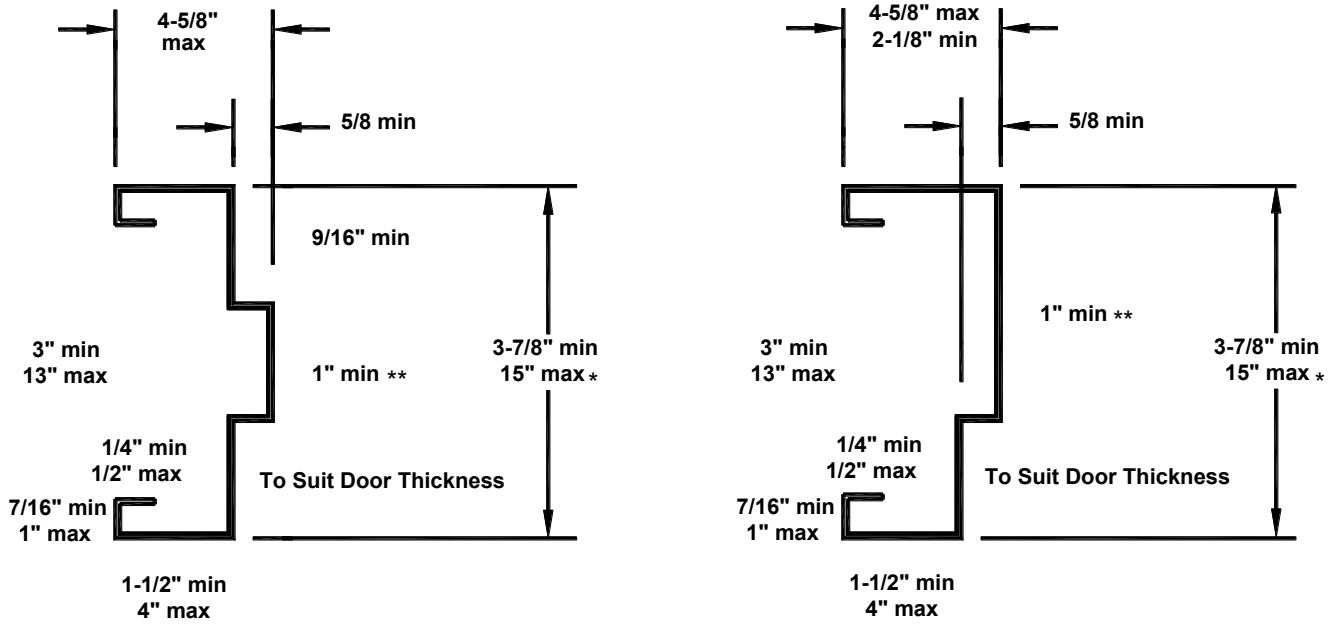
* Jamb depths greater than 14" cannot achieve the 13" maximum throat without the backbends being larger than ½". Frames with mullions have a maximum jamb depth of 10-1/2" and a maximum throat of 9".

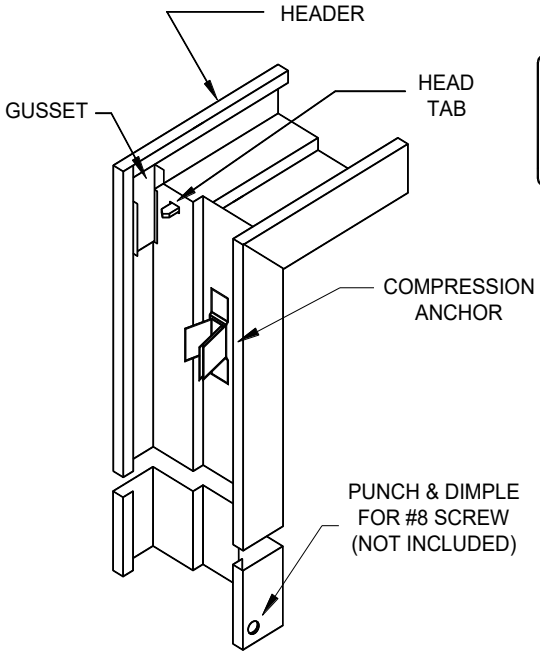
DRYWALL PRESSURE ANCHOR FRAMES

The minimum and maximum dimensions shown below may vary based upon frame elevation, rating, wall type, forming limitations, and for frames other than 16 gauge.

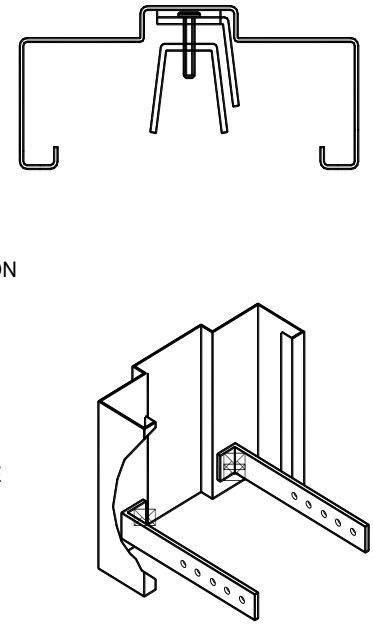
The maximum fire rating for KD DW frames is 90 minutes.

KD DW frame options for Double Egress frames are not eligible for fire ratings.





STANDARD BASE ANCHOR



ALTERNATE BASE ANCHOR

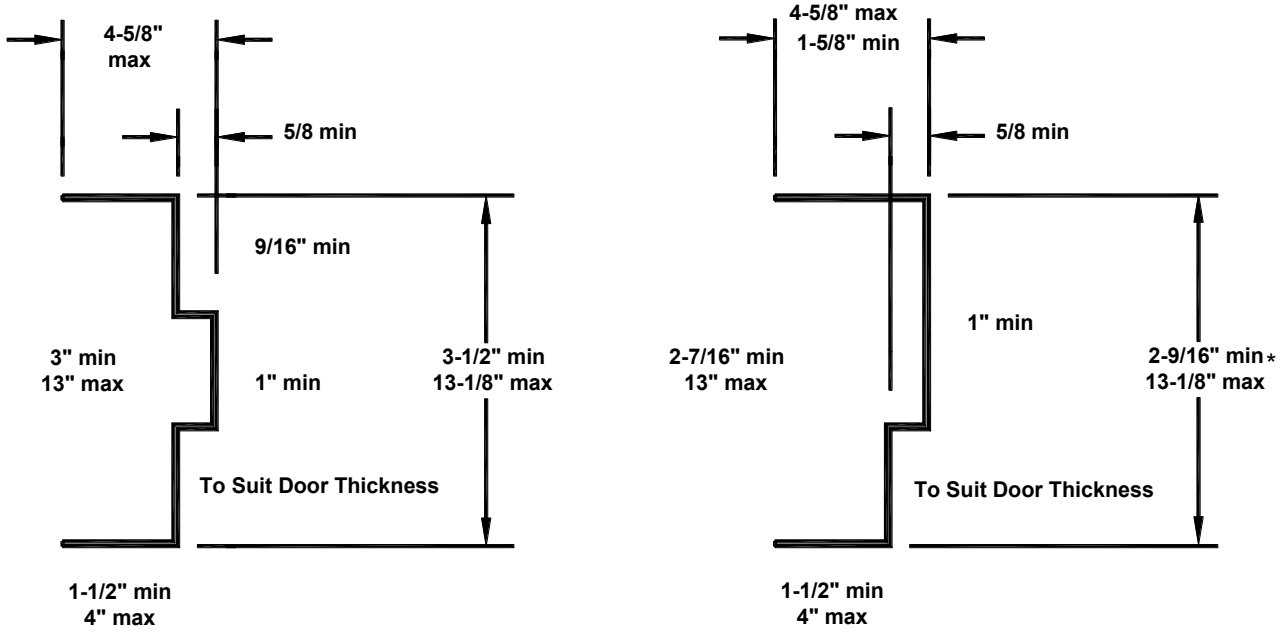
* Jamb depths greater than 14" cannot achieve the 13" maximum throat without the backbends being larger than ½". Frames with mullions have a maximum jamb depth of 10-1/2" and a maximum throat of 9".
** 1" minimum soffit width, but please consult factory if soffit width is less than 1-1/4".

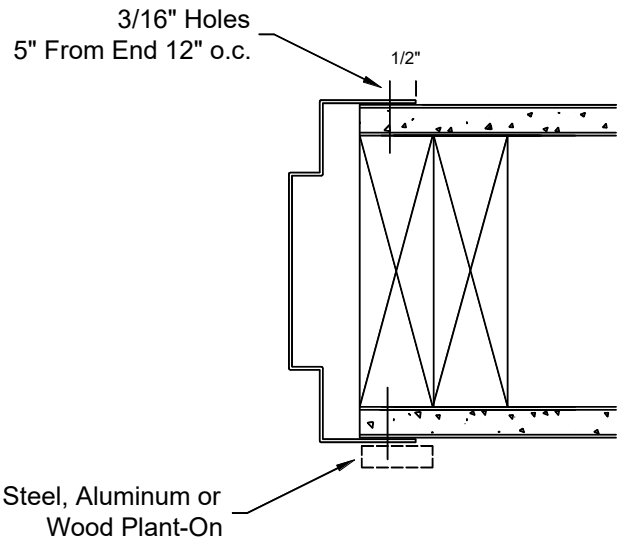
NAIL-IN FACE FRAMES

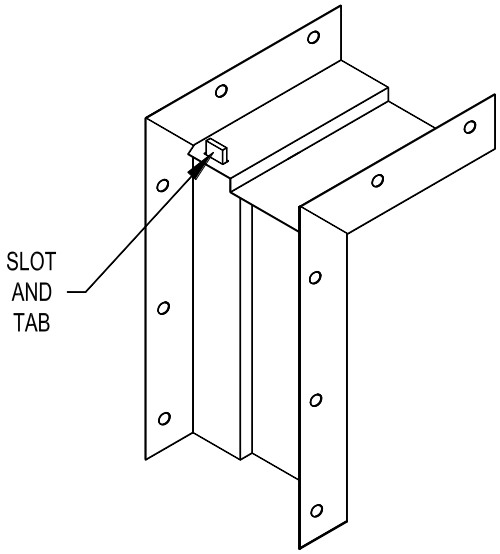
The minimum and maximum dimensions shown below may vary based upon frame elevation, rating, wall type, forming limitations, and for frames other than 16 gauge.

Fire ratings are only available for single openings with limitations as noted on page 106.

Nail-In-Face frames are only available with a KD Corner with the slot and tab detail as shown below.







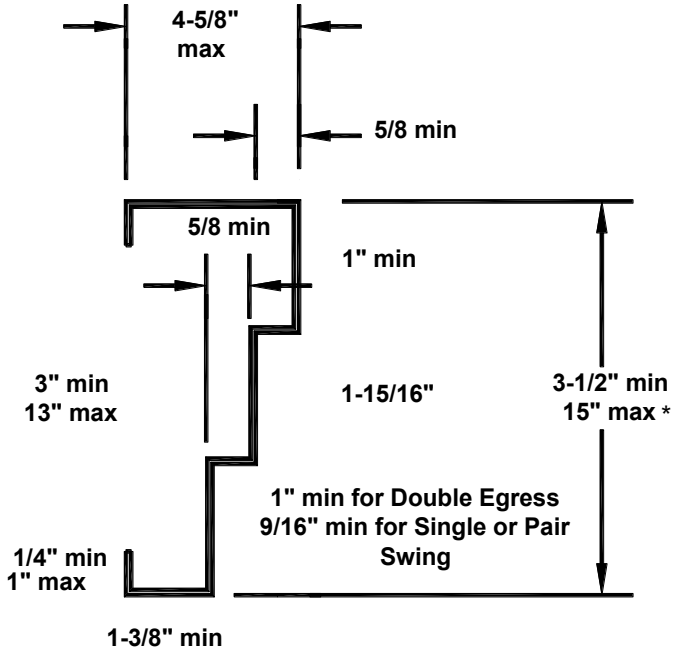
* Jamb depths smaller than 2-15/16" achieve the 1" minimum soffit width without reducing the door rabbet, which limits the permitted door thickness.

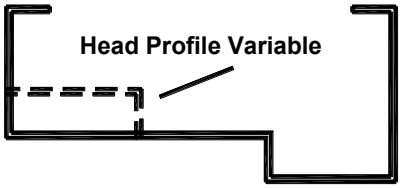
DOUBLE EGRESS / 3-STEP PROFILE

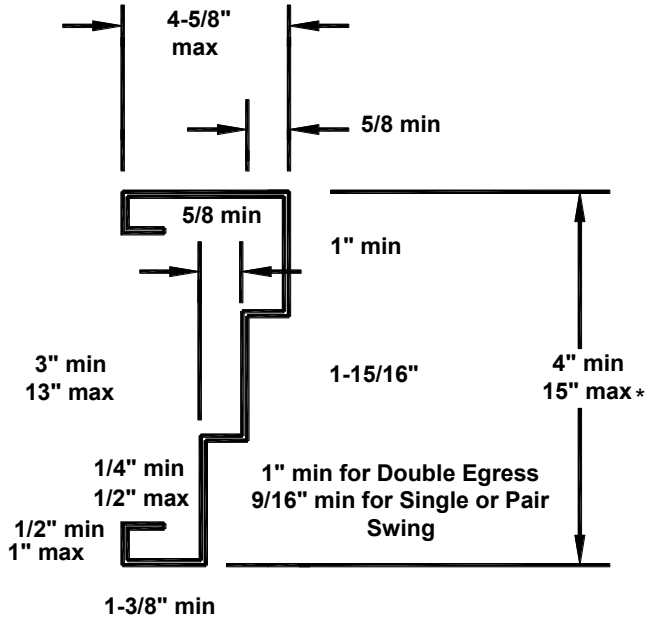
The minimum and maximum dimensions shown below may vary based upon frame elevation, rating, wall type, forming limitations, and for frames other than 16 gauge.

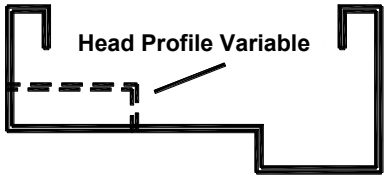
Double egress / 3-step profile frames may have combinations of the return types as shown on pages 96 & 97.

Fire ratings are NOT available for double egress openings with KDDW Corner construction.









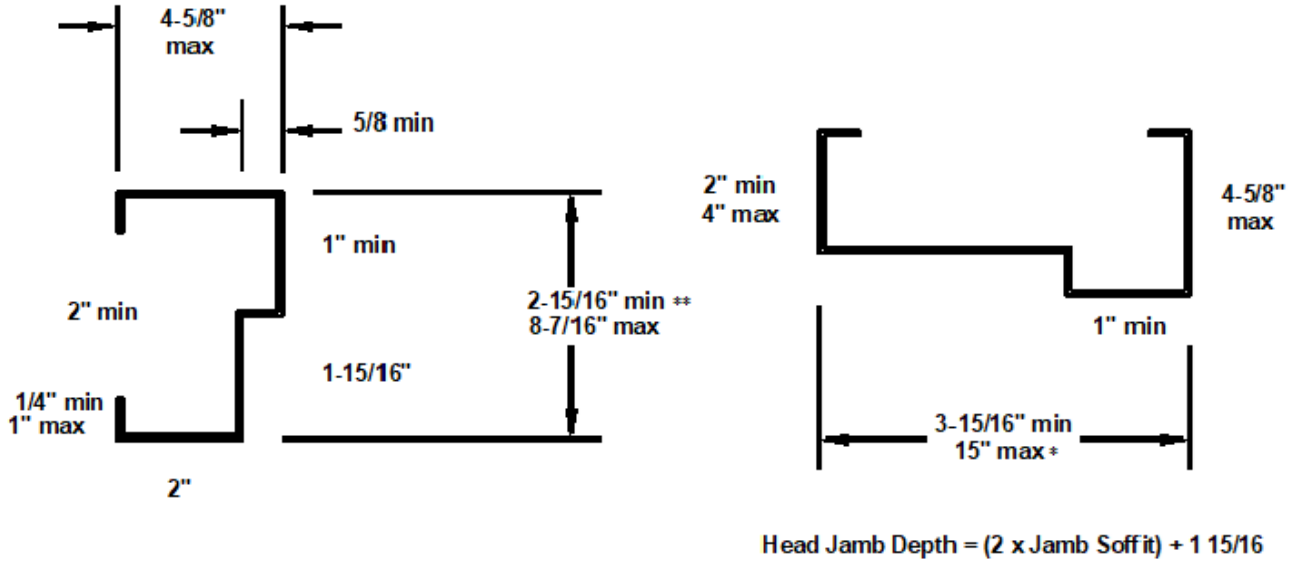
* Jamb depths greater than 14" cannot achieve the 13" maximum throat without the backbends being larger than ½".

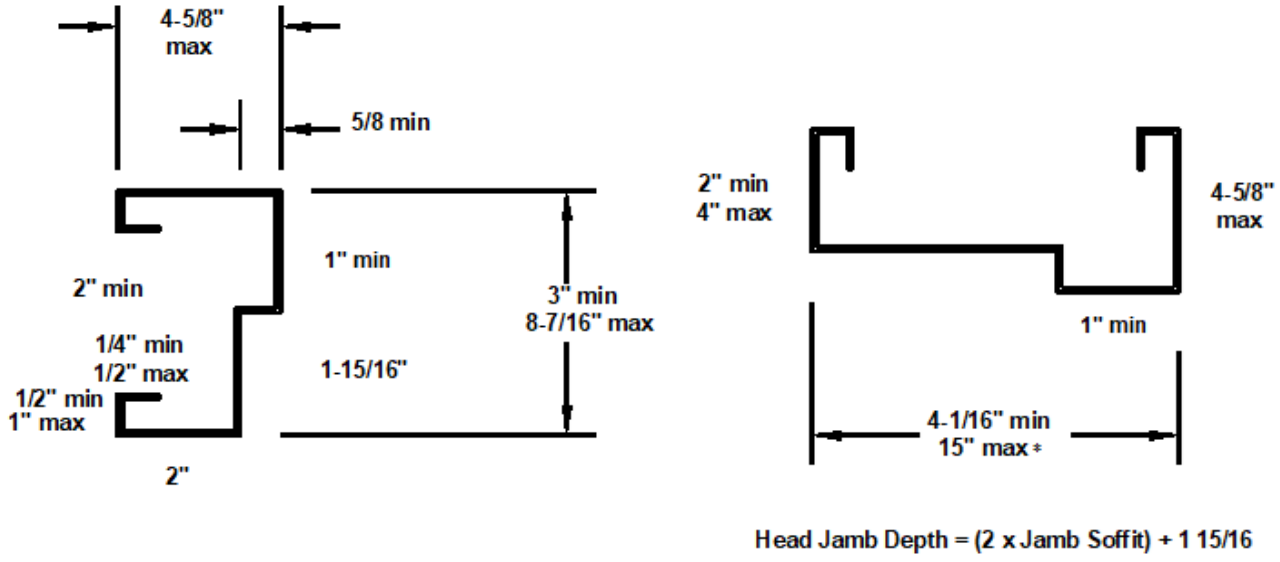
SWING CLEAR DOUBLE EGRESS

The minimum and maximum dimensions shown below may vary based upon frame elevation, rating, wall type, forming limitations, and for frames other than 16 gauge.

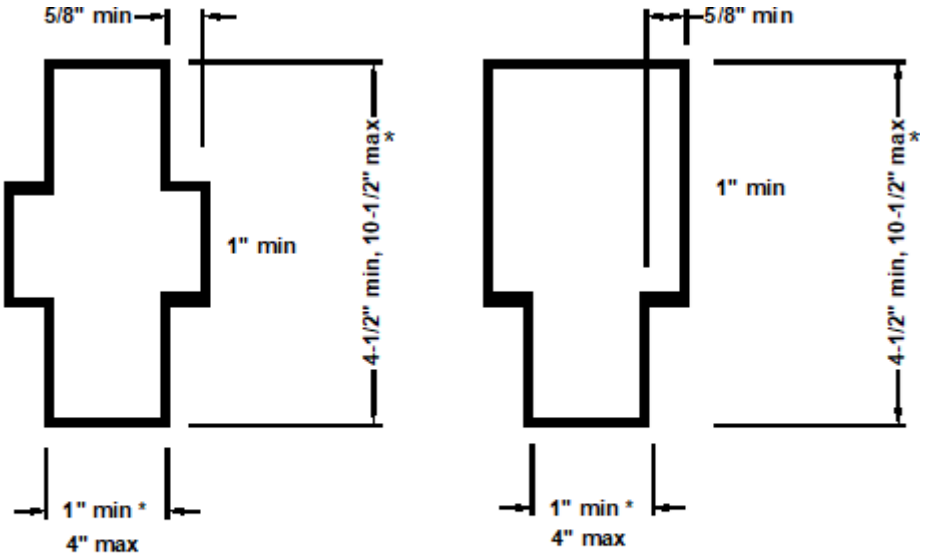
The profiles shown below may have combinations of the return types as shown on pages 96 & 97.

Fire ratings are ONLY available for swing clear double egress openings with Welded Corner construction.



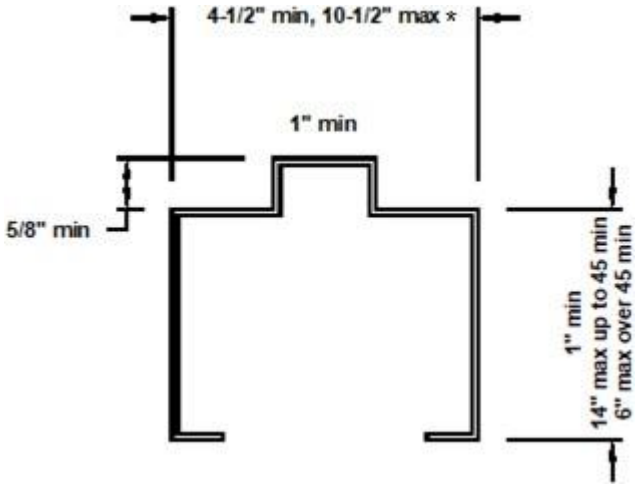


* Jamb depths greater than 14” cannot achieve the 13” maximum throat without the backbends being larger than ½”.
** To achieve the 2” min throat dimension, a 2-15/15” jamb depth profile cannot have ½” returns.



*** 1" minimum at members surrounding glazed openings in sidelight or window frames. 1-1/4" minimum at all transom frames and at members surrounding door openings in sidelight frames.**

Mullions



Sills

* Maximum throat = 9". Jamb depths greater than 10" cannot achieve the 9" maximum throat without the backbends being larger than ½".

ANCHORING AT MULLIONS & SILLS

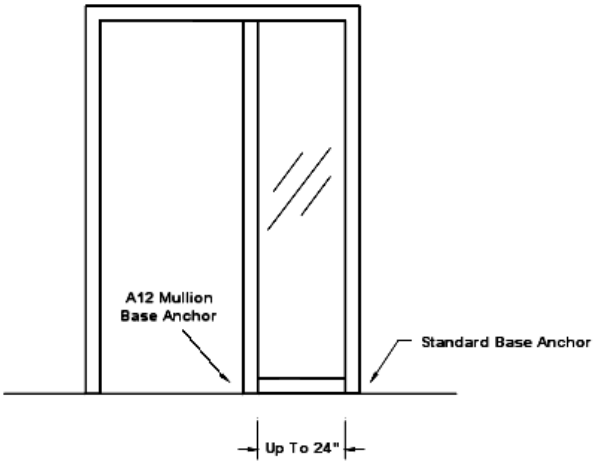
Approved Methods for Anchoring Mullions and Sills

Standard A13 sill base anchors are as shown on page 19. Punch & dimple type sill anchors are available upon request.

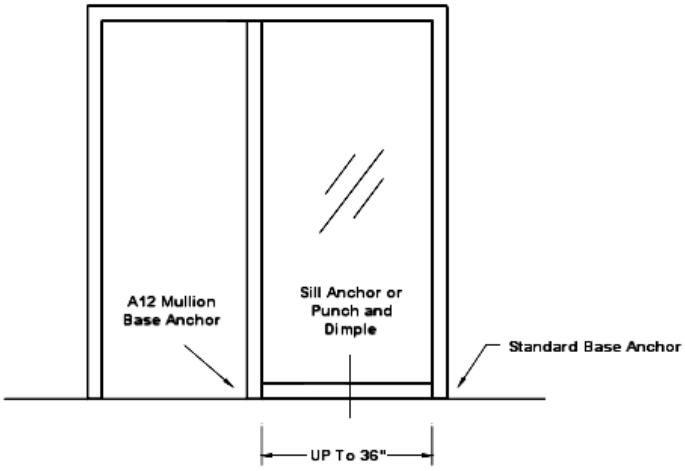
The required spacing for punch & dimple type sill anchors is one anchor for every foot with a minimum of two anchors for sills exceeding 36” in length.

Punch and dimple type sill anchors are to be spaced evenly unless the frame elevation dictates otherwise.

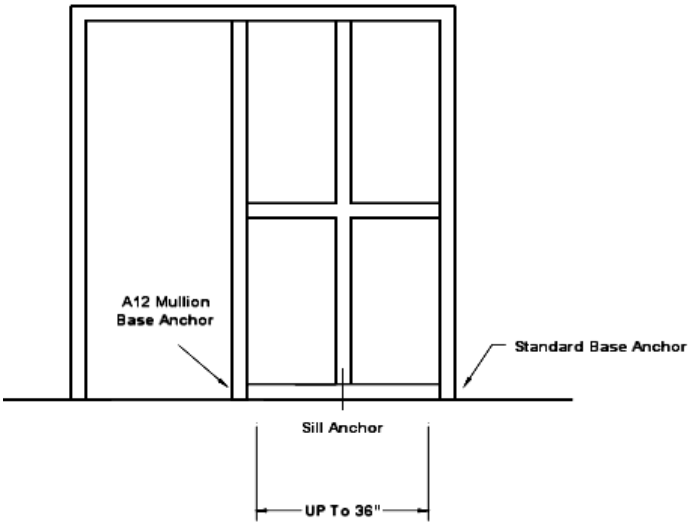
This can include an A12 mullion base anchor and a standard jamb base anchor for the appropriate wall type.



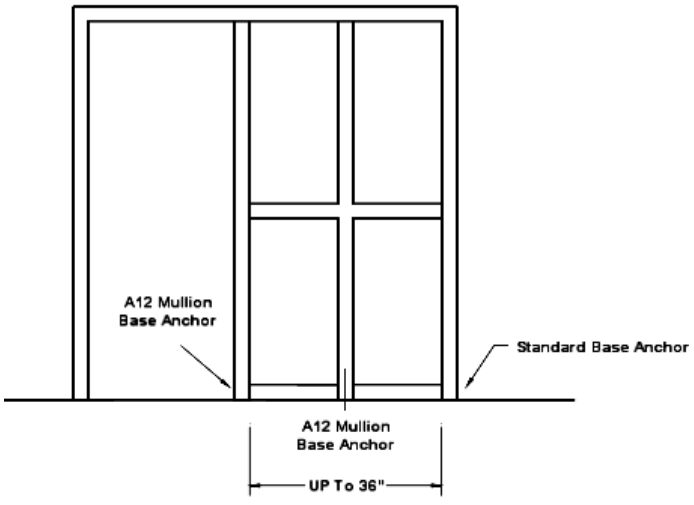
2 Anchors Total



3 Anchors Total



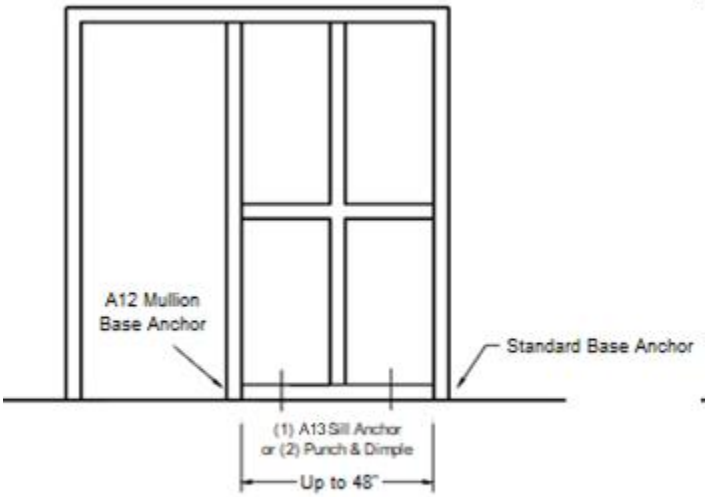
3 Anchors Total



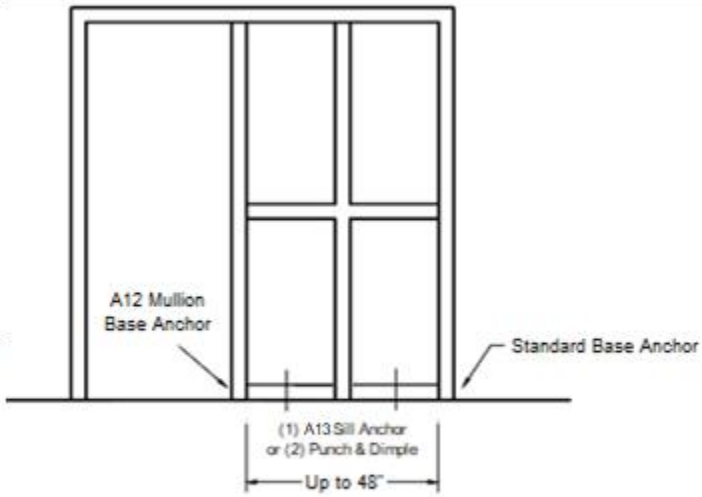
3 Anchors Total

Mullions may be run to the top of the sill or to the floor. Mullion or sill anchors to be supplied accordingly.

ANCHORING AT MULLIONS & SILLS



4 Anchors Total
(When using Punch & Dimple type sill anchors)



4 Anchors Total
(When using Punch & Dimple type sill anchors)

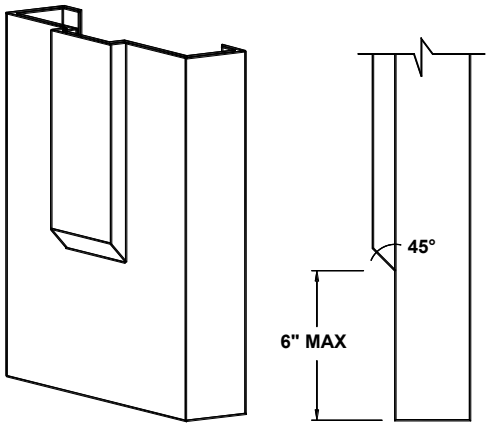
(A12) Mullion Base Anchor Installation:

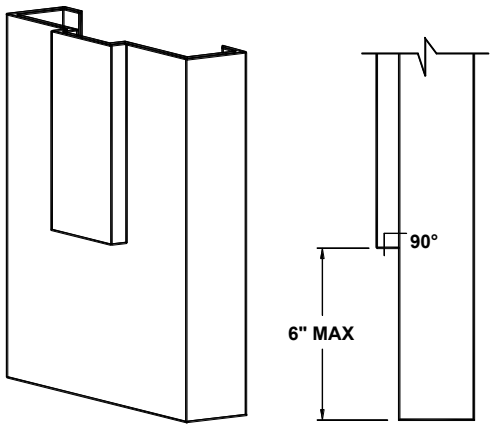
- A Mullion Base Anchor is designed to be centered in the face of the mullion as well as the jamb depth, and the fit should be tight against both dimensions.
- The Mullion Base Anchor should be placed on the floor taking care to ensure alignment with other base anchors as well as with the bottom of the frame.
- Fasten the anchor to the floor using expansion bolts or Rawlplugs.
- Slip mullion over the anchor.

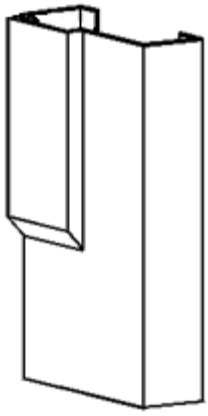
(A13) Sill Base Anchor Installation:

- A Sill Base Anchor is designed to be centered within the length of the sill as well as the throat depth, and the fit should be tight against the throat.
- The Sill Base Anchor should be placed on the floor taking care to ensure alignment with other base anchors as well as with the bottom of the frame.
- Fasten the anchor to the floor using expansion bolts or Rawlplugs.
- Slip sill over the anchor.

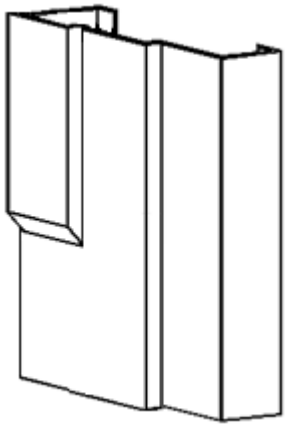
HOSPITAL / TERMINATED STOPS







HOSPITAL STOP
ON SINGLE RABBET



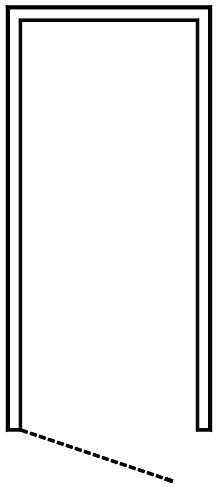
HOSPITAL STOP
ON DOUBLE EGRESS

Hospital stops available up to 180 min frames

Maximum 6” above the floor line, 45° and 90° only

Not permitted for lead lined frames

SINGLE SWING FRAME LIMITATIONS



Standard Profile Frames -

Construction:	KD-Complete or Welded		Welded
Maximum Door Opening Size:	4’-0” x 10’-0”		5’-0” x 12’-0”
Maximum Rating Steel/Wood Stud:	3 Hour		1-1/2 Hour
Maximum Rating Masonry:	3 Hour		1-1/2 Hour
Minimum Jamb Depth:	(See Pages 96-97)		4”
Maximum Jamb Depth:	15” *		15” *
Maximum Throat Depth:	13”		13”
Minimum Gage:	16	***OR***	16
Maximum Gage:	12		12
Minimum Face:	1” for masonry; 1-1/4” for drywall		1-1/2”

Drywall Pressure Anchor Frames -

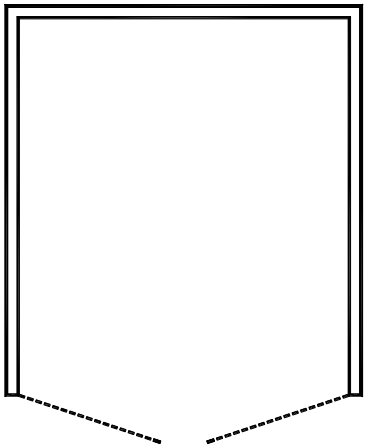
Nail-In Face Frames -

Construction:	KDDW	KD (See Page 99)
Maximum Door Opening Size:	4’-0” x 9’-0”	3’-6” x 7’-6”
Maximum Rating Steel/Wood Stud:	1-1/2 Hour	1-1/2 Hour
Maximum Rating Masonry:	N/A	N/A
Minimum Jamb Depth:	3-7/8”	2-9/16”
Maximum Jamb Depth:	15” *	13-1/8” **
Maximum Throat Depth:	13”	13”
Minimum Gage:	16	16
Maximum Gage:	14	12

* Jamb depths greater than 14” cannot achieve the 13” maximum throat without the backbends being larger than ½”.

** 13-1/8” maximum jamb depth is based upon a 16-gauge frame with the maximum 13” throat. Consult factory for 14- or 12-gauge frame limits.

PAIR SWING FRAME LIMITATIONS



Standard Profile Frames -

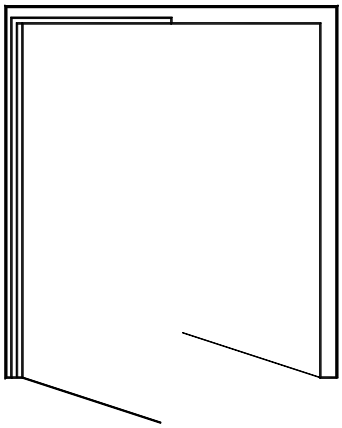
Construction:	KD-Complete or Welded		Welded
Maximum Door Opening Size:	8’-0” x 10’-0”		10’-0” x 12’0”
Maximum Rating Steel/Wood Stud:	3 Hour		1-1/2 Hour
Maximum Rating Masonry:	3 Hour		1-1/2 Hour
Minimum Jamb Depth:	(See Pages 96-97)		4”
Maximum Jamb Depth:	15” *		15” *
Maximum Throat Depth:	13”		13”
Minimum Gage:	16	***OR***	16
Maximum Gage:	12		12
Minimum Face:	1” for masonry; 1-1/4” for drywall		1-1/2”

Drywall Pressure Anchor Frames -

Construction:	KDDW
Maximum Door Opening Size:	8’-0” x 9’-0”
Maximum Rating Steel/Wood Stud:	1-1/2 Hour
Maximum Rating Masonry:	N/A
Minimum Jamb Depth:	3-7/8”
Maximum Jamb Depth:	15” *
Maximum Throat Depth:	13”
Minimum Gage:	16
Maximum Gage:	14

* Jamb depths greater than 14” cannot achieve the 13” maximum throat without the backbends being larger than ½”.

DOUBLE EGRESS FRAME LIMITATIONS



Double Egress Frames -

Construction:	KD-Complete ***		Welded
Maximum Door Opening Size:	8'-0" x 8'-0"		8'-0" x 10'-0"
Maximum Rating Steel/Wood Stud:	1-1/2 Hour		3 Hour
Maximum Rating Masonry:	1-1/2 Hour		3 Hour
Minimum Jamb Depth:	3-15/16" *		3-15/16" *
Maximum Jamb Depth:	15" **		15" **
Maximum Throat Depth:	13"		13"
Minimum Gage:	16	***OR***	16
Maximum Gage:	12		12

Swing Clear Double Egress Frames -

Construction:	Welded
Maximum Door Opening Size:	8'-0" x 8'-0"
Maximum Rating Steel/Wood Stud:	1-1/2 Hour
Maximum Rating Masonry:	1-1/2 Hour
Minimum Jamb Depth:	2-15/16" Legs, 3-15/16" Head ****
Maximum Jamb Depth:	8-7/16" Legs, 15" Head **
Maximum Throat Depth:	13"
Minimum Gage:	16
Maximum Gage:	12

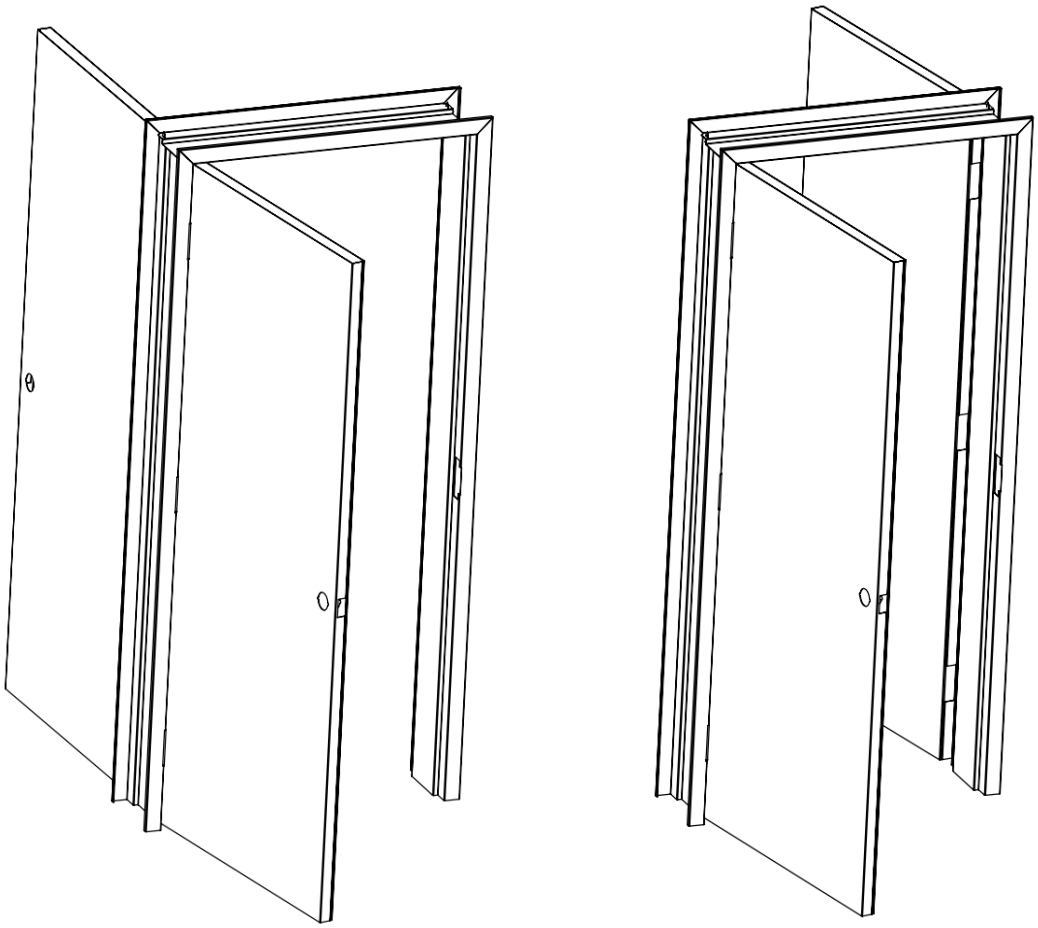
* Jamb depths smaller than 5" cannot achieve the 3" minimum throat with larger backbends on both sides of the profile.

** Jamb depths greater than 14" cannot achieve the 13" maximum throat without the backbends being larger than ½".

*** Double egress frames with KDDW corner construction and pressure anchors are not eligible for fire ratings.

**** Reference section details on page 100.

COMMUNICATING FRAME LIMITATIONS

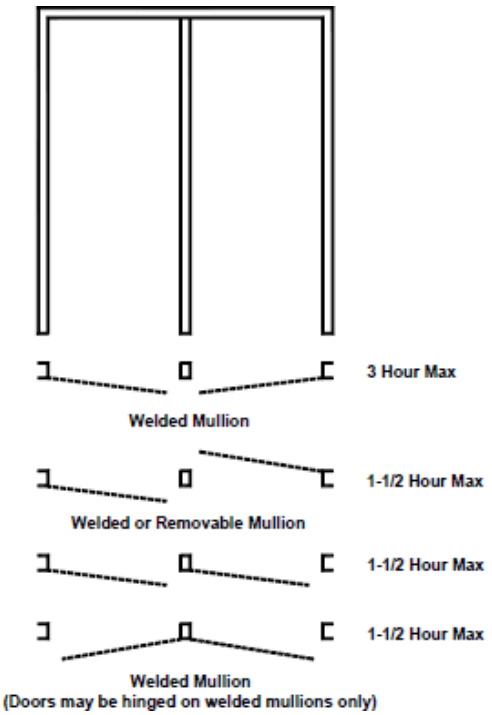


Standard Profile Frames -

Construction:	KD-Complete or Welded		Welded
Maximum Door Opening Size:	4’-0” x 10’-0”		5’-0” x 12’-0”
Maximum Rating Steel/Wood Stud:	3 Hour		1-1/2 Hour
Maximum Rating Masonry:	3 Hour		1-1/2 Hour
Minimum Jamb Depth:	(See Pages 96-97)		4”
Maximum Jamb Depth:	15” *		15” *
Maximum Throat Depth:	13”		13”
Minimum Gage:	16	***OR***	16
Maximum Gage:	12		12
Minimum Face:	1” for masonry; 1-1/4” for drywall		1-1/2”

*Jamb depths greater than 14” cannot achieve the 13” maximum throat without the returns being larger than ½”
** KDDW corner construction and pressure anchors are not eligible for fire ratings. A KD drywall hinge jamb should not hold 2 doors, even for non-rated applications (i.e. diagram on left is not permitted for KDDW)

MULTIPLE OPENING FRAME LIMITATIONS



(2) SINGLE OPENINGS SEPARATED BY (1) CENTER MULLION (“*PAIRS WITH CENTER MULLION*”)

Standard Profile Frames –

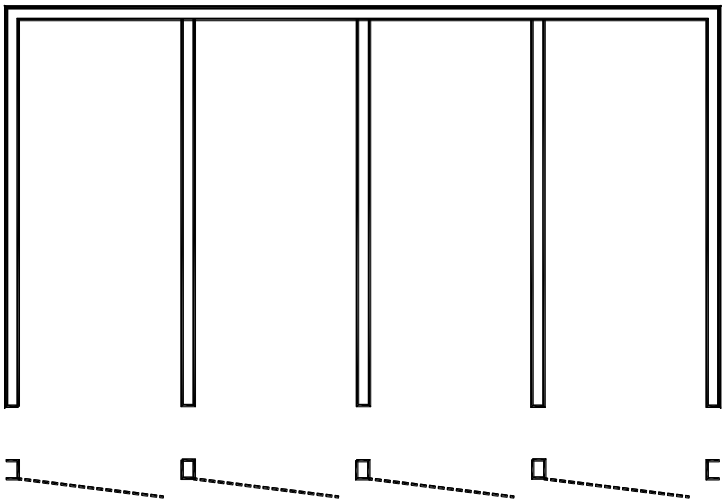
Construction:	KD-Complete (only with Removable Mullion) or Welded
Maximum Door Opening Size:	4’-0” x 8’-0” (each opening)
Maximum Rating Steel/Wood Stud:	1-1/2 Hour
Maximum Rating Masonry:	3 Hour (Welded Construction Only)
Minimum Jamb Depth:	4-1/2” *
Maximum Jamb Depth:	10-1/2” **
Minimum Gage:	16
Maximum Gage:	12
Mullion:	Removable or Welded

Drywall Pressure Anchor Frames –

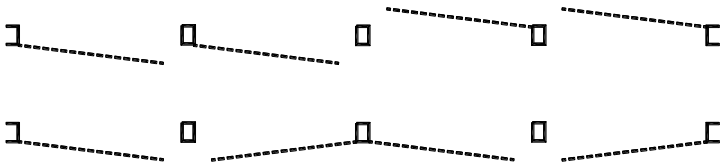
Construction:	KDDW Only
Maximum Door Opening Size:	4’-0” x 8’-0” (each opening)
Maximum Rating Steel/Wood Stud:	1-1/2 Hour
Maximum Rating Masonry:	N/A
Minimum Jamb Depth:	5-3/4” (4-3/4” minimum throat)
Maximum Jamb Depth:	10-1/2” **
Minimum Gage:	16
Maximum Gage:	14
Mullion:	Removable Only

* Minimum throat = 4”. Jamb depths less than 5” cannot achieve the 4” minimum throat with larger backbends on both sides of the profile.
** Maximum throat = 9”. Jamb depths greater than 10” cannot achieve the 9” maximum throat without the backbends being larger than ½”.

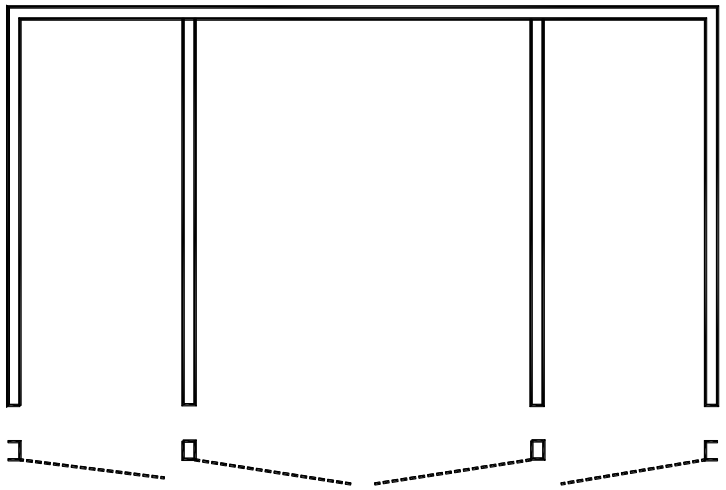
MULTIPLE OPENING FRAME LIMITATIONS



Welded Mullions
(Doors may be hinged on welded mullions only)



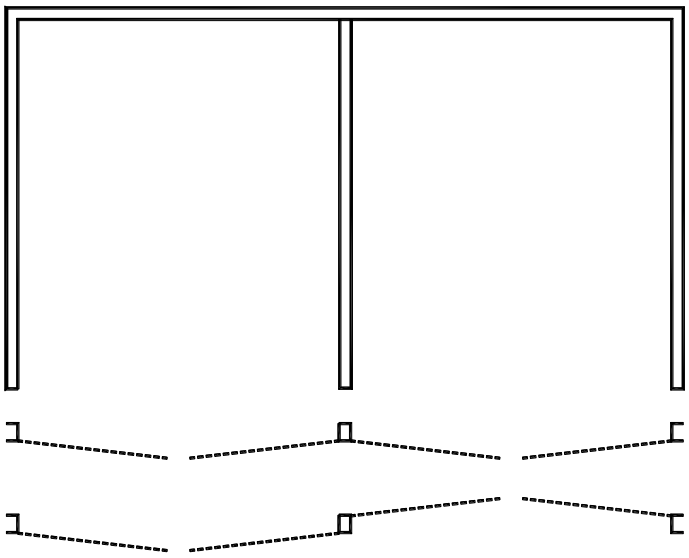
Welded or Removable Mullions
(Doors may be hinged on welded mullions only)



Welded Mullions
(Doors may be hinged on welded mullions only)

See next page for details and for additional elevation options.

MULTIPLE OPENING FRAME LIMITATIONS



Welded Mullions
(Doors may be hinged on welded mullions only)

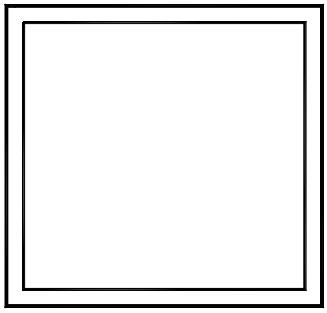
See previous page for additional elevation options.

Standard Profile Frames -

Construction:	Welded
Maximum Overall Frame Size:	12’-8” x 8’-4” Outside Dimensions *
Maximum Door Opening Sizes:	4’-0” x 8’-0” singles, 8’-0” x 8’-0” pairs
Maximum Rating Steel/Wood Stud:	1-1/2 Hour
Maximum Rating Masonry:	1-1/2 Hour
Minimum Jamb Depth:	5” (4-3/4” minimum throat)
Maximum Jamb Depth:	10-1/2” **
Minimum Gage:	16
Maximum Gage:	12
Mullion:	Removable or Welded

* Overall frame dimension limitations will include the nail flanges when present.
** Maximum throat = 9”. Jamb depths greater than 10” cannot achieve the 9” maximum throat without the backbends being larger than ½”.

SINGLE BORROWED LITE LIMITATIONS



¼" Safety Rated Wired Glass

Construction:	Welded
Maximum Clear Glass Width:	54" (20 - 45 min)
Maximum Clear Glass Height:	54" (20 - 45 min)
Maximum Clear Glass Area:	1296 sq inches (20 - 45 min)
Minimum Jamb Depth:	4-1/2" *
Minimum Throat Depth:	4"
Maximum Jamb Depth:	15" **
Maximum Throat Depth:	13"
Minimum Face:	1"
Minimum Gage:	16
Maximum Gage:	14
Glass Stop:	Min. 18 Gage 5/8" x 5/8" with #8 screws 4" from ends and 12" on center

Pilkington Wired Glass with Pemko FG3000 Fireglaze (or equal)

Construction:	Welded
Maximum Clear Glass Width:	98" (20 - 45 min)
Maximum Clear Glass Height:	106" (20 - 45 min)
Maximum Clear Glass Area:	4704 sq inches (20 - 45 min)
Minimum Jamb Depth:	5-3/4"
Minimum Throat Depth:	3-3/4"
Maximum Jamb Depth:	15" **
Maximum Throat Depth:	13"
Minimum Face:	2"
Minimum Gage:	16
Maximum Gage:	14
Glass Stop:	Min. 18 Gage 5/8" x 5/8" with #8 screws 4" from ends and 9" on center

20 minute without hose stream.

* Jamb depths less than 6" cannot achieve the 4" minimum throat with larger backbends on both sides of the profile.
** Jamb depths greater than 14" cannot achieve the 13" maximum throat without the backbends being larger than ½".

SINGLE BORROWED LITE LIMITATIONS

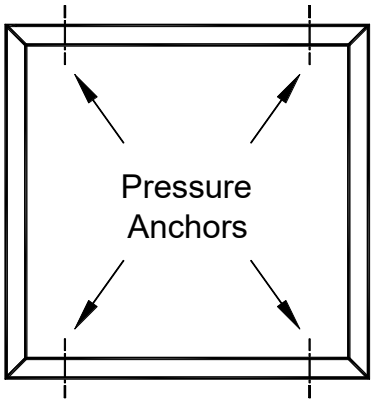
FireLite® (or equal)

Construction:	Welded
Maximum Clear Glass Width:	95” (20 - 60 min)
	56½” (90 min)
Maximum Clear Glass Height:	95” (20 - 60 min)
	56½” (90 min)
Maximum Clear Glass Area:	3325 sq inches (20 - 60 min)
	2627 sq inches (90 min)
Minimum Jamb Depth:	4-3/4” *
Minimum Throat Depth:	3-3/4”
Maximum Jamb Depth:	15” **
Maximum Throat Depth:	13”
Minimum Face:	1”
Minimum Gage:	16
Maximum Gage:	14
Glass Stop:	Min. 18 Gage 5/8” x 5/8” with
	#8 screws 4” from ends and 12” on center

20 minute without hose stream.

* Jamb depths less than 5-3/4” cannot achieve the 3-3/4” minimum throat with larger backbends on both sides of the profile.
** Jamb depths greater than 14” cannot achieve the 13” maximum throat without the backbends being larger than ½”.

KDDW BORROWED LITE LIMITATIONS



Drywall Pressure Anchor Frames -

¼" Safety Rated Wired Glass

Construction:	KDDW
Maximum Clear Glass Width:	54" (20 - 45 min)
Maximum Clear Glass Height:	54" (20 - 45 min)
Maximum Clear Glass Area:	1296 sq inches (20 - 45 min)
Minimum Jamb Depth:	4-1/2" *
Minimum Throat Depth:	4"
Maximum Jamb Depth:	15" **
Maximum Throat Depth:	13"
Minimum Face:	1-1/2"
Minimum Gage:	16
Maximum Gage:	14
Glass Stop:	Min. 18 Gage 5/8" x 5/8" with #8 screws 4" from ends and 12" on center

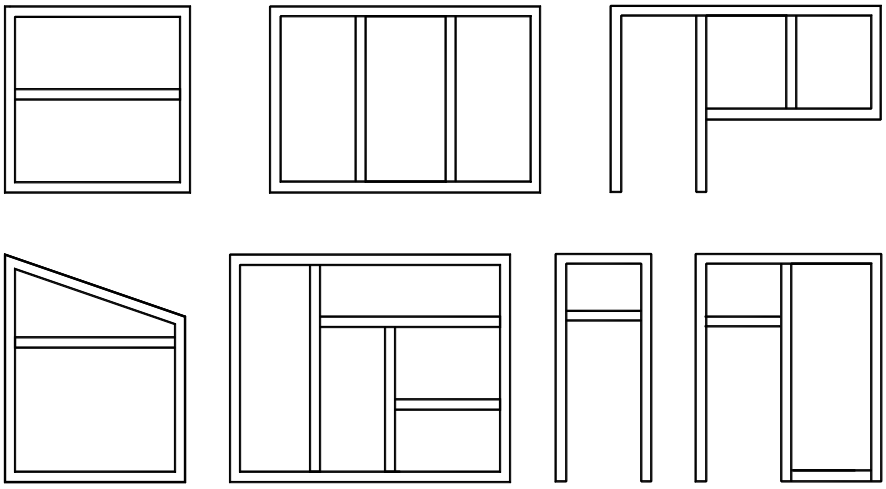
FireLite® (or equal)

Construction:	KDDW
Maximum Clear Glass Width:	54" (20 - 45 min)
Maximum Clear Glass Height:	96" (20 - 45 min)
Maximum Clear Glass Area:	3325 sq inches (20 - 45 min)
Minimum Jamb Depth:	4-3/4" ***
Minimum Throat Depth:	3-3/4"
Maximum Jamb Depth:	15" **
Maximum Throat Depth:	13"
Minimum Face:	1-1/2"
Minimum Gage:	16
Maximum Gage:	14
Glass Stop:	Min. 18 Gage 5/8" x 5/8" with #8 screws 4" from ends and 12" on center

20 minute without hose stream.

* Jamb depths less than 6" cannot achieve the 4" minimum throat with larger backbends on both sides of the profile.
** Jamb depths greater than 14" cannot achieve the 13" maximum throat without the backbends being larger than ½".
*** Jamb depths less than 5-3/4" cannot achieve the 3-3/4" minimum throat with larger backbends on both sides of the profile.

SIDELITE & MULTI-LITE FRAME LIMITATIONS



¼" Safety Rated Wired Glass

Construction:	Welded
Maximum Overall Frame Size:	174"x 120" *
Maximum Door Opening Width:	48" (singles); 96" (pairs; 48" per leaf)
Maximum Door Opening Height:	120"
Maximum Clear Glass Width:	54" (20 - 45 min)
Maximum Clear Glass Height:	54" (20 - 45 min)
Maximum Clear Glass Area:	1296 sq inches (20 - 45 min)
Minimum Jamb Depth:	4-1/2" (3-3/4" minimum throat) **
Maximum Jamb Depth:	10-1/2" (9" maximum throat) ***
Minimum Face:	1"
Minimum Gage:	16
Maximum Gage:	14
Glass Stop:	Min. 18 Gage 5/8" x 5/8" with #8 screws 4" from ends and 12" on center

20 minute without hose stream.

Max 45 Minute Rating with Mechanical Field Splice (See Page 124)

* Overall frame dimension limitations will include the nail flanges when present.

** Jamb depths less than 4-3/4" cannot achieve the 3-3/4" minimum throat with larger backbends on both sides of the profile.

*** Jamb depths greater than 10-1/2" cannot be used for a frame with a mullion. Only single lite borrowed lite frames can exceed a 10-1/2" jamb depth. Maximum throat = 9". Jamb depths greater than 10" cannot achieve the 9" maximum throat without the backbends being larger than ½".

SIDELITE & MULTI-LITE FRAME LIMITATIONS

Pilkington Wired Glass with Pemko FG3000 Fireglaze (or equal)

Construction:	Welded
Maximum Overall Frame Size:	174”x 136” Drywall 174”x 144” Masonry
Maximum Door Opening Width:	48” (singles); 96” (pairs; 48” per leaf)
Maximum Door Opening Height:	120”
Maximum Clear Glass Width:	98” (20 - 45 min)
Maximum Clear Glass Height:	106” (20 - 45 min)
Maximum Clear Glass Area:	4704 sq inches (20 - 45 min)
Minimum Jamb Depth:	5-3/4” (4-3/4” minimum throat)
Maximum Jamb Depth:	10-1/2” (9” maximum throat) *
Minimum Face:	2”
Minimum Gage:	16
Maximum Gage:	14
Glass Stop:	Min. 18 Gage 5/8” x 5/8” with #8 screws 4” from ends and 9” on center

FireLite® (or equal)

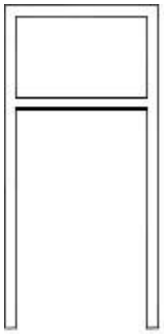
Construction:	Welded
Maximum Overall Frame Size:	174”x 120”
Maximum Door Opening Width:	48” (singles); 96” (pairs; 48” per leaf)
Maximum Door Opening Height:	120”
Maximum Clear Glass Width:	95” (20 - 60 min) 56½” (90 min)
Maximum Clear Glass Height:	95” (20 - 60 min) 56½” (90 min)
Maximum Clear Glass Area:	3325 sq inches (20 - 60 min) 2627 sq inches (90 min)
Minimum Jamb Depth:	4-3/4” (3-3/4” minimum throat)
Maximum Jamb Depth:	10-1/2” (9” maximum throat) *
Minimum Face:	1”
Minimum Gage:	16
Maximum Gage:	14
Glass Stop:	Min. 18 Gage 5/8” x 5/8” with #8 screws 4” from ends and 12” on center

20 minute without hose stream.

Max 45 Minute Rating with Mechanical Field Splice (See Page 124)

* Jamb depths greater than 10-1/2” cannot be used for a frame with a mullion. Only single lite borrowed lite frames can exceed a 10-1/2” jamb depth. Maximum throat = 9”. Jamb depths greater than 10” cannot achieve the 9” maximum throat without the backbends being larger than ½”.

TRANSOM FRAME WITH PANEL AND MULLION



Panels over 48" width to be secured with spring bolts (field or factory install) or tack welds (factory install) max 6" from end, 12" on center – both sides as well as top and bottom.

Otherwise:

Panel under 18" height - (2) spring bolts per side (field or factory install), or
(2) tack welds per side (factory install)
Panel 18"-36" height - (3) spring bolts per side (field or factory install), or
(3) tack welds per side (factory install)
Panel over 36" height - (4) spring bolts per side (field or factory install), or
(4) tack welds per side (factory install)

Construction:	Welded *	
Maximum Door Opening Width:	48" (singles); 96" (pairs; 48" per leaf)	
Maximum Door Opening Height:	120"	
Maximum Panel Opening Width:	96"	(90-minute maximum)
	48"	(180-minute)
Maximum Panel Opening Height:	54"	(single openings)
	48"	(pair openings)
Maximum Combined * Opening Height:	144"	(90-minute maximum)
	120"	(180-minute)
Panel Core:	Honeycomb or Steel Stiffened for Combined ** Heights up to 120"	
	Steel stiffened only for Combined ** Heights over 120" and up to 144"	
Minimum Jamb Depth:	4-1/2"	
Maximum Jamb Depth:	10-1/2" ***	
Minimum Face:	1"	
Minimum Gage:	16	
Maximum Gage:	14	

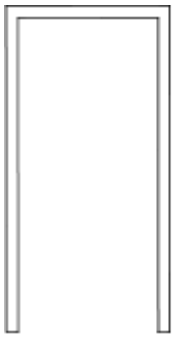
Fire rated frames with panels only (no lites) will include a "FIRE DOOR FRAME WITH TRANSOM / SIDE PANEL" label

Fire rated frames with lites and panels will include a "FIRE DOOR FRAME WITH LITES" label (maximum 90-minute rated). These frames must meet BOTH the limitations as noted on this page for a frame with panels AND the limitations corresponding to the fire rated glass being used.

Optional 3/8" thick "sandwich panel" **** can be included for up to 90-minute rated frames with sizes not to exceed the limitations of 90-minute FireLite® glass. These sandwich panels will be factory installed with standard "glass stop", but no glazing tape is required.

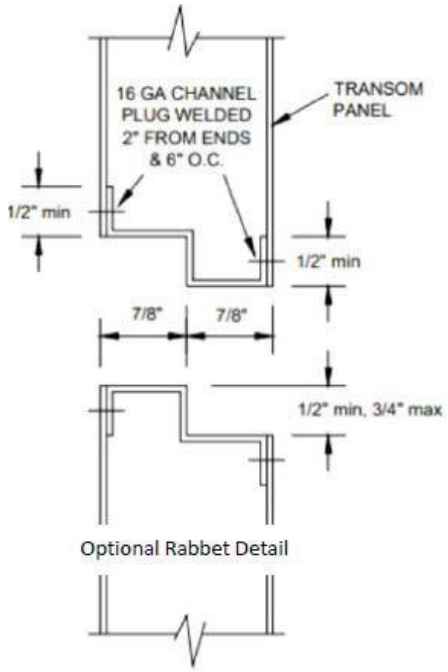
* Frames with a removable transom mullion and panel (see pages 35 & 36) are limited to single openings only with a maximum 90-minute rating and a maximum combined opening height of door, transom bar mullion face, and panel of 120". No vertical transom mullions permitted when using removable transom parts.
** Combined opening height of door, transom bar mullion face, and panel.
*** Maximum throat = 9". Jamb depths greater than 10" cannot achieve the 9" maximum throat without the backbends being larger than ½".
**** Sandwich panel to be ¼" gypsum board glued between 16 GA steel sheets. 3/8" thick panel with 3/8" spacing between fixed stop and "glass stop" (glazing tape not required).

TRANSOM FRAME WITH PANEL & NO MULLION (SINGLES ONLY)



Panel under 18" height - (2) spring bolts per side (field or factory install), or
(2) tack welds per side (factory install)
Panel 18"-36" height - (3) spring bolts per side (field or factory install), or
(3) tack welds per side (factory install)
Panel over 36" height - (4) spring bolts per side (field or factory install), or
(4) tack welds per side (factory install)

Astragal or Rabbet Required (Up to 120" Opening Height)
Astragal Required (Over 120" Opening Height)

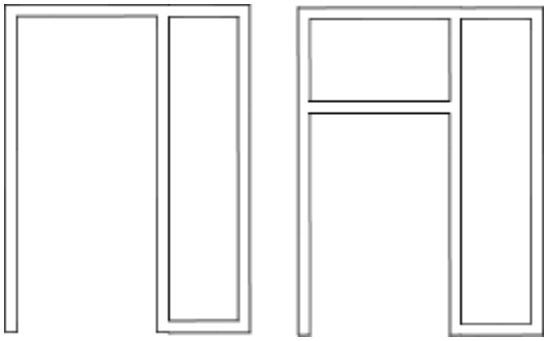


Construction:	Welded *
Maximum Door Opening Width:	48" (singles); N/A (pairs) **For Single Openings Only**
Maximum Door Opening Height:	120"
Maximum Panel Opening Height:	54"
Maximum Combined * Opening Height:	144" (90-minute maximum)
	120" (180-minute)
Door ***and*** 1-3/4" Thick Panel Core:	Honeycomb or Steel Stiffened for Combined ** Heights up to 120"
	Steel stiffened only for Combined ** Heights over 120" and up to 144"
Minimum Jamb Depth:	4-1/2"
Minimum Jamb Depth:	4-1/2"
Maximum Jamb Depth:	10-1/2"
Minimum Face:	1"
Minimum Gage:	16
Maximum Gage:	14

* Frames with a removable transom panel (see page 36) are limited to a maximum 90-minute rating and a maximum combined opening height of door and panel of 120", & a welded astragal is required on the panel (optional rabbet detail cannot be used for a removable transom panel).*

** Combined opening height of door and panel.

DOOR FRAME WITH FIXED SIDE PANEL



Construction:	Welded
Maximum Overall Frame Size:	174”x 120” * (up to 180-minute)
Maximum Door Opening Width:	48” (singles); 96” (pairs; 48” per leaf)
Maximum Door Opening Height:	118”
Maximum Side Panel Opening Width:	Not to exceed approved opening width of a door with the same core
Maximum Side Panel Opening Height:	Not to exceed approved opening height of a door with the same core
1-3/4” Thick Panel Core:	Honeycomb or Steel Stiffened
Minimum Jamb Depth:	4-1/2”
Maximum Jamb Depth:	10-1/2” **
Minimum Face:	1”
Minimum Gage:	16
Maximum Gage:	14
1-3/4” Thick Side Panel Installation:	Factory Welded Only

Fire rated frames with panels only (no lites) will include a “FIRE DOOR FRAME WITH TRANSOM / SIDE PANEL” label

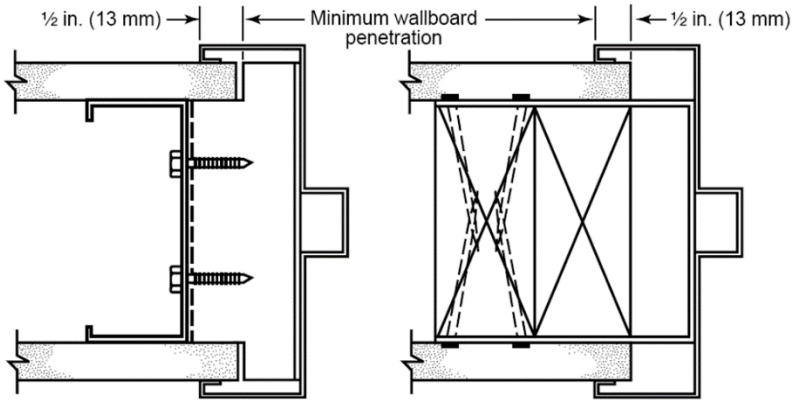
Fire rated frames with lites and panels will include a “FIRE DOOR FRAME WITH LITES” label (maximum 90-minute rated). These frames must meet BOTH the limitations as noted on this page for a frame with panels AND the limitations corresponding to the fire rated glass being used.

Frames with panels in the transom as well as a side panel are subject to the limitations as noted on the previous two pages for the transom panels.

Optional 3/8” thick “sandwich panel” *** can be included for up to 90-minute rated frames with sizes not to exceed the limitations of 90-minute FireLite® glass. These sandwich panels will be factory installed with standard “glass stop”, but no glazing tape is required.

* Overall frame dimension limitations will include the nail flanges when present.
** Maximum throat = 9”. Jamb depths greater than 10” cannot achieve the 9” maximum throat without the backbends being larger than ½”.
*** Sandwich panel to be ¼” gypsum board glued between 16 GA steel sheets. 3/8” thick panel with 3/8” spacing between fixed stop and “glass stop” (glazing tape not required).

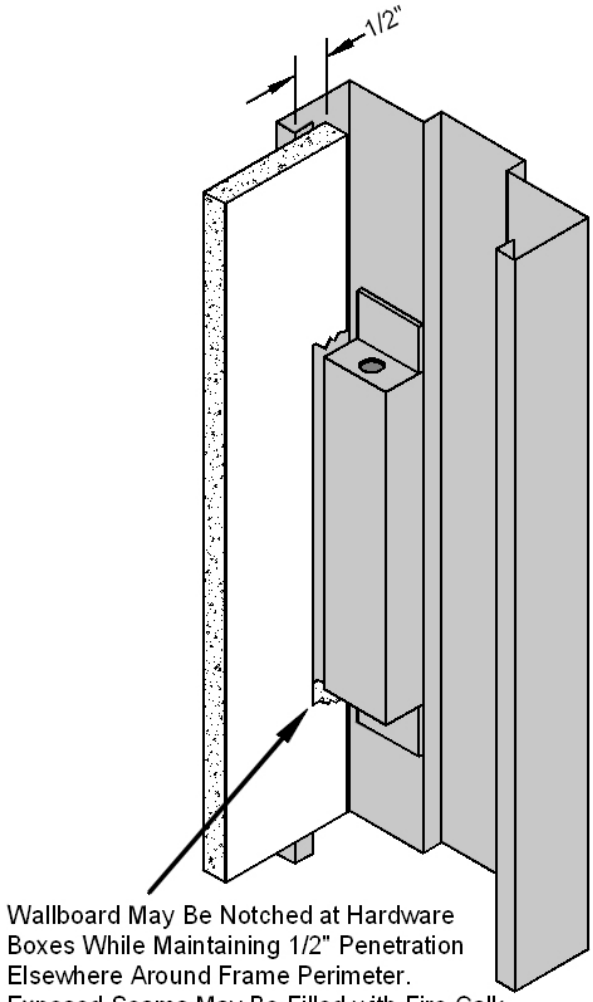
WALLBOARD PENETRATION AT FIRE-RATED FRAMES



Pressed steel door frame with steel stud anchors

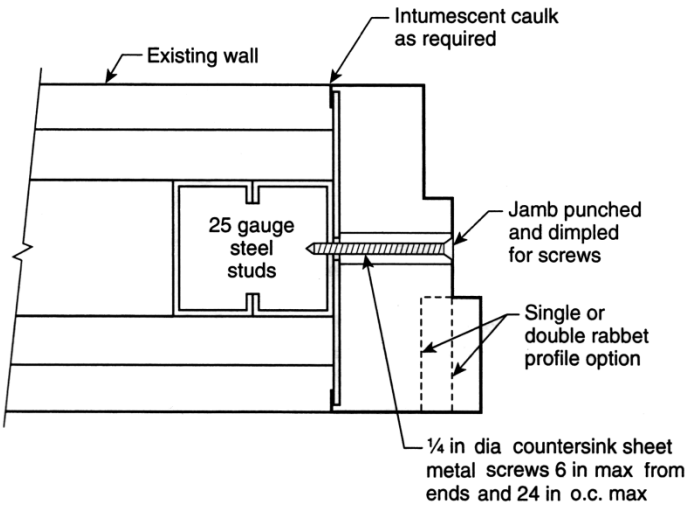
Pressed steel door frame with wood stud anchors

(Per NFPA 80)



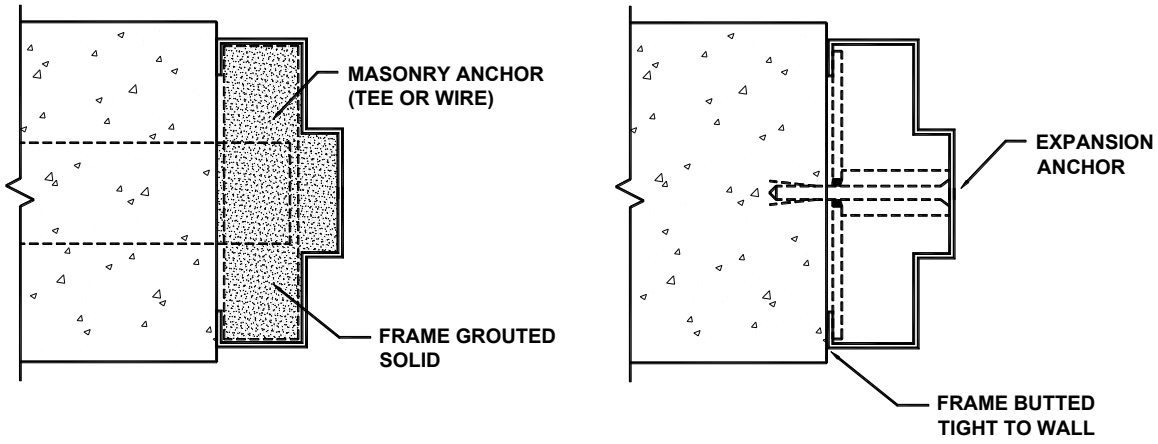
Wallboard May Be Notched at Hardware Boxes While Maintaining 1/2" Penetration Elsewhere Around Frame Perimeter. Exposed Seams May Be Filled with Fire Calk.

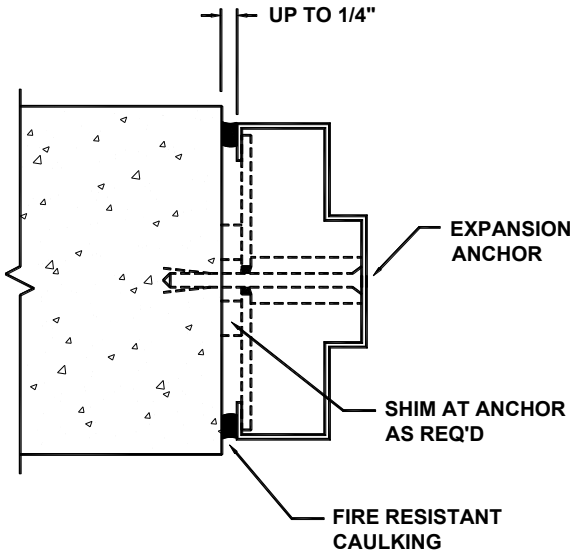
(Per ITS/Warnock Hersey)



Stud Walls with No Wallboard Penetration Required. (Per NFPA 80)

INSTALLATION OF BUTTED MASONRY FRAMES

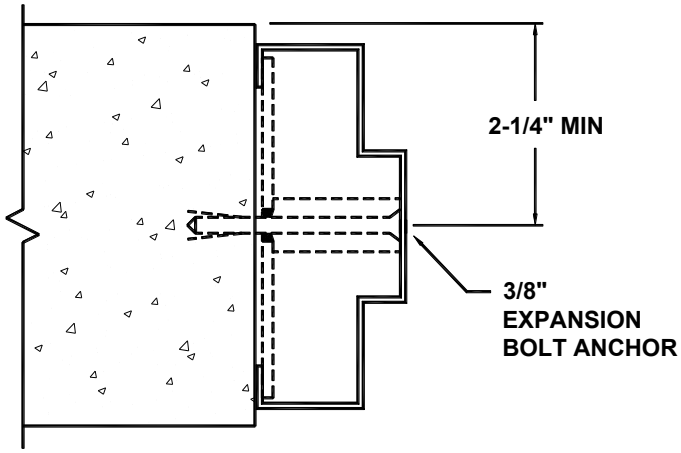




RATED P&D FRAMES- EXPANSION BOLTS/ SHEET METAL SCREWS

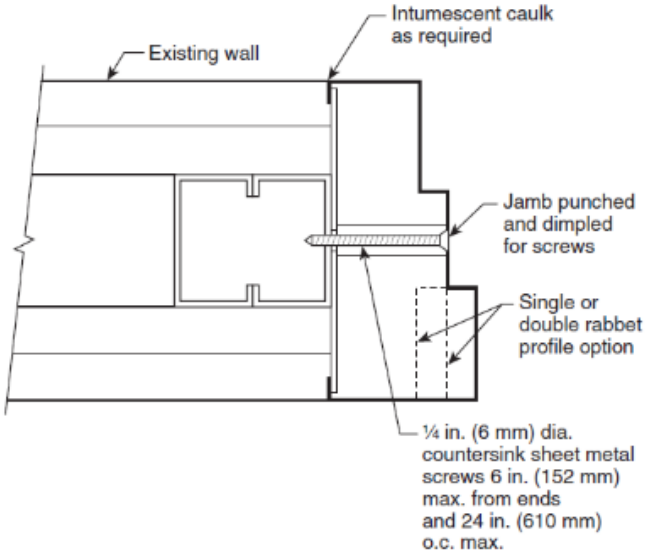
If expansion bolt anchors are to be used, the soffit shall be provided with 7/16” dimpled holes located not more than **6” from the top and bottom** with intermediate holes spaced no more than **26” on center**. The installer is to provide each hole with a minimum 3/8” roundhead bolt.

DCI’s standard is for the bottom bolt to act as the base anchor. However, adjustable type base anchors may also be used upon request.



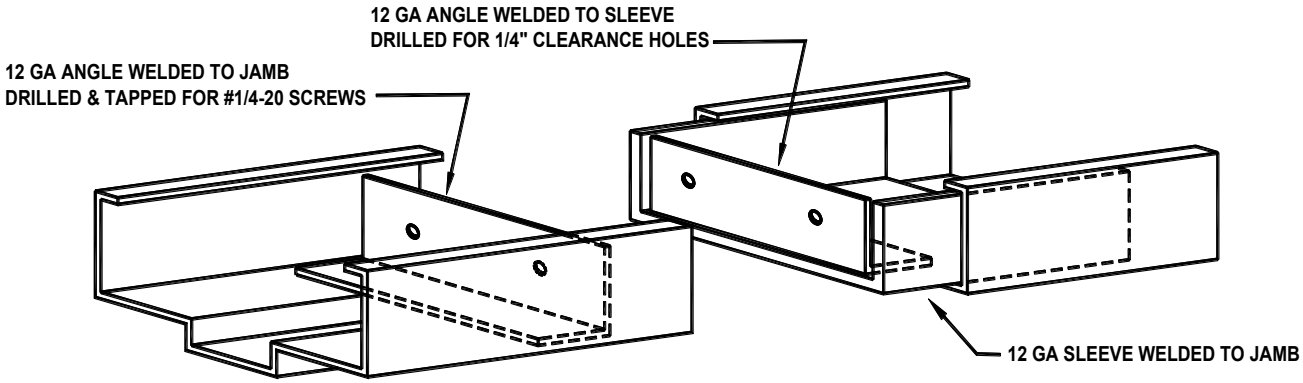
No expansion bolt anchor shall be set closer to the edge of the wall opening than six times the diameter of the anchor or closer to another anchor than eight times the diameter of the anchor.

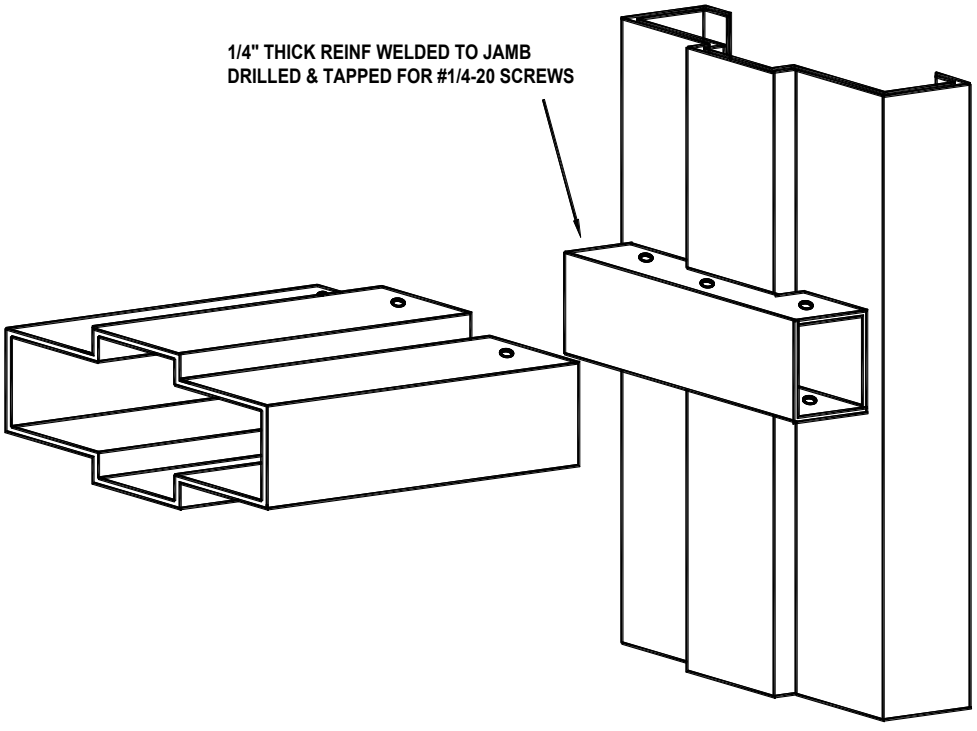
If 1/4” sheet metal screws are to be used, then the dimpled holes are to be located not more than **6” from the top and bottom** with intermediate holes spaced no more than **24” on center**. The installer is to provide each hole with a minimum 1/4” sheet metal screw.



MECHANICAL FIELD SPLICE

Mechanical Field Splices for Oversized Frames
(45 Minute Maximum Rating)





FIELD SPLICE FOR WELDING AVAILABLE FOR FRAMES THAT ARE NOT FIRE RATED

FIRE-RATED SINGLE DOOR TYPES

DOOR TYPE		RATINGS (MINUTES)	CORES[1] H	P	S	TRP-250	TRG-250	TRS-450	Lead Lined	STC
			MAXIMUM OPENING SIZE							
	SINGLE	20 45	4-0x10-0	4-0x8-0	5-0x12-0				4-0x10-0	
	SINGLE	60 90	4-0x10-0	4-0x8-0	5-0x12-0	4-0x8-0	4-0x9-0	5-0x12-0	4-0x10-0	
	SINGLE	180	4-0x10-0	4-0x8-0	4-0x10-0	4-0x8-0			4-0x10-0	
	SINGLE LOUVER[3]	45 60 90	4-0x10-0	4-0x8-0	5-0x12-0					
	SINGLE w/ LITE[3]	20 45	4-0x10-0	4-0x8-0	5-0x12-0				4-0x10-0	
	SINGLE w/ LITE[3]	60 90	4-0x10-0	4-0x8-0	5-0x12-0	4-0x8-0	4-0x9-0	5-0x12-0	4-0x10-0	
	DUTCH[4] DOOR	20 45 60 90 180	4-0x10-0		4-0x10-0					
	STC-48[4] SINGLE w/ LITE	20 45 60 90								3-2x7-0
	STC-50 SINGLE	20 45 60 90								3-0x7-0
	STC-53 SINGLE	20 45 60 90								3-2x7-0
	1-3/8" THICK SINGLE	20 45 60 90	3-6x7-6		3-6x7-6					
	2" THICK SINGLE	20 45			5-0x12-0				4-0x10-0	
	2" THICK SINGLE	60 90			5-0x12-0			5-0x12-0	4-0x10-0	
	2" THICK SINGLE	180			4-0x10-0				4-0x10-0	
	2-1/4" THICK SINGLE	20 45 60 90 180			4-0x10-0				4-0x10-0	

1. H=HONEYCOMB P=POLYSTYRENE(EPS) S=STEEL STIFFENED TRP-250=STANDARD TEMP RISE CORE TRG-250=SPECIAL TEMP RISE CORE TRS-450=STEEL STIFFENED W/ HIGH DENSITY INSULATION LEAD LINED= RADIATION SHIELDING STC=ACOUSTIC
2. ALL DOORS AVAILABLE 1-3/4" THICK FOR 14, 16, 18 GAUGES
3. SEE LOUVER AND GLAZING SPECIFICATIONS FOR MAXIMUM SIZES FOR THOSE COMPONENTS
4. DUTCH DOORS AND STC-48 SINGLES WITH LITES CAN HAVE LITES UP TO 100 SQ INCHES

FIRE-RATED PAIR DOOR TYPES

DOOR TYPE		CORES[1]	H	P	S	TRP-250	TRG-250	TRS-450	Lead Lined	STC
		RATINGS (MINUTES)	MAXIMUM OPENING SIZE							
	PAIR	20 45	8-0x10-0	6-0x8-0	10-0x12-0				8-0x10-0	
	PAIR	60 90	8-0x10-0	6-0x8-0	10-0x12-0	8-0x8-0	8-0x9-0	10-0x12-0	8-0x10-0	
	PAIR	180	8-0x10-0		8-0x10-0	8-0x8-0			8-0x10-0	
	DOUBLE EGRESS	20 45	8-0x10-0		8-0x10-0				8-0x10-0	
	DOUBLE EGRESS	60 90	8-0x10-0		8-0x10-0	8-0x8-0			8-0x10-0	
	DOUBLE EGRESS	180	8-0x10-0		8-0x10-0				8-0x10-0	
	PAIR LOUVER[3]	45 60 90	8-0x10-0	6-0x8-0	10-0x12-0					
	PAIR w/ LITE[3]	20 45	8-0x10-0	6-0x8-0	10-0x12-0				8-0x10-0	
	PAIR w/ LITE[3]	60 90	8-0x10-0	6-0x8-0	10-0x12-0	8-0x8-0	8-0x9-0	10-0x12-0	8-0x10-0	
	STC-48 PAIR	20 45 60 90								6-4x7-0
	2" THICK PAIR	20 45			10-0x12-0				8-0x10-0	
	2" THICK PAIR	60 90			10-0x12-0			10-0x12-0	8-0x10-0	
	2" THICK PAIR	180			8-0x10-0				8-0x10-0	
	2-1/4" THICK PAIR	20 45 60 90 180			8-0x10-0				8-0x10-0	

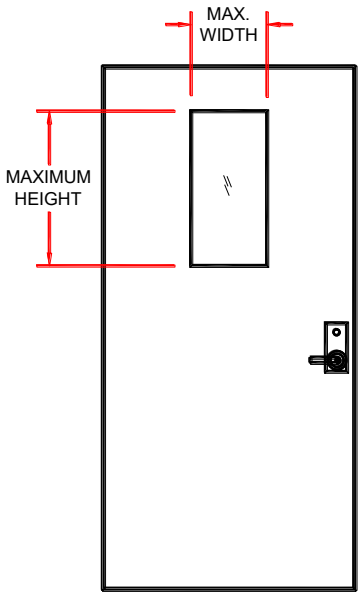
1. H=HONEYCOMB P=POLYSTYRENE(EPS) S=STEEL STIFFENED TRP-250=STANDARD TEMP RISE CORE TRG-250=SPECIAL TEMP RISE CORE TRS-450=STEEL STIFFENED W/ HIGH DENSITY INSULATION LEAD LINED= RADIATION SHIELDING STC=ACOUSTIC
2. ALL DOORS AVAILABLE 1-3/4" THICK FOR 14, 16, 18 GAUGES
3. SEE LOUVER AND GLAZING SPECIFICATIONS FOR MAXIMUM SIZES FOR THOSE COMPONENTS

RATED DOORS WITH LITES

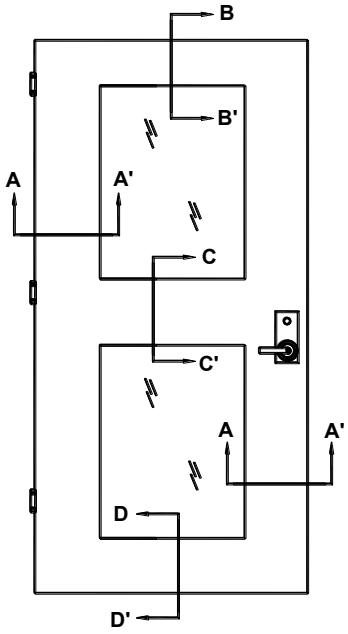
Maximum Glass Opening Sizes Permitted for Doors

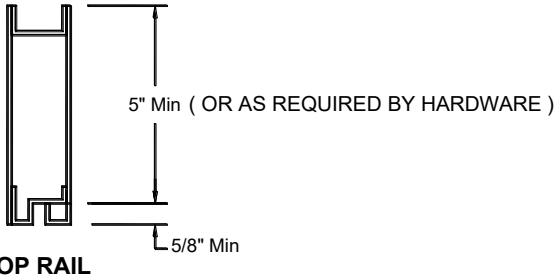
(Dimensions Listed Are All Visible Glass)

RATING (MINUTES)	MAX WIDTH	MAX HEIGHT	MAX AREA PER LITE	MAX AREA PER LEAF	MAX WIDTH	MAX HEIGHT	MAX AREA PER LITE	MAX AREA PER LEAF	MAX WIDTH	MAX HEIGHT	MAX AREA PER LITE	MAX AREA PER LEAF
20	36	54	1296	2400	34	84	2856	2856	36	89	3204	3204
45	36	54	1296	2400	34	84	2856	2856	36	89	3204	3204
60	12	33	100	100	12	46	552	1716	36	89	3204	3204
90	12	33	100	100	12	46	552	1716	36	56-1/2	2034	2034
180	-	-	-	-	-	-	-	-	12	33	100	100
60 TR	12	33	100	100	12	33	100	100	12	33	100	100
90 TR	12	33	100	100	12	33	100	100	12	33	100	100
180 TR	-	-	-	-	-	-	-	-	-	-	-	-
	SAFETY RATED WIRE GLASS				PILKINGTON GLASS w/ PEMKO FIREGLAZE (OR EQUAL)				FIRELITE (OR EQUAL)			

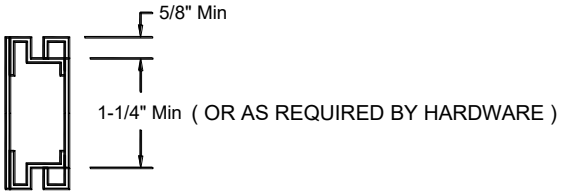


RATED DOORS WITH LITES

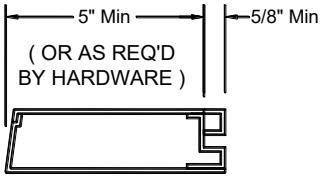




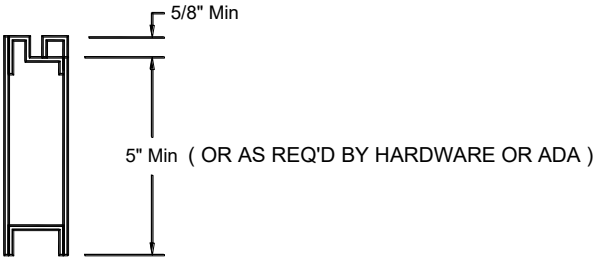
TOP RAIL
Section B-B'



INTERMEDIATE RAILS
Section C-C'



STILES
Section A-A'



BOTTOM RAIL
Section D-D'

OVERSIZED FIRE DOORS & FRAMES

ALL OVERSIZED FIRE DOORS

1. A door is considered oversized if the nominal width exceeds 4’0” (48”) and/or the nominal height exceeds 10’0” (120”).
2. An oversized fire door cannot exceed a 5’0” (60”) nominal width and/or a 12’0” (144”) nominal height.
3. Must have a Steel Stiffened Core (TRS core is an option).
4. Panic Hardware is not permitted except as detailed in the note at the bottom of the page. *
5. All latches (whether for a lock, mortise fire exit hardware device, flush bolt or surface bolt) require a ¾” throw.
6. The Fire Pin Preps noted below are to be located a maximum of 6” from the lock edge at the top of the door.
7. 90-minute maximum fire rating.

OVERSIZED FIRE DOORS FOR SINGLE OPENINGS

1. A Fire Pin Prep is required for all oversized single doors.
2. Only cylindrical locks and mortise locks are permitted. *

OVERSIZED FIRE DOORS FOR PAIR OPENINGS

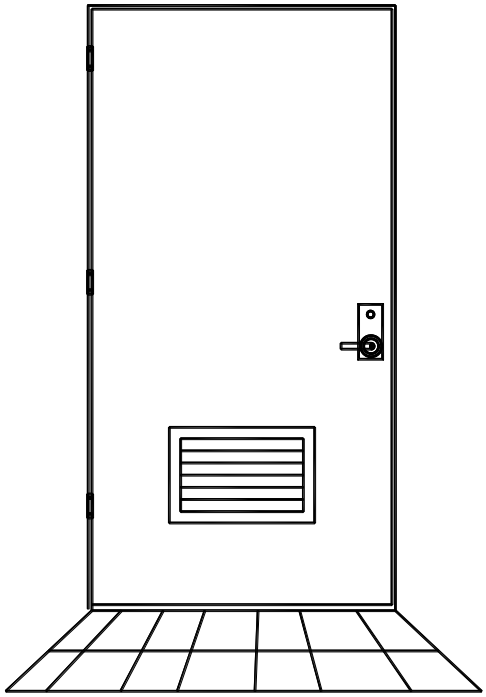
1. A welded-on flat bar astragal must be provided for all pairs having at least one oversized fire door.
2. Even if the active leaf itself is not oversized, the active for any pair opening with at least one oversized door will require a Fire Pin Prep.
3. Only cylindrical locks and mortise locks are permitted for the active leaf. *
4. Inactive leaf is permitted to have listed flush or surface bolts (automatic, manual, or self-latching types). Bolts up to a maximum 60” length are permitted. A fire pin CANNOT be substituted in place of the bottom bolt.

FRAMES FOR OVERSIZED FIRE DOORS

1. Must be welded. Neither a KD-Complete nor a KDDW corner prep can be used.
2. Can ship with mechanical field splice joint if required or requested.
3. Must include strike for the Fire Pin Prep aligned with the Door Prep as indicated above for each door opening type.
4. Double Egress Not Permitted. Sidelite Not Permitted.
5. Transom Frames permitted. | 4’0” maximum transom opening height. | Combined height of door and transom not to exceed 12’0”. | Must include fixed header mullion. | Transom can contain listed glass within visible lite dimension limitations of the glass or infill panels or 1-3/4” thick hollow metal panels. | Door(s) & hollow metal transom panel(s) must be steel stiffened (TRS is NOT an option for the panels).

* A mortise fire exit hardware device can be used at single openings or at the active leaf of a pair, but there are limitations to consider:
  1) The device must qualify as a panic device.
     a) This means that the actuating mechanism for the device must span at least half the width of the door.
     b) This panic device qualification requirement will disqualify touchpad devices at a much smaller door leaf width than for crossbar devices.
  2) For pair openings, the device must be approved for use with the hardware preps as detailed above for the inactive leaf.
     a) This information is typically found in the catalog details for the device.

RATED DOORS WITH LOUVERS



Louvers are <u>NOT</u> permitted in the following :

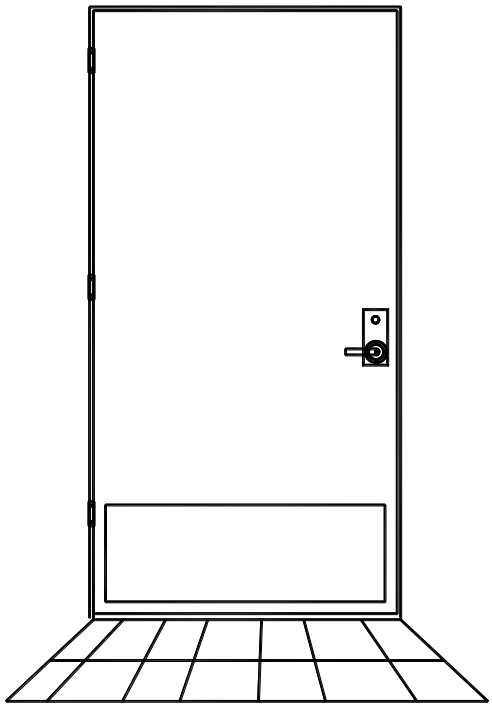
- **20 Minute Doors**
- **3 Hour Doors**
- **Rated Doors with Exit Devices Unless Permitted by Local Code**
- **Rated Doors with Lites**
- **The Top Half of a Rated Door (No Part of Louver can be higher than 40” Above Finished Floor)**
- **Temperature Rise Rated Doors**
- **Transoms of Rated Frames**
- **No DCI Integral Louvers in Rated Doors**

Louvers are restricted to use in 45, 60 and 90 min doors where smoke control is not a consideration. They must be listed Fusible Link types and may not exceed 24"x 24".

Per DCI Installation Instructions, the "S" indicated on the label for this door is voided by the use of louvers.

All cutouts for louvers will be channeled by DCI.

RATED DOORS WITH PROTECTION PLATES



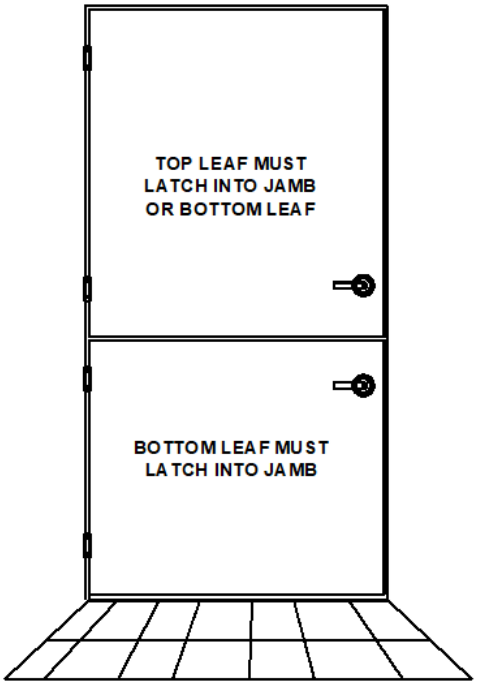
Per HMMA 850 & NFPA 80:

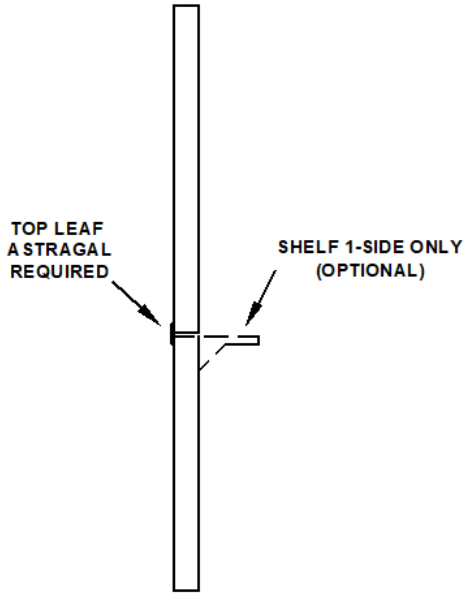
"The top of the plate shall be a maximum of 16 inches above the bottom of the door unless otherwise tested and labeled. Such plates may be on both faces. No other plates shall be installed."

Exception:

Life Safety Code NFPA 101 permits the following for Health Care Facilities only:
"Nonrated, factory- or field-applied protective plates, unlimited in height, shall be permitted."

RATED DUTCH DOORS





Rated up to 3 hours

Lite up to 100 sq in. allowed

4’-0” x 10’-0” maximum opening size

Dead latch is permitted in top leaf when not in a means of egress.

Dead bolt (in addition to latches) is permitted when not in a means of egress.

RATED DOORS WITH FLAT BAR ASTRAGALS

1. Overlapping flat bar astragals are not required on standard pairs of doors or double egress doors up to 8'-0"x 10'-0" in size and up to a 3-hour fire rating.

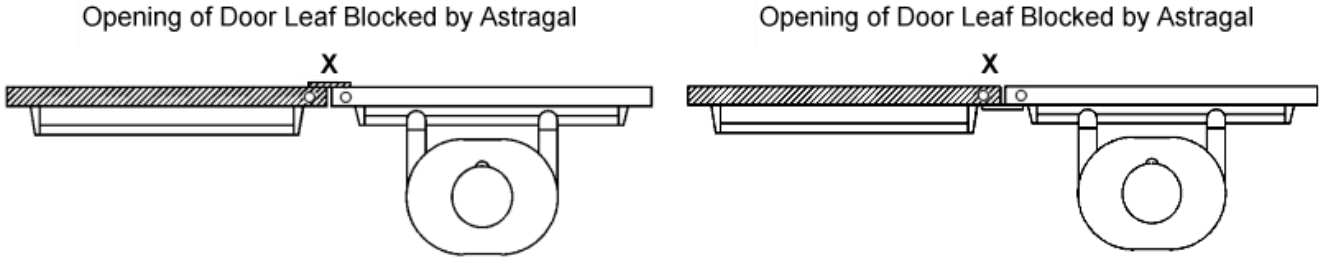
2. Astragals are NOT permitted on pairs of doors with vertical rod exit devices on both leaves, or when rim panic devices are used on both leaves with a hardware mullion.

3. Overlapping flat bar astragals ARE required on all standard pairs over 10'-0" high or when one leaf exceeds 4’-0” in width. (Therefore, exit devices may not be used on standard pairs of doors over 10'-0" high or when either door leaf is over 4’-0” wide)

4. The customer may need to provide additional gasketing/smoke-seal type astragals to comply with UL1784 smoke and draft control requirements where required by code.

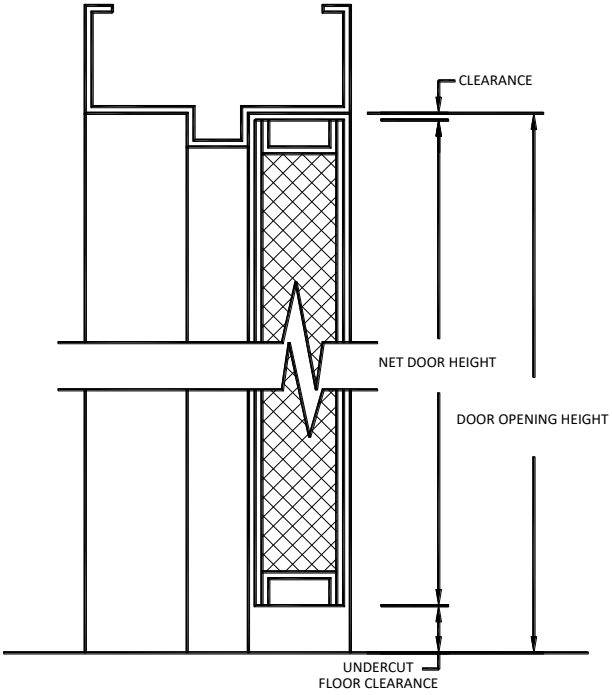
<u>Standard swing pairs of doors with vertical rod devices:</u>

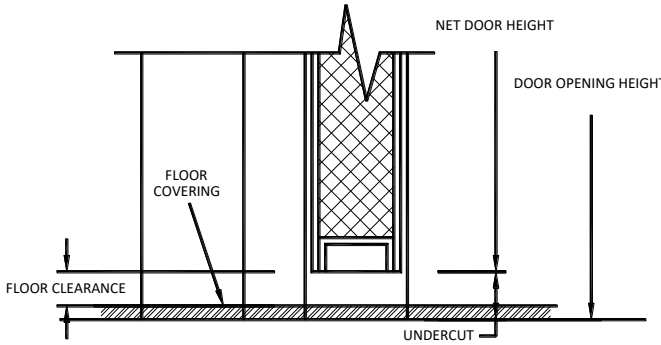
With two vertical rods, an astragal applied to the pull side of one leaf would disable the leaf without the astragal.
An astragal applied to the push side of one leaf would also disable the leaf with the astragal.

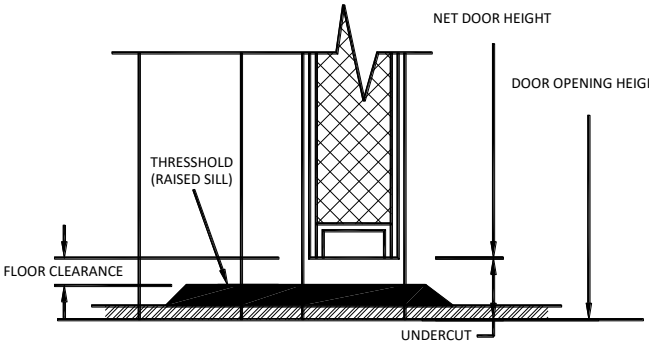


Astragals <u>ARE</u> permitted when a door with a vertical rod device is used with an inactive leaf equipped with flush bolts.
(Where only one leaf is needed to satisfy egress requirements)

UNDERCUTS AT RATED DOORS





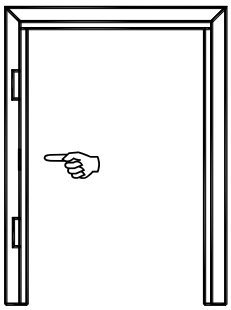


Maximum Clearances allowed:

Clearance Between Bottom of Door and …	
raised sill *	3/4"
floor without sill	3/4"
rigid floor tiles	3/4"

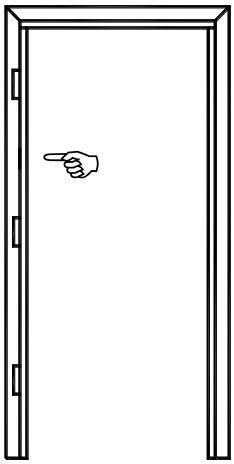
* Life Safety Code, NFPA 101 limits sill height to 1/2" in a means of egress.
ADA limits sill height to 1/2" at all accessible doors.

LABEL PLACEMENT GUIDE FOR FRAMES



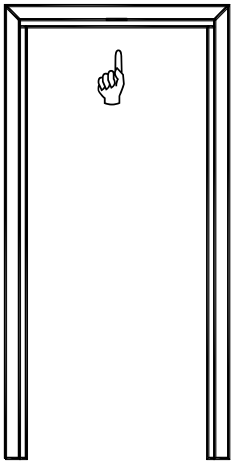
2 HINGES OR PIVOTS

MIDDLE OF HINGE JAMB RABBET
(PLUS OR MINUS 8 INCHES)



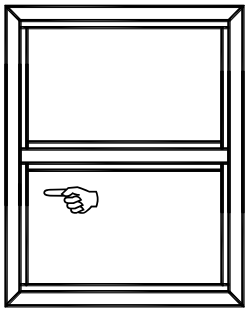
3 OR 4 HINGES OR PIVOTS

HINGE JAMB RABBET AT 52” TO
68” FROM BOTTOM OF FRAME



NO HINGES (CONTINUOUS OR
SURFACE HINGES)

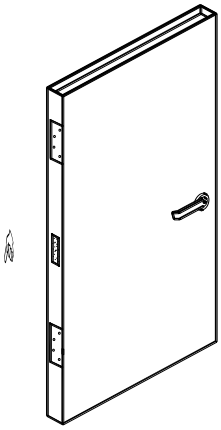
DOOR SIDE RABBET OF HEAD
CENTERED IN WIDTH AT SINGLES
CENTERED OVER INACTIVE LEAF AT PAIRS



BORROWED LITE

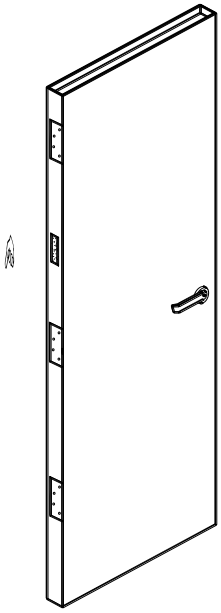
ANYWHERE VISIBLE -
PREFERABLY ON SIDE JAMB,
ON RABBET OPPOSITE GLASS,
ABOUT HALFWAY UP

LABEL PLACEMENT GUIDE FOR DOORS



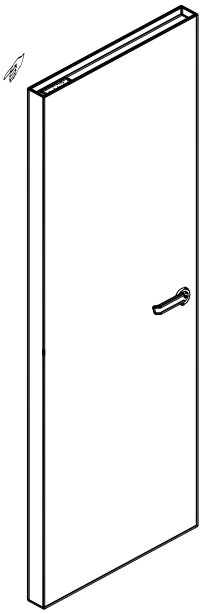
2 HINGES OR PIVOTS

MIDDLE OF HINGE EDGE
(PLUS OR MINUS 8 INCHES)



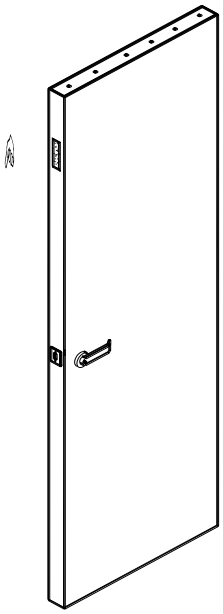
3 OR 4 HINGES OR PIVOTS

HINGE EDGE 52” TO 68”
FROM BOTTOM OF DOOR



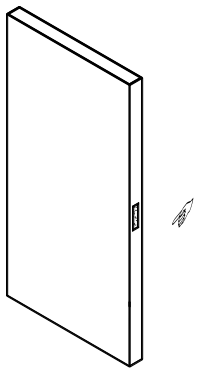
CONTINUOUS OR SURFACE
HINGES OR POCKET PIVOTS

TOP OF DOOR AS TIGHT AS
POSSIBLE TO THE HINGE EDGE
FOR EITHER STANDARD CHANNEL
OR FOR FLUSH AND SEAL



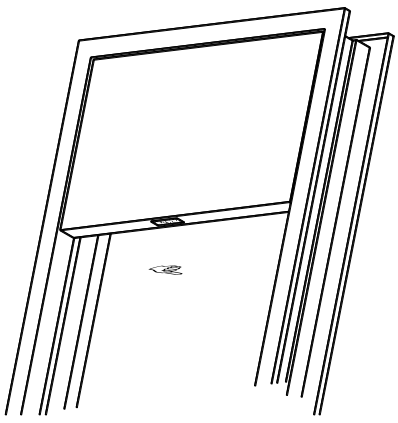
DOOR WITH REMOVABLE TOP CAP
OR DUTCH DOOR SHELF
& CONTINUOUS OR SURFACE
HINGES OR POCKET PIVOTS

LOCK EDGE AT 6” DOWN



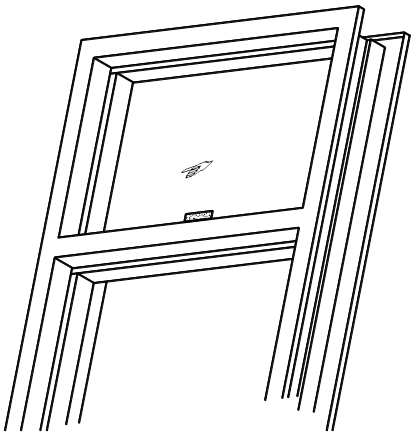
LOOSE PANEL

ON ANY EDGE



WELDED IN FRAME
NO TRANSOM BAR

ON BOTTOM EDGE
CENTERED IN WIDTH
(PLUS OR MINUS 8 INCHES)



WELDED IN FRAME
WITH TRANSOM BAR

ON FACE OF PANEL
STOP SIDE - CENTERED IN WIDTH
(PLUS OR MINUS 8 INCHES)

PERMITTED FIELD PREPS

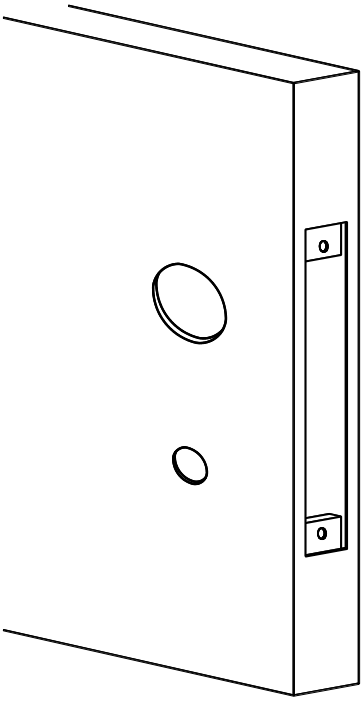
FIELD PREPS PERMITTED ON RATED HOLLOW METAL FRAMES AND DOORS

THE FOLLOWING IS PERMITTED PER NFPA 80 (2025 EDITION) SECTIONS 5.1.4.2.2 – 5.1.4.2.2.2

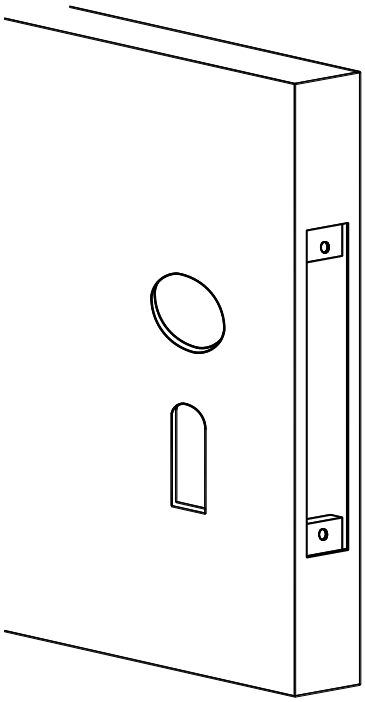
FOR JOB SITE PREPARATION OF SURFACE-APPLIED HARDWARE, FUNCTION HOLES FOR MORTISE LOCKS, AND HOLES FOR LABELED VIEWERS, AND PROTECTION PLATES:

SURFACE-APPLIED HARDWARE SHALL BE APPLIED TO THE FACE OF THE DOOR OR FRAME WITHOUT REMOVING MATERIAL OTHER THAN DRILLING **ROUND HOLES**.

THE HOLES SHALL NOT BE PERMITTED TO EXCEED A DIAMETER OF 1" WITH THE EXCEPTION OF CYLINDERS.

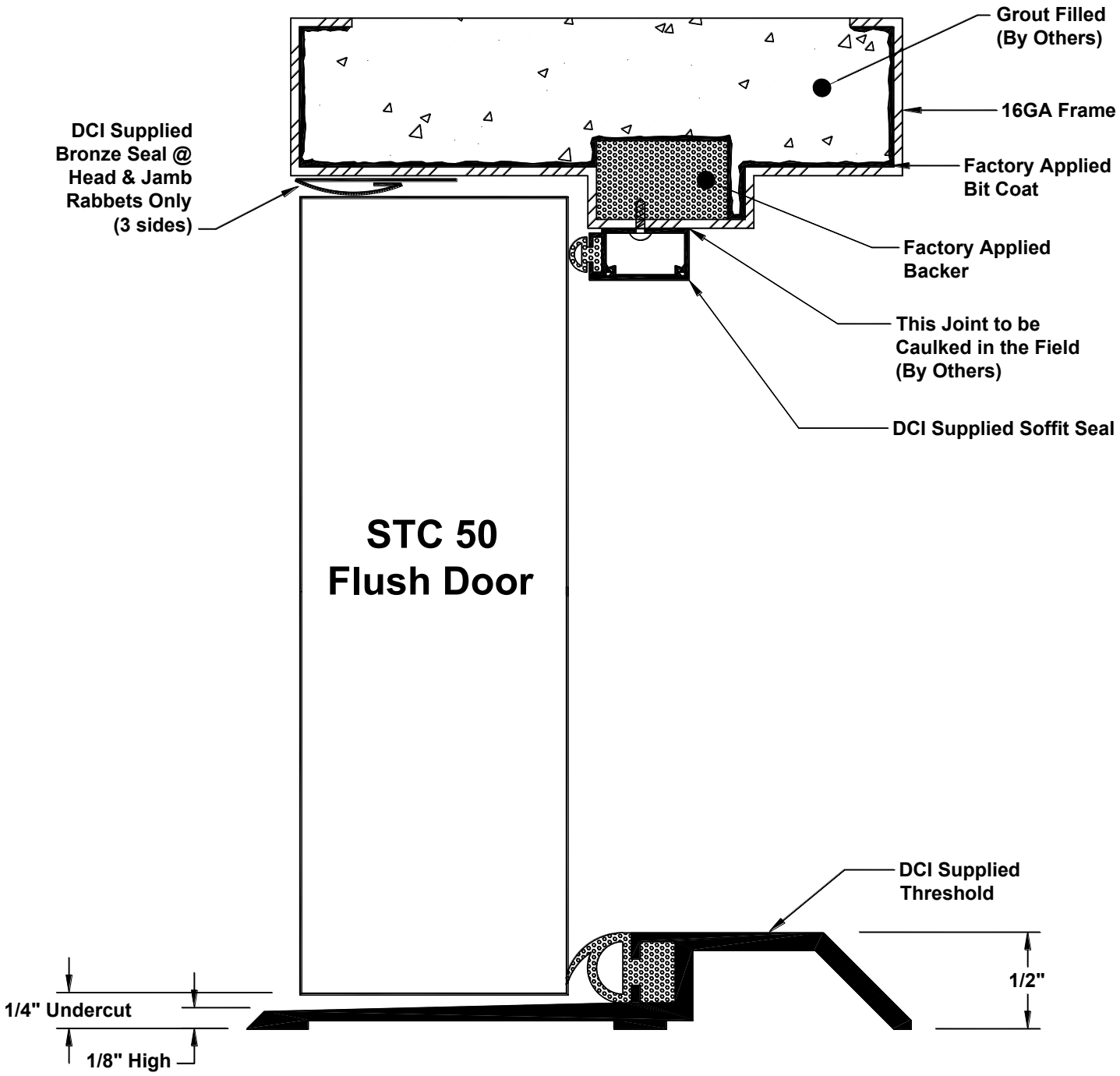


PERMITTED



NOT PERMITTED

STC 50 FLUSH DOOR ASSEMBLY

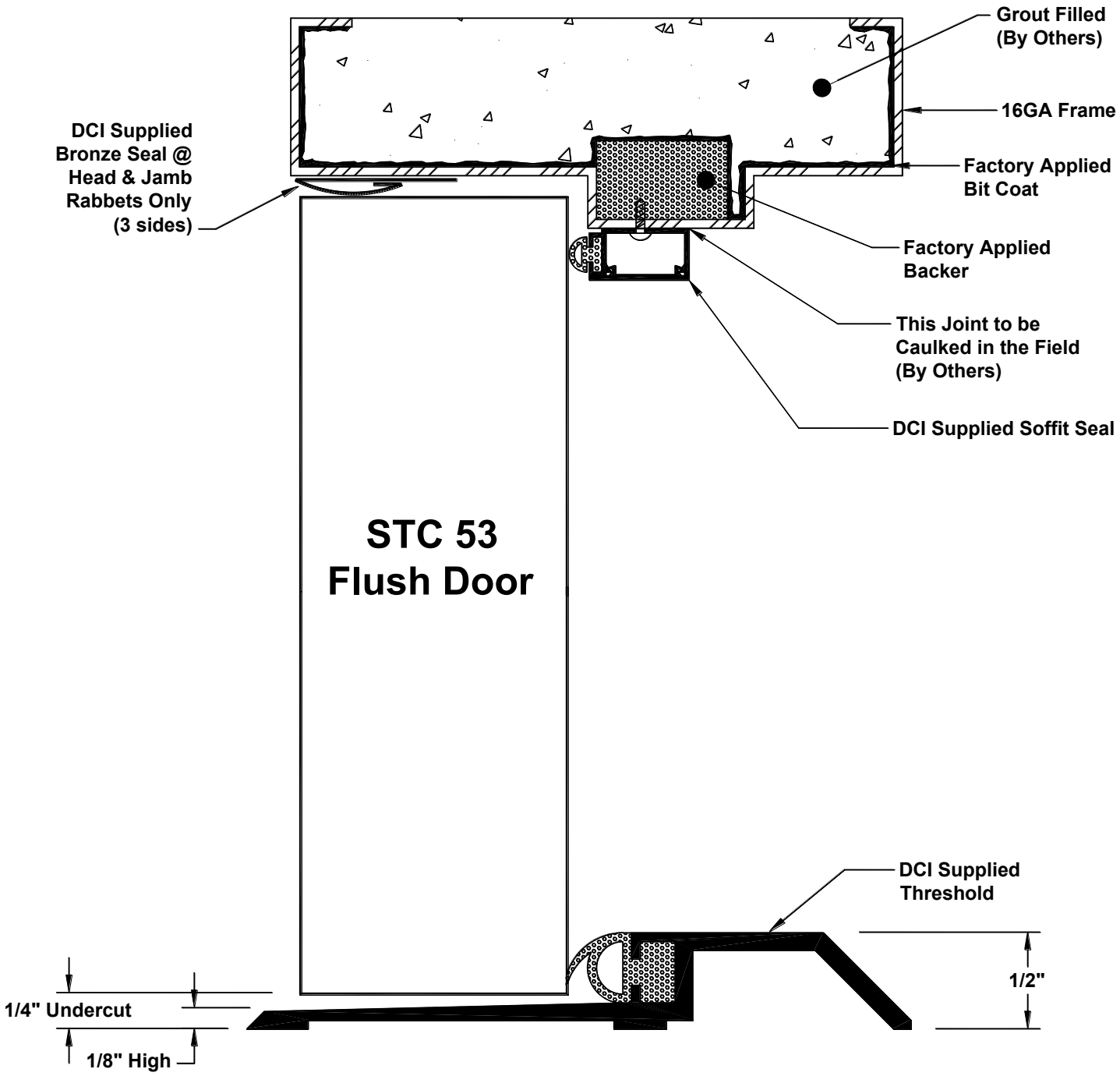


Tested as 3’0”x7’0” with (3) 4-1/2” Hinges, Cylindrical Lock

Fire rated: White Bit Coat, Cork Backer, Max Size: 3’0”x7’0”

Nonrated: Black Bit Coat, Neoprene Backer, Max Size: 4’0”x8’0”

STC 53 FLUSH DOOR ASSEMBLY

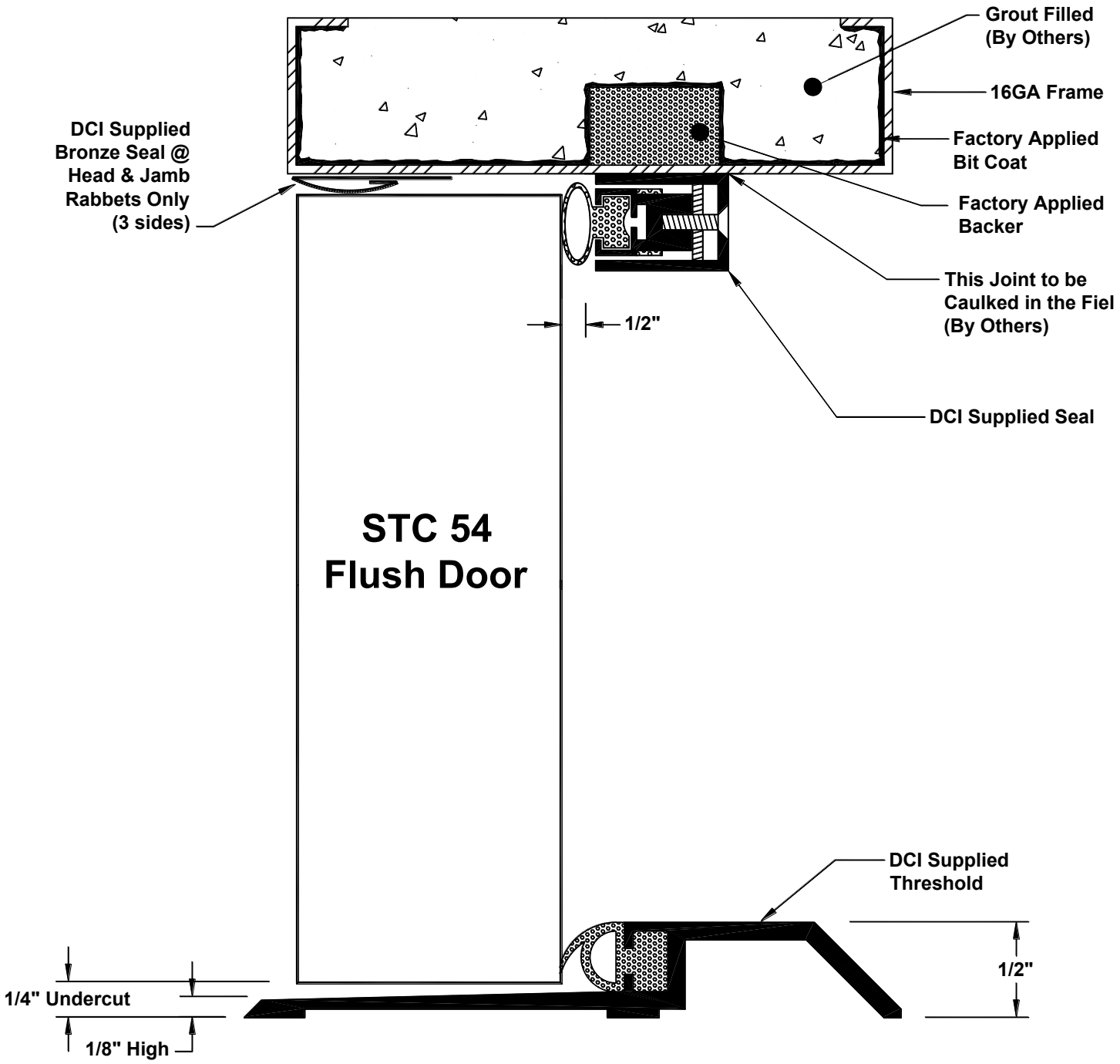


Tested as 3’0”x7’0” with (3) 4-1/2” Hinges, Cylindrical Lock

Fire rated: White Bit Coat, Cork Backer, Max Size: 3’2”x7’0”

Nonrated: Black Bit Coat, Neoprene Backer, Max Size: 4’0”x8’0”

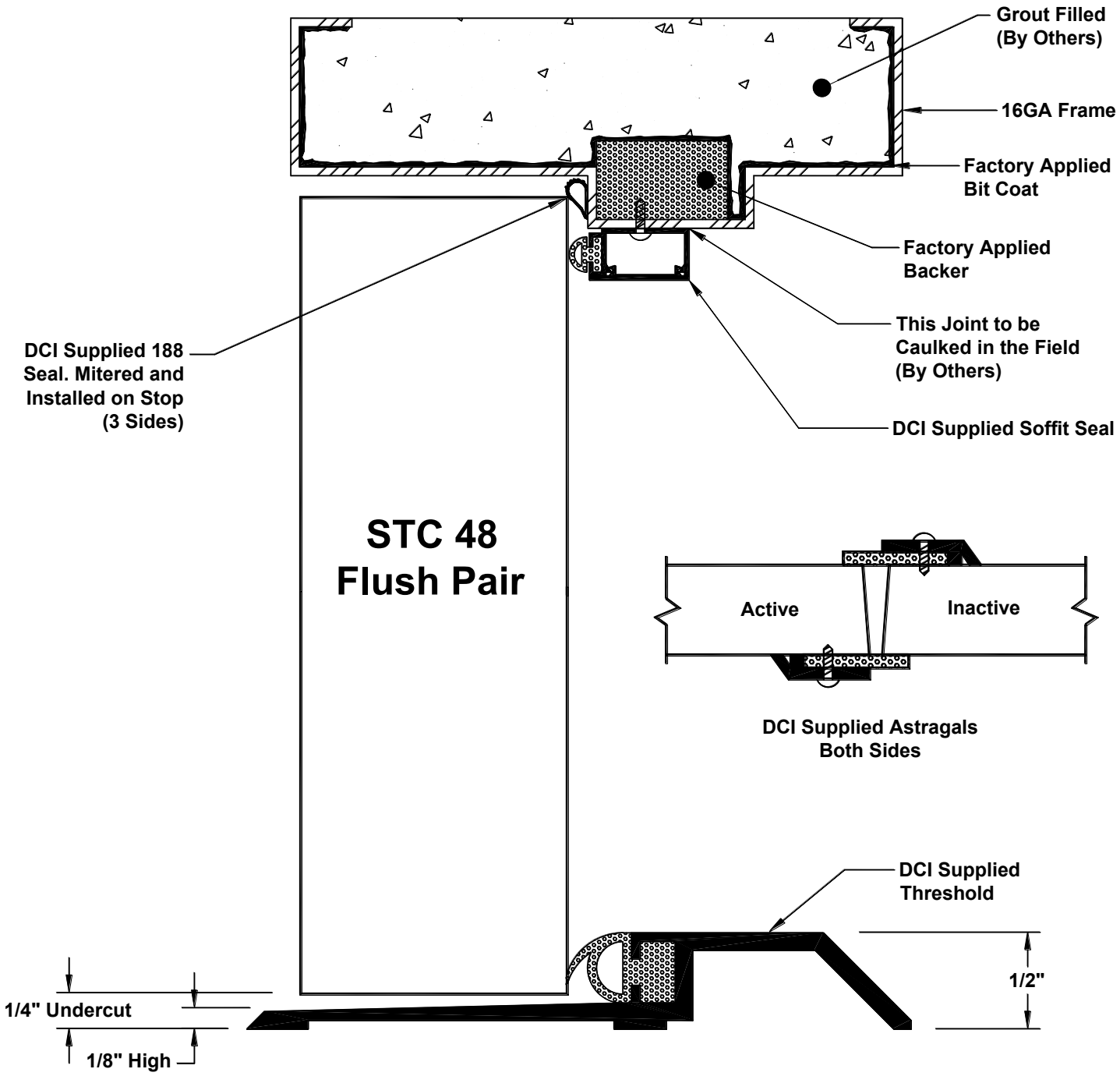
STC 54 FLUSH DOOR ASSEMBLY



Tested as 3’0”x7’0” with (3) 4-1/2” Hinges, Cylindrical Lock

Nonrated: Black Bit Coat, Neoprene Backer, Max Size: 4’0”x8’0”

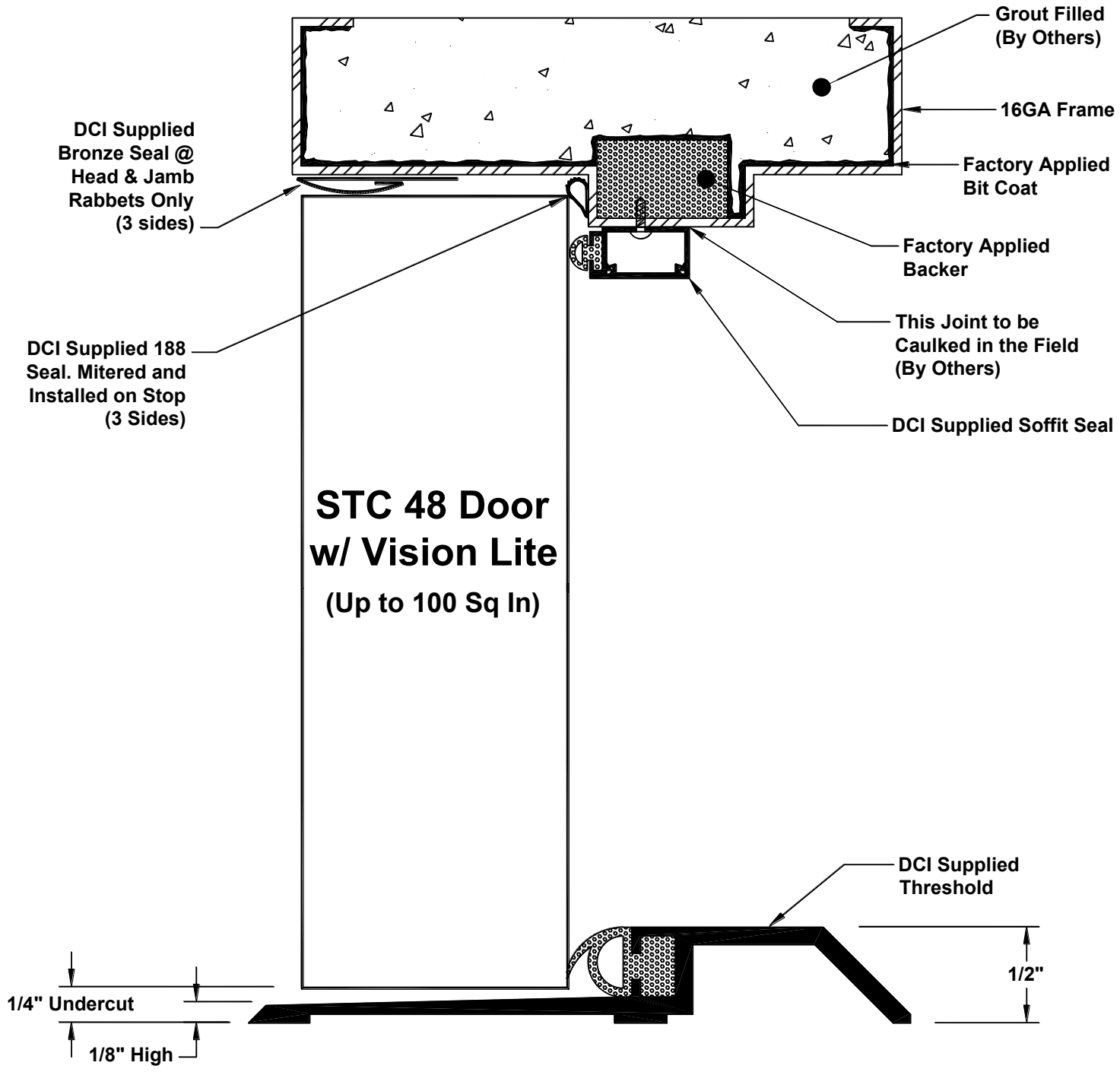
STC 48 FLUSH DOOR PAIR ASSEMBLY



Tested as 6'0"x7'0" with (6) 4-1/2" Hinges, Cylindrical Lock, (2) 6" Manual Flush Bolts

Fire rated: White Bit Coat, Cork Backer, Max Size: 6'4"x7'0"
Nonrated: Black Bit Coat, Neoprene Backer, Max Size: 8'0"x8'0"

STC 48 DOOR ASSEMBLY WITH LITE

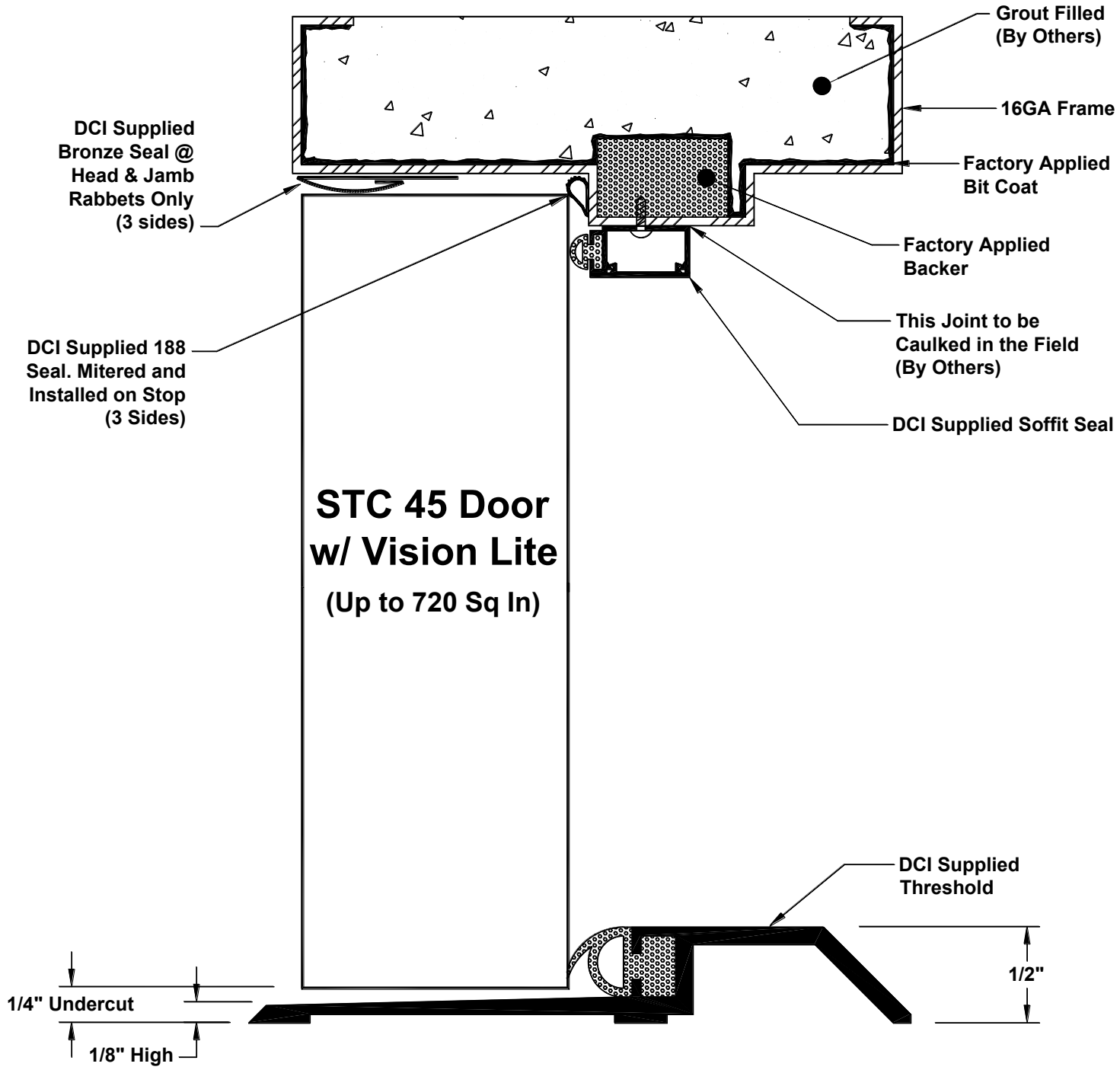


Tested as 3'0"x7'0" with (3) 4-1/2" Hinges, Cylindrical Lock

Fire rated: White Bit Coat, Cork Backer, Max Size: 3'2"x7'0"

Nonrated: Black Bit Coat, Neoprene Backer, Max Size: 4'0"x8'0"

STC 45 DOOR ASSEMBLY WITH LITE

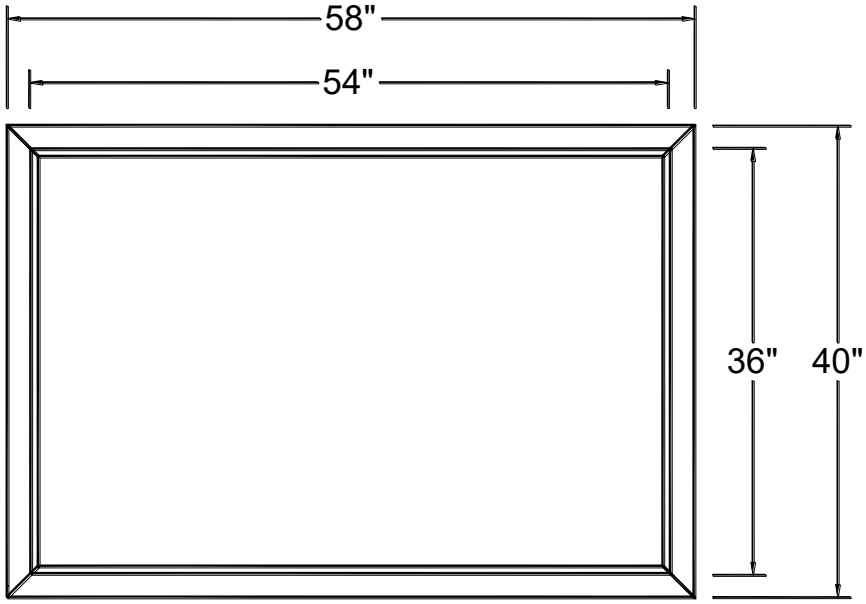


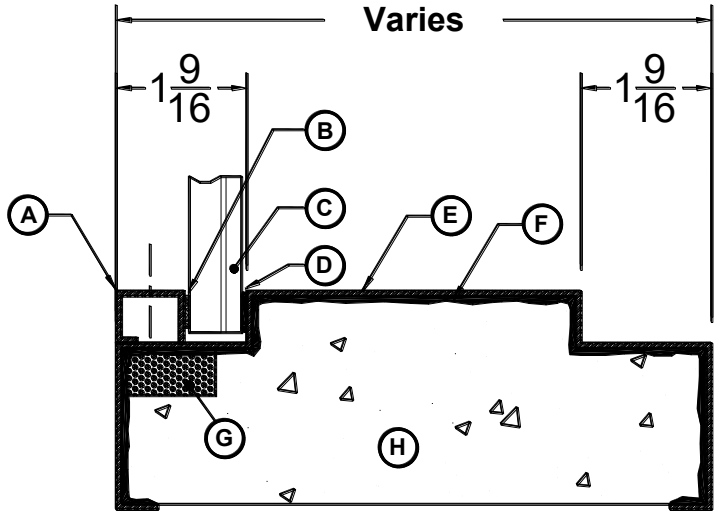
Tested as 3’0”x7’0” with (3) 4-1/2” Hinges, Cylindrical Lock

Nonrated: Black Bit Coat, Neoprene Backer, Max Size: 4’0”x8’0”

STC 40 WINDOW ASSEMBLY

Maximum Size Tested:

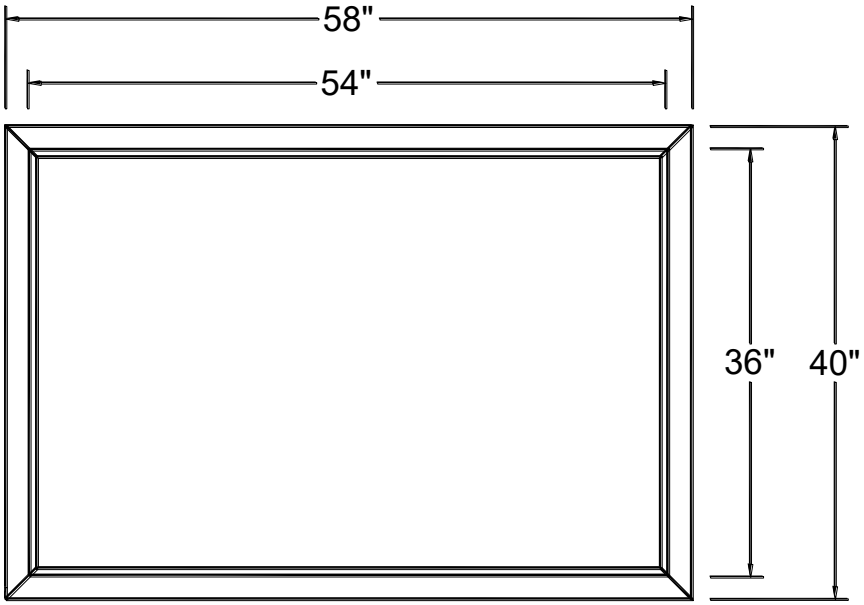


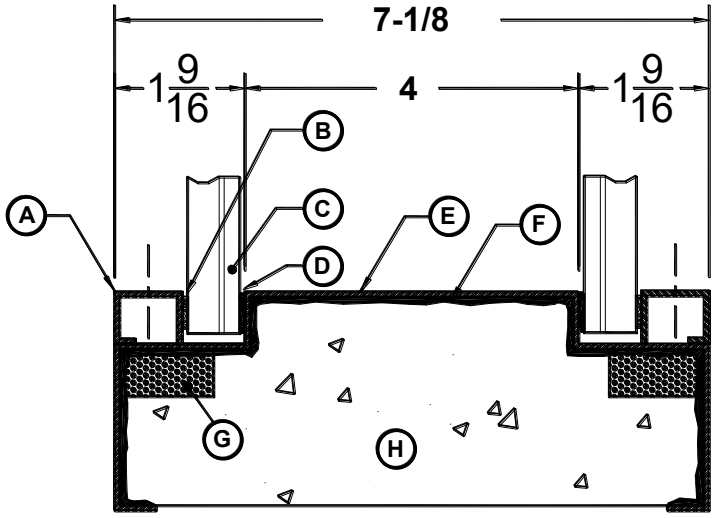


A. 13/16” x 5/8” Glass Stop
B. CR Laurence 1/8" x 3/8" Norseal® V730 Acoustical Sealant Tape (By Others)
C. 5/8" Lam Glass (3/8" + 3/16" with .060" Interlayer) - (By Others)
D. Stay-Soft Glazing Rope - 1/4" Diameter (By Others)
E. 16 GA Frame
F. Factory Applied Bit Coat (Black)
G. Factory Applied Backer (Neoprene)
H. Grout Filled (By Others)

STC 52 WINDOW ASSEMBLY

Maximum Size Tested:

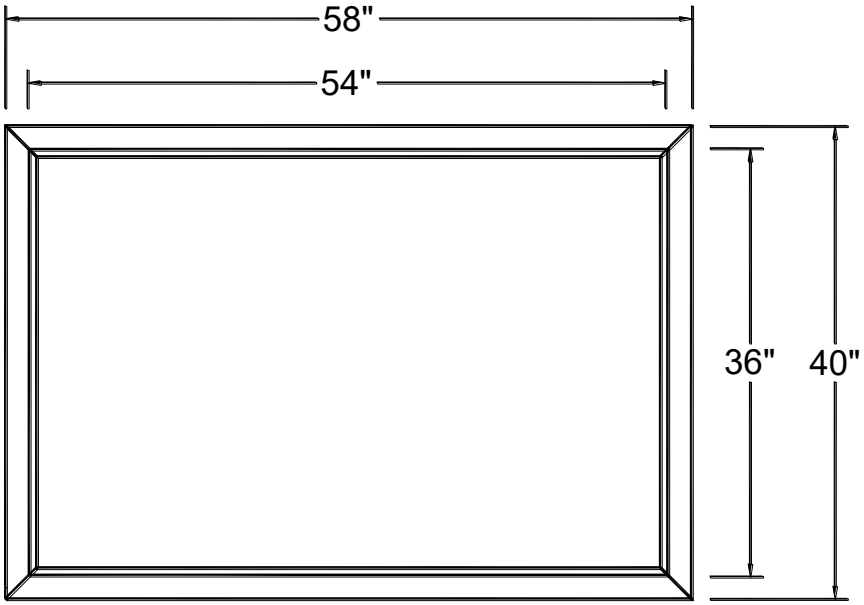


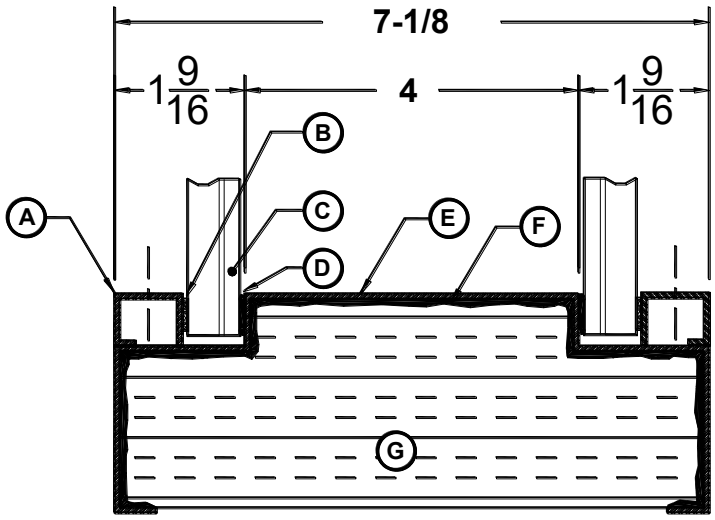


A. 13/16” x 5/8” Glass Stop
B. CR Laurence 1/8" x 3/8" Norseal® V730 Acoustical Sealant Tape (By Others)
C. 5/8" Lam Glass (3/8" + 3/16" with .060" Interlayer) - (By Others)
D. Stay-Soft Glazing Rope - 1/4" Diameter (By Others)
E. 16 GA Frame
F. Factory Applied Bit Coat (Black)
G. Factory Applied Backer (Neoprene)
H. Grout Filled (By Others)

STC 49 WINDOW ASSEMBLY

Maximum Size Tested:

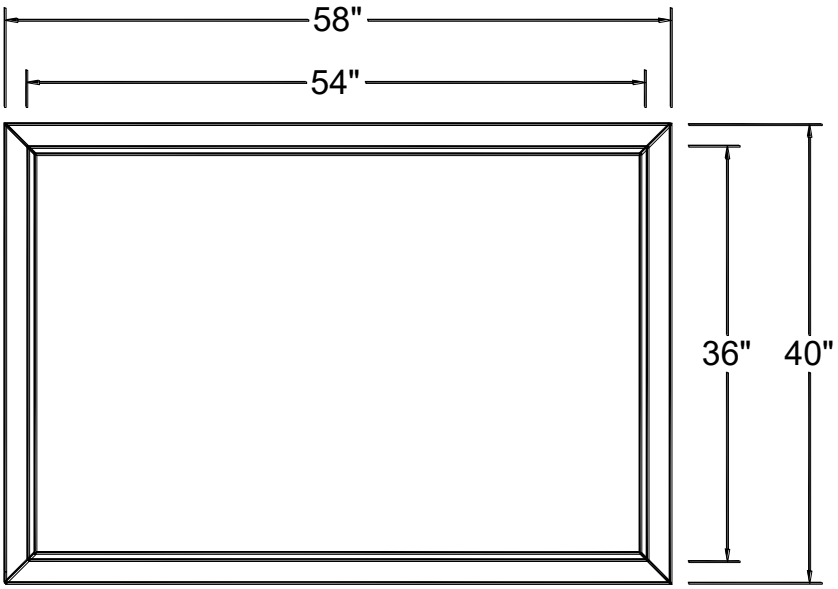


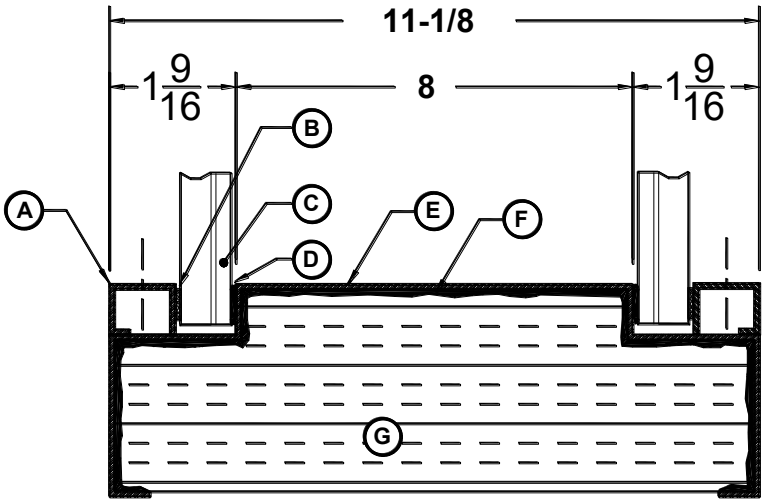


A. 13/16” x 5/8” Glass Stop
B. CR Laurence 1/8" x 3/8" Norseal® V730 Acoustical Sealant Tape (By Others)
C. 5/8" Lam Glass (3/8" + 3/16" with .060" Interlayer) - (By Others)
D. Stay-Soft Glazing Rope - 1/4" Diameter (By Others)
E. 16 GA Frame
F. Factory Applied Bit Coat (Black)
G. Factory Installed Insulation

STC 53 WINDOW ASSEMBLY

Maximum Size Tested:

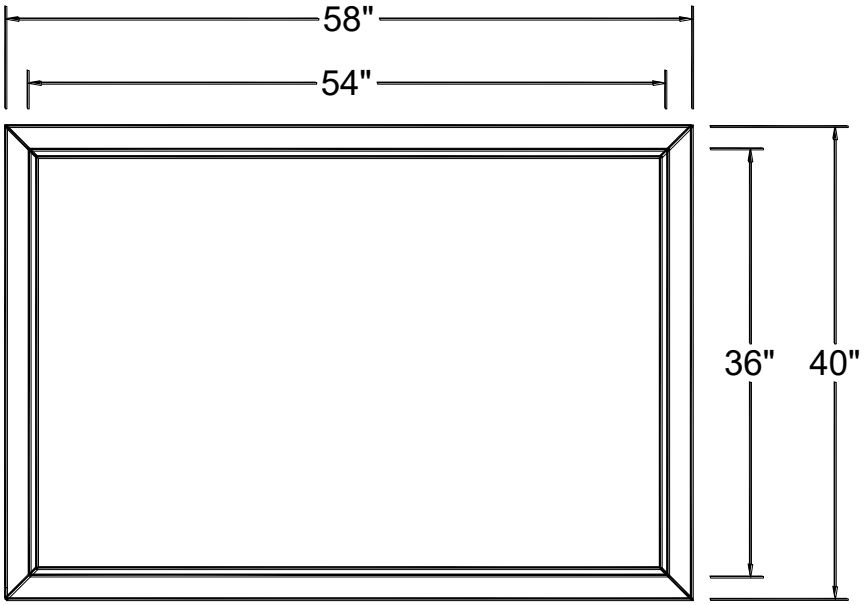


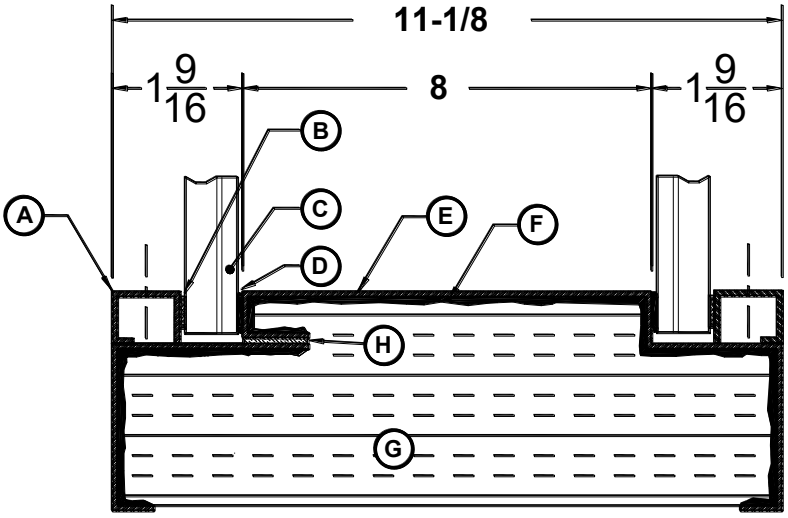


A. 13/16” x 5/8” Glass Stop
B. CR Laurence 1/8" x 3/8" Norseal® V730 Acoustical Sealant Tape (By Others)
C. 5/8" Lam Glass (3/8" + 3/16" with .060" Interlayer) - (By Others)
D. Stay-Soft Glazing Rope - 1/4" Diameter (By Others)
E. 16 GA Frame
F. Factory Applied Bit Coat (Black)
G. Factory Installed Insulation

STC 54 WINDOW ASSEMBLY

Maximum Size Tested:

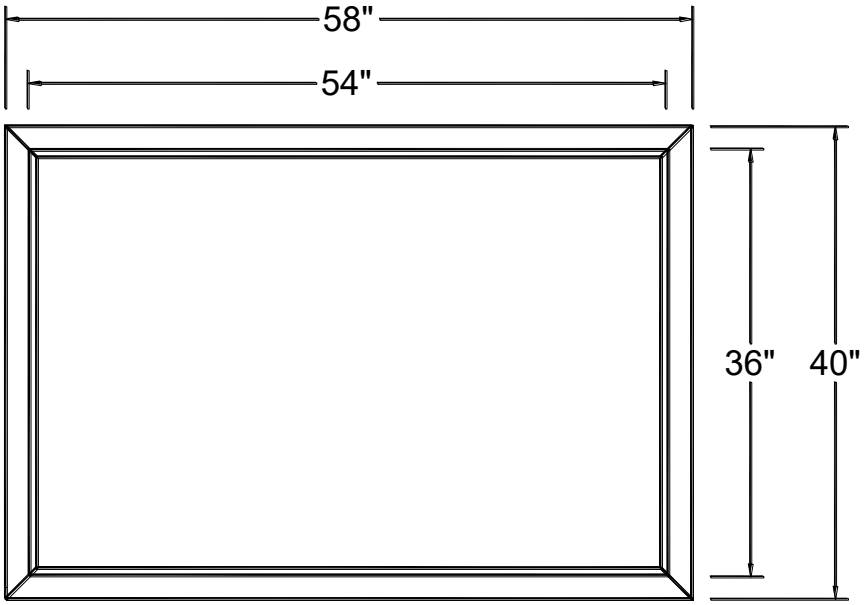


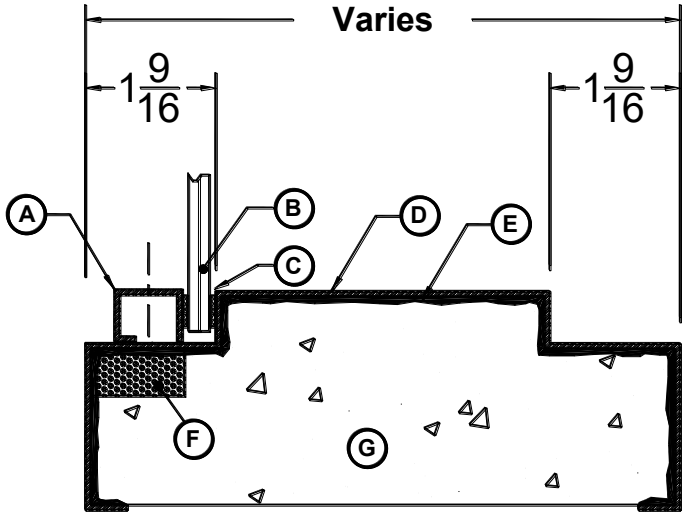


A. 13/16” x 5/8” Glass Stop
B. CR Laurence 1/8" x 3/8" Norseal® V730 Acoustical Sealant Tape (By Others)
C. 5/8" Lam Glass (3/8" + 3/16" with .060" Interlayer) - (By Others)
D. Stay-Soft Glazing Rope - 1/4" Diameter (By Others)
E. 16 GA Frame
F. Factory Applied Bit Coat (Black)
G. Factory Installed Insulation
H. Sound/Thermal Break

STC 37 FIRE RATED WINDOW ASSEMBLY

Maximum Size Tested:

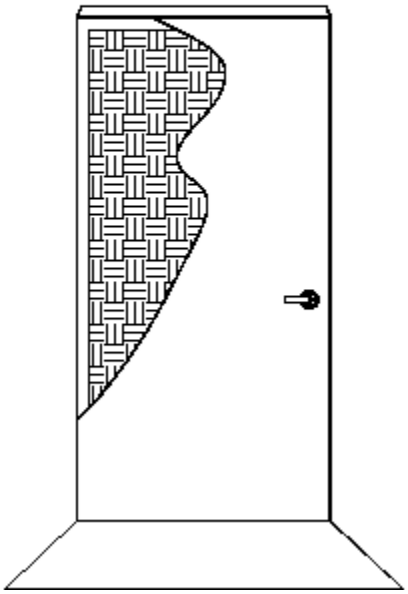


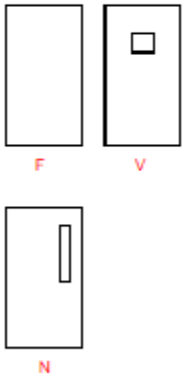


A. 13/16” x 5/8” Glass Stop
B. 5/16” FireLite Plus® Glazing (By Others)
C. Rated Glazing Tape (By Others)
D. 16 GA Frame
E. Factory Applied Bit Coat (White)
F. Factory Applied Backer (Cork)
G. Grout Filled (By Others)

SPECIALTY PRODUCTS: BULLET RESISTANT DOOR

STANDARD SPECIFICATION
MATERIAL: A40 GALVANNEAL
GAUGE: 16
UL 752 LEVELS 1 - 8 AVAILABLE

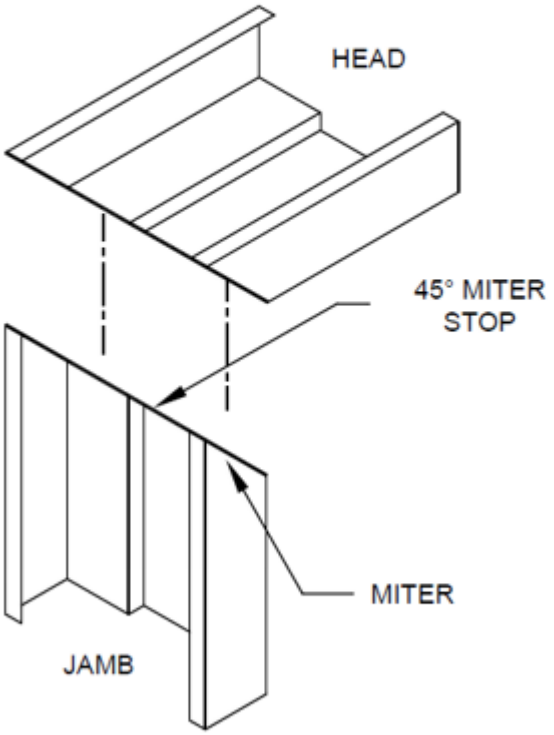




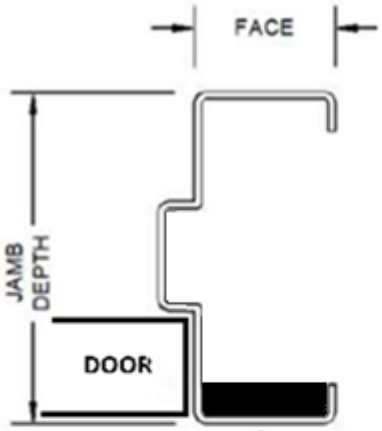
1.) 1-3/4" STANDARD DOOR THICKNESS.
2.) A60 GALVANNEAL AND G90 GALVANIZED AVAILABLE.
3.) 14 GAUGE AVAILABLE.
4.) PROPRIETARY BULLET RESISTANT CORE.
5.) 18 GA FLUSH TOP AND BOTTOM REVERSED CHANNELS WELDED TO FACE SHEETS.
6.) CONTINUOUS WELD SEAMLESS EDGE STANDARD.
7.) HARDWARE REINFORCEMENTS:
    3/16" FOR HINGES
    12 GA FOR LOCKS
    14 GA FOR CLOSER
9.) TESTED AS SINGLE SWING ONLY. UP TO 4'-0" x 9'-0". UL752 ELEVENTH EDITION
10.) LEVELS I – III
    APPROXIMATELY 9 LBS/SQ FT
    CHANNELED CUTOUTS AVAILABLE FOR LITE KITS BY OTHERS (UP TO 100 SQ INCHES OF VISIBLE GLASS)
    HARDWARE PREPS MAY INCLUDE CYLINDRICAL, MORTISE LOCKS, SURFACE MOUNTED EXIT DEVICES, MORTISE EXIT DEVICES, OR ELECTRIFIED HARDWARE
11.) LEVELS IV – VIII
    APPROXIMATELY 22 LBS/SQ FT
    FLUSH DOORS ONLY
    HARDWARE PREPS MAY INCLUDE CYLINDRICAL LOCKS OR SURFACE MOUNTED EXIT DEVICES ONLY
12.) NOT AVAILABLE AS A FIRE RATED PRODUCT
13.) NOT AVAILABLE AS DUTCH DOOR

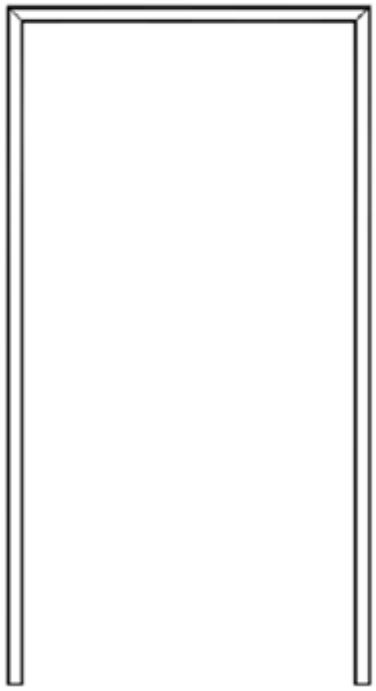
SPECIALTY PRODUCTS:
BULLET RESISTANT FRAME

CONSTRUCTION: FULLY WELDED
ANCHORS: AS REQUIRED
FINISH: RUST INHIBITIVE BAKED-ON GREY PRIMER



CONTINUOUSLY WELDED RETURN, FACE, RABBET, AND STOP INSIDE OF MITER. GRIND FACE AND FINISH SMOOTH

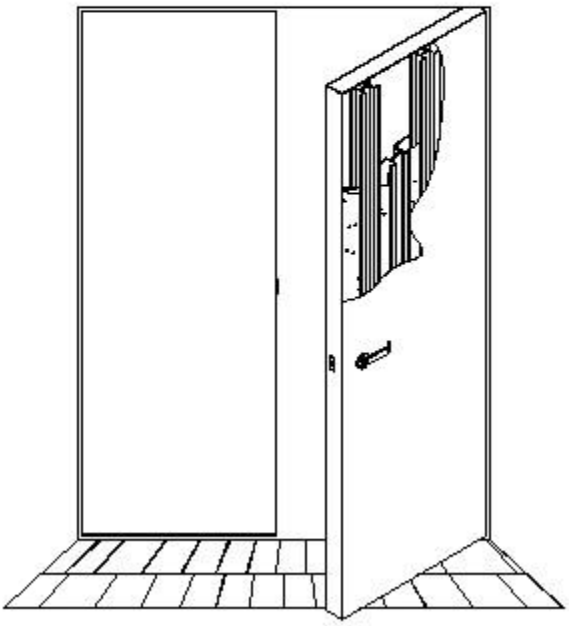


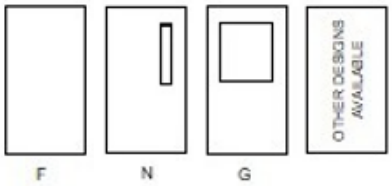


1.) 14GA STANDARD FOR LEVELS I-IV. 12GA STANDARD FOR LEVELS V-VIII
2.) A60 GALVANNEAL AND G90 GALVANIZED AVAILABLE.
3.) CONTINUOUSLY WELDED ONLY. NO KNOCKDOWN OR FACE WELDED CONSTRUCTION.
4.) TESTED AS SINGLE SWING ONLY. UP TO 4'-0" x 9'-0". UL752 ELEVENTH EDITION
5.) LEVELS I-III: MINIMUM 11/16” RETURNS ON DOOR SIDE; SINGLE RETURNS ONLY ON DOOR SIDE
LEVEL IV: MINIMUM 11/16” RETURNS ON BOTH SIDES; SINGLE RETURNS ONLY ON BOTH SIDES
LEVELS V-VIII: MINIMUM 5/8” RETURNS ON DOOR SIDE; SINGLE RETURNS ONLY ON DOOR SIDE
6.) LEVEL IV-VIII: AUXILIARY OR HIGH FREQUENCY HINGE REINFORCEMENTS REQUIRED DUE TO WEIGHT OF DOOR
7.) LEVEL IV-VIII: MOST ELECTRIC HARDWARE NOT ALLOWED
8.) NOT AVAILABLE AS A FIRE RATED PRODUCT

SPECIALTY PRODUCTS:
LEAD-LINED DOOR

STANDARD SPECIFICATION
MATERIAL: A40 GALVANNEAL
GAUGE: 18

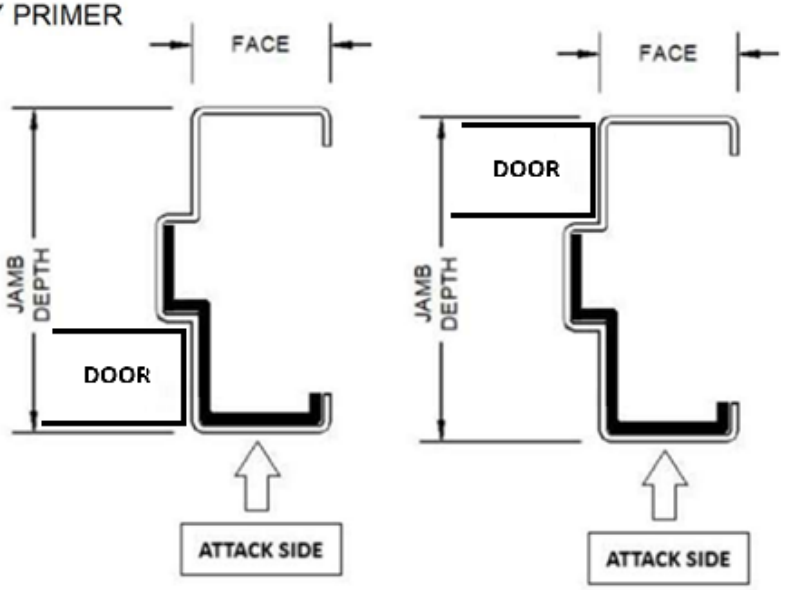


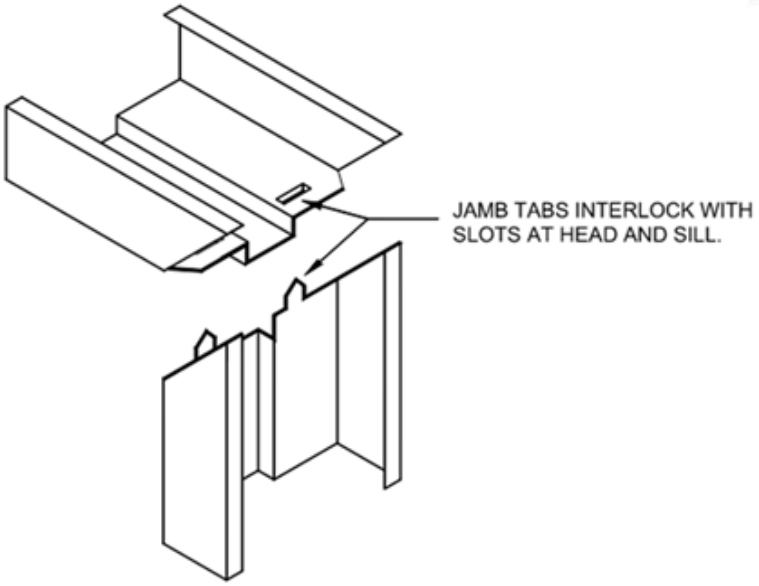


1.) 1-3/4" STANDARD DOOR THICKNESS. OTHER THICKNESSES AVAILABLE.
2.) A60 GALVANNEAL AND G90 GALVANIZED AVAILABLE.
3.) 16, 14, AND 12 GAUGE AVAILABLE.
4.) 20 GA HAT-STYLE STIFFENERS 6" ON CENTER WITH FIBERGLASS INSULATION BETWEEN STIFFENERS AND LEAD-LINING CENTERED IN THE DOOR
5.) 18 GA TOP AND BOTTOM REVERSED CHANNELS SPOTWELDED TO FACE SHEETS. OTHER GAUGES AVAILABLE.
6.) CONTINUOUS WELD SEAMLESS EDGE STANDARD.
7.) HARDWARE REINFORCEMENTS:
    3/16" FOR HINGES
    12 GA FOR LOCKS
    14 GA FOR CLOSER
8.) NO INTEGRAL LITES, LOUVERS, CVR, MORTISED DOOR BOTTOMS, CONCEALED CLOSERS, RECESSED PANIC DEVICES
9.) 99.9% PURE LEAD MEETING FEDERAL SPECIFICATIONS QQ-L-201F GRADE C AND ASTM B749
10.) WHI LABELS AVAILABLE

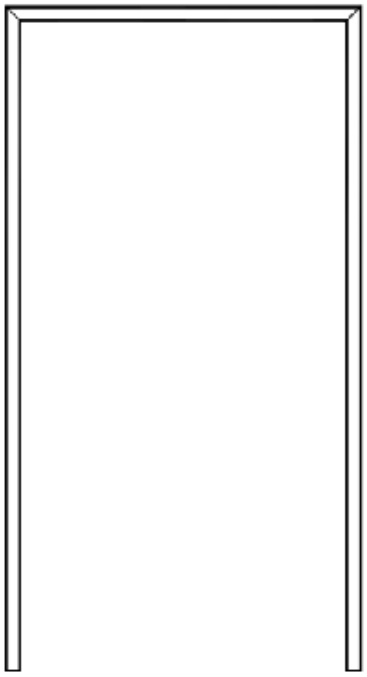
SPECIALTY PRODUCTS:
LEAD-LINED FRAME

CONSTRUCTION: FULLY WELDED
ANCHORS: AS REQUIRED
FINISH: RUST INHIBITIVE BAKED-ON GREY PRIMER





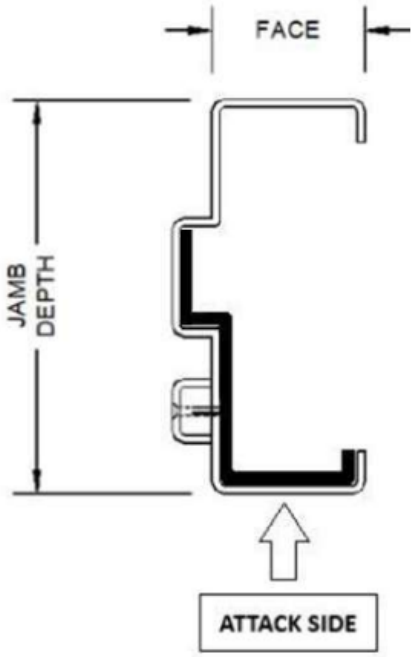
MACHINE-MITERED AND FULLY WELDED. (CONTINUOUSLY WELDED AT FACE AND RETURN. FACE GROUND SMOOTH AND FINISHED.)



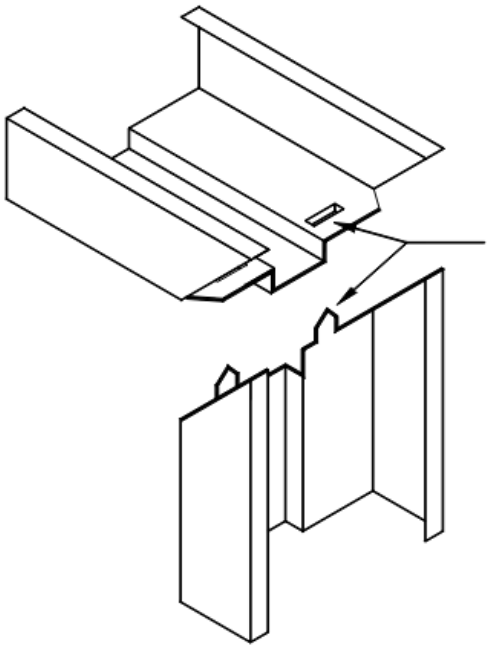
1.) 16GA STANDARD. 12GA AND 14GA AVAILABLE
2.) AVAILABLE FACE WELDED OR CONTINUOUSLY WELDED. NO KNOCKDOWN CONSTRUCTION
3.) FIRE RATING AVAILABLE (WITH NO MULLIONS)
4.) NO INTERSECTING MULLIONS (UNABLE TO ACHIEVE FULL LEAD COVERAGE)
5.) 99.9% PURE LEAD MEETING FEDERAL SPECIFICATIONS QQ-L-201F GRADE C AND ASTM B749
6.) LEAD COVERAGE INCLUDES UP TO 1-1/2" OF SOFFIT WHEN RADIATION ON DOOR SIDE
LEAD COVERAGE INCLUDES FULL WIDTH OF SOFFIT WHEN RADIATION ON OPPOSITE DOOR SIDE

SPECIALTY PRODUCTS:
LEAD-LINED BORROWED LITE

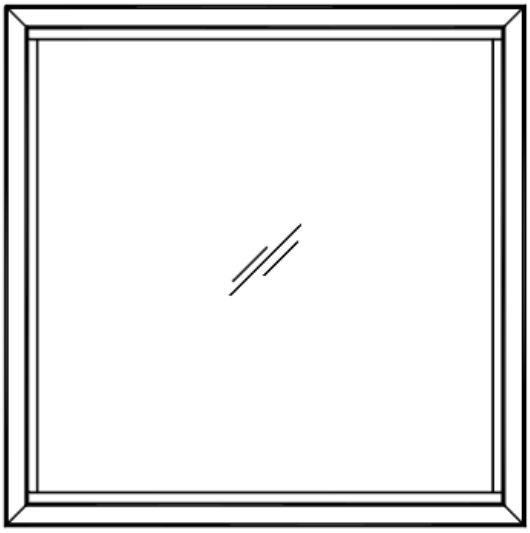
CONSTRUCTION: FULLY WELDED
GLASS STOP: 5/8 x 5/8 x 18 GA.
ANCHORS: AS REQUIRED
FINISH: RUST INHIBITIVE BAKED-ON GREY PRIMER



GLASS INSTALLED ON RADIATION SOURCE SIDE FOR SUFFICIENT RADIATION COVERAGE



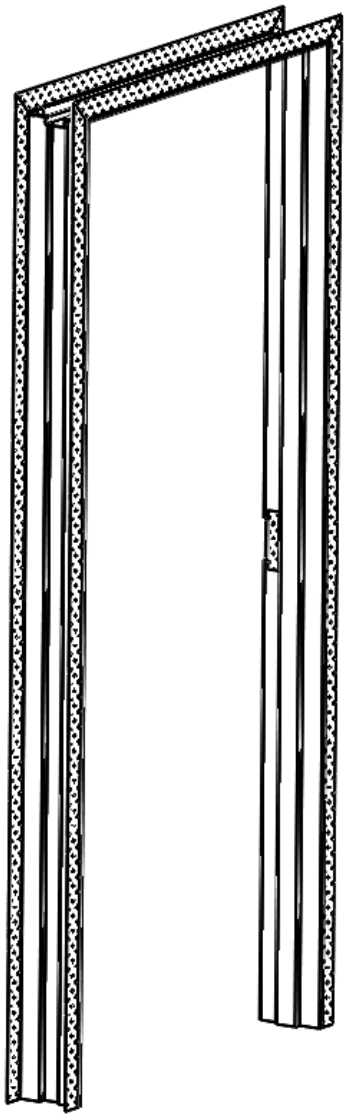
JAMB TABS INTERLOCK WITH SLOTS AT HEAD AND SILL.

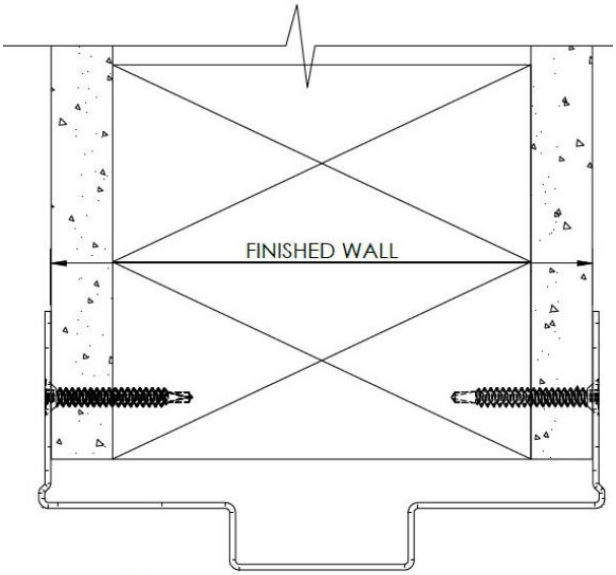


MACHINE-MITERED AND FULLY WELDED. (CONTINUOUSLY WELDED AT FACE AND RETURN. FACE GROUND SMOOTH AND FINISHED.)

1.) 16GA STANDARD. 12GA AND 14GA AVAILABLE
2.) FIRE RATING AVAILABLE (WITH NO MULLIONS)
3.) NO INTERSECTING MULLIONS (UNABLE TO ACHIEVE FULL LEAD COVERAGE)
4.) 99.9% PURE LEAD MEETING FEDERAL SPECIFICATIONS QQ-L-201F GRADE C AND ASTM B749

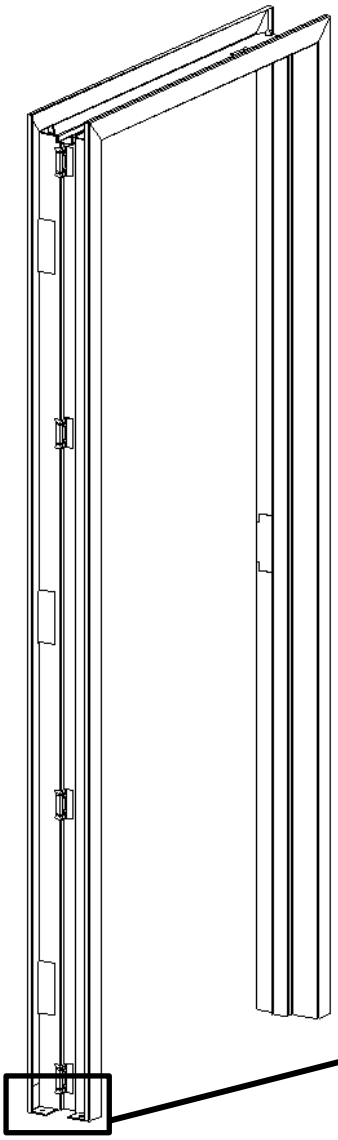
SPECIALTY PRODUCTS:
FLUSH FRAMES



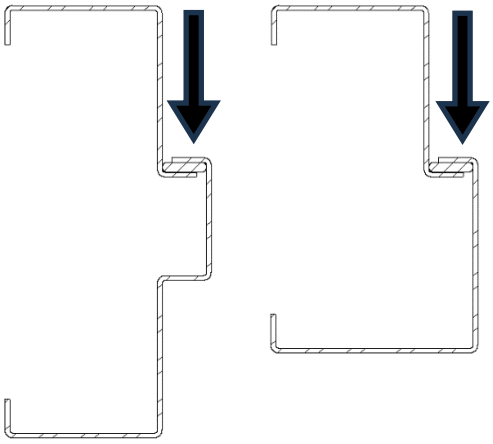


1.) DOUBLE RABBET AND CASED OPEN PROFILES AVAILABLE FOR SINGLE AND PAIRED OPENINGS
2.) A40 GALVANNEAL AND A60 GALVANNEAL AVAILABLE.
3.) 18 OR 20 GAUGE RECOMMENDED. 16 GAUGE ALSO AVAILABLE
4.) 3-SIDED AND 4-SIDED FRAMES AVAILABLE
5.) AVAILABLE KNOCK-DOWN ONLY. WELDED NOT AVAILABLE
6.) WHI LABELS AVAILABLE:
SINGLE 3-SIDED OPENINGS ONLY, MAX 4'x8', 18 OR 20 GAUGE, 2" FACES ONLY, DOUBLE RABBET
7.) JAMB DEPTH IS FINISHED WALL + 1/4":
NOT RATED OR 60-MINUTE RATED: 3-3/4"-9-1/2"; 90-MINUTE RATED: 6-1/4"-9-1/2"
8.) MAX OPENING SIZE: NOT RATED: 8'x10'; RATED: 4'x8'
9.) NO RETURNS, NO ANCHORS
10.) FIRE RATED FRAME INSTALLED WITH #6 MINIMUM DRYWALL SCREWS SPACED MAX 12" ON CENTER AND 2" FROM ENDS. TWO SCREWS REQUIRED ABOVE AND BELOW EACH HINGE AND ABOVE AND BELOW EACH LATCH

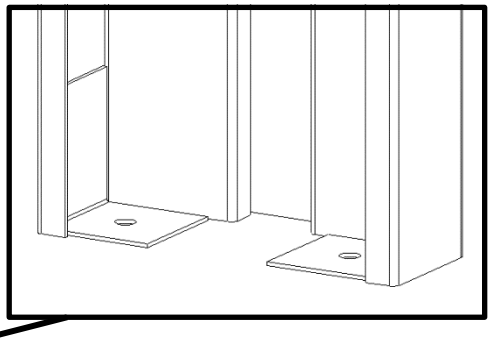
SPECIALTY PRODUCTS:
THERMAL BREAK FRAMES



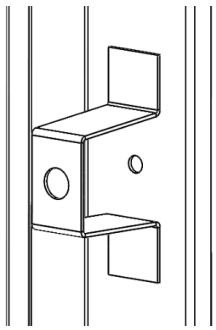
PVC THERMAL STRIP
AT DOOR SIDE OF STOP;
STOP MUST BE 5/8”



SPLIT BASE ANCHOR STANDARD

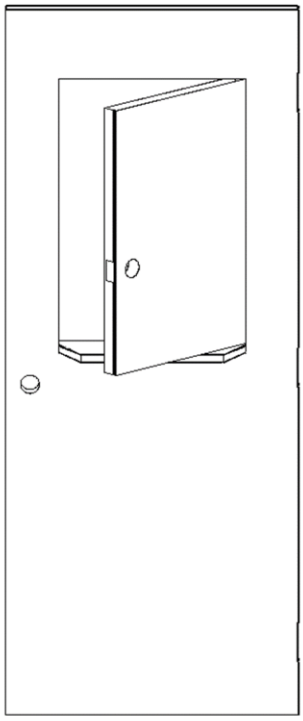


A7H HAT ANCHOR
OPTION FOR P&D ANCHORING

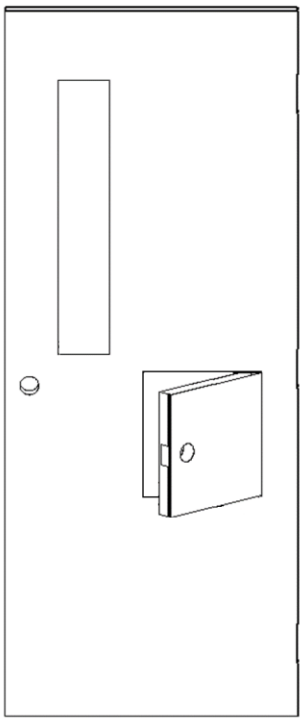


1.) SINGLE OR DOUBLE RABBET PROFILES AVAILABLE FOR SINGLE AND PAIRED OPENINGS
2.) A40 GALVANNEAL STANDARD. A60 GALVANNEAL AND G90 GALVANIZED AVAILABLE.
3.) 16 GAUGE STANDARD. 14 GAUGE AVAILABLE.
4.) 3-SIDED, 4-SIDED, AND BORROWED LITE FRAMES AVAILABLE (NO MULLIONS)
5.) THERMAL PERFORMANCE FOR 3-SIDED THERMAL BREAK ASSEMBLY TESTED PER ASTM C1363:
THERMAL TRANSMITTANCE U-FACTOR = 0.37 (STANDARD FRAME U-FACTOR: 0.44)
THERMAL RESISTANCE R-VALUE = 2.68 (STANDARD FRAME R-VALUE: 2.28)
6.) THERMAL BREAK PERFORMANCE MAY BE DIMINISHED BY ANCHORS THAT BRIDGE ACROSS BOTH SIDES OF THE THERMAL BREAK. RECOMMENDED ANCHOR TYPES INCLUDE: A1, A6, A7H, A11.
7.) AVAILABLE WELDED ONLY. KNOCK-DOWN NOT AVAILABLE
8.) WHI LABELS NOT AVAILABLE
9.) MINIMUM 1-1/8” SOFFIT, MINIMUM JAMB DEPTH FOR SINGLE RABBET IS 3-1/16”, MINIMUM JAMB DEPTH FOR EQUAL RABBET IS 5”. MINIMUM JAMB DEPTH FOR UNEQUAL RABBET IS 4-1/2”

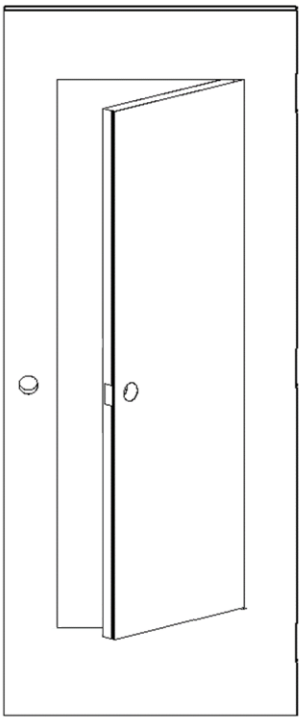
SPECIALTY PRODUCTS:
WICKET DOORS



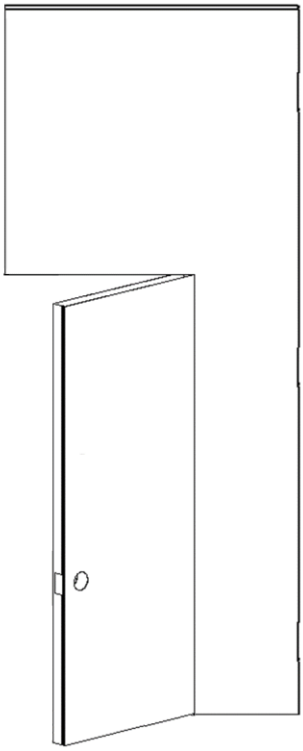
PASS THROUGH WITH SHELF



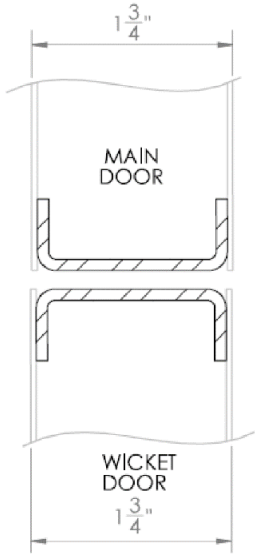
SMALL PASS THROUGH



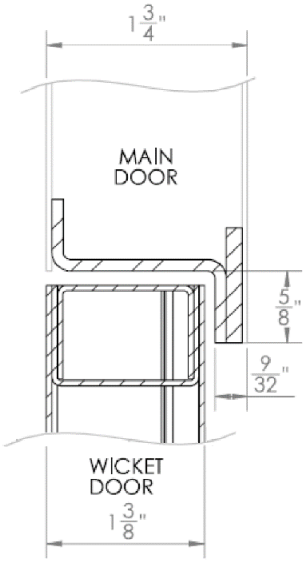
STEP THROUGH



CORNER STEP THROUGH



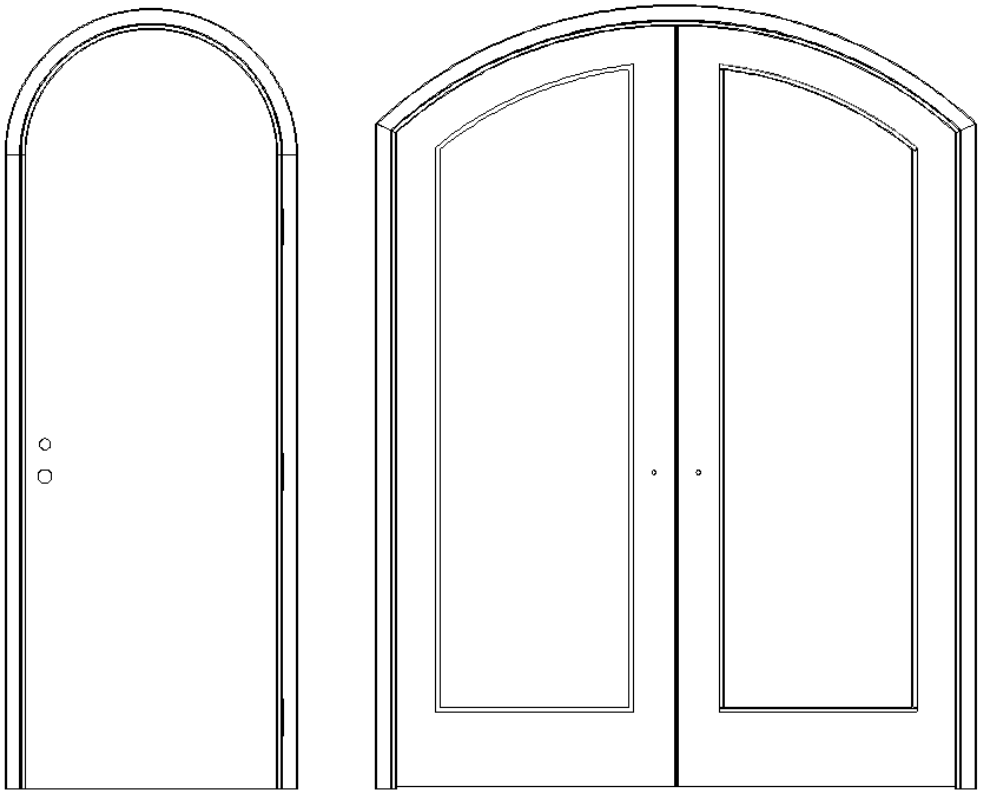
STANDARD FLUSH

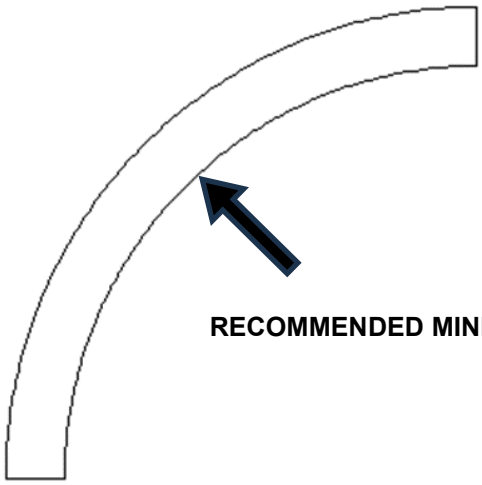


INTEGRAL STOP OPTION

1.) SWING OF BOTH DOORS MUST BE SPECIFIED (ALL WICKET DOORS SHOWN ABOVE ARE RHR VIEWED FROM SECURE SIDE)
2.) 1-3/4" DOOR THICKNESS STANDARD. 1-3/8" DOOR THICKNESS WITH INTEGRAL STOP OPTION AVAILABLE.
3.) INTEGRAL LITES AND CHANNELED CUTOUTS AVAILABLE FOR 1-3/4" AND THICKER DOORS. CHANNELED CUTOUTS ONLY FOR DOORS THINNER THAN 1-3/4
4.) WHI LABELS NOT AVAILABLE

SPECIALTY PRODUCTS: RADIUS DOORS/FRAMES



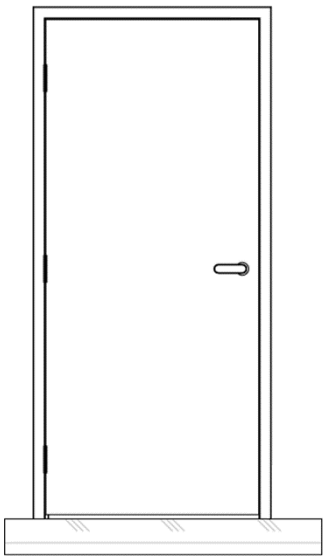


RECOMMENDED MINIMUM 18” INNER RADIUS[1]

1.) SMALLER RADIUS POSSIBLE IF POTENTIAL THINNING/WAVINESS IS ACCEPTABLE, 12” MINIMUM RADIUS POSSIBLE WITH FLAT TOP PROFILE
2.) A40 GALVANNEAL, A60 GALVANNEAL, AND G90 GALVANIZED AVAILABLE.
3.) 14 AND 16 GAUGE AVAILABLE.
4.) RADIUS FRAMES AVAILABLE AS WELDED OR KD CONSTRUCTION. WELDED STANDARD
5.) RADIUS DOORS AVAILABLE WITH FLUSH, FLUSH AND SEALED, AND RECESSED TOP CHANNELS. FLUSH STANDARD
6.) RADIUS DOORS AVAILABLE WITH ANY CORE. HONEYCOMB STANDARD
7.) WHI LABELS NOT AVAILABLE

3’0” x 7’0” SINGLE SWING-SEATED DOOR (18 GAUGE)
TESTED TO ASTM F2927-12

Blast - H777
Peak Pressure=5.01, psi Impulse=32 psi-msec
Charge Weight=I, Standoff=179ft OR
Charge Weight=II, Standoff=89ft
Standard Honeycomb
Frame 16ga P&D W/ Pipe and Strap Anchors 24”
ASTM Damage Level Cat. 2
Single outswing only with Grade 1 Cyl Lock



3’0” x 7’0” SINGLE SWING OR 6’0” x 7’0” PAIR SWING- SEATED DOOR (16 GAUGE)
PROTECTION LEVEL = LOW (AS DEFINED BY UFC4-010-01)

Blast - S777OH/BL777P
Peak Pressure=5.77 psi, Impulse=29.69 psi-msec OR
Peak Pressure=4.76 psi, Impulse=41.11 psi-msec OR
Charge Weight=I, Standoff=148ft OR
Charge Weight=II, Standoff=82ft
Hat Style Steel Stiffened w/18ga stiffeners
14ga Lock and Hinge Channel
16ga recess channels w/18ga flush & seal channels
Frame 14ga P&D W/ Pipe and Strap Anchors 10"OC
seated only (normally outswing)

APPROVED HARDWARE:

-GRADE 1 CYL LOCK

-SARGENT 8200 MORT LOCK W/DEADBOLT FUNCTION

-SARGENT WS8900 MORT LOCK W/DEADBOLT FUNCTION

-SARGENT 8600 CVR

-SARGENT HC8700 SVR

-YALE 7170 SVR

-YALE 7150 RIM PANIC

-YALE M200FWS REMOVABLE MULLION

-YALE 988Y SURFACE BOLTS

DCI BLAST DOORS ACHIEVED GSA TS-01 LEVEL C (PEAK PRESSURE / IMPULSE = 4 PSI / 28 PSI-MSEC)
NO LITES OR LOUVERS
FIRE RATING UP TO 3 HR AVAILABLE
REFER TO SDI-133 FOR FURTHER INFO ON SPECIFYING BLAST PRODUCTS

SPECIALTY PRODUCTS: WINDSTORM & TORNADO DOORS/FRAMES

3’0” x 7’0” SINGLE SWING OR 6’0” x 7’0” PAIR SWING- (16 GAUGE; OUTSWING ONLY; FIRE RATING AVAILABLE)

Windstorm – S777OH or W777/W777P

ANSI A250.13 wind-storm testing is for 10 second load includes impact debris (9.1lb missile @ 50 f/s)

These doors are for +/-70 PSF, equivalent to 165 MPH wind load – Outswing only.

Hat Style Steel Stiffened w/ 18ga stiffeners
14ga Lock and Hinge channels
16ga recess channels w/18ga flush and seal channels

Frame 14ga P&D W/ Pipe and Strap Anchors 10” OC

APPROVED HARDWARE:

-GRADE 1 CYL LOCK

-SARGENT 8200 MORT LOCK W/DEADBOLT FUNCTION

-SARGENT WS8900 MORT LOCK W/DEADBOLT FUNCTION

-SARGENT 8600 CVR

-SARGENT HC8700 SVR

-YALE 7170 SVR

-YALE 7150 RIM PANIC

-YALE M200FWS REMOVABLE MULLION

-YALE 988Y SURFACE BOLTS

3’0” x 7’0” OR 4’0” x 8’0” SINGLE SWING- (14 GAUGE; INSWING OR OUTSWING)

Tornado –W777

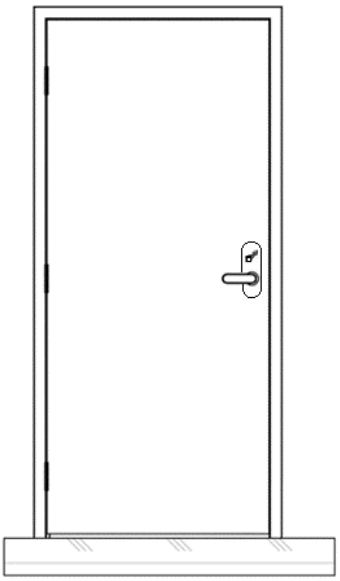
FEMA 361 testing:
These doors are for +/-305 PSF, equivalent to 345 MPH wind load – Inswing or Outswing
includes impact debris (15lb missile @ 100 MPH)

14ga Steel Stiffened w/ 14ga C-stiffeners on both skins
W/ 10 lb insulation
12ga Lock and Hinge channels
14ga recess channels w/12ga flush and seal channels

Frame 14ga P&D W/ Pipe and Strap Anchors 16” OC
12 ga reinforcements (7ga hinge reinforcement)

APPROVED HARDWARE:

-SECURITECH 4340 3-POINT LATCHING W/ MORTISE DEADBOLT(4340-SS320-105) + CYLINDRICAL LOCKSET



NO LITES OR LOUVERS
REFER TO SDI-136 FOR FURTHER INFO ON SPECIFYING WINDSTORM PRODUCTS